© RTCM – Not for reproduction or redistribution

RTCM 10403.4
RTCM Paper 2023-SC104-1342-STD
With Amendment 1
RTCM Paper 2024-SC104-1400

RTCM STANDARD 10403.4

DIFFERENTIAL GNSS (GLOBAL NAVIGATION SATELLITE SYSTEMS) SERVICES – VERSION 3

DEVELOPED BY
RTCM SPECIAL COMMITTEE NO. 104

November 01, 2024

COPYRIGHT © 2024 RTCM

Radio Technical Commission for Maritime Services
1150 18th Street NW Suite 910
Washington, DC 20036
USA
E-Mail: info@rtcm.org
Web Site: www.rtcm.org

© RTCM – Not for reproduction or redistribution

The Radio Technical Commission for Maritime Services (RTCM) is an incorporated non-profit organization, with participation in its work by international representatives from both government and non-government organizations. The RTCM does not work to induce sales, it does not test or endorse products, and it does not monitor or enforce the use of its standards.

The RTCM does not engage in the design, sale, manufacture or distribution of equipment or in any way control the use of this standard by any manufacturer, service provider, or user. Use of, and adherence to, this standard is entirely within the control and discretion of each manufacturer, service provider, and user.

For information on RTCM Documents or on
participation in development of future RTCM documents contact:

Radio Technical Commission for Maritime Services
1150 18TH Street NW, Suite 910, Washington, DC 20036, USA

Telephone: +1-703-527-2000
Telefax: +1-904-410-2109
E-Mail: info@rtcm.org

© RTCM – Not for reproduction or redistribution

RTCM 10403.4
RTCM Paper 2023-SC104-1342-STD
With Amendment 1
RTCM Paper 2024-SC104-1400

RTCM STANDARD 10403.4

DIFFERENTIAL GNSS (GLOBAL NAVIGATION SATELLITE SYSTEMS) SERVICES – VERSION 3

DEVELOPED BY
RTCM SPECIAL COMMITTEE NO. 104

November 01, 2024

COPYRIGHT © 2024 RTCM

Radio Technical Commission for Maritime Services
1150 18th Street NW Suite 910
Washington, DC 20036
USA
E-Mail: info@rtcm.org
Web Site: www.rtcm.org

PREFACE

RTCM 10403.4 has been developed by RTCM Special Committee 104 for the communication of data related to GNSS services. It replaces and augments a previous standard, RTCM 10402.3.

The messages in this standard have undergone testing for validity and interoperability and are permanent. Future modifications of the standard may change the meaning of reserved bits or provide additional clarifying text, but no changes will be made in the data fields. Changes will require new messages to be developed. As new messages and capabilities have been demonstrated through validity and interoperability testing, they will be incorporated into future revisions of the Version 3 standard, either as amendments to an existing revision or as a new revision. Amendments will be made available electronically to those who have recently purchased the standard. Periodically, accumulated Amendments will be incorporated into a complete revision of standard 10403.x.

RTCM SC-104 believes that Standard 10403.4 for DGNSS services is essential in supporting highly accurate differential and kinematic positioning as well as a wide range of navigation applications worldwide throughout the next decade.RTCM draws attention to the possibility that the implementation of this standard may involve the use of (a) patent(s). RTCM takes no position concerning the evidence, validity or applicability of any claimed patent rights in respect thereof.

As of the date of publication of this standard, RTCM had not received notice of (a) patent(s), which may be required to implement this document. However, implementers are cautioned that this may not represent the latest information which may be obtained from the latest version of the applicable standard. RTCM shall not be held responsible for identifying any or all such patent rights or claims.

© RTCM – Not for reproduction or redistribution

© RTCM – Not for reproduction or redistribution

AMENDMENT 1

RTCM RECOMMENDED STANDARDS
FOR
DIFFERENTIAL GNSS
(GLOBAL NAVIGATION SATELLITE SYSTEMS)
SERVICES VERSION 3

RTCM 10403.4 (Paper 2023-SC104-1342-STD) - RTCM Recommended Standards for Differential GNSS (Gloval Navigation Satellite Systems) Service Version 3.4, dated December 01, 2023, is revised to account for QZSS Network RTK Residual Message and non-technical editorial as follows:

1. Added RTCM updated patent statement to the Preface
2. Added a list of tables
3. Added a list of figures
4. Updated *Table 1-1: Version 3 Revision (3.4) and Amendment (A.1) Summary*
5. Updated Section 1.3 to reference Galileo OS SIS ICD Version 2.1.
6. Updated *Table 3.1-1 Message Groups*
7. Updated *Table 3.2-1 Message Type Table*
8. In *Table 3.4 1 Data Fields*:
    a. Added clarification for DF218
    b. Added clarification for DF221
    c. Changed range of values from 0-604800 to 0-604799 in DF571, DF573, and DF575
    d. Added data field for QZSS Residuals Epoch Time (TOW): DF575
    e. Added data field for QZSS Number of Satellite Signals Processed: DF576
9. Added *Table 3.5-46 Header Part of the QZSS Network RTK Residual Message Type 1305*
10. Added *Table 3.5-47 Satellite-Specific Part of the QZSS Network RTK Residual Message Type 1305*
11. Incremented all the table numbers from 3.5-48 to 3.5-127
12. Added a sentence in section *3.6 Proprietary Messages* with guidance to find the list of proprietary messages on the RTCM website.
13. Added a proprietary number in *Table 3.6-1 Assigned Proprietary Message Types* for CM Intelligent Mobility Network Co., Ltd and Point One Navigation.

© RTCM – Not for reproduction or redistribution

Contents

LIST OF TABLES ..... **VI**
LIST OF FIGURES ..... **XI**
1 INTRODUCTION AND SCOPE **1**
1.1 Introduction **1**
1.2 Scope **3**
1.3 Normative References **3**
1.4 Patent Policy **4**
1.5 List of Abbreviations **6**
2 APPLICATION LAYER **9**
3 PRESENTATION LAYER **11**
3.1 Information **11**
3.1.1 Version 3 Database Architecture ***11***
3.1.2 Message Groups ***11***
3.1.3 Operation with Multiple Services ***20***
3.1.4 Reference Receiver Time and Observations ***21***
3.1.5 Network RTK Corrections ***22***
3.1.6 Proper Handling of Antenna Phase Center Variation Corrections ***23***
3.1.7 Scheduling of Network RTK Messages ***24***
3.1.8 Handling of Quarter-Cycle Carrier Phase Shifts ***25***
3.2 Message Type Summary **28**
3.3 Data Types **35**
3.4 Data Fields **38**
3.5 Messages **115**
3.5.1 Experimental Messages ***115***
3.5.2 Legacy RTK Observable Messages ***115***
3.5.3 System Parameter Messages ***124***
3.5.4 Stationary Antenna Reference Point Messages ***125***
3.5.5 Receiver and Antenna Description Messages ***127***
3.5.6 Master Auxiliary Concept Network RTK Correction Messages ***129***
3.5.7 Unicode Text String Message ***140***
3.5.8 Coordinate Information and Transformation Messages ***142***
3.5.9 Network RTK Residual Error Messages ***171***
3.5.10 State Space Messages ***177***
3.5.11 FKP Network RTK Correction Messages ***198***
3.5.12 Multiple Signal Messages Observable Messages ***202***
3.5.13 GLONASS Bias Information Message ***239***
3.5.14 Ephemeris Messages ***241***
3.6 Proprietary Messages **257**
4 TRANSPORT LAYER **261**
4.1 Description **261**
4.2 Example **262**
5 DATA LINK LAYER **264**

6 PHYSICAL LAYER 265
ANNEX A (INFORMATIVE) SUGGESTIONS AND EXAMPLES FOR NETWORK OPERATION 1
A.1 How to Use Network ID's and Subnetwork ID's 1
A.2 A Scheduling Example for RTK Networks 4
ANNEX B (INFORMATIVE) PATENTS 1
B.1 Policy 1
B.2 Essential Patented Technologies 1
B.3 Non-essential Patented Technologies 1

© RTCM – Not for reproduction or redistribution

LIST OF TABLES

Table 1-1: Version 3 Revision (3.x) and Amendment (A.x) Summary 1
Table 3.1-1 Message Groups 12
Table 3.1-2 Message Types Supporting Different Service Levels..... 15
Table 3.1-3 High Update Rate, for Ease of Comparison Between Different Data Streams. 25
Table 3.1-4 Update Rate for Typical Operation 25
Table 3.1-5 GNSS Receiver Manufacturers – Handling of Quarter-Cycle Phase Shifts 26
Table 3.2-1 Message Type Table 28
Table 3.3-1 Data Type Table..... 35
Table 3.4-1 Data Fields 38
Table 3.4-2 Lock Time Indicator, Data Fields DF013, DF019, DF043, DF049 111
Table 3.4-3 SBAS PRN Codes, Data Fields DF009, DF038..... 112
Table 3.4-4 Carrier Smoothing Interval of Code Phase, DF008 and DF037..... 112
Table 3.4-5 GLONASS Carrier Frequencies in L1 and L2 Bands..... 113
Table 3.4-6 GLONASS Satellite Frequency Channel Number (DF419)..... 114
Table 3.5-1 Contents of the Message Header, Types 1001, 1002, 1003, 1004: GPS RTK Messages..... 115
Table 3.5-2 Contents of the Satellite-Specific Portion of a Type 1001 Message, Each Satellite – GPS Basic RTK, L1 Only 116
Table 3.5-3 Contents of the Satellite-Specific Portion of a Type 1002 Message, Each Satellite – GPS Extended RTK, L1 Only 117
Table 3.5-4 Contents of the Satellite-Specific Portion of a Type 1003 Message, Each Satellite – GPS Basic RTK, L1 & L2..... 118
Table 3.5-5 Contents of the Satellite-Specific Portion of a Type 1004 Message, Each Satellite – GPS Extended RTK, L1 & L2 119
Table 3.5-6 Contents of the Message Header, Types 1009 through 1012: GLONASS RTK Messages..... 120
Table 3.5-7 Contents of the Satellite-Specific Portion of a Type 1009 Message, Each Satellite – GLONASS Basic RTK, L1 Only..... 121
Table 3.5-8 Contents of the Satellite-Specific Portion of a Type 1010 Message, Each Satellite – GLONASS Extended RTK, L1 Only..... 121
Table 3.5-9 Contents of the Satellite-Specific Portion of a Type 1011 Message, Each Satellite – GLONASS Basic RTK, L1 & L2 122
Table 3.5-10 Contents of the Satellite-Specific Portion of a Type 1012 Message, Each Satellite – GLONASS Extended RTK, L1 & L2 123
Table 3.5-11 Contents of the Type 1013 Message, System Parameters 124
Table 3.5-12 Contents of the Type 1005 Message – Stationary Antenna Reference Point, No Height Information 125
Table 3.5-13 Contents of the Type 1006 Message – Stationary Antenna Reference Point, with Height Information..... 126
Table 3.5-14 Contents of the Type 1007 Message – Antenna Descriptor 128
Table 3.5-15 Contents of the Type 1008 Message – Antenna Descriptor and Serial Number 128

© RTCM – Not for reproduction or redistribution

Table 3.5-16 Contents of the Type 1033 Message – Receiver and Antenna Descriptors.... 129
Table 3.5-17 Contents of the Network Auxiliary Station Data Message 1014 133
Table 3.5-18 Contents of Header Network RTK -- Messages 1015, 1016 or 1017 133
Table 3.5-19 Contents of Data Block for GPS Ionospheric Correction Differences 1015.. 134
Table 3.5-20 Contents of Data Block for GPS Geometric Correction Differences 1016.... 134
Table 3.5-21 Contents of Data Block for GPS Combined Geometric and Ionospheric Correction Differences..... 135
Table 3.5-22 Contents of Header for GLONASS Network RTK Messages 1037, 1038, or 1039..... 138
Table 3.5-23 Contents of Data Block for Ionospheric Correction Difference Message 1037 139
Table 3.5-24 Contents of Data Block for Geometric Correction Difference Message 1038139
Table 3.5-25 Contents of Data Block for Combined Geometric and Ionospheric Correction Differences Message 1039..... 140
Table 3.5-26 Contents of the Unicode Text String, Message Type 1029..... 140
Table 3.5-27 Contents of Message Type 1300, Service-CRS Message 144
Table 3.5-28 Header Part of the RTCM-CRS Message 1302..... 146
Table 3.5-29 Database-Link Block of the RTCM-CRS Message 1302..... 146
Table 3.5-30 Contents of the Helmert / Abridged Molodenski Message Type 1021..... 161
Table 3.5-31 Contents of the Molodenski-Badekas Transformation Message Type 1022 . 163
Table 3.5-32 Contents of the Residual Message, Ellipsoidal Grid Representation, Message Type 1023..... 165
Table 3.5-33 Contents of the Residual Message, Plane Grid Representation, Message Type 1024..... 166
Table 3.5-34 Contents of the Projection Message Type 1025 (Projection Types except LCC2SP, OM) 167
Table 3.5-35 Contents of the Projection Message 1026 (Projection Type LCC2SP - Lambert Conic Conformal (2 SP)) 167
Table 3.5-36 Contents of the Projection Message 1027 (Projection Type OM - Oblique Mercator) 168
Table 3.5-37 Contents of the Time-Dependent Linear Expression Helmert Transformation Message 1301 169
Table 3.5-38 Header Part of the GPS Network RTK Residual Message Type 1030 172
Table 3.5-39 Satellite-Specific Part of the GPS Network RTK Residual Message Type 1030 172
Table 3.5-40 Header Part of the GLONASS Network RTK Residual Message Type 1031173
Table 3.5-41 Satellite-Specific Part of the GLONASS Network RTK Residual Message Type 1031..... 173
Table 3.5-42 Header Part of the BDS Network RTK Residual Message Type 1303 174
Table 3.5-43 Satellite-Specific Part of the BDS Network RTK Residual Message Type 1303 174
Table 3.5-44 Header Part of the Galileo Network RTK Residual Message Type 1304..... 175
Table 3.5-45 Satellite-Specific Part of the Galileo Network RTK Residual Message Type 1304..... 175
Table 3.5-46 Header Part of the QZSS Network RTK Residual Message Type 1305..... 176

© RTCM – Not for reproduction or redistribution

Table 3.5-47 Satellite-Specific Part of the QZSS Network RTK Residual Message Type 1305 ... 176
Table 3.5-48 Physical Reference Station Position Information Message Type 1032 ... 177
Table 3.5-49 Header Part of the SSR GPS Orbit Correction Message 1057 ... 185
Table 3.5-50 Satellite-Specific Part of the SSR GPS Orbit Correction Message 1057 ... 186
Table 3.5-51 Header Part of the SSR GPS Clock Correction Message 1058 ... 186
Table 3.5-52 Satellite-Specific Part of the of the SSR GPS Clock Correction Message 1058 ... 187
Table 3.5-53 Header Part of the SSR GPS Satellite Code Bias Message 1059 ... 187
Table 3.5-54 Satellite-Specific Part of the of the SSR GPS Satellite Code Bias Message 1059 ... 188
Table 3.5-55 Code-Specific Part of the of the SSR GPS Satellite Code Bias Message 1059 ... 188
Table 3.5-56 Header Part of the SSR GPS Combined Orbit and Clock Correction Message 1060 ... 188
Table 3.5-57 Satellite-Specific Part of the SSR GPS Combined Orbit and Clock Correction Message 1060 ... 189
Table 3.5-58 Header Part of the SSR GPS URA Message 1061 ... 190
Table 3.5-59 Satellite-Specific Part of the SSR GPS URA Message 1061 ... 190
Table 3.5-60 Header Part of the SSR GPS High-Rate Clock Correction Message 1062 ... 190
Table 3.5-61 Satellite-Specific Part of the SSR GPS High-Rate Clock Message 1062 ... 191
Table 3.5-62 Header Part of the SSR GLONASS Orbit Correction Message 1063 ... 191
Table 3.5-63 Satellite-Specific Part of the SSR GLONASS Orbit Correction Message 1063 ... 192
Table 3.5-64 Header Part of the SSR GLONASS Clock Correction Message 1064 ... 193
Table 3.5-65 Satellite-Specific Part of the of the SSR GLONASS Clock Correction Message 1064 ... 193
Table 3.5-66 Header Part of the SSR GLONASS Satellite Code Bias Message 1065 ... 194
Table 3.5-67 Satellite-Specific Part of the of the SSR GLONASS Satellite Code Bias Message 1065 ... 194
Table 3.5-68 Code-Specific Part of the of the SSR GLONASS Satellite Code Bias Message 1065 ... 195
Table 3.5-69 Header Part of the SSR GLONASS Combined Orbit and Clock Correction Message 1066 ... 195
Table 3.5-70 Satellite-Specific Part of the SSR GLONASS Combined Orbit and Clock Correction Message 1066 ... 195
Table 3.5-71 Header Part of the SSR GLONASS URA Message 1067 ... 196
Table 3.5-72 Satellite-Specific Part of the SSR GLONASS URA Message 1067 ... 197
Table 3.5-73 Header Part of the SSR GLONASS High-Rate Clock Correction Message 1068 ... 197
Table 3.5-74 Satellite-Specific Part of the SSR GLONASS High-Rate Clock Correction Message 1068 ... 198
Table 3.5-75 Header Part of the GPS Network FKP Gradient Message 1034 ... 200
Table 3.5-76 Satellite-Specific Part of the GPS Network FKP Gradient Message 1034 ... 201
Table 3.5-77 Header Part of the GLONASS Network FKP Gradient Message 1035 ... 201

© RTCM – Not for reproduction or redistribution

Table 3.5-78 Satellite-Specific Part of the Network FKP GLONASS Gradient Message 1035 202
Table 3.5-79 MSM Message Types 205
Table 3.5-80 Contents of the MSM Message Types 207
Table 3.5-81 Reference Signal for Phase Alignment 209
Table 3.5-82 GNSS Phaserange Lock Time Indicator (DF402) 210
Table 3.5-83 GNSS Phaserange Lock-Time Indicator with Extended Range and Resolution (DF407) 212
Table 3.5-84 Computing GNSS Phaserange Lock-Time Indicator from Actual Lock Time (DF407) 214
Table 3.5-85 Content of an MSM Message, and Sequence of Blocks 216
Table 3.5-86 Content of MSM Header 217
Table 3.5-87 Content of Satellite Data for MSM1, MSM2, MSM3 218
Table 3.5-88 Content of Satellite Data for MSM4 and MSM6 218
Table 3.5-89 Content of Satellite Data for MSM5 and MSM7 219
Table 3.5-90 Content of Signal Data for MSM1 219
Table 3.5-91 Content of Signal Data for MSM2 220
Table 3.5-92 Content of Signal Data for MSM3 220
Table 3.5-93 Content of Signal Data for MSM4 221
Table 3.5-94 Content of Signal Data for MSM5 221
Table 3.5-95 Content of Signal Data for MSM6 222
Table 3.5-96 Content of Signal Data for MSM7 223
Table 3.5-97 GPS Message Types 225
Table 3.5-98 GPS Satellite ID mapping 225
Table 3.5-99 GPS Signal ID mapping 226
Table 3.5-100 Matching Between DF016 and GPS MSM Signal ID 227
Table 3.5-101 GLONASS Message Types 227
Table 3.5-102 GNSS Time for GLONASS MSMs 228
Table 3.5-103 GLONASS Satellite ID mapping 228
Table 3.5-104 GLONASS Signal ID mapping 228
Table 3.5-105 Galileo Message Types 229
Table 3.5-106 Galileo Satellite ID mapping 230
Table 3.5-107 Galileo Signal ID mapping 230
Table 3.5-108 SBAS Message Types 231
Table 3.5-109 SBAS Satellite ID mapping 232
Table 3.5-110 SBAS Signal ID Mapping 232
Table 3.5-111 QZSS Message Types 233
Table 3.5-112 QZSS Satellite ID mapping 233
Table 3.5-113 QZSS Signal ID Mapping 234
Table 3.5-114 BeiDou Message Types 236
Table 3.5-115 BeiDou Satellite ID Mapping 236
Table 3.5-116 BeiDou Signal ID Mapping 237
Table 3.5-117 NavIC/IRNSS Message Types 238
Table 3.5-118 NavIC/IRNSS Satellite ID Mapping 238
Table 3.5-119 NavIC/IRNSS Signal ID Mapping 239
Table 3.5-120 Contents of GLONASS L1 and L2 Code-Phase Biases Message 1230 239

© RTCM – Not for reproduction or redistribution

Table 3.5-121 Contents of GPS Satellite Ephemeris Data, Message Type 1019 242
Table 3.5-122 Contents of GLONASS Satellite Ephemeris Data, Message Type 1020 244
Table 3.5-123 Contents of Galileo F/NAV Satellite Ephemeris Data, Message Type 1045 246
Table 3.5-124 Contents of Galileo I/NAV Satellite Ephemeris Data, Message Type 1046. 248
Table 3.5-125 Contents of QZSS Satellite Ephemeris Data, Message Type 1044 251
Table 3.5-126 Contents of BDS Satellite Ephemeris Data, Message Type 1042 253
Table 3.5-127 Contents of NavIC/IRNSS Satellite Ephemeris Message Type 1041 255
Table 3.6-1 Assigned Proprietary Message Types 258
Table 4.1-1 Frame Structure..... 261

© RTCM – Not for reproduction or redistribution

LIST OF FIGURES

Figure 3.5-1 Steps in the Coordinate Transformation Process..... 147
Figure 3.5-2 Definition of Translation and Rotations..... 149
Figure 3.5-3 Area of Validity 157
Figure 3.5-4 Grid Definition for Residual Messages..... 158
Figure 3.5-5 Raster Points Used for Bi-Linear Interpolation 159
Figure 3.5-6 Raster Points Used for the Bi-quadratic Interpolation..... 160
Figure 3.5-7 Raster Points Used for the Bi-Spline Interpolation. 160
Figure 3.5-8 Radial, along-track and cross-track orbit components 179
Figure 3.5-9 Sketch of Concept of SSR Update Interval to Ensure the Consistency of Data for Five Different SSR Parameters. 184
Figure 3.5-10 Loss of Lock detection algorithm..... 212
Figure 3.5-11 Loss of Lock detection algorithm for extended range and resolution lock time indicator 213
Figure A.1-1. Network Example 1
Figure A.1-2. Example of Multiple Solutions 2

© RTCM – Not for reproduction or redistribution

© RTCM – Not for reproduction or redistribution

© RTCM – Not for reproduction or redistribution

1 INTRODUCTION AND SCOPE

1.1 Introduction

GPS, GLONASS, BeiDou, and Galileo are satellite-based positioning systems that are currently providing global service 24 hours each day. Collectively, these systems are called Global Navigation Satellite Systems (GNSSs). We also include as GNSSs systems that are regional at the time of this writing, but which provide GNSS-like features in the regions they serve, such as QZSS and NavIC. GNSSs typically provide navigation and positioning services having accuracies in the range of 5-40 meters (2drms). Differential operation provides meter-level accuracy, while Real-Time Kinematic (RTK) operation provides decimeter accuracy or better.

The RTCM Special Committee 104 (SC-104), Differential GNSS Service, has examined the technical and institutional issues and formulated recommendations on the data format and content that are designed to support the most stringent applications in an efficient manner. The Committee has attempted to accommodate the widest possible user community, including not only maritime users, but land-based and airborne users as well. Radio location, surveying, and radio navigation applications are supported.

The RTCM 10403 standard series (i.e., Version 3 or Version 3.4 in the case of this specific revision) describes messages and techniques for supporting GNSS operation with one reference station or a network of reference stations. It can also accommodate augmentation systems that utilize geostationary satellites with transponders operating in the same frequency bands. Generically these are called Satellite-Based Augmentation Systems (SBASs), and they have been designed to be interoperable. The first to be implemented is the Wide-Area Augmentation System (WAAS), which has been developed by the U.S. Federal Aviation Administration to supplement the GPS. Subsequently, other SBAS have been developed including, but not limited to EGNOS, MSAS, and GAGAN.

A summary of the version history of this document is shown in Table 1-1.

Table 1-1: Version 3 Revision (3.x) and Amendment (A.x) Summary

Version	Messages Defined	Change Description
3.0	1001-1013	RTK messages for single or dual-frequency GPS and GLONASS operation.
3.1	1001-1029 4088-4095	GPS Network RTK GPS and GLONASS Ephemeris messages Transformation messages UNICODE message
3.1 A.1	1001-1029 4001-4087	Improved messages descriptions. Addition of proprietary messages
3.1 A.2	1001-1033 4001-4095	Network RTK residual messages Physical reference station position message (for VRS) Receiver and antenna description messages
3.1 A.3	1001-1033	Handling of quarter-cycle phase shifts

© RTCM – Not for reproduction or redistribution

Version	Messages Defined	Change Description
	4001-4095	
3.1 A.4	1001-1039 4001-4095	GPS and GLONASS FKP GLONASS MAC
3.1 A.5	1001-1039 1057-1068 4001-4095	State space representation (SSR; PPP) messages
3.2	1001-1039 1057-1068 1071-1230 4001-4095	Multiple signal messages (MSM) GLONASS bias information message
3.2 A.1	1001-1039 1045 1057-1068 1071-1230 4001-4095	Added Galileo F/NAV Satellite Ephemeris Data (1045) BDS MSM (1121-1127)
3.2 A.2	1001-1039 1044-1045 1057-1068 1071-1230 4001-4095	Added QZSS Ephemeris (1044) QZSS MSM (1111-1117)
3.3	1-100 1001-1039 1042 1044-1046 1057-1068 1071-1230 4001-4095	Added Experimental messages (1 – 100) BDS Ephemeris (1042) Galileo I/NAV Ephemeris (1046) SBAS MSM (1101-1107)
3.3 A1	1-100 1001-1039 1041-1042 1044-1046 1057-1068 1071-1230 4001-4095	Added NavIC/IRNSS Ephemeris (1041) NavIC MSM (1131-1137)
3.3 A2	1-100 1001-1039 1041-1042 1044-1046 1057-1068	Added BDS phase 3 signal types for MSM

© RTCM – Not for reproduction or redistribution

Version	Messages Defined	Change Description
	1071-1230	
3.3 A3		Added NavIC S-band signal supported
3.4	1-100 1001-1039 1041-1042 1044-1046 1057-1068 1071-1230 1300-1302 1303-1304	Consolidation of previous 3.3 amendments. Document restructuring. Added CRS messages (1300, 1302) Helmert Transformation, Time-Dependent Linear Expression (1301) BDS and Galileo Network RTK Residual Message Types (1303, 1304)
3.4 A1	1305	Added QZSS Network RTK Residual Message

1.2 Scope

This standard defines a flexible messaging structure to support augmentation of navigation systems. It is the purpose of this structure to provide integrity and capability for existing and future applications in an efficient manner. To promote these qualities this standard has been designed using a layered approach adapted from the Open System Interconnection (OSI) standard reference model.

1. Application Layer
2. Presentation Layer
3. Transport Layer
4. Data Link Layer
5. Physical Layer

Application Layer considerations are briefly discussed in Section 2, which includes instructions on creating and applying data for navigation and positioning applications. Section 3, which comprises the bulk of the document, addresses the Presentation Layer, and describes the messages, the data elements, and the data definitions. The Transport Layer is described in Section 4, and includes the definition of the message frames, the method of implementing variable-length messages, and the Cyclic Redundancy Check (CRC) that provides message integrity. The Data Link Layer is tailored around the Physical Layer, which defines how the data is conveyed at the electrical and mechanical level.

1.3 Normative References

The following documents contain provisions which, through direct reference in this Standard, constitute provisions of this Standard, and are indispensable for its application:

Cabinet Office, Government of Japan. (2022). Quasi-Zenith Satellite System Interface Specification Satellite Positioning, Navigation, and Timing Service. *IS-QZSS-PNT-005*.

Cabinet Office, Government of Japan. Retrieved from http://qzss.go.jp/en/technical/ps-is-qzss/ps-is-qzss.html

China Satellite Navigation Office. (2019). BeiDou Navigation Satellite System Signal In Space Interface Control Document Open Service Signal B1I (Version 3.0). *BDS-SIS-ICD-3.0*. Retrieved from http://en.beidou.gov.cn/SYSTEMS/ICD/

European Union. (2023). European GNSS (Galileo) Open Service Signal-in-Space Interface Control Document. *OS SIS ICD, Issue 2.1*. European Union. Retrieved from https://www.gsc-europa.eu/sites/default/files/sites/all/files/Galileo_OS_SIS_ICD_v2.1.pdf

Global Positioning Systems Directorate Systems Engineering & Integration. (2018). Interface Specification IS-GPS-200, Navstar GPS Space Segment/Navigation User Interfaces. *IS-GPS-200J*. The Interface Control Working Group (ICWG) of the Global Positioning Systems Directorate. Retrieved from https://www.gps.gov/technical/icwg/

Indian Space Research Organization Satellite Centre. (2017). IRNSS SIS ICD for Standard Positioning Service, Version 1.1. *IRNSS-SIS-ICD-v1.1*. Bangalore: Indian Space Research Organization Satellite Centre.

Russian Institute of Space Device Engineering. (2008). GLONASS Interface Control Document Navigational Radiosignal in Bands L1, L2. *GLONASS ICD Edition 5.1*.

1.4 Patent Policy

It is the intent of the Radio Technical Commission for Maritime Services (RTCM) to develop standards that do not require the use of patented technologies as the only means to comply with a standard (essential patents).

With respect to the patented technologies, the RTCM Standards development policies include the following key statements[1]:

> *RTCM adopts the Common Patent Policy for ITU-T/ITU-R/ISO/IEC of 16 December2022 for essential patents. The policy, along with the common guidelines adopted by those organizations is available at* https://www.iec.ch/members_experts/tools/patents/patent_policy.htm
>
> *Patented elements may be included in a standard, as long as the patented item is not essential to comply with the standard. For example, a patented element may be essential to the construction of one type of device which meets the standard, as long as other types of devices can be constructed to meet the standard without using the patented element.*
>
> *In lieu of a patent database, each RTCM standard shall list the patents which have been declared as applicable to that standard. In addition to essential patents, non-essential patents for which RTCM has received information shall also be listed.*

[1] Radio Technical Commission for Maritime Services, Standards Development Policies, Version 2.0 (RTCM paper 226-2013-BD-509), September 23, 2013

© RTCM – Not for reproduction or redistribution

For the purpose of this policy, patented technologies do not include those covered by patents no longer in force.

This document primarily focuses on the information content and the format of messages. Additionally, this standard categorizes some messages as being useful for various applications. There are numerous patents in the GNSS field, and possibly other fields, that could be applicable depending on the implementation. It is incumbent upon the implementer to ensure that if their specific implementation does employ patent related items, the implementer obtains the appropriate releases (e.g., licenses) to use those patented items.

Annex B contains patents known to be related to this standard.

© RTCM – Not for reproduction or redistribution

© RTCM – Not for reproduction or redistribution

1.5 List of Abbreviations

AODC	Age Of Data Clock
AODE	Age Of Data Ephemeris
ARP	Antenna Reference Point
AS	Anti-Spoofing
ASCII	American Standard Code for Information Interchange
BDS	BeiDou Navigation Satellite System
BDT	BeiDou System Time
C/A	Coarse Acquisition
CD	Correction Difference
CDMA	Code Division Multiple Access
CNR	Carrier to Noise Ratio
COM	Center Of Mass
CPB	Code-Phase Bias
CRC	Cyclic Redundancy Check
CRS	Coordinate Reference System
DARC FM	Data Radio Channel Frequency Modulation
DD	Data Dictionary
DF	Data Field
ECEF	Earth Centered Earth Fixed
EGNOS	European Geostationary Navigational Overlay System
EPC	Easting at Projection Center
EPSG	European Petroleum Survey Group
ETRF	European Terrestrial Reference Frame
FDMA	Frequency Division Multiple Access
FE	False Easting
FKP	(German) FlächenKorrekturParameter, area correction parameter
FN	False Northing
GCPCD	Geometric Carrier Phase Correction Difference
GLONASS	Global Navigation Satellite System
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
ICD	Interface Control Document
ICPCD	Ionospheric Carrier Phase Correction Difference
IERS	International Earth Rotation and Reference Systems Service
IGS	International GNSS Service
IOD	Issue of Data
IODC	Issue of Data Clock
IODE	Issue of Data Ephemeris
IRNSS	Indian Regional Navigation Satellite System
ITRF	International Terrestrial Reference Frame
LAAS	Local Area Augmentation System
LaNO	Latitude of Natural Origin
LCC	Lambert Conformal Conic projection (LCC1SP, LCC2SP, LCCW)
LOC	Loss of Continuity

© RTCM – Not for reproduction or redistribution

LoNO	Longitude of Natural Origin
LSB	Least Significant Bit
M	Number of characters in Target Name
MAC	Master Auxiliary Concept
MJD	Modified Julian Day number
MLT	Minimum Lock Time
MMI	Multiple Message Indicator
MSB	Most Significant Bit
MSK	Minimum Shift Keying
MSM	Multiple Signal Message
N	Number of characters in Source Name
NavIC	Navigation with Indian Constellation
NGS	National Geodetic Survey
Nm	Number of Message Types Transmitted
NMEA	National Marine Electronics Association
NPC	Northing at Projection Center
Ns	No. of Satellites
OGP	International Association of Oil and Gas Producers
OM	Oblique Mercator
OSI	Open System Interconnection
ppm	Parts Per Million
PPP	Precise Point Positioning
PRN	Pseudo-Random Noise (used to denote a specific GNSS satellite)
QZSS	Quasi-Zenith Satellite System
QZSST	QZSS Time
RINEX	Receiver INdependent EXchange format
RTCM	Radio Technical Commission for Maritime Services
RTK	Real-Time Kinematic
SBAS	Satellite-Based Augmentation Systems
SMF	Synchronous Message Flag
SPS	Standard Positioning Service
SRP	Satellite Reference Point
SSR	State Space Representation
TM	Transverse Mercator
TMS	Tile Map Service
TOW	Time of Week
UHF	Ultra-High Frequency
URA	User Range Accuracy
UTC	Universal Coordinated Time (Seconds of Day)
UTC(SU)	National time scale of the Russian Federation
VHF	Very High Frequency
VRS	Virtual Reference Station
WAAS	Wide-Area Augmentation System
yr	Year

© RTCM – Not for reproduction or redistribution

This page intentionally left blank.

© RTCM – Not for reproduction or redistribution

2 APPLICATION LAYER

The Application Layer defines how the Version 3 messages can be applied for different end-user applications. The fundamental feature of Differential Service is that it is a broadcast service, not a 2-way data link. As such, information is developed centrally by a Service Provider, who has an institutional or commercial interest in providing a positioning or navigation service. Recently, point-to-multipoint services using cell phones and Internet connections have become popular, but such services primarily support a one-way flow of information.

In general, navigation applications are serviced very well with 1-10 meters horizontal accuracy positioning. An exception is the GNSS-based aircraft landing system, called the Local Area Augmentation System, or LAAS. A separate standard has been developed for this by RTCM's sister organization, RTCA, Inc., which develops aviation standards. Conventional differential GNSS service supports these applications nicely, and they utilize broadcast links with relatively low data rates. These low data rates can be supported by low-frequency broadcasts that are received over large areas, and high accuracy is maintained over hundreds of miles.

As innovative engineers and scientists have found uses for sub-meter accuracy positioning, RTK service has increased in importance. RTK service requires the transmission of significantly more data, so that generally line-of-sight broadcasts and point-to-multipoint services that utilize higher bandwidths are employed. Tropospheric and ionospheric variations cause phase and time delay variations in the GNSS signals that limit the area over which a given accuracy can be achieved. For example, relative positioning accuracies of one centimeter or better using single-frequency GNSS signals can be achieved only over distances of 10 kilometers or so (from reference station to user). Using dual-frequency GNSS signals enables one to estimate the ionospheric effects, and water vapor measurements can be made which improve tropospheric delay estimation, so that using these techniques the range can be extended to 50 kilometers or so in certain parts of the world. Dual-frequency RTK is very common and is supported by this standard. Because RTK provides relative positioning, the knowledge of the absolute (usually fixed) position of the reference station enables the user to achieve high absolute-position accuracies, too.

To achieve the highest accuracy, it is important to account for GNSS antenna variations. Antenna patterns differ slightly from manufacturer to manufacturer and even from model to model. Differential GNSS service supports this by transmitting messages with reference station antenna information. Antenna patterns can also vary between different units of the same model and can vary due to environmental effects, but these can be mitigated by manufacturing design and reference site selection, respectively. Such variations are outside the scope of this document.

The applications of RTK to air, water and land operations are too many to enumerate, but a sampling is useful:

- Marine – Hydrographic surveying, dredge operations, navigation in narrow channels, buoy placement and auditing, even tidal height
- Air – Aerial surveying, landing system testing, calibration of other navigation systems
- Land – Surveying, building and bridge construction, surface mining, agriculture, road construction, asset location and management, and vehicle navigation.

It turns out that the RTK requirements for all these different applications do not vary that much. The broadcast link bandwidth and update rates are primarily determined by the accuracy requirements and the signal blockage environment. Otherwise, the required services are similar for air, land, and sea applications.

© RTCM – Not for reproduction or redistribution

3 PRESENTATION LAYER

3.1 Information

3.1.1 Version 3 Database Architecture

Version 3 is written in a database format, loosely patterned after the NMEA 2000 standard. Whereas the NMEA standard is written for a networked set of different electronic units, the Differential GNSS Version 3 standard is written for a centralized distribution of data. For the Version 3 broadcast, every bit counts in the frequently repeated messages, so while lining up on byte boundaries is desirable, forcing each data field to occupy whole numbers of bytes is not practical.

Also, the NMEA 2000 database has a wide disparity between Data Dictionary (DD) and Data Field (DF) records. In the case of Version 3 broadcasts, there would be little difference. Consequently, rather than utilize both DF and DD tables, these are collapsed into one DF definition. Rather than referring to "Parameter Groups", this document will use the more familiar term "Message Types".

3.1.2 Message Groups

Message types contained in the current Version 3 standard (RTCM 10403.3) have been structured in different groups. For proper operation of a particular service the provider needs to transmit messages from each of several groups, as shown in Table 3.1-1. For example, to provide RTK services the provider must transmit at least one message type from each of the following groups: Observations, Station Coordinates, and Antenna Description. The different message types in each group contain messages with similar information content.

In some cases, the shorter ones contain the minimum needed to provide the service, while the longer message types contain additional information for enhancing the performance of the service. For example, Message Type 1001 contains the shortest version of a message for GPS Observations, namely L1-only observables. For a broadcast link limited in throughput, use of Message Type 1001 might be appropriate. Message Type 1002 contains the Type 1001 information plus additional information that enhances performance. If throughput is not limited and the additional information is available, it is recommended to use the longer version of messages. Similarly, Message Type 1003 provides minimum data for L1/L2 operation, while Message Type 1004 provides the full data content. The shorter observation messages save throughput but contain less information. However, since the additional information in the longer observation messages does not change very often, they could be sent less often.

© RTCM – Not for reproduction or redistribution

© RTCM – Not for reproduction or redistribution

Table 3.1-1 Message Groups

Group Name	Sub-Group Name	Message Type
Experimental messages		1-100
Legacy GPS/GLONASS Observable Messages	GPS L1	1001 1002
	GPS L1 / L2	1003 1004
	GLONASS L1	1009 1010
	GLONASS L1 / L2	1011 1012
MSM Observable Messages	GPS	1071-1077
	GLONASS	1081-1087
	Galileo	1091-1097
	SBAS	1101-1107
	QZSS	1111-1117
	BDS	1121-1127
	NavIC/IRNSS	1131-1137
Station Coordinates		1005 1006 1032
Antenna Description		1007 1008
Receiver and Antenna Description		1033
Network RTK Corrections	Network Auxiliary Station Data Message	1014
	GPS Ionospheric Correction Differences	1015
	GPS Geometric Correction Differences	1016
	Combined GPS Geometric and Ionospheric Correction Differences	1017

© RTCM – Not for reproduction or redistribution

Group Name	Sub-Group Name	Message Type
	GPS Network RTK Residual Message	1030
	GLONASS Network RTK Residual Message	1031
	GPS Network FKP Gradient Message	1034
	GLONASS Network FKP Gradient Message	1035
	GLONASS Ionospheric Correction Differences	1037
	GLONASS Geometric Correction Differences	1038
	Combined GLONASS Geometric and Ionospheric Correction Differences	1039
	BDS Network RTK Residual Message	1303
	Galileo Network RTK Residual Message	1304
	QZSS Network RTK Residual Message	1305
Auxiliary Operation Information	System Parameters	1013
	Satellite Ephemeris Data	1019 1020 1041 1042 1044 1045 1046
	Unicode Text String	1029
	GLONASS bias information	1230
Coordinate Information and Transformation Parameters	Service-CRS	1300
	RTCM-CRS	1302

Group Name	Sub-Group Name	Message Type
	Helmert Transformation, Time-Dependent Linear Expression	1301
	Helmert/Abridged Molodenski Message	1021
	Molodenski-Badekas Message	1022
	Representation Residual Messages	1023 1024
	Projection Parameters Messages	1025 1026 1027
State Space Representation Parameters	GPS Orbit Correction	1057
	GPS Clock Correction	1058
	GPS Code Bias	1059
	GPS Combined Orbit and Clock	1060
	GPS URA	1061
	GPS High-Rate Clock Correction	1062
	GLONASS Orbit Correction	1063
	GLONASS Clock Correction	1064
	GLONASS Code Bias	1065
	GLONASS Combined Orbit and Clock	1066
	GLONASS URA	1067
	GLONASS High-Rate Clock Correction	1068
Proprietary Information		Assigned in the range 4001 – 4095 See Table 3.6-1

Table 3.1-2 shows various levels of RTK services that can be supported, with the message types that support them. It also provides the appropriate set of messages for both the mobile and reference station receivers for each service.

© RTCM – Not for reproduction or redistribution

© RTCM – Not for reproduction or redistribution

Table 3.1-2 Message Types Supporting Different Service Levels

Service	Group	Mobile Receiver	Service Provider Message Type(s)	
		Minimum Decoding Requirement	Minimum Service Operation	Full Service Operation
All RTK and SSR services	Coordinate Information	1300 and 1302	1300	1300 and 1302
Precision GPS, L1 only	Observations (GPS)	1001-1004	1001	1002
	Station Description	1005 and 1006	1005 or 1006	1005 or 1006
	Antenna and Receiver Description	1033 (see note 1)	1033	1033
	Auxiliary Operation Information			1013
Precision GPS RTK, L1 & L2	Observations (GPS)	1003-1004	1003	1004
	Station Description	1005 and 1006	1005 or 1006	1005 or 1006
	Antenna and Receiver Description	1033 (see note 1)	1033	1033
	Auxiliary Operation Information			1013
Precision GLONASS L1 only	Observations (GLONASS)	1009-1012	1009	1010
	Station Description	1005 and 1006	1005 or 1006	1005 or 1006
	Antenna and Receiver Description	1033 (see note 1)	1033	1033
	Auxiliary Operation Information		1230	1013 and 1230
Precision GLONASS RTK	Observations (GLONASS)	1011-1012	1011	1012
	Station Description	1005 and 1006	1005 or 1006	1005 or 1006
	Antenna and Receiver Description	1033 (see note 1)	1033	1033
	Auxiliary Operation Information		1230	1013 and 1230
Precision GPS and GLONASS	Observations (GPS)	1001-1004	1001	1002
	Observations (GLONASS)	1009-1012	1009	1010

© RTCM – Not for reproduction or redistribution

Service	Group	Mobile Receiver	Service Provider Message Type(s)	
		Minimum Decoding Requirement	Minimum Service Operation	Full Service Operation
L1 only	Station Description	1005 and 1006	1005 or 1006	1005 or 1006
	Antenna and Receiver Description	1033 (see note 1)	1033	1033
	Auxiliary Operation Information		1230	1013 and 1230
Precision GPS and GLONASS RTK, L1 & L2	Observations (GPS)	1003-1004	1003	1004
	Observations (GLONASS)	1011-1012	1011	1012
	Station Description	1005 and 1006	1005 or 1006	1005 or 1006
	Antenna and Receiver Description	1033 (see note 1)	1033	1033
	Auxiliary Operation Information		1230	1013 and 1230
Precision GPS Network RTK (MAC)	Observations (GPS)	1003-1004	1003	1004
	Station Description	1005 and 1006	1005 or 1006	1005 or 1006
	Antenna and Receiver Description	1033 (see note 1)	1033	1033
	Auxiliary Operation Information			1013
	Network RTK Corrections (MAC)		1014	1014
			1017	1015 and 1016 or 1017
				1030
Precision GPS Network RTK (FKP)	Observations (GPS)	1003-1004	1003	1004
	Station Description	1005 and 1006	1005 or 1006	1005 or 1006
	Antenna and Receiver Description	1033 (see note 1)	1033	1033

© RTCM – Not for reproduction or redistribution

	Auxiliary Operation Information			1013
	Network RTK Corrections (FKP)		1034	1034
				1030
Precision GLONASS Network RTK (MAC)	Observations (GLONASS)	1011-1012	1011	1012
	Station Description	1005 and 1006	1005 or 1006	1005 or 1006
	Receiver and Antenna Description	1033	1033	1033
	Auxiliary Operation Information		1230	1013 and 1230
	Network RTK Corrections (MAC)		1014	1014
			1039	1037 and 1038 or 1039 1031
Precision GLONASS Network RTK (FKP)	Observations (GLONASS)	1011-1012	1011	1012
	Station Description	1005 and 1006	1005 or 1006	1005 or 1006
	Receiver and Antenna Description	1033	1033	1033
	Auxiliary Operation Information		1230	1013 and 1230
	Network RTK Corrections (FKP)		1035	1035
				1031
Precision GPS and GLONASS Network RTK (MAC)	Observations (GPS)	1003-1004	1003	1004
	Observations (GLONASS)	1011-1012	1011	1012
	Station Description	1005 and 1006	1005 or 1006	1005 or 1006
	Receiver and Antenna Description	1033	1033	1033
	Auxiliary Operation Information		1230	1013 and 1230
	Network RTK Corrections (MAC) - GPS-		1014	1014
			1017	1015 and 1016 or 1017
				1030

© RTCM – Not for reproduction or redistribution

	Network RTK Corrections – (MAC) - GLONASS		1039	1037 and 1038 or 1039
				1031
Precision GPS and GLONASS Network RTK (FKP)	Observations (GPS)	1003-1004	1003	1004
	Observations (GLONASS)	1011-1012	1011	1012
	Station Description	1005 and 1006	1005 or 1006	1005 or 1006
	Receiver and Antenna Description	1033	1033	1033
	Auxiliary Operation Information		1230	1013 and 1230
	Network RTK Corrections (FKP) - GPS		1034	1034 1030
	Network RTK Corrections (FKP) - GLONASS		1035	1035 1031
GPS SSR	Orbit and Clock Corrections	1057 1058 1060 1062	1060	1057 1058 1062
	Bias Correction			1059
	Auxiliary Operation Information			1061
GLONASS SSR	Orbit and Clock Corrections	1063 1064 1066 1068	1066	1063 1064 1068
	Bias Correction			1065
	Auxiliary Operation Information			1067
GPS and GLONASS SSR	Orbit and Clock Corrections	1057 1058 1060 1062 1063 1064 1066 1068	1060 1066	1057 1058 1062 1063 1064 1068
	Bias Correction			1059 1065

© RTCM – Not for reproduction or redistribution

	Auxiliary Operation Information			1061 1067
GNSS CODE-differential operation	Observations (GNSS)	MSM1-MSM7	MSM1	MSM1
	Station Description	1005 and 1006	1005 or 1006	1005 or 1006
	Antenna Description			1007 or 1008 or 1033
	Auxiliary Operation Information			1013
GNSS RTK Standard Precision operation	Observations (GNSS)	MSM1-MSM7	MSM3	MSM5
	Station Description	1005 and 1006	1005 or 1006	1005 or 1006
	Antenna and Receiver Description	1033	1033	1033
	Auxiliary Operation Information		1230 (if GLONASS data is transmitted)	1013 1230 (if GLONASS data is transmitted)
GNSS RTK High Precision operation	Observations (GNSS)	MSM6 and MSM7	MSM6	MSM7
	Station Description	1005 and 1006	1005 or 1006	1005 or 1006
	Antenna and Receiver Description	1033	1033	1033
	Auxiliary Operation Information		1230 (if GLONASS data is transmitted)	1013 1230 (if GLONASS data is transmitted)
GNSS Standard-Precision Data Collection	Observations (GNSS)		MSM5	MSM5
	Station Description		1005 or 1006	1005 or 1006
	Antenna and Receiver Description		1033	1033

	Auxiliary Operation Information		1230 (if GLONASS data is transmitted)	1013 1230 (if GLONASS data is transmitted)
GNSS High-Precision Data Collection	Observations (GNSS)		MSM7	MSM7
	Station Description		1005 or 1006	1005 or 1006
	Antenna and Receiver Description		1033	1033
	Auxiliary Operation Information		1230 (if GLONASS data is transmitted)	1013 1230 (if GLONASS data is transmitted)

Note 1: MT1033 is preferred over MT1008. MT1008 is intended to be used to service legacy equipment.

Service Providers can provide a variety of different services ranging from a basic to a complete service. A basic service would involve, e.g., a GPS single-frequency operation, with no attempt to optimize accuracy or ambiguity resolution time. A complete service would provide dual-frequency operations, possibly involving both GPS and GLONASS, attempting to optimize accuracy, baseline length, and ambiguity resolution time, as well as providing helpful ancillary data for quick startup and post-mission analysis.

Mobile equipment should be designed to decode all the message types in a group, even if all the information is not processed. For example, by decoding a Message Type 1002, the RTK observable data that matches that of Message Type 1001 can be utilized, but the additional information may be ignored. If the mobile equipment only operates on L1, it should still be designed to decode Message Types 1003 and 1004 and to pull out the L1 information.

3.1.3 Operation with Multiple Services

Providing information for multiple GNSSs (e.g., GNSS1=GPS and GNSS2=GLONASS) can be accommodated if guidelines are carefully followed. In particular:

1. The messages for all satellites of a particular system should be grouped in one message, if possible. For example, for GPS L1/L2 operation, each 1003 or 1004 message should contain the data for all GPS satellites that are processed. This ensures that a GPS-only mobile receiver will be certain that all relevant data has been received even if the "Synchronous GNSS Message Flag", which indicates that more GNSS data (e.g., GLONASS) referenced to the same time epoch will be transmitted next, is set to "1".
2. When the "extended" messages, i.e., Message Types 1002, 1004, 1010, and 1012, are transmitted, they should include the entire set of satellites processed.

© RTCM – Not for reproduction or redistribution

© RTCM – Not for reproduction or redistribution

3. For combined GPS/GLONASS operation, GPS data should be transmitted first. This is because it will reduce latency for GPS-only mobile receivers, while combined GPS/GLONASS mobile receivers will suffer no penalty.
4. If the GNSS1 and GNSS2 data are not synchronous (i.e., the observations are not taken within one microsecond of each other), the "Synchronous GNSS Message Flag" should be set to zero for each set.

Now that the GLONASS constellation is complete and the Galileo system should soon be fully operational, these rules may be re-examined and modified.

3.1.4 Reference Receiver Time and Observations

The reference receiver should, if possible, align the measurement epoch times close to the GNSS system time. This is commonly referred to as *Clock Steering*. It ensures that the pseudorange magnitude is bounded and will fit within data field ranges. If clock steering is not possible then the observation shall be adjusted to correct for possible large receiver clock offsets.

To ensure consistency between the observations and the epoch time, the adjustment for receiver clock offset shall be done according to the following equations:

$$p' = p - \delta t\left(c - \dot{\Phi}\right) \ (m) \tag{3.1-1}$$

$$\Phi' = \Phi - \delta t\left(c - \dot{\Phi}\right) \ (m) \tag{3.1-2}$$

$$\dot{\Phi}' = \dot{\Phi} - \dot{\delta t}\left(c - \dot{\Phi}\right) \ (m/s) \tag{3.1-3}$$

In these equations p', Φ', and $\dot{\Phi}'$ are, respectively, the transmitted Pseudorange, Phaserange, and Phaserange rate observations corrected for clock offset. The uncorrected, raw observations are p, Φ, and $\dot{\Phi}$. The corrections require the clock offset, δt, the clock offset rate, $\dot{\delta t}$, and the speed of light, c. The resulting receiver epoch time should align with the GNSS system epoch time to within ±1 µs.

Note that the Phaserange is in meters, has the same sign as the Pseudorange, and changes in the same sense. The Phaserange rate is the derivative of the Phaserange. This means, for example, that for a continually-tracked approaching satellite the Phaserange will decrease in value and the Phaserange rate will be negative. This Phaserange and Phaserange rate convention is used throughout this standard. Raw Receiver observables may use a different convention; care must be taken when converting raw observables to quantities used in this standard.

For combined GNSS operation, if all GNSS observables are measured at the same instant of receiver time (in other words, if GNSS1 and GNSS2 clocks are based on the same oscillator), the clock offset utilized in the formulas above should be identical for the correction of all observations across both satellite systems and frequencies. The relations of differences between different clock biases in the observations are maintained in their original form. In this case, "Single Receiver Oscillator Indicator" (DF142) should contain "1". Also, "Synchronous GNSS Message Flag" (DF005) should indicate that GNSS measurements are synchronous as described in point 3.1.3. Some reference station installations may not allow for identical clock offsets over all the satellite systems tracked (for example, if two or more independent receiver boards produce the observations). Correspondingly, the "Single Receiver Oscillator Indicator" (DF142) should be set to "0". However, in such a case all GNSSs might be still synchronous, indicating that the

observations have been obtained within one microsecond. The "Synchronous GNSS Message Flag" (DF005) should identify the proper state. It should be noted that the conditions for DF005 and DF142 refer to the configuration of the reference station equipment, thus do not change during the transmission of a data stream.

3.1.5 Network RTK Corrections

Different concepts to provide corrections from Network RTK are supported. There are

- Non-Physical Reference Station
- FKP: range correction gradients
- MAC: Master Auxiliary Concept

3.1.5.1 Non-physical reference station

The Non-Physical or Computed Reference Station is typically calculated based on information from a network of reference stations. Different approaches have been established over years. The Non-Physical or Computed Reference Stations are sometimes trademarked and may not be compatible. Examples of these names are “Virtual Reference Stations”, “Pseudo-Reference Stations”, and “Individualized Reference Stations”.

3.1.5.2 FKP range correction gradients

With the concept of FKP (from the German word “Flächenkorrekturparameter”, meaning Area Correction Parameters) horizontal gradients of distance-dependent errors like ionosphere, troposphere and orbits are derived from a network of GNSS reference stations and transmitted to a rover together with raw or correction data of a corresponding reference station.

3.1.5.3 Master Auxiliary Concept (MAC)

The fundamental functionality of networking software that combines the information of several permanent reference stations is the determination of integer ambiguities between the reference stations. The resulting integer ambiguities may be used for reducing the original raw reference station observations. This manipulation of the raw observations leaves the general properties of the carrier phase observations (troposphere, ionosphere, phase center variations, etc.) untouched, since only integer numbers have been introduced. This process is named “integer ambiguity-leveling” and the resulting observations of permanent reference stations are “(integer) ambiguity-leveled”.

An application accessing ambiguity-leveled observations of a single reference station will not see any difference. The modeling requirements within the application are identical. However, when an application uses the observations of more than one reference station, the application will no longer have to account for integer ambiguities between the reference stations on the same ambiguity level. Roving user equipment receiving observations of more than one reference station on the same ambiguity level and utilizing the observations in its positioning algorithm may switch from one reference station to another without reinitialization of its filter.

In order to preserve throughput, Network RTK messages utilize data fields that extend the approach described above: the raw observations are reduced by the geometric representation of the satellite and receiver distance; and inter-reference station single differences are used (see Annex A). Master-Auxiliary Network RTK Corrections (MAC) are designed as additional information

© RTCM – Not for reproduction or redistribution

© RTCM – Not for reproduction or redistribution

for improved performance and precision. A service provider utilizing the network capability will broadcast Precision GPS and GLONASS RTK messages for the Master Reference Station, but it will broadcast Auxiliary Reference Station information as well. Until a future version of revision of this standard is published, service providers are advised to limit the data stream to information associated with one single Master Reference Station and its associated Auxiliary Reference Stations. Participating mobile receivers must be designed to accept and process the Network RTK Corrections. Mobile equipment operating close to the Master Reference Station may be designed to use the Observation, Station Description, and Antenna Description information of the Master reference station exclusively.

3.1.6 Proper Handling of Antenna Phase Center Variation Corrections

Antennas designed for precise RTK operation account for antenna phase center offsets and variations in the centimeter-range. These offsets and variations can be corrected within precise RTK equipment using calibration information. Antenna model type calibrations are available from several sources (e.g., IGS, and NGS). High precision applications occasionally use individual antenna calibrations. Also, within permanent reference station networks, individually calibrated antennas are increasingly being used. The proper handling of dissimilar antennas is a pressing issue for the interoperability of RTK network data. Therefore, for Network RTK operation adjustments may be made to raw observations for the Master Reference Stations (message 1004, for example) for antenna biases (phase center offsets and phase center variations). When corrections of antenna phase center variations are required, one should ensure that consistent sets are used throughout the application. The best way to ensure a consistent set of antenna phase center variations is to use only information from a single source (e.g., IGS, NGS) and ensure that the same form of representation is used consistently throughout each application (note the difference between absolute and relative representations). Note that reference station network software and rover firmware are different applications and thus may use different representations. It is recommended that published antenna parameters be used as they are. It is crucial to avoid mixing different forms of representation, and/or "fine-tuning" given sets of information by assembling a new set out of different sources (e.g., mixing offsets of one calibration with phase center variations with another calibration for one antenna). Offsets and phase center variations comprise a self-consistent set of information for a particular antenna. Both parts of the information are correlated with each other. The shape of one particular antenna phase pattern may be represented in principle by an indefinite number of different consistent sets of information (e.g., the introduction of a different value in the offset will be compensated by the antenna phase center variations).

In the event that it is necessary to change Master Reference Stations within a Precision Network RTK operation, a bias error could occur in the rover position as a consequence of using inconsistent phase center correction sets at the rover (e.g., obtained from different sources). Furthermore, achieving consistency of antenna correction models within large network setups would require storing antenna phase center corrections for dozens of Master Reference Stations in order to allow use of the most accurate information that would be obtained from individually calibrated antennas. There is another approach to achieving consistent operation of user equipment, which is recommended here: namely, the observation data messages (i.e., 1004) for all Master Reference Stations of a homogenous network should be referenced to a single antenna (preferably, the ADVNULLANTENNA). The modification of the observation information with respect to antenna phase center variations must be indicated in the disseminated data stream using antenna descriptor messages (i.e. 1008). The antenna descriptor field must then state the descriptor of the antenna

(e.g., ADVNULLANTENNA). Note that the reduction to the ADVNULLANTENNA is defined through the correction of the antenna phase center offsets and variations based on the absolute antenna correction representation.

3.1.7 Scheduling of Network RTK Messages

Scheduling of the Network RTK messages is a crucial procedure in the rover application. In general, the concept chosen for Network RTK messages accommodates several different schemes. To achieve best interoperability, some guidelines are necessary that limit the scheduling but not the resulting performance.

The recommended guidelines for scheduling are:

- First, dissemination of raw observation message (1003 or 1004, for example) containing Master reference station data at a high data rate (0.5 – 2 Hz) immediately when information is available (low latency).
- Next, dissemination of ionospheric (dispersive) and geometric (non-dispersive) Correction Difference messages for all Auxiliary Reference Stations ((1015 and 1016) or 1017 for GPS and (1037 and 1038) or 1039 for GLONASS) at identical epoch times. The chosen epoch time should be identical to the epoch time of the respective raw observations of the Master Reference Station. The update rate may be identical or at a lower data rate than for raw observations. For operation with Correction Difference messages (1015 and1016) or (1037 and 1038), the epoch time should be identical for both messages. The maximum interval should not exceed 15 seconds. When Correction Differences are updated at a lower rate than the Master Reference Station observations, both the ionospheric and the geometric carrier phase components may be filtered to reduce the effect of noise.
- Network Auxiliary Station Data (1014): The complete set of station information messages for all Master and Auxiliary Reference Stations within the data stream may be distributed over time in order to optimize throughput. The dissemination should be completed after a maximum time span of 15 seconds (optimization of start-up time of rover operation).
- Other messages with additional information as needed for proper rover operation (see Table 3.1-2) should be transmitted as required.
- For combined GPS/GLONASS operation, GPS data should be transmitted first.

Scheduling schemes within these bounds are recommended for best operation of a Network RTK provider service with Network RTK messages.

These recommended guidelines are based on the scheduling used during interoperability testing, using two different update rates. These rates were chosen to represent typical RTK operations in the field and are described in Table 3.1-3 and Table 3.1-4. Other update rates can be employed, but a Service Provider should be aware that these are the only ones that were tested for interoperability.

© RTCM – Not for reproduction or redistribution

© RTCM – Not for reproduction or redistribution

Table 3.1-3 High Update Rate, for Ease of Comparison Between Different Data Streams

Group Name		Message Type	Update Rate
Observations (GPS)		1004	1 Hz
Station Description		1005 or 1006	As typical in an RTK operation
Antenna description		1007 or 1008	As typical in an RTK operation
Network RTK	Network Auxiliary Station Data	1014	
Network RTK	GPS Ionospheric Correction Difference	1015	1 Hz
Network RTK	GPS Geometric Correction Difference	1016	1 Hz

Table 3.1-4 Update Rate for Typical Operation

Group name		Message Type	Update rate
Observations (GPS)		1004	1 Hz
Station Description		1005 or 1006	As typical in an RTK operation
Antenna description		1007 or 1008	As typical in an RTK operation
Network RTK	Network Auxiliary Station Data	1014	
Network RTK	GPS Ionospheric Correction Difference	1015	Update completed every 10 seconds
Network RTK	GPS Geometric Correction Difference	1016	Update completed every 10 seconds

3.1.8 Handling of Quarter-Cycle Carrier Phase Shifts

Some GNSS receiver manufacturers have implemented carrier phase encoding of Version 3 messages 1001, 1002, 1003, 1004, 1009, 1010, 1011, 1012 in a way that carrier phase observations are in phase for all carrier phases of a specific frequency, i.e., they correct for quarter cycle phase shifts. Others retain the quarter cycle offset between the carrier phase observations in the data.

Table 3.1-5 gives the status of implementation as provided by the individual manufacturers.

Table 3.1-5 GNSS Receiver Manufacturers – Handling of Quarter-Cycle Phase Shifts

	Geo++	Javad	Leica	Ashtech	NavCom	NovAtel	Septen-trio	Trimble	Topcon
GPS L1CA	No correction	No correction	No correction	No correction	No correction	No correction	No correction	No correction	No correction
GPS L1P	No correction			0.25 cycles x λ_{L1}) ** added					
GPS L2P	No correction	No correction	No correction	No correction	No correction	No correction		No correction	No correction
GPS L2Y	No correction	No correction	No correction	No correction	No correction	No correction	No correction	No correction	No correction
GPS L2C	No correction	No correction	No correction	(0.25 cycles x λ_{L2}) ** added	No correction	No correction	No correction	(0.25 cycles x λ_{L2}) ** added	No correction
GLN L1CA	No correction	No correction	No correction	No correction	No correction	No correction	No correction	No correction	No correction
GLN L1P	No correction	No correction			No correction	No correction	No correction	0.25 cycles x λ_{L1}) ** subtracted	No correction
GLN L2P	No correction	No correction	No correction	No correction	No correction	No correction	No correction	No correction	No correction
GLN L2CA	No correction	No correction	No correction	(0.25 cycles x λ_{L2}) ** added	No correction	No correction	No correction	(0.25 cycles x λ_{L2}) ** added	No correction

** λ_{L1} = L1 wavelength and λ_{L2} = L2 wavelength

To avoid inconsistencies between different manufacturers a two-bit indicator was introduced in messages 1005 and 1006 indicating whether the carrier phases are corrected for satellite induced quarter-cycle phase shifts (provided carrier phase for different signals on the same frequency are available), or whether the carrier phases are not corrected. This indicator provides information only for Message Types 1001, 1002, 1003, 1004, 1009, 1010, 1011, 1012. If the Quarter Cycle carrier phase Indicator is zero, there is no information on the correction of quarter cycle shifts.

For Message Types 1001, 1002, 1003, 1004, 1009, 1010, 1011, 1012, each manufacturer shall maintain the phase relationship conventions specified in Table 3.1-5. This ensures backwards compatibility with legacy receivers that do not make use of the Quarter Cycle Indicator. Manufacturers not listed in this table shall not apply 1/4 cycle shifts. To augment the information in the Type 1033 message, the phase relationships shall be indicated using the Quarter Cycle Indicator in the Type 1005 and 1006 messages (see Table 3.5-12 and Table 3.5-13). However, the use of the Quarter Cycle Indicator shall not contradict the information provided in Table 3.1-5.

© RTCM – Not for reproduction or redistribution

For the network Message Types 1015-1017 it is necessary that in the case of mixed phase observations derived from a common frequency (e.g., L2C and L2P for GPS), the network software must ensure that the observations are processed with a consistent phase relationship.

Firmware updates for any manufacturer should include implementation of Message Type 1033 (to indicate the receiver type) and should also implement the definition of 2 bits in 1005 and 1006.

The phase alignments in the Multiple Signal Messages (Section 3.5.12) are explicitly specified by Table 3.5-81. The corrections in Table 3.1-5 are not to be applied to the MSM messages.

© RTCM – Not for reproduction or redistribution

3.2 Message Type Summary

Table 3.2-1 lists the currently standardized messages.

Note that messages must contain a whole number of bytes. If the number of bytes is fractional then zero-valued fill bits must be used to complete the last byte at the end of the message data before the CRC to maintain the last byte boundary. Thus, the total number of bytes must be the next full integer if fill bits are needed. For example, 55.125 computed bytes means 56 bytes total.

Table 3.2-1 Message Type Table

Message Type	Message Name	No. of Bytes **	Notes
1-100	Experimental Messages		
1001	L1-Only GPS RTK Observables	8.00+7.25*Ns	
1002	Extended L1-Only GPS RTK Observables	8.00+9.25*Ns	
1003	L1&L2 GPS RTK Observables	8.00+12.625*Ns	
1004	Extended L1&L2 GPS RTK Observables	8.00+15.625*Ns	
1005	Stationary RTK Reference Station ARP	19	
1006	Stationary RTK Reference Station ARP with Antenna Height	21	
1007	Antenna Descriptor	5-36	
1008	Antenna Descriptor & Serial Number	6-68	
1009	L1-Only GLONASS RTK Observables	7.625+8*Ns	
1010	Extended L1-Only GLONASS RTK Observables	7.625+9.875*Ns	
1011	L1&L2 GLONASS RTK Observables	7.625+13.375*Ns	
1012	Extended L1&L2 GLONASS RTK Observables	7.625+16.25*Ns	
1013	System Parameters	8.75+3.625*Nm	Nm = Number of Message Types Transmitted

© RTCM – Not for reproduction or redistribution

Message Type	Message Name	No. of Bytes **	Notes
1014	Network Auxiliary Station Data	14.625	
1015	GPS Ionospheric Correction Differences	9.5 + 3.5*Ns	
1016	GPS Geometric Correction Differences	9.5 + 4.5*Ns	
1017	GPS Combined Geometric and Ionospheric Correction Differences	9.5 + 6.625*Ns	
1018	RESERVED for Alternative Ionospheric Correction Difference Message		
1019	GPS Ephemerides	61	One message per satellite
1020	GLONASS Ephemerides	45	One message per satellite
1021	Helmert / Abridged Molodenski Transformation Parameters	51.5+N+M	N = Number of characters in Source Name M = Number of characters in Target Name
1022	Molodenski-Badekas Transformation Parameters	64.625+N+M	N = Number of characters in Source Name M = Number of characters in Target Name
1023	Residuals, Ellipsoidal Grid Representation	72.25	
1024	Residuals, Plane Grid Representation	73.75	
1025	Projection Parameters, Projection Types other than Lambert Conic Conformal (2 SP) and Oblique Mercator	24.5	
1026	Projection Parameters, Projection Type LCC2SP (Lambert Conic Conformal (2 SP))	29.25	
1027	Projection Parameters, Projection Type OM (Oblique Mercator)	32.25	
1028	*(Reserved for Global to Plate-Fixed Transformation)*		

© RTCM – Not for reproduction or redistribution

Message Type	Message Name	No. of Bytes **	Notes
1029	Unicode Text String	9+N	N = Number of UTF-8 Code Units (Octets)
1030	GPS Network RTK Residual Message	$7+6.125*N_S$	
1031	GLONASS Network RTK Residual Message	6.625+6.125*Ns	
1032	Physical Reference Station Position Message	19.5	
1033	Receiver and Antenna Descriptors	9+M+N+I+J+K	M = Number of characters in antenna serial number N = Number of characters in antenna descriptor I = Number of characters in receiver descriptor J = Number of characters in firmware descriptor K = Number of characters in receiver serial number
1034	GPS Network FKP Gradient	6.125+8.25*Ns	
1035	GLONASS Network FKP Gradient	5.75+8.25*Ns	
1037	GLONASS Ionospheric Correction Differences	9.125+3.5*Ns	
1038	GLONASS Geometric Correction Differences	9.125+4.5*Ns	
1039	GLONASS Combined Geometric and Ionospheric Correction Differences	9.125+6.625*Ns	
1041	NavIC/IRNSS Satellite Ephemeris Message	61	One message per satellite
1042	BDS Satellite Ephemeris Data	64	One message per satellite
1044	QZSS Ephemerides	61	One message per satellite
1045	Galileo F/NAV Satellite Ephemeris Data	62	One message per satellite
1046	Galileo I/NAV Satellite Ephemeris Data	63	One message per satellite
1057	SSR GPS Orbit Correction	8.5+16.875*Ns	
1058	SSR GPS Clock Correction	8.375+9.5*Ns	

© RTCM – Not for reproduction or redistribution

Message Type	Message Name	No. of Bytes **	Notes
1059	SSR GPS Code Bias	8.375+1.375*Ns +2.375 ∑ NCB	NCB = No. of Code Biases per individual Satellite
1060	SSR GPS Combined Orbit and Clock Corrections	8.5+25.625*Ns	
1061	SSR GPS URA	8.375+1.5*Ns	
1062	SSR GPS High-Rate Clock Correction	8.375+3.5*Ns	
1063	SSR GLONASS Orbit Correction	8.125+16.75*Ns	
1064	SSR GLONASS Clock Correction	8+9.375*Ns	
1065	SSR GLONASS Code Bias	8+1.250*Ns +2.375 ∑ NCB	NCB = No. of Code Biases per individual Satellite
1066	SSR GLONASS Combined Orbit and Clock Corrections	8.125+25.5*Ns	
1067	SSR GLONASS URA	8+1.375*Ns	
1068	SSR GLONASS High-Rate Clock Correction	8+3.375*Ns	
1070	Reserved MSM		
1071	GPS MSM1	See Table 3.5-79	
1072	GPS MSM2	See Table 3.5-79	
1073	GPS MSM3	See Table 3.5-79	
1074	GPS MSM4	See Table 3.5-79	
1075	GPS MSM5	See Table 3.5-79	
1076	GPS MSM6	See Table 3.5-79	
1077	GPS MSM7	See Table 3.5-79	
1078	Reserved MSM		
1079	Reserved MSM		

© RTCM – Not for reproduction or redistribution

Message Type	Message Name	No. of Bytes **	Notes
1080	Reserved MSM		
1081	GLONASS MSM1	See Table 3.5-79	
1082	GLONASS MSM2	See Table 3.5-79	
1083	GLONASS MSM3	See Table 3.5-79	
1084	GLONASS MSM4	See Table 3.5-79	
1085	GLONASS MSM5	See Table 3.5-79	
1086	GLONASS MSM6	See Table 3.5-79	
1087	GLONASS MSM7	See Table 3.5-79	
1088	Reserved MSM		
1089	Reserved MSM		
1090	Reserved MSM		
1091	Galileo MSM1	See Table 3.5-79	
1092	Galileo MSM2	See Table 3.5-79	
1093	Galileo MSM3	See Table 3.5-79	
1094	Galileo MSM4	See Table 3.5-79	
1095	Galileo MSM5	See Table 3.5-79	
1096	Galileo MSM6	See Table 3.5-79	
1097	Galileo MSM7	See Table 3.5-79	
1098	Reserved MSM		
1099	Reserved MSM		
1100	Reserved MSM		
1101	SBAS MSM1	See Table 3.5-79	

© RTCM – Not for reproduction or redistribution

Message Type	Message Name	No. of Bytes **	Notes
1102	SBAS MSM2	See Table 3.5-79	
1103	SBAS MSM3	See Table 3.5-79	
1104	SBAS MSM4	See Table 3.5-79	
1105	SBAS MSM5	See Table 3.5-79	
1106	SBAS MSM6	See Table 3.5-79	
1107	SBAS MSM7	See Table 3.5-79	
1108	Reserved MSM		
1109	Reserved MSM		
1110	Reserved MSM		
1111	QZSS MSM1	See Table 3.5-79	
1112	QZSS MSM2	See Table 3.5-79	
1113	QZSS MSM3	See Table 3.5-79	
1114	QZSS MSM4	See Table 3.5-79	
1115	QZSS MSM5	See Table 3.5-79	
1116	QZSS MSM6	See Table 3.5-79	
1117	QZSS MSM7	See Table 3.5-79	
1118	Reserved MSM		
1119	Reserved MSM		
1120	Reserved MSM		
1121	BeiDou MSM1	See Table 3.5-79	
1122	BeiDou MSM2	See Table 3.5-79	
1123	BeiDou MSM3	See Table 3.5-79	

© RTCM – Not for reproduction or redistribution

Message Type	Message Name	No. of Bytes **	Notes
1124	BeiDou MSM4	See Table 3.5-79	
1125	BeiDou MSM5	See Table 3.5-79	
1126	BeiDou MSM6	See Table 3.5-79	
1127	BeiDou MSM7	See Table 3.5-79	
1128	Reserved MSM		
1129	Reserved MSM		
1130	Reserved MSM		
1131	NavIC/IRNSS MSM1	See Table 3.5-79	
1132	NavIC/IRNSS MSM2	See Table 3.5-79	
1133	NavIC/IRNSS MSM3	See Table 3.5-79	
1134	NavIC/IRNSS MSM4	See Table 3.5-79	
1135	NavIC/IRNSS MSM5	See Table 3.5-79	
1136	NavIC/IRNSS MSM6	See Table 3.5-79	
1137	NavIC/IRNSS MSM7	See Table 3.5-79	
1138	Reserved MSM		
1139	Reserved MSM		
1140	Reserved MSM		
1141-1229	Reserved MSM		
1230	GLONASS L1 and L2 Code-Phase Biases	4+2*N	N = Number of Code-Phase Biases (max 4)
1300	Service-CRS	4.125 + 1*N	N = Number of characters in Service-CRS Name
1301	15 Parameter Transformation	45+N+M	N = Number of characters in Source CRS Name M = Number of characters in Target CRS Name

© RTCM – Not for reproduction or redistribution

Message Type	Message Name	No. of Bytes **	Notes
1302	RTCM-CRS	3.25 + 1*N + I*(0.625+1*M)	N = Number of characters in RTCM-CRS Name I = Number of Database-Links M = Number of characters resulting from I Database-Link Counters
1303	BDS Network RTK Residual Message	7+6.125*Ns	
1304	Galileo Network RTK Residual Message	7+6.125*Ns	
1305	QZSS Network RTK Residual Message	7+5.875*Ns	
4001-4095	Proprietary Messages		These message types are assigned to specific companies for the broadcast of proprietary information. See Section 3.6.

Ns = No. of Satellites

3.3 Data Types

The data types used are shown in Table 3.3-1.

Table 3.3-1 Data Type Table

Data Type	Description	Range	Data Type Notes
bit(n)	bit field	0 or 1, each bit	Reserved bits set to “0”
char8(n)	8-bit characters, ISO 8859-1 (not limited to ASCII)	character set	Reserved or unused characters: [0x00]
int8	8 bit 2’s complement integer	± 127	-128 indicates data not available
int9	9 bit 2’s complement integer	± 255	-256 indicates data not available
int10	10 bit 2’s complement integer	± 511	-512 indicates data not available
int14	14 bit 2’s complement integer	± 8191	-8192 indicates data not available

© RTCM – Not for reproduction or redistribution

Data Type	Description	Range	Data Type Notes
int15	15 bit 2's complement integer	± 16,383	-16,384 indicates data not available
int16	16 bit 2's complement integer	± 32,767	-32,768 indicates data not available
int17	17 bit 2's complement integer	± 65,535	-65,536 indicates data not available
int19	19 bit 2's complement integer	± 262,143	-262,144 indicates data not available
int20	20 bit 2's complement integer	± 524, 287	-524,288 indicates data not available
int21	21 bit 2's complement integer	± 1,048,575	-1,048,576 indicates data not available
int22	22 bit 2's complement integer	± 2,097,151	-2,097,152 indicates data not available
int23	23 bit 2's complement integer	± 4,194,303	-4,194,304 indicates data not available
int24	24 bit 2's complement integer	± 8,388,607	-8,388,608 indicates data not available
int25	25 bit 2's complement integer	± 16,777,215	-16,777,216 indicates data not available
int26	26 bit 2's complement integer	± 33,554,431	-33,554,432 indicates data not available
int27	27 bit 2's complement integer	±67,108,863	-67,108,864 indicates data not available
int30	30 bit 2's complement integer	± 536,870,911	-536,870,912 indicates data not available
int32	32 bit 2's complement integer	± 2,147,483,647	-2,147,483,648 indicates data not available
int34	34 bit 2's complement integer	± 8,589,934,591	-8,589,934,592 indicates data not available
int35	35 bit 2's complement integer	± 17,179,869,183	-17,179,869,184 indicates data not available
int38	38 bit 2's complement integer	± 137,438,953,471	-137,438,953,472 indicates data not available
uint2	2 bit unsigned integer	0 to 3	
uint3	3 bit unsigned integer	0 to 7	
uint4	4 bit unsigned integer	0 to 15	
uint5	5 bit unsigned integer	0 to 31	
uint6	6 bit unsigned integer	0 to 63	
uint7	7 bit unsigned integer	0 to 127	

© RTCM – Not for reproduction or redistribution

Data Type	Description	Range	Data Type Notes
uint8	8 bit unsigned integer	0 to 255	
uint9	9 bit unsigned integer	0 to 511	
uint10	10 bit unsigned integer	0 to 1023	
uint11	11 bit unsigned integer	0 to 2047	
uint12	12 bit unsigned integer	0 to 4095	
uint14	14 bit unsigned integer	0 to 16,383	
uint16	16 bit unsigned integer	0 to 65,535	
uint17	17 bit unsigned integer	0 to 131,071	
uint18	18 bit unsigned integer	0 to 262,143	
uint20	20 bit unsigned integer	0 to 1,048,575	
uint23	23 bit unsigned integer	0 to 8,388,607	
uint24	24 bit unsigned integer	0 to 16,777,215	
uint25	25 bit unsigned integer	0 to 33,554,431	
uint26	26 bit unsigned integer	0 to 67,108,863	
uint27	27 bit unsigned integer	0 to 134,217,727	
uint30	30 bit unsigned integer	0 to 1,073,741,823	
uint32	32 bit unsigned integer	0 to 4,294,967,295	
uint35	35 bit unsigned integer	0 to 34,359,738,367	
uint36	36 bit unsigned integer	0 to 68,719,476,735	
intS5	5 bit sign-magnitude integer	± 15	*See Note 1*
intS11	11 bit sign-magnitude integer	± 1023	*See Note 1*
intS22	22 bit sign-magnitude integer	± 2,097,151	*See Note 1*
intS24	24 bit sign-magnitude integer	± 8,388,607	*See Note 1*

© RTCM – Not for reproduction or redistribution

Data Type	Description	Range	Data Type Notes
intS27	27 bit sign-magnitude integer	± 67,108,863	*See Note 1*
intS32	32 bit sign-magnitude integer	± 2,147,483,647	*See Note 1*
utf8(N)	Unicode UTF-8 Code Unit	00h to FFh	8-bit value that contains all or part of a Unicode UTF-8 encoded character

Note 1. Sign-magnitude representation records the number's sign and magnitude. MSB is 0 for positive numbers and 1 for negative numbers. The rest of the bits are the number's magnitude. For example, for 8-bit words the representations of the numbers "-5" and "+5" in a binary form are 10000101 and 00000101, respectively. Negative zero is not used.

3.4 Data Fields

The data fields used are shown in Table 3.4-1. Each Data Field (DF) uses one of the Data Types of Table 3.3-1. Note that the Data Field ranges may be less than the maximum possible range allowed by the Data Type.

Table 3.4-1 Data Fields

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF001	Reserved			bit(n)	All bits in reserved bit fields shall be set to "0". Because reserved fields may be given concrete definitions in future versions of this standard, decoders should not rely on any reserved bit field having all zeros. If reserved bit fields are defined in the future, then all zero combinations of these bits will not have a meaningful value.
DF002	Message Number	0-4095		uint12	
DF003	Reference Station ID	0-4095		uint12	The Reference Station ID is determined by the service provider. Its primary purpose is to link all message data to their unique source. It is useful in distinguishing between desired and undesired data in cases where more than one service may be using the same data link frequency. It is

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					also useful in accommodating multiple reference stations within a single data link transmission. In reference network applications the Reference Station ID plays an important role, because it is the link between the observation messages of a specific reference station and its auxiliary information contained in other messages for proper operation. Thus, the Service Provider should ensure that the Reference Station ID is unique within the whole network, and that ID's should be reassigned only when absolutely necessary. Service Providers may need to coordinate their Reference Station ID assignments with other Service Providers in their region in order to avoid conflicts. This may be especially critical for equipment accessing multiple services, depending on their services and means of information distribution.
DF004	GPS Epoch Time (TOW)	0-604,799,999 ms	1 ms	uint30	GPS Epoch Time is provided in milliseconds from the beginning of the GPS week.
DF005	Synchronous GNSS Message Flag			bit(1)	0 - No further GNSS observables referenced to the same Epoch Time will be transmitted. This enables the receiver to begin processing the data immediately after decoding the message. 1 - The next message will contain observables of another GNSS source referenced to the same Epoch Time. Note: "Synchronous" here means that the measurements are taken within one microsecond of each other
DF006	No. of GPS Satellite Signals Processed	0-31		uint5	The Number of GPS Satellite Signals Processed refers to the number of satellites in the message. It does not necessarily equal the number of satellites visible to the Reference Station.
DF007	GPS Divergence-free Smoothing Indicator			bit(1)	0 - Divergence-free smoothing not used 1 - Divergence-free smoothing used

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF008	GPS Smoothing Interval	See Table 3.4-4		bit(3)	The GPS Smoothing Interval is the integration period over which reference station pseudorange code phase measurements are averaged using carrier phase information. Divergence-free smoothing may be continuous over the entire period the satellite is visible.
DF009	GPS Satellite ID	1-63 (See Table 3.4-3)		uint6	A GPS Satellite ID number from 1 to 32 refers to the PRN code of the GPS satellite. Satellite IDs higher than 32 are assigned to signals from Satellite-Based Augmentation Systems (SBASs). SBAS PRN codes from 120 to 138 are assigned Satellite IDs from 40 to 58, so that the SBAS PRN codes are derived from the Satellite ID code by adding 80. SBAS or other PRN codes larger than 138 shall not be encoded into this field. Due to this limitation new encoding implementations should use the MSM messages for SBAS. The SBAS MSM messages can encode the full range of SBAS PRNs (see field DF394).
DF010	GPS L1 Code Indicator			bit(1)	The GPS L1 Code Indicator identifies the code being tracked by the reference station. Civil receivers can track the C/A code, and optionally the P code, while military receivers can track C/A, and can also track P and Y code, whichever is broadcast by the satellite 0 - C/A Code 1 - P(Y) Code Direct
DF011	GPS L1 Pseudorange	0-299,792.46 m	0.02 m	uint24	The GPS L1 Pseudorange field provides the raw L1 pseudorange measurement at the reference station in meters, modulo one light-millisecond (299,792.458 meters). The GPS L1 pseudorange measurement is reconstructed by the user receiver from the L1 pseudorange field by: (GPS L1 pseudorange measurement) = (GPS L1 pseudorange field) modulo (299,792.458 m) + integer as determined from the user receiver's estimate of the reference station range, or as provided by the extended data set.

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					A field value greater than the valid field maximum E4 B937h (299,792.46 m) indicates that the L1 pseudorange is invalid. A field value of 8 0000h (10,485.76 m) also indicates that the L1 pseudorange is invalid. This value, within the valid range of the field, was erroneously designated as an invalid value in earlier versions of this standard. This designation cannot be changed without harming existing implementations. In the rare case of having to encode a pseudorange of 10,485.76 m implementations should add or subtract 0.02 m to shift the encoded value from the invalid value. For all applications using pseudoranges such a shift should be inconsequential. Implementations should not include invalid L1 pseudoranges when encoding measurements.
DF012	GPS L1 Phaserange – L1 Pseudorange	±262.1435 m (See Data Field Note)	0.0005 m	int20	The GPS L1 Phaserange – L1 Pseudorange field provides the information necessary to determine the L1 phase measurement. Note that the Phaserange defined here has the same sign as the pseudorange. The Phaserange has much higher resolution than the pseudorange, so that providing this field is just a numerical technique to reduce the length of the message. At start up and after each cycle slip, the initial ambiguity is reset and chosen so that the L1 Phaserange should match the L1 Pseudorange as closely as possible (i.e., within 1/2 L1 cycle) while not destroying the integer nature of the original carrier phase observation. The Full GPS L1 Phaserange is constructed as follows (all quantities in units of meters): (Full L1 Phaserange) = (L1 pseudorange as reconstructed from L1 pseudorange field) + (GPS L1 Phaserange – L1 Pseudorange field) Certain ionospheric conditions might cause the GPS L1 Phaserange – L1 Pseudorange to diverge over time across the range limits defined. Under these circumstances the computed value needs to be adjusted (rolled over) by the

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					equivalent of 1500 cycles to bring the value back within the range. A field value of 8 0000h indicates that the L1 Phaserange is invalid. Encoders setting the L1 pseudorange (DF011) invalid shall also set this field invalid. If the decoded L1 pseudorange (DF011) is invalid, then this field shall be considered invalid regardless of its value. See comments in sections 3.1.6 and 3.5.1 for the correction of antenna phase center variations in Network RTK applications.
DF013	GPS L1 Lock-Time Indicator	See Table 3.4-2		uint7	The GPS L1 Lock-Time Indicator provides a measure of the amount of time that has elapsed during which the Reference Station receiver has maintained continuous lock on that satellite signal. If a cycle slip occurs during the previous measurement cycle, the lock indicator will be reset to zero.
DF014	GPS Integer L1 Pseudorange Modulus Ambiguity	0-76,447,076.790 m	299,792.458 m	uint8	The GPS Integer L1 Pseudorange Modulus Ambiguity represents the integer number of full pseudorange modulus divisions (299,792.458 m) of the raw L1 pseudorange measurement.
DF015	GPS L1 CNR	0-63.75 dB-Hz	0.25 dB-Hz	uint8	The GPS L1 CNR measurements provide the reference station's estimate of the carrier-to-noise ratio of the satellite's signal in dB-Hz. 0 - the CNR measurement is not computed.
DF016	GPS L2 Code Indicator			bit(2)	The GPS L2 Code Indicator depicts which L2 code is processed by the reference station, and how it is processed. 0 - C/A or L2C code 1 - P(Y) code direct 2 - P(Y) code cross-correlated 3 - Correlated P/Y

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					The GPS L2 Code Indicator refers to the method used by the GPS reference station receiver to recover the L2 pseudorange. The GPS L2 Code Indicator should be set to "0" (C/A or L2C code) for any of the L2 civil codes. It is assumed here that a satellite will not transmit both C/A code and L2C code signals on L2 simultaneously, so that the reference station and user receivers will always utilize the same signal. The code indicator should be set to "1" if the satellite's signal is correlated directly, i.e., either P code or Y code depending whether anti-spoofing (AS) is switched off or on. The code indicator should be set to "2" when the reference station receiver L2 pseudorange measurement is derived by adding a cross-correlated pseudorange measurement (Y2-Y1) to the measured L1 C/A code. The code indicator should be set to "3" when the GPS reference station receiver is using a proprietary method that uses only the L2 P(Y) code signal to derive L2 pseudorange.
DF017	GPS L2-L1 Pseudorange Difference	±163.82 m (See Data Field Note)	0.02 m	int14	The GPS L2-L1 Pseudorange Difference field is utilized, rather than the full L2 pseudorange, to reduce the message length. The receiver must reconstruct the L2 pseudorange by using the following formula: (GPS L2 pseudorange measurement) = (GPS L1 pseudorange as reconstructed from L1 pseudorange field) + (GPS L2-L1 pseudorange field) A field value of 2000h indicates that the L2 Pseudorange is invalid. Encoders setting the L1 pseudorange (DF011) invalid shall also set this field invalid. If the decoded L1 pseudorange (DF011) is invalid, then this field shall be considered invalid regardless of its value.
DF018	GPS L2 Phaserange – L1 Pseudorange	±262.1435 m	0.0005 m	int20	The GPS L2 Phaserange - L1 Pseudorange field provides the information necessary to determine the L2 phase measurement. Note that the Phaserange defined here has the same sign as the pseudorange. The Phaserange has much

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
		(See Data Field Note)			higher resolution than the pseudorange, so that providing this field is just a numerical technique to reduce the length of the message. At start up and after each cycle slip, the initial ambiguity is reset and chosen so that the L2 Phaserange should match the L1 Pseudorange as closely as possible (i.e., within 1/2 L2 cycle) while not destroying the integer nature of the original carrier phase observation. The Full GPS L2 Phaserange is constructed as follows (all quantities in units of meters): (Full L2 Phaserange) = (L1 pseudorange as reconstructed from L1 pseudorange field) + (GPS L2 Phaserange – L1 Pseudorange field) Certain ionospheric conditions might cause the GPS L2 Phaserange – L1 Pseudorange to diverge over time across the range limits defined. Under these circumstances the computed value needs to be adjusted (rolled over) by the equivalent of 1500 cycles in order to bring the value back within the range. A field value of 8 0000h indicates that the L2 Phaserange is invalid. See also comments in sections 3.1.6 and 3.5.1 for correction of antenna phase center variations in Network RTK applications. Encoders setting the L1 pseudorange (DF011) invalid shall also set this field invalid. If the decoded L1 pseudorange (DF011) is invalid, then this field shall be considered invalid regardless of its value.
DF019	GPS L2 Lock-Time Indicator	See Table 3.4-2		uint7	The GPS L2 Lock-Time Indicator provides a measure of the amount of time that has elapsed during which the Reference Station receiver has maintained continuous lock on that satellite signal. If a cycle slip occurs during the previous measurement cycle, the lock indicator will be reset to zero.

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF020	GPS L2 CNR	0-63.75 dB-Hz	0.25 dB-Hz	uint8	The GPS L2 CNR measurements provide the reference station's estimate of the carrier-to-noise ratio of the satellite's signal in dB-Hz. 0 - the CNR measurement is not computed
DF021	Reserved			bit(6)	
DF022	GPS Indicator			bit(1)	0 - No GPS service supported 1 - GPS service supported
DF023	GLONASS Indicator			bit(1)	0 - No GLONASS service supported 1 - GLONASS service supported
DF024	Galileo Indicator			bit(1)	0 - No Galileo service supported 1 - Galileo service supported
DF025	Antenna Ref. Point ECEF-X	±13,743,895.3471 m	0.0001 m	int38	The antenna reference point X-coordinate.
DF026	Antenna Ref. Point ECEF-Y	±13,743,895.3471 m	0.0001 m	int38	The antenna reference point Y-coordinate.
DF027	Antenna Ref. Point ECEF-Z	±13,743,895.3471 m	0.0001 m	int38	The antenna reference point Z-coordinate.
DF028	Antenna Height	0-6.5535 m	0.0001 m	uint16	The Antenna Height field provides the height of the Antenna Reference Point above the marker used in the survey campaign.
DF029	Descriptor Counter	0-31		uint8	The Descriptor Counter defines the number of characters (bytes) to follow in DF030, Antenna Descriptor
DF030	Antenna Descriptor			char8 (n)	Alphanumeric characters. IGS limits the number of characters to 20 at this time, but this DF allows more characters for future extension.
DF031	Antenna Setup ID	0-255		uint8	0 - Use standard IGS Model 1-255 - Specific Antenna Setup ID#

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					The Antenna Setup ID is a parameter for use by the service provider to indicate the particular reference station-antenna combination. The number should be increased whenever a change occurs at the station that affects the antenna phase-center variations. While the Antenna Descriptor and the Antenna Serial Number give an indication of when the installed antenna has been changed, it is envisioned that other changes could occur. For instance, the antenna may have been repaired, or the surrounding of the antenna may have been changed and the provider of the service may want to make the user station aware of the change. Depending on the change of the phase center variations due to a setup change, a change in the Antenna Setup ID would mean that the user should check with the service provider to see if the antenna phase center variation in use is still valid. Of course, the provider must make appropriate information available to the users.
DF032	Serial Number Counter	0-31		uint8	The Serial Number Counter defines the number of characters (bytes) to follow in Antenna Serial Number
DF033	Antenna Serial Number			char8 (n)	Alphanumeric characters. The Antenna Serial Number is the individual antenna serial number as issued by the manufacturer of the antenna. A possible duplication of the Antenna Serial Number is not possible, because together with the Antenna Descriptor only one antenna with the particular number will be available. In order to avoid confusion, the Antenna Serial Number should be omitted when the record is used together with reverse reduction to model-type calibration values, because it cannot be allocated to a real physical antenna.
DF034	GLONASS Epoch Time (t_k)	0-86,399,999ms	1 ms	uint27	GLONASS Epoch Time of measurement is defined by the GLONASS ICD as UTC(SU) + 3.0 hours. The addition of positive leap second to GLONASS system time may happen at 10,800,000 epoch. In this case, two different epochs of UTC time shall be mapped into the same GLONASS epoch (e.g., UTC epochs 23h59m60s and

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					00h00m00s are mapped into 10,800,000 GLONASS epoch), i.e., during the period of leap-second insertion and one second after it this data field shall utilize the same values to refer to epochs both during and after the leap second. For example, at a 2-Hz update rate the epoch sequence during the addition of a positive leap second would be: …, [10,799,000], [10,799,500], [10,800,000], [10,800,500], [10,800,000], [10,800,500], [10,801,000], [10,801,500]… Clients (e.g., rover receivers) can identify correct epochs by means of additional information or computation.
DF035	No. of GLONASS Satellite Signals Processed	0-31	1	uint5	The Number of GLONASS Satellite Signals Processed refers to the number of satellites in the message. It does not necessarily equal the number of satellites visible to the Reference Station.
DF036	GLONASS Divergence-free Smoothing Indicator			bit(1)	0 - Divergence-free smoothing not used 1 - Divergence-free smoothing used
DF037	GLONASS Smoothing Interval	See Table 3.4-4		bit(3)	The GLONASS Smoothing Interval is the integration period over which reference station pseudorange code phase measurements are averaged using carrier phase information. Divergence-free smoothing may be continuous over the entire period the satellite is visible.
DF038	GLONASS Satellite ID (Satellite Slot Number)	0-63 (See Table 3.4-3)		uint6	A GLONASS Satellite ID number from 1 to 24 refers to the slot number of the GLONASS satellite. A Satellite ID of zero indicates that the slot number is unknown. Satellite IDs higher than 32 are assigned for signals from Satellite-Based Augmentation Systems (SBASs). SBAS PRN codes from 120 to 138 are assigned Satellite IDs from 40 to 58, so that the SBAS PRN codes are derived from the Satellite ID code by adding 80. SBAS or other PRN codes larger than 138 shall not be encoded into this field. Due to this limitation new encoding implementations should use

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					the MSM messages for SBAS. The SBAS MSM messages can encode the full range of SBAS PRNs (see field DF394). Note: Until the correct Satellite Slot Number is determined by means of decoding the GLONASS-M navigation data word "n" or by any other method, a zero value for DF038 may be used in message types 1009-1012 and 1020. In these messages the Satellite Frequency Channel (field DF040) allows for unambiguous identification of the satellite.
DF039	GLONASS L1 Code Indicator			bit(1)	0 - C/A Code; 1 - P Code
DF040	GLONASS Satellite Frequency Channel Number	0-20 (See Table 3.4-5)	1	uint5	The GLONASS Satellite Frequency Channel Number identifies the frequency of the GLONASS satellite. By providing both the Slot ID and the Frequency Code of the satellite, the user instantly knows the frequency of the satellite without requiring an almanac. 0 - –07 1 - –06 … 19 - +12 20 - +13
DF041	GLONASS L1 Pseudorange	0-599,584.92 m	0.02 m	uint25	The GLONASS L1 Pseudorange field provides the raw L1 pseudorange measurement at the reference station in meters, modulo two light-milliseconds (599,584.916 meters). The L1 pseudorange measurement is reconstructed by the user receiver from the L1 pseudorange field by: (L1 pseudorange measurement) = (L1 pseudorange field) modulo (599,584.916 m) + integer as determined from the user receiver's estimate of the reference station range, or as provided by the extended data set.

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF042	GLONASS L1 Phaserange – L1 Pseudorange	±262.1435 m (See Data Field Note)	0.0005 m	int20	The GLONASS L1 Phaserange – L1 Pseudorange field provides the information necessary to determine the L1 phase measurement. Note that the Phaserange defined here has the same sign as the pseudorange. The Phaserange has much higher resolution than the pseudorange, so that providing this field is just a numerical technique to reduce the length of the message. At start up and after each cycle slip, the initial ambiguity is reset and chosen so that the L1 Phaserange should match the L1 Pseudorange as closely as possible (i.e., within 1/2 L1 cycle) while not destroying the integer nature of the original carrier phase observation. The Full GLONASS L1 Phaserange is constructed as follows (all quantities in units of meters): (Full L1 Phaserange) = (L1 pseudorange as reconstructed from L1 pseudorange field) + (GLONASS L1 Phaserange – GLONASS L1 Pseudorange field) Certain ionospheric conditions might cause the GLONASS L1 Phaserange – L1 Pseudorange to diverge over time across the range limits defined. Under these circumstances the computed value needs to be adjusted (rolled over) by the equivalent of 1500 cycles in order to bring the value back within the range. 80000h - invalid carrier phase measurement. This indication may be used at low signal levels where carrier tracking is temporarily lost, but code tracking is still possible.
DF043	GLONASS L1 Lock-Time Indicator	See Table 3.4-2		uint7	The GLONASS L1 Lock-Time Indicator provides a measure of the amount of time that has elapsed during which the Reference Station receiver has maintained continuous lock on that satellite signal. If a cycle slip occurs during the previous measurement cycle, the lock indicator will be reset to zero.

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF044	GLONASS Integer L1 Pseudorange Modulus Ambiguity	0-76,147,284.332 m	599,584.916 m	uint7	The GLONASS Integer L1 Pseudorange Modulus Ambiguity represents the integer number of full pseudorange modulus divisions (599,584.916 m) of the raw L1 pseudorange measurement
DF045	GLONASS L1 CNR	0-63.75 dB-Hz	0.25 dB-Hz	uint8	The GLONASS L1 CNR measurements provide the reference station's estimate of the carrier-to-noise ratio of the satellite's signal in dB-Hz. 0 - the CNR measurement is not computed.
DF046	GLONASS L2 Code Indicator			bit(2)	The GLONASS L2 Code Indicator depicts which L2 code is processed by the reference station. 0 - C/A code 1- P code 2 - Reserved 3 - Reserved
DF047	GLONASS L2-L1 Pseudorange Difference	±163.82 m (See Data Field Note)	0.02 m	int14	The GLONASS L2-L1 Pseudorange Difference field is utilized, rather than the full L2 pseudorange, in order to reduce the message length. The receiver must reconstruct the L2 pseudorange by using the following formula: (GLONASS L2 pseudorange measurement) = (L1 pseudorange as reconstructed from L1 pseudorange field) + (L2-L1 pseudorange field) 2000h (-163.84) – there is no valid L2 code available, or the value exceeds the allowed range.
DF048	GLONASS L2 Phaserange – L1 Pseudorange	±262.1435 m (See Data Field Note)	0.0005 m	int20	The GLONASS L2 Phaserange - L1 Pseudorange field provides the information necessary to determine the L2 phase measurement. Note that the Phaserange defined here has the same sign as the pseudorange. The Phaserange has much higher resolution than the pseudorange, so that providing this field is just a numerical technique to reduce the length of the message. At start up and after each cycle slip, the initial ambiguity is reset and chosen so that the L2 Phaserange should match the L1 Pseudorange as closely as

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					possible (i.e., within 1/2 L2 cycle) while not destroying the integer nature of the original carrier phase observation. The Full GLONASS L2 Phaserange is constructed as follows (all quantities in units of meters): (Full L2 Phaserange) = (L1 pseudorange as reconstructed from L1 pseudorange field) + (GLONASS L2 Phaserange – L1 Pseudorange field) Certain ionospheric conditions might cause the GLONASS L2 Phaserange – L1 Pseudorange to diverge over time across the range limits defined. Under these circumstances the computed value needs to be adjusted (rolled over) by the equivalent of 1500 cycles to bring the value back within the range. 80000h - invalid carrier phase measurement. This indication may be used at low signal levels where carrier tracking is temporarily lost, but code tracking is still possible.
DF049	GLONASS L2 Lock-Time Indicator	See Table 3.4-2		uint7	The GLONASS L2 Lock-Time Indicator provides a measure of the amount of time that has elapsed during which the Reference Station receiver has maintained continuous lock on that satellite signal. If a cycle slip occurs during the previous measurement cycle, the lock indicator will be reset to zero.
DF050	GLONASS L2 CNR	0-63.75 dB-Hz	0.25 dB-Hz	uint8	The GLONASS L2 CNR measurements provide the reference station's estimate of the carrier-to-noise ratio of the satellite's signal in dB-Hz. 0 – The CNR measurement is not computed.
DF051	Modified Julian Day (MJD) Number	0-65,535 d	1 d	uint16	Modified Julian Day number (MJD) is the continuous count of day numbers since November 17, 1858 midnight. For example, the first day in GPS week 0 has MJD 44244. The full MJD number shall always be transmitted. At this point in time the rollover of the MJD is quite far away, but experience with the Y2K problem showed that the actual life of software and applications can be considerably longer

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					than expected. Therefore, it is foreseen to have a rollover of the MJD in calendar year 2038. At day 65,536 MJD the counter will start at 0 again.
DF052	Seconds of Day (UTC)	0-86,400 s	1 s	uint17	Seconds Of Day (UTC) are the seconds of the day counted from midnight Greenwich time. GPS seconds of week have to be adjusted for the appropriate number of leap seconds. 86,400 - a leap second has been issued.
DF053	Number of Message ID Announcements to Follow (N_m)	0-31	1	uint5	The Number of Message ID Announcements to follow informs the receiver of the number of message types and the frequency of their broadcast by the reference station. When the value is zero, it implies that either no message ID announcements are present, or that more than 31 message ID announcements follow. The actual number of message ID announcements present in a Type 1013 can be determined by considering the message length field found in the frame structure, using the formula: Actual number of message ID = floor((8*Message_Length - 70) / 29)
DF054	Leap Seconds, GPS-UTC	0-254 s	1 s	uint8	See the GPS/SPS Signal Specification, available from the U.S. Coast Guard Navigation Information Service. 255 - the value is not provided.
DF055	Message ID	0-4095	1	uint12	Each announcement lists the Message ID as transmitted by the reference station.
DF056	Message Sync Flag			bit(1)	0 - Asynchronous – not transmitted on a regular basis 1 - Synchronous – scheduled for transmission at regular intervals
DF057	Message Transmission Interval	0-6,553.5 s	0.1 s	uint16	Each announcement lists the Message Transmission Interval as transmitted by the reference station. If asynchronous, the transmission interval is approximate. If the transmission interval is not known the value of zero shall be sent.

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF058	Number of Auxiliary Stations Transmitted	0-31	1	uint5	Number of Auxiliary Reference Stations Transmitted in conjunction with designated Master Reference Station. Defines the number of different messages received of one type.
DF059	Network ID	0-255	1	uint8	Network ID defines the network and the source of the particular set of reference stations that their observation information belongs to. The service provider should ensure that the Network ID is unique in the region serviced. The Network ID indicates an area and its reference stations where the service providers will provide a homogenous solution with leveled integer ambiguities between its reference stations. In general, the area indicated by Network ID will comprise one subnetwork with a unique Subnetwork ID. (See description on how to use Network IDs and Subnetwork IDs in Section 3.5.6.).
DF060	Master Reference Station ID	0-4095	1	uint12	The Master Reference Station must have the identical ID as one of the reference stations used within the same data stream for providing observation or correction information. The Master Auxiliary Concept allows in principle for several Master Reference Stations in the same data stream. Every Master Reference Station would require a separate raw observation message transmitted for the identical reference station. However, for the current version of the standard it is recommended to have only one Master Reference Station in a data stream (see also Section 3.1.5).
DF061	Auxiliary Reference Station ID	0-4095	1	uint12	Station ID to identify Auxiliary Reference Station used to derive attached information.
DF062	Aux-Master Delta Latitude	±13.1071 °	25 x 10^{-6} °	int20	Delta value in latitude of Antenna Reference Point of "Auxiliary Reference Station minus Master Reference Station" in geographical coordinates based on GRS80 ellipsoid parameters for the same ECEF system as used in message 1005 or 1006 within the same data stream. Note: In severe ionospheric conditions it may not be

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					possible to provide complete service over the entire allowed range, because Ionospheric Correction Differences may exceed the allowed range of DF069.
DF063	Aux-Master Delta Longitude	±26.2142 °	25 x 10^{-6} °	int21	Delta value in longitude of Antenna Reference Point of "Auxiliary Reference Station minus Master Reference Station" in geographical coordinates based on GRS80 ellipsoid parameters for the same ECEF system as used in message 1005 or 1006 within the same data stream. Note: In severe ionospheric conditions it may not be possible to provide complete service over the entire allowed range, because Ionospheric Correction Differences may exceed the allowed range of DF069.
DF064	Aux-Master Delta Height	±4194.303 m	1 mm	int23	Delta value in ellipsoidal height of Antenna Reference Point of "Auxiliary Reference Station minus Master Reference Station" in geographical coordinates based on GRS80 ellipsoid parameters for the same ECEF system as used in message 1005 or 1006 within the same data stream. Note: In severe ionospheric conditions it may not be possible to provide complete service over the entire allowed range, because Ionospheric Correction Differences may exceed the allowed range of DF069.
DF065	GPS Epoch Time (GPS TOW)	0-603,799.9 s	0.1 s	uint23	Epoch time of observations used to derive correction differences
DF066	Multiple Message Indicator	0-1	1	bit(1)	1 - Messages with the same Message Number and Epoch Time will be transmitted in sequence. 0 - Last message of a sequence.
DF067	Number of GPS Satellites	0-15	1	uint4	Number of correction differences for GPS satellites contained in message. Only one message per Auxiliary-Master Reference Station pair and Epoch Time is allowed. Each message shall contain respective correction differences for all satellites tracked at the relevant Master-Auxiliary Reference Station combination

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF068	GPS Satellite ID	1-32	1	uint6	
DF069	GPS Ionospheric Carrier Phase Correction Difference	±32.767 m	0.5 mm	int17	Ionospheric Carrier Phase Correction Difference (ICPCD) is the Correction Difference for the ionospheric part calculated based on integer-leveled L1 and L2 correction differences (L1CD and L2CD). $ICPCD = \frac{f_2^2}{f_2^2 - f_1^2} L1CD - \frac{f_2^2}{f_2^2 - f_1^2} L2CD$ L1CD, L2CD, and ICPCD are presented in meters. (See discussion of L1 and L2 corrections in Section 3.5.6.) In extreme conditions the value of this field may lie outside the specified range. If this happens, the data block for the specific satellite containing this field (Table 3.5-19 and Table 3.5-21) should not be transmitted.
DF070	GPS Geometric Carrier Phase Correction Difference	±32.767 m	0.5 mm	int17	Geometric Carrier Phase Correction Difference (GCPCD) is the Correction Difference for the geometric part calculated based on integer-leveled L1 and L2 correction differences (L1CD and L2CD). $GCPCD = \frac{f_1^2}{f_1^2 - f_2^2} L1CD - \frac{f_2^2}{f_1^2 - f_2^2} L2CD$ L1CD, L2CD, and ICPCD are presented in meters. (See discussion of L1 and L2 corrections in Section 3.5.6.)
DF071	GPS IODE	0-255	1	bit(8)	IODE value of broadcast ephemeris used for calculation of Correction Differences. See GPS SPS Signal Spec, 2.4.4.2
DF072	Subnetwork ID	0-15		uint4	Subnetwork ID identifies the subnetwork of a network identified by Network ID. In general, the area indicated by Network ID will consist of one subnetwork. The Subnetwork ID indicates the actual solution number of Integer Ambiguity Level (see the description of Integer Ambiguity Level in Section 3.5.6). If one network has only

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					one subnetwork, this indicates that an ambiguity level throughout the whole network is established. In case of problems, it might not be possible to have one homogenous integer ambiguity-leveled solution throughout the whole network. The solution might break up into several homogeneous solutions, which can be indicated and distinguished by separate Subnetwork IDs. Every independent homogeneous integer ambiguity-leveled solution needs to have an independent Subnetwork ID. Master Reference Stations with different Subnetwork IDs indicate that no hand-over from one to another Master Reference Station is possible since the solutions are not consistent and have no common stations. (See description on how to use Network IDs and Subnetwork IDs in Section 3.5.6. or Annex A.1.) Note: Subnetwork ID's greater than "0" should be utilized only if the associated messages for Master Reference Station observations (1001 through 1004 for GPS or 1009 through 1012 for GLONASS) are brought to the same Ambiguity Level. It is recommended that this field be set to "0" for now. In the future a Subnetwork ID of "0" will indicate that corresponding raw data streams are not ambiguity leveled. Note: For RTCM 10403 of the standard, only one Master Reference Station with its associated Auxiliary Stations should be used in a single data stream. The result of this restriction is that Subnetwork ID's may not be needed. Future versions are expected to support Subnetwork ID's.
DF073	RESERVED for Provider ID	0-255		uint8	Unique ID identifying a service provider for a region. Service providers have to make sure that they are using a unique ID that is not used by another service provider in the region.
DF074	GPS Ambiguity Status Flag	0-3		bit(2)	0 - Reserved for future use (artificial observations) 1 - Correct Integer Ambiguity Level for L1 and L2

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					2 - Correct Integer Ambiguity Level for L1-L2 widelane 3 - Uncertain Integer Ambiguity Level. Only a likely guess is used. (See the description of Correct Integer Ambiguity Level and Ambiguity Status Flag in Section 3.5.6.)
DF075	GPS Non-Sync Count	0-7		uint3	Whenever an unrecoverable cycle slip occurs this count shall be increased. The counter shall not be increased more than once per minute. (See the discussion of cycle slips and ambiguity levels in Section 3.5.6)
DF076	GPS Week number	0-1023 weeks	1 week	uint10	Roll-over every 1024 weeks starting from Midnight on the night of January 5/Morning January 6, 1980
DF077	GPS SV Acc. (URA)		N/A	bit(4)	Units: meters; see GPS SPS Signal Spec, 2.4.3.2
DF078	GPS CODE ON L2	0-3	1	bit(2)	00 - Reserved 01 - P code ON 10 - C/A code ON 11 - L2C ON
DF079	GPS IDOT	*See Note 1*	2^{-43} semi-circles/s	int14	
DF080	Reserved				
DF081	GPS t_{oc}	0-604,784 s	2^{4} s	uint16	
DF082	GPS a_{f2}	*See Note 1*	2^{-55} s/s^{2}	int8	
DF083	GPS a_{f1}	*See Note 1*	2^{-43} s/s	int16	
DF084	GPS a_{f0}	*See Note 1*	2^{-31} s	int22	
DF085	GPS IODC	0-1023	1	uint10	The 8 LSB's of IODC contain the same bits and sequence as those in IODE; see GPS SPS Signal Spec, 2.4.3.4.
DF086	GPS C_{rs}	*See Note 1*	2^{-5} m	int16	

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF087	GPS Δn (DELTA n)	*See Note 1*	2^{-43} semi-circles/s	int16	
DF088	GPS M_0	*See Note 1*	2^{-31} semi-circles	int32	
DF089	GPS C_{uc}	*See Note 1*	2^{-29} rad	int16	
DF090	GPS Eccentricity (e)	0-0.03	2^{-33}	uint32	
DF091	GPS C_{us}	*See Note 1*	2^{-29} rad	int16	
DF092	GPS $A^{1/2}$	*See Note 1*	2^{-19} $m^{1/2}$	uint32	
DF093	GPS t_{oe}	0-604,784 s	2^4 s	uint16	
DF094	GPS C_{ic}	*See Note 1*	2^{-29} rad	int16	
DF095	GPS Ω_0 (OMEGA)$_0$	*See Note 1*	2^{-31} semi-circles	int32	
DF096	GPS C_{is}	*See Note 1*	2^{-29} rad	int16	
DF097	GPS i_0	*See Note 1*	2^{-31} semi-circles	int32	
DF098	GPS C_{rc}	*See Note 1*	2^{-5} m	int16	
DF099	GPS ω (Argument of Perigee)	*See Note 1*	2^{-31} semi-circles	int32	
DF100	GPS OMEGADOT (Rate of Right Ascension)	*See Note 1*	2^{-43} semi-circles/s	int24	
DF101	GPS t_{GD}	*See Note 1*	2^{-31} s	int8	

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF102	GPS SV HEALTH	See GPS SPS Signal Spec, 2.4.5.3	1	uint6	MSB: 0 - all NAV data are OK; 1 - some or all NAV data are bad. See GPS SPS Signal Spec, 2.4.3.3.
DF103	GPS L2 P data flag	Subframe 1, Word 4, Bit 1	1	bit(1)	0 - L2 P-Code NAV data ON 1 - L2 P-Code NAV data OFF
DF104	GLONASS almanac health			bit(1)	C_n word
DF105	GLONASS almanac health availability indicator			bit(1)	0 - GLONASS almanac health is not available; 1 - GLONASS almanac health is available;
DF106	GLONASS P1			bit(2)	GLONASS P1 word
DF107	GLONASS t_k	bits 11-7: 0-23 bits 6-1: 0-59 bit 0 (LSB): 0-1		bit(12)	Time referenced to the beginning of GLONASS subframe within the current day. The integer number of hours elapsed since the beginning of the current day occupies 5 MSB. The integer number of minutes occupies the next six bits. The number of thirty-second intervals occupies the LSB.
DF108	GLONASS MSb of B_n word			bit(1)	The ephemeris health flag.
DF109	GLONASS P2			bit(1)	
DF110	GLONASS t_b	15-1425 min	15 min	uint7	Time to which GLONASS navigation data are referenced.
DF111	GLONASS $x_n(t_b)$, first derivative	±4.3 km/s	2^{-20} km/s	intS24	GLONASS ECEF-X component of satellite velocity vector in PZ-90 datum
DF112	GLONASS $x_n(t_b)$	±27000 km	2^{-11} km	intS27	GLONASS ECEF-X component of satellite coordinates in PZ-90 datum
DF113	GLONASS $x_n(t_b)$, second derivative	$\pm 6.2*10^{-9}$ km/s	2^{-30} km/s^2	intS5	GLONASS ECEF-X component of satellite acceleration in PZ-90 datum

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF114	GLONASS $y_n(t_b)$, first derivative	±4.3 km/s	2^{-20} km/s	intS24	GLONASS ECEF-Y component of satellite velocity vector in PZ-90 datum
DF115	GLONASS $y_n(t_b)$	±27000 km	2^{-11} km	intS27	GLONASS ECEF-Y component of satellite coordinates in PZ-90 datum
DF116	GLONASS $y_n(t_b)$, second derivative	$\pm 6.2*10^{-9}$ km/s	2^{-30} km/s^2	intS5	GLONASS ECEF-Y component of satellite acceleration in PZ-90 datum
DF117	GLONASS $z_n(t_b)$, first derivative	±4.3 km/s	2^{-20} km/s	intS24	GLONASS ECEF-Z component of satellite velocity vector in PZ-90 datum
DF118	GLONASS $z_n(t_b)$	±27000 km	2^{-11} km	intS27	GLONASS ECEF-Z component of satellite coordinates in PZ-90 datum
DF119	GLONASS $z_n(t_b)$, second derivative	$\pm 6.2*10^{-9}$ km/s	2^{-30} km/s^2	intS5	GLONASS ECEF-Z component of satellite acceleration in PZ-90 datum
DF120	GLONASS P3			bit(1)	
DF121	GLONASS $\gamma_n(t_b)$	$\pm 2^{-30}$	2^{-40}	intS11	GLONASS relative deviation of predicted satellite carrier frequency from nominal value
DF122	GLONASS-M P	0-3		bit(2)	
DF123	GLONASS-M l_n (third string)			bit(1)	GLONASS-M l_n word extracted from third string of the subframe
DF124	GLONASS $\tau_n(t_b)$	$\pm 2^{-9}$ s	2^{-30} s	intS22	GLONASS correction to the satellite time relative to GLONASS system time
DF125	GLONASS-M $\Delta\tau_n$	$\pm 13.97*10^{-9}$ s	2^{-30} s	intS5	GLONASS time difference between navigation RF signal transmitted in L2 sub-band and navigation RF signal transmitted in L1 sub-band
DF126	GLONASS E_n	0-31 d	1 d	uint5	The age of GLONASS navigation data
DF127	GLONASS-M P4			bit(1)	

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF128	GLONASS-M F_T	0-15		uint4	GLONASS-M predicted satellite user range accuracy at time t_b
DF129	GLONASS-M N_T	1-1461 d	1 d	uint11	GLONASS calendar number of day within four-year interval starting from the 1-st of January in a leap year. Note. For GLONASS satellites this data field (if it is not equal to zero) may contain computed calendar number of day that corresponds to the parameter t_b.
DF130	GLONASS-M M	0-3		bit(2)	Type of GLONASS satellite. 01 - GLONASS-M satellite type; All GLONASS-M data fields are valid. 00 – Not GLONASS-M satellite type; All GLONASS-M data fields are invalid, thus they may contain arbitrary values.
DF131	GLONASS The Availability of Additional Data			bit(1)	The rest of the parameters of GLONASS ephemeris message contain data (data fields DF132-DF136) extracted from the fifth string of the subframe. These parameters do not belong to predefined ephemeris data; nevertheless they can be useful for positioning and timing. 1 - The additional parameters are in the message. 0 - DF132-DF136 may contain arbitrary values.
DF132	GLONASS N^A	1-1461 d	1 d	uint11	GLONASS calendar number of day within the four-year period to which τ_c is referenced.
DF133	GLONASS τ_c	$\pm(1-2^{-31})$ s	2^{-31} s	intS32	Difference between GLONASS system time and UTC(SU). This parameter is referenced to the beginning of day N^A.
DF134	GLONASS-M N_4	1-31	4 yr interval	uint5	GLONASS four-year interval number starting from 1996
DF135	GLONASS-M τ_{GPS}	$\pm 1.9*10^{-3}$ s	2^{-30} s	intS22	Correction to GPS system time relative to GLONASS system time.
DF136	GLONASS-M l_n (fifth string)			bit(1)	GLONASS-M l_n word extracted from fifth string of the subframe

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF137	GPS Fit Interval	Subframe 2, Word 10, Bit 17	1	bit(1)	0 - curve-fit interval is 4 hours 1 - curve-fit is greater than 4 hours
DF138	Number of Characters to Follow	0-127 characters	1 character	uint7	Provides the number of fully formed Unicode characters in the message text. It is not necessarily the number of bytes in the string.
DF139	Number of UTF-8 Code Units	0-255 bytes	1 byte	uint8	
DF140	UTF-8 Character Code Units			utf8(n)	
DF141	Reference-Station Indicator			bit(1)	0 - Real, Physical Reference Station 1 - Non-Physical or Computed Reference Station Note: A Non-Physical or Computed Reference Station is typically calculated based on information from a network of reference stations. Different approaches have been established over the years. The Non-Physical or Computed Reference Stations are sometimes trademarked and may not be compatible. Examples of these names are “Virtual Reference Stations”, “Pseudo-Reference Stations”, and “Individualized Reference Stations”.
DF142	Single Receiver Oscillator Indicator			bit(1)	0 - All raw data observations in messages 1001-1004 and 1009-1012 may be measured at different instants. This indicator should be set to “0” unless all the conditions for “1” are clearly met. 1 - All raw data observations in messages 1001-1004 and 1009-1012 are measured at the same instant, as described in Section 3.1.4.
DF143	Source-Name Counter	0–31		uint5	The Source-Name Counter defines the number of characters (bytes) to follow in Source-Name
DF144	Source Name			char8(N)	Name of source coordinate-system. If available, the EPSG identification code for the CRS has to be used. Otherwise, service providers should try to introduce unknown CRSs

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					into the EPSG database or could use other reasonable names.
DF145	Target-Name Counter	0–31		uint5	The Target-Name Counter defines the number of characters (bytes) to follow in Target-Name
DF146	Target-Name			char8(N)	Name of target coordinate-system. If available, the EPSG identification code for the CRS has to be used. Otherwise, service providers should try to introduce unknown CRSs into the EPSG database or could use other reasonable names.
DF147	System Identification Number	0–255		uint8	A unique system identification number has to be used for all messages related to the same sets of CRSs. This is necessary if transformation information for more than one set of CRSs should be transferred within one data stream.
DF148	Utilized Transformation Message Indicator			bit(10)	This data field says which messages are assigned to Transformation messages for the system identification number mentioned under DF147. (0 – Message not utilized, 1 - Message utilized) bit 0 (LSB) – Message 1023 bit 1 – Message 1024 bit 2 – Message 1025 bit 3 – Message 1026 bit 4 – Message 1027 bits 5-9 – Reserved, must be 0
DF149	Plate Number	0–31		uint5	0: unknown plate 1: AFRC - Africa 2: ANTA - Antarctica 3: ARAB - Arabia 4: AUST - Australia 5: CARB - Caribbean 6: COCO - Cocos 7: EURA - Eurasia 8: INDI - India

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					9: NOAM - N. America 10: NAZC - Nazca 11: PCFC - Pacific 12: SOAM - S. America 13: JUFU - Juan de Fuca 14: PHIL - Philippine 15: RIVR - Rivera 16: SCOT - Scotia 17 to 31: Reserved
DF150	Computation Indicator	0–15		uint4	Transformation method to be used: 0 - standard seven parameter, approximation 1 - standard seven parameter, strict formula 2 - Molodenski, abridged 3 - Molodenski-Badekas 4 to 15: Reserved
DF151	Height Indicator	0–3		uint2	0 - Geometric heights result 1 - Physical heights result If physical heights are derived via Helmert/Molodenski transformation: $H = h_{Target} - \text{Mean } \Delta H + \Delta H \text{ (Grid interpolation)}$ 2 - Physical heights result Height definition is in Source System; for instance, if a geoid model is involved: $H = h_{Source} - \text{Mean } \Delta H - \Delta H \text{ (Grid interpolation)}$ 3 - Reserved
DF152	Φ_V	±324000 ″	2 ″	int19	Area of validity (Ref.: Figure 3.5-3) of the Helmert/Molodenski transformation: Latitude Erof Origin in arc seconds Coordinates defined in Source-System
DF153	Λ_V	±648000 ″	2 ″	int20	Area of validity (Ref.: Figure 3.5-3) the Helmert/Molodenski transformation: Longitude of Origin in arc seconds

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					Coordinates defined in Source-System
DF154	$\Delta\Phi_V$	0-32766 ″	2 ″	uint14	Area of validity (Ref.: Figure 3.5-3) of the Helmert/Molodenski transformation: Area Extension to North and to South in arc seconds Delta Coordinates defined in Source-System 0 – undefined
DF155	$\Delta\Lambda_V$	0-32766 ″	2 ″	uint14	Area of validity (Ref.: Figure 3.5-3) of the Helmert/Molodenski transformation: Area Extension to East and to West in arc seconds Delta Coordinates defined in Source-System 0 - undefined
DF156	*dX*	±4194.303 m	0.001 m	int23	Translation in X (*dX*, *dY*, *dZ*): Translation vector, to be added to the point's position vector in the source coordinate reference system in order to transform from source coordinate reference system to target coordinate reference system; also: the coordinates of the origin of source coordinate reference system in the target frame.
DF157	*dY*	±4194.303 m	0.001 m	int23	Translation in Y
DF158	*dZ*	±4194.303 m	0.001 m	int23	Translation in Z
DF159	R_1	±42,949.67294 ″	0.00002 ″	int32	Rotation around the X-axis in arc seconds (*RX*, *RY*, *RZ*): Rotations to be applied to the coordinate reference frame. The sign convention is such that a positive rotation of the frame about an axis is defined as a clockwise rotation of the coordinate reference frame when viewed from the origin of the Cartesian coordinate reference system in the positive direction of that axis, that is a positive rotation about the Z-axis only from source coordinate reference system to target coordinate reference system will result in a smaller longitude value for the point in the target coordinate reference system.

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF160	R_2	±42,949.67294 ″	0.00002 ″	int32	Rotation around the Y-axis in arc seconds
DF161	R_3	±42,949.67294 ″	0.00002 ″	int32	Rotation around the Z-axis in arc seconds
DF162	*dS*	±167.77215 ppm	0.00001 ppm	int25	dS is the scale correction expressed in parts per million (ppm).
DF163	X_P	±17,179,869.183 m	0.001 m	int35	X Coordinate for Molodenski-Badekas rotation point (X_p, Y_p, Z_p): Coordinates of the point about which the coordinate reference frame is rotated, given in the source Cartesian coordinate reference system. Must always be the same within the area of a service provider
DF164	Y_P	±17,179,869.183 m	0.001 m	int35	Y Coordinate for Molodenski-Badekas rotation point Must always be the same within the area of a service provider
DF165	Z_P	±17,179,869.183 m	0.001 m	int35	Z Coordinate for Molodenski-Badekas rotation point Must always be the same within the area of a service provider
DF166	add a_S	0-16,777.215 m	0.001 m	uint24	Semi-major axis of source system ellipsoid a_S = 6370000 + add a_S 0 - undefined If add a_S, add b_S, add a_T or add b_T is 0 (undefined) then only the 7-parameter transformation could be performed. The conversion to ellipsoidal coordinates, the projection and the local transformation (Helmert) have to be omitted.
DF167	add b_S	0-33,554.431 m	0.001 m	uint25	Semi-minor axis of source system ellipsoid b_S = 6350000 + add b_S 0 - undefined (see add a_S)
DF168	add a_T	0-16,777.215 m	0.001 m	uint24	Semi-major axis of target system ellipsoid a_T = 6370000 + add a_T 0 - undefined (see add a_S)

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF169	add b_T	0-33,554.431 m	0.001 m	uint25	Semi-minor axis of target system ellipsoid b_T = 6350000 + add b_T 0 - undefined (see add a_S)
DF170	Projection Type	0-63		uint6	Projection type 0 - unknown projection type 1 - TM - Transverse Mercator (OGP 1.4.5.1, EPSG dataset coordinate operation method code 9807) 2 - TMS - Transverse Mercator (South Orientated) (OGP 1.4.5.3, EPSG dataset coordinate operation method code 9808) 3 - LCC1SP - Lambert Conic Conformal (1SP) (OGP 1.4.1.2, EPSG dataset coordinate operation method code 9801) 4 - LCC2SP - Lambert Conic Conformal (2SP) (OGP 1.4.1.1, EPSG dataset coordinate operation method code 9802) 5 - LCCW - Lambert Conic Conformal (West Orientated) (OGP 1.4.1.3, EPSG dataset coordinate operation method code 9826) 6 - CS - Cassini-Soldner (OGP 1.4.4, EPSG dataset coordinate operation method code 9806) 7 - OM - Oblique Mercator (OGP 1.4.6, EPSG dataset coordinate operation method code 9815) 8 - OS - Oblique Stereographic (OGP 1.4.7.1, EPSG dataset coordinate operation method code 9809) 9 - MC - Mercator (OGP 1.4.3, EPSG dataset coordinate operation method code 9804 or 9805) 10 - PS - Polar Stereographic

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					(OGP 1.4.7.2, EPSG dataset coordinate operation method code 9810) 11 - DS - Double Stereographic 12 to 63 - Reserved If the Projection type is 0 (unknown) then only the 7-parameter transformation and the interpolations for $\delta\varphi_i$, $\delta\lambda_i$ δh_i (1023) could be performed. The Projection and interpolations for δN_i, δE_i δh_i (1024) have to be omitted.
DF171	LaNO	±90.0000 °	0.000000011 °	int34	Latitude of natural origin (TM, TMS, LCC1SP, LCCW, CS, OS, PS, DS)
DF172	LoNO	±180.0000 °	0.000000011 °	int35	Longitude of natural origin (TM, TMS, LCC1SP, LCCW, CS, OS, MC, PS, DS)
DF173	add SNO	0-10737.41823 ppm	0.00001 ppm	uint30	Scale factor at natural origin (TM, TMS, LCC1SP, LCCW, OS, PS, DS) SNO = 993000 + add SNO [ppm]
DF174	FE	0-68,719,476.735 m	0.001 m	uint36	False Easting (TM, TMS, LCC1SP, LCCW, CS, OS, MC, PS, DS) (Contains zone term if exists)
DF175	FN	±17,179,869.183 m	0.001 m	int35	False Northing (TM, TMS, LCC1SP, LCCW, CS, OS, MC, PS, DS)
DF176	LaFO	±90.0000 °	0.000000011 °	int34	Latitude of false origin (LCC2SP)
DF177	LoFO	±180.0000 °	0.000000011 °	int35	Longitude of false origin (LCC2SP)
DF178	LaSP1	±90.0000 °	0.000000011 °	int34	Latitude of 1st standard parallel (LCC2SP)
DF179	LaSP2	±90.0000 °	0.000000011 °	int34	Latitude of 2nd standard parallel (LCC2SP)
DF180	EFO	0-68,719,476.735 m	0.001 m	uint36	Easting of false origin (LCC2SP)

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF181	NFO	±17,179,869.183 m	0.001 m	int35	Northing of false origin (LCC2SP) (Contains zone term if exists)
DF182	Rectification Flag	0-1		bit(1)	0 - Not rectified (OM) 1 - Rectified Oblique Mercator projection
DF183	LaPC	±90.0000 °	0.000000011 °	int34	Latitude of projection center (OM)
DF184	LoPC	±180.0000 °	0.000000011 °	int35	Longitude of projection center (OM)
DF185	AzIL	0-360 °	0.000000011 °	uint35	Azimuth of initial line (OM)
DF186	Diff ARSG	±0.369098741 °	0.000000011 °	int26	Difference from azimuth of initial line to angle from rectified to skew grid ARSG = AzIL + Diff ARSG (OM)
DF187	Add SIL	0-10,737.41823 ppm	0.00001 ppm	uint30	Scale factor on initial line (OM) SIL = 993000 + add SIL [ppm]
DF188	EPC	0-68,719,476.735 m	0.001 m	uint36	Easting at projection center (OM) (Contains zone term if exists)
DF189	NPC	±17,179,869.183 m	0.001 m	int35	Northing at projection center (OM)
DF190	Horizontal Shift Indicator	0-1		bit(1)	0 - no horizontal shift 1 - apply horizontal shift
DF191	Vertical Shift Indicator	0-1		bit(1)	0 - no vertical shift 1 - apply vertical shift
DF192	Φ_0	±324000 ″	0.5 ″	int21	Latitude of Origin of the grids in arc seconds (Figure 3.5-4) Coordinates defined in Target-System. In this context "Target system" means directly after utilizing Helmert or Molodenski transformation (1021 or 1022).
DF193	Λ_0	±648000 ″	0.5 ″	int22	Longitude of Origin of the grids in arc seconds (Figure 3.5-4)

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					Coordinates defined in Target-System. In this context "Target system" means directly after utilizing Helmert or Molodenski transformation (1021 or 1022).
DF194	$\Delta\varphi$	0-2047.5 ″	0.5 ″	uint12	Grid area extension North to South in arc seconds (Figure 3.5-4) Delta Coordinates defined in Target-System. In this context "Target system" means directly after utilizing Helmert or Molodenski transformation (1021 or 1022). 0 – undefined
DF195	$\Delta\lambda$	0-2047.5 ″	0.5 ″	uint12	Grid area extension East to West in arc seconds (Figure 3.5-4) Delta Coordinates defined in Target-System. In this context "Target system" means directly after utilizing Helmert or Molodenski transformation (1021 or 1022). 0 – undefined
DF196	Mean $\Delta\varphi$	±0.127 ″	0.001 ″	int8	Mean offset for all 16 grid points.
DF197	Mean $\Delta\lambda$	±0.127 ″	0.001 ″	int8	Mean offset for all 16 grid points.
DF198	Mean ΔH	±163.83 m	0.01 m	int15	Mean height offset for all 16 grid points to cover all possible geoid heights. If "Height Indicator" = 2 - defined in Source CRS else - defined in Target CRS
DF199	$\delta\varphi_i$	±0.00765 ″	0.00003 ″	int9	Residual in latitude for point i (Figure 3.5-4) - only for small areas - defined in Target CRS
DF200	$\delta\lambda_i$	±0.00765 ″	0.00003 ″	int9	Residual in longitude for point i (Figure 3.5-4) - only for small areas - defined in Target CRS

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF201	δh_i	±0.255 m	0.001 m	int9	Residual in height for point i (Figure 3.5-4) - only for small areas If "Height Indicator" = 2 - defined in Source CRS else - defined in Target CRS
DF202	N_0	±167,772,150 m	10 m	int25	Northing of Origin of the grids in meters (Figure 3.5-4) Coordinates defined in local system after projection
DF203	E_0	0-671,088,630 m	10 m	uint26	Easting of Origin of the grids in meters (Figure 3.5-4) Coordinates defined in local system after projection
DF204	ΔN	0-40,950 m	10 m	uint12	Grid area extension North to South in meters (Figure 3.5-4) Delta Coordinates defined in local system after projection 0 – undefined
DF205	ΔE	0-40,950 m	10 m	uint12	Grid area extension East to West in meters (Figure 3.5-4) Delta Coordinates defined in local system after projection 0 – undefined
DF206	Mean ΔN	±5.11 m	0.01 m	int10	Mean local Northing offset for all 16 grid points.
DF207	Mean ΔE	±5.11 m	0.01 m	int10	Mean local Easting offset for all 16 grid points.
DF208	Mean Δh	±163.83 m	0.01 m	int15	Mean local height offset for all 16 grid points to cover all possible geoid heights. If "Height Indicator" = 2 - defined in Source CRS else defined in local system after projection
DF209	δN_i	±0.255 m	0.001 m	int9	Residual in local Northing for point i (Figure 3.5-4) - only for small areas

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF210	δE_i	±0.255 m	0.001 m	int9	Residual in local Easting for point i (Figure 3.5-4) - only for small areas
DF211	δh_i	±0.255 m	0.001 m	int9	Residual in height for point i (Figure 3.5-4) - only for small areas
DF212	Horizontal Interpolation Method Indicator	0-3		uint2	Defining horizontal interpolation method to be used (Figures 3.5-5 through 3.5-7) 0 - Bi-linear 1 - Bi-quadratic 2 - Bi-spline 3 - Reserved
DF213	Vertical Interpolation Method Indicator	0-3		uint2	Defining vertical interpolation method to be used (Figures 3.5-5 through 3.5-7) 0 - Bi-linear 1 - Bi-quadratic 2 - Bi-spline 3 - Reserved
DF214	Horizontal Helmert/ Molodenski Quality Indicator	0-7		uint3	Maximum approximation error after application of Helmert/Molodenski transformation within the 'area of validity'. The quality could be further improved by application of information in the residual message (grid residuals). 0 - Unknown quality 1 - Quality better 21 mm 2 - Quality 21 to 50 mm 3 - Quality 51 to 200 mm 4 - Quality 201 to 500 mm 5 - Quality 501 to 2000 mm 6 - Quality 2001 to 5000 mm 7 - Quality worse than 5001 mm
DF215	Vertical Helmert/ Molodenski Quality Indicator	0-7		uint3	Maximum approximation error after application of Helmert/Molodenski transformation within the 'area of validity'. The quality could be further improved by

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					application of information in the residual message (grid residuals). 0 - Unknown quality 1 - Quality better than 21 mm 2 - Quality 21 to 50 mm 3 - Quality 51 to 200 mm 4 - Quality 201 to 500 mm 5 - Quality 501 to 2000 mm 6 - Quality 2001 to 5000 mm 7 - Quality worse than 5001 mm
DF216	Horizontal Grid-Quality Indicator	0-7		uint3	Maximum horizontal case within the given area after applying the grid residuals. Replaces the Helmert/Molodenski Quality 0 - Unknown quality 1 - Quality 0 to 10 mm 2 - Quality 11 to 20 mm 3 - Quality 21 to 50 mm 4 - Quality 51 to 100 mm 5 - Quality 101 to 200 mm 6 - Quality 201 to 500 mm 7 - Quality worse than 501 mm
DF217	Vertical Grid-Quality Indicator	0-7		uint3	Maximum vertical case within the given area after applying the grid residuals. Replaces the Helmert/Molodenski Quality 0 - Unknown quality 1 - Quality 0 to 10 mm 2 - Quality 11 to 20 mm 3 - Quality 21 to 50 mm 4 - Quality 51 to 100 mm 5 - Quality 101 to 200 mm 6 - Quality 201 to 500 mm 7 - Quality worse than 501 mm

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF218	s_{oc}	0-127 mm	0.5 mm	uint8	Constant term of standard deviation (1 sigma) for non-dispersive interpolation residuals. The maximum value (127.5 mm) indicates that the computed residual error value is unknown or exceeds the data field range.
DF219	s_{od}	0-5.11 PPM	0.01 PPM	uint9	Distance dependent term of standard deviation (1 sigma) for non-dispersive interpolation residuals
DF220	s_{oh}	0-5.1 PPM	0.1 PPM	uint6	Height-dependent term of standard deviation (1 sigma) for non-dispersive interpolation residuals. The complete standard deviation for the expected non-dispersive interpolation residual is computed from DF218, DF219 and DF220 using the formula: $s_0 = \sqrt{s_{0c}^2 + s_{0d}^2 \cdot d_{\mathrm{Re}\,f}^2 + s_{0h}^2 \cdot dh_{\mathrm{Re}\,f}^2}$ [mm] where $d_{\mathrm{Re}\,f}$ is the distance of the rover from the nearest physical reference station in [km] and $\lvert dh_{\mathrm{Re}\,f} \rvert$ is the absolute value of the height difference between nearest physical reference station and rover in [km].
DF221	s_{Ic}	0-511 mm	0.5 mm	uint10	Constant term of standard deviation (1 sigma) for dispersive interpolation residuals (as affecting GPS L1 frequency). The maximum value (511.5 mm) indicates that the computed residual error value is unknown or exceeds the data field range.
DF222	s_{Id}	0-10.23 PPM	0.01 PPM	uint10	Distance-dependent term of standard deviation (1 sigma) for dispersive interpolation residuals as affecting 1575.42 MHz. The complete standard deviation for the expected dispersive interpolation residual is computed from DF221 and DF222 using the formula:

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					$s_I(1575.42\text{ MHz}) = \sqrt{s_{Ic}^2 + s_{Id}^2 \cdot d_{Ref}^2}$ [mm] where d_{Ref} is the distance of the rover from the nearest physical reference station in kilometres. For frequencies other than 1575.42 MHz, the standard deviation can be scaled by the ratio of squared frequencies: $s_I(f_i) = s_I(1575.42\ MHz) \cdot \frac{(1575.42\ MHz)^2}{(f_i)^2}$ [mm]
DF223	N-Refs	0-127		uint7	Number of reference stations used to derive residual statistics (1 to 127, use 127 for 127 or more stations). The number of reference stations should never be zero. If zero is encountered the rover should ignore the message.
DF224	GPS Residuals Epoch Time (TOW)	0-604799 s	1 s	uint20	
DF225	GLONASS Residuals Epoch Time (tk)	0-86399 s	1 s	uint17	Seconds since beginning of GLONASS day. The addition of positive leap second to GLONASS system time may happen at 10,800 epoch. In this case, two different epochs of UTC time shall be mapped into the same GLONASS epoch (UTC epochs 23h59m60s and 00h00m00s are mapped into 10,800 GLONASS epoch). For example, the epoch sequence (with 1 Hz update rate) during the addition of a positive leap second would be: …, [10,799], [10,800], [10,800], [10,801], … Service providers can either avoid generation of corrections referenced to such epochs or ensure that one second mismatch in computation of time interval (with respect to GLONASS epoch time) leads to allowable degradation of performances.
DF226	Physical Reference Station ID	0-4095		uint12	The Physical Reference Station ID specifies the station ID of a real reference station, when the data stream itself is based on a non-physical reference station. Consequently, for the Physical Reference Station ID the same notes apply as for DF003.

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF227	Receiver Type Descriptor Counter	0-31		uint8	Number of characters in the name of the receiver type
DF228	Receiver Type Descriptor			char8 (n)	
DF229	Receiver Firmware Version Counter	0-31		uint8	Number of characters in the name of the receiver firmware
DF230	Receiver Firmware Version			char8 (n)	
DF231	Receiver Serial Number Counter	0-31		uint8	Number of characters in the name of the receiver serial number
DF232	Receiver Serial Number			char8 (n)	
DF233	GLONASS NW Epoch Time	0-86,399.9 s	0.1 s	uint20	Epoch time of observations used to derive correction differences. The addition of positive leap second to GLONASS system time may happen at 10,800,0 epoch. In this case, two different epochs of UTC time shall be mapped into the same GLONASS epoch (e.g., UTC epochs 23h59m60s and 00h00m00s are mapped into 10,800,0 GLONASS epoch), i.e., during the period of leap second insertion and one second after it this data field shall utilize the same values to refer to epochs both during and after the leap second. For example, the epoch sequence (with 1 Hz update rate) during the addition of a positive leap second would be: …, [10,799,0], [10,800,0], [10,800,0], [10,801,0], … Service providers can either avoid generation of correction differences referenced to such epochs or ensure that one second mismatch in computation of time interval (with respect to GLONASS epoch time) leads to allowable degradation of performances.

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF234	Number of GLONASS Data Entries	0–15		uint4	The number of data entries (total number of satellites for which given data blocks are available).
DF235	GLONASS Ambiguity Status Flag	0–3		bit(2)	0 - Reserved for future use (artificial observations) 1 - Correct Integer Ambiguity Level for L1 and L2 2 - Correct Integer Ambiguity Level for L1-L2 widelane 3 - Uncertain Integer Ambiguity Level. Only a likely guess is used. (See the description of Correct Integer Ambiguity Level and Ambiguity Status Flag in Sections 3.5.6 and 3.5.13)
DF236	GLONASS Non-Sync Count	0–7		uint3	Whenever an unrecoverable cycle slip occurs this count must be increased. The counter shall not be increased more than once per minute. (See the discussion of cycle slips and ambiguity levels in Section 3.5.6 and 3.5.13)
DF237	GLONASS Ionospheric Carrier Phase Correction Difference	±32.767 m	0.5 mm	int17	The GLONASS Ionospheric Carrier Phase Correction Difference (ICPCDR) is the correction difference for ionospheric part calculated from integer-leveled L1 and L2 correction differences (L1CDR and L2CDR). $ICPCDR = \frac{f_2^2}{f_2^2 - f_1^2} L1CDR - \frac{f_2^2}{f_2^2 - f_1^2} L2CDR$ L1CDR, L2CDR, and ICPCDR are presented in meters. The terms f_1 and f_2 represent the GLONASS L1 and L2 carrier frequencies respectively. Note that the coefficient $\frac{f_2^2}{f_2^2 - f_1^2}$ is constant for all GLONASS frequency channel numbers. In extreme conditions the value of this field may lie outside the specified range. If this happens, the data block for the specific satellite containing this field (Table 3.5-23 and Table 3.5-25) shall not be transmitted. (See discussion of L1 and L2 corrections in Section 3.5.13).

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF238	GLONASS Geometric Carrier Phase Correction Difference	±32.767 m	0.5 mm	int17	The GLONASS Geometric Carrier Phase Correction Difference (GCPCDR) is the correction difference for geometric part calculated from integer leveled L1 and L2 correction differences (L1CDR and L2CDR). $GCPCDR = \frac{f_1^2}{f_1^2 - f_2^2} L1CDR - \frac{f_2^2}{f_1^2 - f_2^2} L2CDR$ L1CDR, L2CDR, and GCPCDR are presented in meters. The terms f_1 and f_2 represent the GLONASS L1 and L2 carrier frequencies respectively. Note that the coefficients $\frac{f_1^2}{f_1^2 - f_2^2}$ and $\frac{f_2^2}{f_1^2 - f_2^2}$ are constant for all GLONASS frequency channel numbers. (See discussion of L1 and L2 corrections in Section 3.5.13).
DF239	GLONASS IOD	0-255		bit(8)	Issue Of Data (IOD) of GLONASS broadcast ephemeris. Bits 0-5 are the 6 LSB's of the t_b field in the current ephemeris (see DF110). Bits 6-7 are set to zero in this version. If these bits are not equal to zero, the given satellite should be excluded from positioning.
DF240	GPS FKP Epoch Time	0–604799 s	1 s	uint20	Seconds since the beginning of the GPS week.
DF241	GLONASS FKP Epoch Time	0-86399 s	1 s	uint17	Seconds since beginning of GLONASS day. The addition of positive leap second to GLONASS system time may happen at 10,800 epoch. In this case, two different epochs of UTC time shall be mapped into the same GLONASS epoch (UTC epochs 23h59m60s and 00h00m00s are mapped into 10,800 GLONASS epoch). For example, the epoch sequence (with 1 Hz update rate) during the addition of a positive leap second would be: …, [10,799], [10,800], [10,800], [10,801], … Service providers can either avoid generation of corrections referenced to such epochs or

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					ensure that one second mismatch in computation of time interval (with respect to GLONASS epoch time) leads to allowable degradation of performances.
DF242	N0: Geometric Gradient in North (ppm)	±20.47 PPM	0.01 PPM	int12	Gradient (FKP) of the geometric (non-dispersive) error components in South-North direction in parts per million of the south-north distance to the reference station. 800h - Invalid.
DF243	E0: Geometric gradient in east (ppm)	±20.47 PPM	0.01 PPM	int12	Gradient (FKP) of the geometric (non-dispersive) error components in West-East direction in parts per million of the west-east distance to the reference station. Note: A bit pattern equivalent to 800h in this field indicates the Geometric Gradient in East is invalid.
DF244	NI: Ionospheric gradient in north (ppm)	±81.91 PPM	0.01 PPM	int14	Gradient (FKP) of the ionospheric (dispersive) error component in South-North direction. Note: A bit pattern equivalent to 2000h in this field indicates the Ionospheric Gradient in North is invalid.
DF245	EI: Ionospheric gradient in east (ppm)	±81.91 PPM	0.01 PPM	int14	Gradient (FKP) of the ionospheric (dispersive) error component in West-East direction. Note: A bit pattern equivalent to 2000h in this field indicates the Ionospheric Gradient in East is invalid.
DF246-DF247					RESERVED
DF248	Galileo Epoch Time (TOW)	0-604,799,999 ms	1 ms	uint30	Galileo Epoch Time is provided in milliseconds from the beginning of the Galileo week.
DF249-DF251					RESERVED

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF252	Galileo Satellite ID	0-63		uint6	The Galileo SVID parameter is coded with 6 bits. However, the maximum constellation which can be accommodated within the I/NAV and F/NAV frames is 36 satellites (3 planes of 12 satellites each).
DF286	Galileo SISA (E1,E5b)			bit(8)	SISA index associated with a SISA value as defined below. Signal In Space Accuracy (SISA(E1,E5b)) is a prediction of the minimum standard deviation (1-sigma) of the unbiased Gaussian distribution which overbounds the Signal-In-Space Error (SISE) predictable distribution for all possible user locations within the satellite coverage area. When no accurate prediction is available (SISA = NAPA), this is an indicator of a potential anomalous SIS. The SISA Index shall be encoded according to the values stated in the following table: 0 - 49 - 0 cm to 49 cm with 1 cm resolution 50 - 74 - 50 cm to 0.98 m with 2 cm resolution 75 - 99 - 1 m to 2 m with 4 cm resolution 100 - 125 - 2 m to 6 m with 16 cm resolution 126 - 254 - Spare 255 - No Accuracy Prediction Available (NAPA)
DF287	Galileo E1-B Signal Health Status			bit(2)	The E1-B Signal Health Status Bit Values are: 0 - Signal OK 1 - Signal out of service 2 - Signal in Extended Operations Mode 3 - Signal Component currently in Test See also note in DF290.
DF288	Galileo E1-B Data Validity Status			bit(1)	The navigation data validity status transmitted on E1-B. The signal health status and data validity status refer to the transmitting satellite. These status flags are used as service performance level notification (e.g., notification of satellite non availability). See also note in DF290.

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF289	Galileo Week number	0–4095 weeks	1 week	uint12	Galileo week number. Roll-over every 4096 weeks (about 78 years). Galileo System Time (GST) is defined in OS SIS ICD. See Galileo Signal Specification. The GST start epoch is defined as 13 seconds before midnight between 21st August and 22nd August 1999, i.e., GST was equal to 13 seconds at 22nd August 1999 00:00:00 UTC. As GST is a continuous time scale, and UTC is corrected periodically with an integer number of leap seconds, the Galileo navigation message contains all necessary parameters to convert between GST and UTC.
DF290	Galileo IODnav	0-1023	1	uint10	Issue of Data for navigation data (IODnav). Each IODnav has an associated reference time parameter disseminated within the batch. Note: The broadcast group delay, navigation data validity status and signal health status are not identified by any Issue of Data value - unitless
DF291	Galileo SV SISA			bit(8)	
DF292	Galileo $\dot{i}$ (IDOT)	*See Note 1*	2^{-43}	int14	Rate of inclination. Unit: semi-circles/s
DF293	Galileo toc	0-604,740	60	uint14	Clock reference time. Galileo System Time (GST) is defined in OS SIS ICD. See Galileo Signal Specification. The GST start epoch is defined as 13 seconds before midnight between 21st August and 22nd August 1999, i.e., GST was equal to 13 seconds at 22nd August 1999 00:00:00 UTC. As GST is a continuous time scale, and UTC is corrected periodically with an integer number of leap seconds, the Galileo navigation message contains all necessary parameters to convert between GST and UTC.
DF294	Galileo af2	*See Note 1*	2^{-59}	int6	Clock correction. Unit: s/s2

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF295	Galileo af1	*See Note 1*	2^{-46}	int21	Clock correction. Unit: s/s
DF296	Galileo af0	*See Note 1*	2^{-34}	int31	Clock correction. Unit: seconds
DF297	Galileo Crs	*See Note 1*	2^{-5}	int16	Amplitude of the sine harmonic correction term to the orbit radius. Unit: meters
DF298	Galileo Δn	*See Note 1*	2^{-43}	int16	Mean motion difference from computed value. Unit: semi-circles/s
DF299	Galileo M0	*See Note 1*	2^{-31}	int32	Mean anomaly at reference time. Unit: semi-circles
DF300	Galileo Cuc	*See Note 1*	2^{-29}	int16	Amplitude of the cosine harmonic correction term to the argument of latitude. Unit: radians
DF301	Galileo e	*See Note 1*	2^{-33}	uint32	Eccentricity - unitless
DF302	Galileo Cus	*See Note 1*	2^{-29}	int16	Amplitude of the sine harmonic correction term to the argument of latitude. Unit: radians
DF303	Galileo $A^{1/2}$	*See Note 1*	2^{-19} $m^{1/2}$	uint32	
DF304	Galileo toe	0-604,740	60	uint14	Ephemeris reference time (time of ephemeris). Galileo System Time (GST) is defined in OS SIS ICD, See Galileo Signal Specification. The GST start epoch is defined as 13 seconds before midnight between 21st August and 22nd August 1999, i.e., GST was equal to 13 seconds at 22nd August 1999 00:00:00 UTC. As GST is a continuous time scale, and UTC is corrected periodically with an integer number of leap seconds, the Galileo navigation message contains all necessary parameters to convert between GST and UTC.
DF305	Galileo Cic	*See Note 1*	2^{-29}	int16	Amplitude of the cosine harmonic correction term to the angle of inclination. Unit: radians

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF306	Galileo Ω0	*See Note 1*	2^{-31}	int32	Longitude of ascending node of orbital plane at weekly epoch. Unit: semi-circles
DF307	Galileo Cis	*See Note 1*	2^{-29}	int16	Amplitude of the sine harmonic correction term to the angle of inclination. Unit: radians
DF308	Galileo i0	*See Note 1*	2^{-31}	int32	Inclination angle at reference time. Unit: semi-circles
DF309	Galileo Crc	*See Note 1*	2^{-5}	int16	Amplitude of the cosine harmonic correction term to the orbit radius. Unit: meters
DF310	Galileo ω	*See Note 1*	2^{-31}	int32	Argument of Perigee. Unit: semi-circles
DF311	Galileo $\dot{\Omega}_0$ (OMEGADOT)	*See Note 1*	2^{-43}	int24	Rate of right ascension. Unit: semi-circles/sec
DF312	Galileo BGD (E1/E5a)	*See Note 1*	2^{-32}	int10	E1/E5a Broadcast Group Delay
DF313	Galileo BGDE5b/E1	*See Note 1*	2^{-32}	int10	Broadcast Group Delay E5b/E1 (reserved)
DF314	E5a SIGNAL Health Status			bit(2)	The Signal Health Status Bit Values are: 0 - Signal OK 1 - Signal out of service 2 - Signal in Extended Operations Mode 3 - Signal Component currently in Test
DF315	E5a Data Validity Status			bit(1)	The navigation data validity status transmitted on E5a. The signal health status and data validity status refer to the transmitting satellite. These status flags are used as service performance level notification (e.g., notification of satellite non availability).

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF316	Galileo SOL NAV Signal Health Status (SOLHS)			bit(2)	The E5b Signal Health Status Bit Values are: 0 - Signal OK 1 - Signal out of service 2 - Signal in Extended Operations Mode 3 - Signal Component currently in Test See also note in DF290.
DF317	Galileo SOL NAV Data Validity Status (SOLDVS)			bit(1)	The navigation data validity status transmitted on E5b. The signal health status and data validity status refer to the transmitting satellite. These status flags are used as service performance level notification (e.g., notification of satellite non availability). See also note in DF290.
DF318-DF363					RESERVED
DF364	Quarter-Cycle Indicator			bit(2)	The Quarter-Cycle Indicator denotes whether different carrier phase signals tracked on the same frequency have a common phase, i.e., whether or not the fractional Phaseranges of two signals on the same frequency show a quarter-cycle difference (see also section 3.1.7 for further explanation). The definition of the indicator relates exclusively to the correction status of the quarter cycle, and applies to Messages Types 1001, 1002, 1003, 1004, 1009, 1010, 1011, 1012. Other possible corrections cannot be indicated by this indicator. 00 - Correction status unspecified 01 - Phaseranges in Message Types 1001, 1002, 1003, 1004, 1009, 1010, 1011, 1012 are corrected in such a way that whenever Phaseranges for different signals on the same frequency are present in these messages, they are guaranteed to be in phase and thus shall show no Quarter-Cycle bias between them (see Table 3.1-5 for details on the adjustments made). Double

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					differences of Phaseranges tracked with different signals shall show no Quarter-Cycle differences. 10 - Phase observations are not corrected. Double differences may show Quarter-Cycle differences for Phaseranges based on different signals on the same frequency. Processing will require appropriate corrections. 11 – Reserved
DF365	Delta Radial	±209.7151 m	0.1 mm	int22	Radial orbit correction for broadcast ephemeris. The reference time t0 is Epoch Time (DF385, DF386) plus ½ SSR Update Interval. The reference time t0 for SSR Update Interval "0" is Epoch Time.
DF366	Delta Along-Track	±209.7148 m	0.4 mm	int20	Along-Track orbit correction for broadcast ephemeris. See note on reference time t0 of DF365.
DF367	Delta Cross-Track	±209.7148 m	0.4 mm	int20	Cross-Track orbit correction for broadcast ephemeris. See note on reference time t0 of DF365.
DF368	Dot Delta Radial	±1.048575 m/s	0.001 mm/s	int21	Velocity of Radial orbit correction for broadcast ephemeris. See note on reference time t0 of DF365.
DF369	Dot Delta Along-Track	±1.048572 m/s	0.004 mm/s	int19	Velocity of Along-Track orbit correction for broadcast ephemeris. See note on reference time t0 of DF365.
DF370	Dot Delta Cross-Track	±1.048572 m/s	0.004 mm/s	int19	Velocity of Cross-Track orbit correction for broadcast ephemeris. See note on reference time t0 of DF365.
DF375	Satellite Reference Datum	0-1	N/A	bit(1)	Orbit corrections refer to Satellite Reference Datum: 0 - ITRF 1 - Regional

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF376	Delta Clock C0	±209.7151 m	0.1 mm	int22	C0 polynomial coefficient for correction of broadcast satellite clock. The reference time t0 is Epoch Time (DF385, DF386) plus ½ SSR Update Interval. The reference time t0 for SSR Update Interval "0" is Epoch Time.
DF377	Delta Clock C1	±1.048575 m/s	0.001 mm/s	int21	C1 polynomial coefficient for correction of broadcast satellite clock. See note on reference time t0 of DF376.
DF378	Delta Clock C2	±1.34217726 m/s^2	0.00002 mm/s^2	int27	C2 polynomial coefficient for correction of broadcast satellite clock. See note on reference time t0 of DF376.
DF379	No. of Code Biases Processed	0-31	1	uint5	Number of Code Biases for one individual satellite
DF380	GPS Signal and Tracking Mode Identifier	0-31	1	uint5	Indicator to specify the GPS signal and tracking mode: 0 - L1 C/A 1- L1 P 2- L1 Z-tracking and similar (AS on) 3 - Reserved 4 - Reserved 5 - L2 C/A 6 - L2 L1(C/A)+(P2-P1) (semi-codeless) 7 - L2 L2C (M) 8 - L2 L2C (L) 9 - L2 L2C (M+L) 10 - L2 P 11 - L2 Z-tracking and similar (AS on) 12 - Reserved 13 - Reserved 14 - L5 I 15 - L5 Q >15 - Reserved

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF381	GLONASS Signal and Tracking Mode Identifier	0-31	1	uint5	Indicator to specify the GLONASS signal and tracking mode: 0 - G1 C/A 1 - G1 P 2 - G2 C/A (GLONASS M) 3 - G2 P >3 -Reserved
DF382	RESERVED for Galileo Signal and Tracking Mode Identifier	0-31	1	uint5	Indicator to specify the Galileo signal and tracking mode: 0 - E1 A PRS 1 - E1 B I/NAV OS/CS/SoL 2 - E1 C no data 3 - Reserved 4 - Reserved 5 - E5a I F/NAV OS 6 - E5a Q no data 7 - Reserved 8 - E5b I I/NAV OS/CS/SoL 9 - E5b Q no data 10 - Reserved 11 - E5 I 12 - E5 Q 13 - Reserved 14 - E6 A PRS 15 - E6 B C/NAV CS 16 - E6 C no data >16 - Reserved
DF383	Code Bias	±81.91 m	0.01 m	int14	Code Bias for specified Signal
DF384	GLONASS Satellite ID	1–24	1	uint5	GLONASS Satellite ID's only
DF385	GPS Epoch Time 1s	0–604799 s	1 s	uint20	Full seconds since the beginning of the GPS week

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF386	GLONASS Epoch Time 1s	0–86399 s	1 s	uint17	Seconds since beginning of GLONASS day. The addition of positive leap second to GLONASS system time may happen at 10,800 epoch. In this case, two different epochs of UTC time shall be mapped into the same GLONASS epoch (UTC epochs 23h59m60s and 00h00m00s are mapped into 10,800 GLONASS epoch). For example, the epoch sequence (with 1 Hz update rate) during the addition of a positive leap second would be: …, [10,799], [10,800], [10,800], [10,801], … Service providers can either avoid generation of corrections referenced to such epochs or ensure that one second mismatch in computation of time interval (with respect to GLONASS epoch time) leads to allowable degradation of performances.
DF387	No. of Satellites	0–63	1	uint6	Number of satellites
DF388	Multiple Message Indicator	0–1	1	bit(1)	Indicator for transmitting messages with the same Message Number and Epoch Time: 0 - last message of a sequence 1 - multiple messages transmitted
DF389	SSR URA	bits 5–3: 0–7 bits 2–0: 0–7		bit(6)	SSR User Range Accuracy (URA) (1 sigma) for a range correction computed from complete SSR set as disseminated by RTCM SSR messages. The URA is represented by a combination of URA_CLASS and URA_VALUE. The 3 MSb define the URA_CLASS with a range of 0-7 and the 3 LSb define the URA_VALUE with a range of 0-7. The URA is computed by: $\text{URA [mm]} \leq 3^{\text{URA_CLASS}}\left(1+\frac{\text{URA_VALUE}}{4}\right) - 1\ \text{[mm]}$ Special cases are: 000 000 - URA undefined/unknown, SSR corrections for the corresponding satellite may not be reliable. 111 111 - URA > 5466.5 mm

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF390	High-Rate Clock Correction	±209.7151 m	0.1 mm	int22	High-Rate Clock correction to be added to the polynomial clock correction (see DF376, DF377, DF378)
DF391	SSR Update Interval	0-15	1	bit(4)	SSR Update Interval. The SSR Update Intervals for all SSR parameters start at time 00:00:00 of the GPS time scale. A change of the SSR Update Interval during the transmission of a SSR data stream should ensure consistent data for a rover. The supported SSR Update Intervals are: 0 - 1 s 1 - 2 s 2 - 5 s 3 - 10 s 4 - 15 s 5 - 30 s 6 - 60 s 7 - 120 s 8 - 240 s 9 - 300 s 10 - 600 s 11 - 900 s 12 - 1800 s 13 - 3600 s 14 - 7200 s 15 - 10800 s Note that the update intervals are aligned to the GPS time scale for all GNSS in order to allow synchronous operation for multiple GNSS services. This means that the update intervals may not be aligned to the beginning of the day for another GNSS. Due to the leap seconds this is generally the case for GLONASS.
DF392	GLONASS Issue Of Data (IOD)	0-255		bit(8)	Issue Of Data of GLONASS broadcast ephemeris If bit 7 is 0 (bit 7 is MSB): Bits 0-6 represent the 7 bits of the GLONASS tb field in the current ephemeris (see DF110)

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					If bit 7 is 1: Reserved for future use. Data should not be used. This can be the case for pre-GLONASS-M satellite if ephemeris changed within tb interval.
DF393	MSM Multiple message bit			bit(1)	1 indicates that more MSMs follow for given physical time and reference station ID 0 indicates that it is the last MSM for given physical time and reference station ID
DF394	GNSS Satellite mask			bit(64)	The sequence of bits that specifies those GNSS satellites for which there is available data in this message. The Most Significant Bit (MSB), or the first encoded bit corresponds to GNSS satellite with ID=1, the second bit corresponds to GNSS satellite with ID=2, etc. And the Least Significant Bit (LSB), or the last encoded bit corresponds to GNSS satellite with ID=64. Exact mapping of actual GNSS satellites (PRNs for GPS, “slot numbers” for GLONASS, etc.) to satellite mask IDs is specific for each GNSS (see corresponding tables in MSM description for each GNSS). Some ID values may refer to specific satellites, while some ID values may be indicated as “Reserved” in this standard. These IDs may be used in the future for other satellites and thus decoding software shall ensure that it does not skip these bits, but decodes complete GNSS Satellite mask, decodes corresponding observables as if they refer to known satellites, but should refrain from using them, unless new satellite mapping table becomes available to map corresponding ID to a specific satellite. If any data for satellite with ID=n follows, then the corresponding bit (bit number n) is set to 1. If data for satellite with ID=m does not follow, then corresponding bit (bit number m) is set to 0.
DF395	GNSS Signal mask			bit(32)	The sequence of bits that specifies those GNSS signals for which there is available data in this message. Each bit

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					corresponds to a particular signal (observable) type for a given GNSS. The Most Significant Bit (MSB), or the first encoded bit corresponds to a signal with ID=1, the second bit corresponds to a signal with ID=2, etc. And Least Significant Bit (LSB), or the last encoded bit corresponds to signal with ID=32. Exact mapping of actual signal identifier (in correspondence with RINEX 3.01 signal naming convention) to signal mask IDs is specific for each GNSS (see corresponding tables in MSM description for each GNSS). Some ID values may refer to specific signals, while some ID values may be indicated as “Reserved” in this standard. These IDs may be used in the future for other signals and thus decoding software shall ensure, that it does not skip these bits, but decodes complete GNSS Signal mask, decodes corresponding observables as if they refer to known signals, but should refrain from using them, unless new signal mapping table becomes available to map corresponding ID to a specific signal. If a signal (observable) with ID=n is available for at least one transmitted satellite, then the corresponding bit (number n) is set to 1, otherwise the corresponding bit is set to 0.
DF396	GNSS Cell mask			bit(X)	A two-dimensional table, which determines signal availability for each transmitted satellite. This field is of variable size: X=Nsig*Nsat, where Nsat is the number of satellites (the number of those bits set to 1 in Satellite Mask, DF394) and Nsig is the number of available signals (the number of those bits set to 1 in Signal Mask, DF395). The first row of this rectangular table corresponds to the signal with the smallest ID for which the corresponding bit in the Signal Mask is set to 1. The second row corresponds to the signal with the second smallest ID for which the corresponding bit in Signal Mask is set to 1. The last row

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					corresponds to the signal with the highest ID for which the corresponding bit in the Signal Mask is set to 1. The first column of this rectangular table corresponds to the satellite with the smallest ID for which the corresponding bit in the Satellite Mask is set to 1. The second column corresponds to the satellite with the second smallest ID for which the corresponding bit in the Satellite Mask is set to 1. The last column corresponds to the satellite with the highest ID, for which the corresponding bit in the Satellite Mask is set to 1. If observable data for a given satellite and a given signal follows then the corresponding cell in this table is set to 1, otherwise it is set to 0. This bit table is packed by columns, starting from the column corresponding to smallest satellite ID. The size of each column is Nsig bits, and it is packed starting from the cell corresponding to the smallest signal ID. Each cell of the table is represented by one bit, which is set to 1 or 0, according to the value in the cell.
DF397	The number of integer milliseconds in GNSS Satellite rough range	0–254 ms	1 ms	uint8	Rough range can be used to restore complete observables for a given satellite. Rough range takes 18 bits which are split between two fields (DF397 and DF398). This field contains the integer number of milliseconds in the satellite rough range. If this field is not transmitted (MSM1, MSM2, MSM3) then it is the responsibility of decoding equipment to restore it using rough reference station position and ephemeris data. A bit pattern equivalent to FFh (255 ms) indicates invalid value.

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF398	GNSS Satellite rough range modulo 1 millisecond	0 to $(1 – 2^{-10})$ ms	2^{-10} ms	uint10	See the note above. Allows restoring full rough range with accuracy 1/1024 ms (about 300 m).
DF399	GNSS Satellite rough Phaserange Rate	±8191 m/s	1 m/s	int14	Phaserange Rate has the same sign as the mathematical derivative of the Phaserange. Similar to ranges, the full Phaserange Rate observable for a particular signal can be constructed by the sum of the rough Phaserange Rate (unique for given satellite) and the fine Phaserange Rate (unique for each particular signal corresponding to given satellite). A bit pattern equivalent to 2000h (-8192 m/s) indicates invalid value.
DF400	GNSS signal fine Pseudorange	$\pm(2^{-10}–2^{-24})$ ms (Approx: ±292 m)	2^{-24} ms (Approx: 0.018 m)	int15	Specific for each signal of given Satellite. Being added to fields DF397 and DF398 allows getting the full Pseudorange observable corresponding to given signal. A bit pattern equivalent to 4000h ($–2^{-10}$ ms) indicates invalid value.
DF401	GNSS signal fine Phaserange data	$\pm(2^{-8}–2^{-29})$ ms (Approx: ±1171 m)	2^{-29} ms (Approx: 0.0006 m)	int22	Similar to DF400, but refers to the Phaserange. The proper integer number of cycles have been removed from the original full carrier to match it to the corresponding Pseudorange at the start of Phaserange generation. This integer number is kept constant for the epochs that follow until a cycle slip is detected, after which a new integer number of cycles must be determined. In this case the associated GNSS Phaserange Lock Time Indicator (DF402) must be reset to zero. **Note that the Phaserange defined here has the same sign as the Pseudorange.** Certain ionospheric conditions (or incorrect initialization) may cause the difference between the Phaserange and the Pseudorange (Phaserange - Pseudorange) to diverge over time, which may cause the value to exceed the defined range

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					limit. Under these circumstances the above mentioned "integer number of cycles" shall be reinitialized. In this case the associated GNSS Phaserange Lock Time Indicator (DF402) shall be reset to zero. A bit pattern equivalent to 200000 h (-2^{-8} m) indicates invalid value.
DF402	GNSS Phaserange Lock Time Indicator	See Table 3.5-82		uint4	The Lock Time Indicator provides a measure of the amount of time during which the receiver has maintained continuous lock on that satellite signal. If a cycle slip occurs during the previous measurement cycle, the lock indicator will be reset to zero.
DF403	GNSS signal CNR	1–63 dB-Hz	1 dB-Hz	uint6	The GNSS CNR measurement provides an estimate of the carrier-to-noise ratio of the satellite's signal in dB-Hz. A value of "0" means that the CNR measurement is not computed, or not available. Availability or unavailability of CNR does not affect validity of corresponding observables.
DF404	GNSS signal fine Phaserange Rate	±1.6383 m/s	0.0001 m/s	int15	Fine Phaserange Rate for a given signal. Full Phaserange Rate is the sum of this field and the Satellite Rough Phaserange Rate (DF399). A bit pattern equivalent to 4000h (-1.6384 m/s) indicates invalid value.
DF405	GNSS signal fine Pseudorange with extended resolution	$\pm(2^{-10}–2^{-29})$ ms (Approx: ±292 m)	2^{-29} ms (Approx: 0.0006 m)	int20	Same as DF400, but with extended resolution except the bit pattern equivalent to 80000h (-2^{-10} ms) indicates an invalid value instead of the invalid pattern defined for DF400.
DF406	GNSS signal fine Phaserange data with extended resolution	$\pm(2^{-8}–2^{-31})$ ms (Approx: ±1171 m)	2^{-31} ms (Approx: 0.00014 m)	int24	Same as DF401, but with extended resolution except the bit pattern equivalent to 800000h (-2^{-8} ms) indicates an invalid value instead of the invalid pattern defined for DF401.

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF407	GNSS Phaserange Lock Time Indicator with extended range and resolution.	See Table 3.5-83 and Table 3.5-84		uint10	Same as DF402, but with extended range and higher resolution
DF408	GNSS signal CNR with extended resolution	0.0625–63.9375 dB-Hz	2^{-4} dB-Hz	uint10	Same as DF403, but with extended resolution. A value "0" indicates that the CNR measurement has not been computed or is not available. Availability or unavailability of the CNR does not affect validity of other observables.
DF409	IODS – Issue Of Data Station	0-7	1	uint3	This field is reserved to be used to link MSM with future site-description (receiver, antenna description, etc.) messages. A value of "0" indicates that this field is not utilized.
DF410	Reserved				
DF411	Clock Steering Indicator			uint2	0 – clock steering is not applied In this case receiver clock must be kept in the range of ±1 ms (approximately ±300 km) 1 – clock steering has been applied In this case receiver clock must be kept in the range of ±1 microsecond (approximately ±300 meters). 2 – unknown clock steering status 3 – reserved
DF412	External Clock Indicator			uint2	0 – internal clock is used 1 – external clock is used, clock status is "locked" 2 – external clock is used, clock status is "not locked", which may indicate external clock failure and that the transmitted data may not be reliable.

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					3 – unknown clock is used
DF413	IOD SSR	0-15	1	uint4	A change of Issue Of Data SSR is used to indicate a change in the SSR generating configuration, which may be relevant for rover operation.
DF414	SSR Provider ID	0-65535	1	uint16	The SSR Provider ID is a globally unique number assigned by RTCM to identify an SSR provider. Numbers can be requested by contacting RTCM by email at info@rtcm.org. 0 to 255 - reserved for experimental services 256 to 65535 - unique SSR Provider ID
DF415	SSR Solution ID	0-15	1	uint4	SSR Solution ID indicates different SSR services of one SSR provider
DF416	GLONASS Day Of Week	0-7	1	uint3	0 – Sunday 1 – Monday 2 – Tuesday 3 – Wednesday 4 – Thursday 5 – Friday 6 – Saturday 7 – The day of week is not known
DF417	GNSS Smoothing Type Indicator			bit(1)	1 – Divergence-free smoothing is used 0 – Other type of smoothing is used
DF418	GNSS Smoothing Interval	See Table 3.4-4		bit(3)	The GNSS Smoothing Interval is the integration period over which the pseudorange code phase measurements are averaged using carrier phase information. Divergence-free smoothing may be continuous over the entire period for which the satellite is visible. Notice: A value of zero indicates no smoothing is used.

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF419	GLONASS Satellite Frequency Channel Number	See Table 3.4-6	1	uint(4)	The GLONASS Satellite Frequency Channel Number identifies the frequency of the GLONASS satellite.
DF420	Half-cycle ambiguity indicator			bit(1)	0 – No half-cycle ambiguity. 1 – Half-cycle ambiguity When transmitting Phaserange with unresolved polarity encoding software shall set this bit to 1. Receiving software that is not capable of handling half-cycle ambiguities shall skip such Phaserange observables. If polarity resolution forced Phaserange to be corrected by half-a-cycle, then the associated GNSS Phaserange Lock Time Indicator (DF402, DF407) must be reset to zero, indicating that despite continuous tracking the final Phaserange experienced non-continuity.
DF421	GLONASS Code-Phase Bias Indicator			bit(1)	0 = The GLONASS Pseudorange and Phaserange observations in the data stream are not aligned to the same measurement epoch. 1 = The GLONASS Pseudorange and Phaserange observations in the data stream are aligned to the same measurement epoch.
DF422	GLONASS FDMA Signals Mask			bit(4)	Bitmask, where the MSB corresponds to the presence or absence of DF423, continuing to the LSB or last bit corresponding to the absence or presence of DF426. If the bit is set to zero then the corresponding field is absent from the message. If the bit is set to one then the corresponding field is present in the message.
DF423	GLONASS L1 C/A Code-Phase Bias	±655.34 m	0.02 m	int16	The GLONASS L1 C/A Code-Phase Bias represents the offset between the L1 C/A Pseudorange and L1 Phaserange measurement epochs in meters. If DF421 is set to 0, the measurement epoch of the GLONASS L1 Phaserange measurements may be aligned using:

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					Aligned GLONASS L1 Phaserange = Full GLONASS L1 Phaserange (see DF042) + GLONASS L1 C/A Code-Phase Bias. If DF421 is set to 1, the measurement epoch of the GLONASS L1 Phaserange measurements may be unaligned using: Unaligned GLONASS L1 Phaserange = Full GLONASS L1 Phaserange (see DF042) – GLONASS L1 C/A Code-Phase Bias. A bit pattern equivalent to 8000h (-655.36 m) in this field indicates invalid value (unknown or falling outside DF-range)
DF424	GLONASS L1 P Code-Phase Bias	±655.34 m	0.02 m	int16	The GLONASS L1 P Code-Phase Bias represents the offset between the L1 P Pseudorange and L1 Phaserange measurement epochs in meters. If DF421 is set to 0, the measurement epoch of the GLONASS L1 Phaserange measurements may be aligned using: Aligned GLONASS L1 Phaserange = Full GLONASS L1 Phaserange (see DF042) + GLONASS L1 P Code-Phase Bias. If DF421 is set to 1, the measurement epoch of the GLONASS L1 Phaserange measurements may be unaligned using: Unaligned GLONASS L1 Phaserange = Full GLONASS L1 Phaserange (see DF042) – GLONASS L1 P Code-Phase Bias. A bit pattern equivalent to 8000h (-655.36 m) in this field indicates invalid value (unknown or falling outside DF-range)
DF425	GLONASS L2 C/A Code-Phase Bias	±655.34 m	0.02 m	int16	The GLONASS L2 C/A Code-Phase Bias represents the offset between the L2 C/A Pseudorange and L2 Phaserange measurement epochs in meters.

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					If DF421 is set to 0, the measurement epoch of the GLONASS L2 Phaserange measurements may be aligned to the Pseudorange measurements using: Aligned GLONASS L2 Phaserange = Full GLONASS L2 Phaserange (see DF048) + GLONASS L2 C/A Code-Phase Bias. If DF421 is set to 1, the measurement epoch of the GLONASS L2 Phaserange measurements may be unaligned using: Unaligned GLONASS L2 Phaserange = Full GLONASS L2 Phaserange (see DF048) – GLONASS L2 C/A Code-Phase Bias. A bit pattern equivalent to 8000h (-655.36 m) in this field indicates invalid value (unknown or falling outside DF-range)
DF426	GLONASS L2 P Code-Phase Bias	±655.34 m	0.02 m	int16	The GLONASS L2 P Code-Phase Bias represents the offset between the L2 P Pseudorange and L2 Phaserange measurement epochs in meters. If DF421 is set to 0, the measurement epoch of the GLONASS L2 Phaserange measurements may be aligned to the Pseudorange measurements using: Aligned GLONASS L2 Phaserange = Full GLONASS L2 Phaserange (see DF048) + GLONASS L2 P Code-Phase Bias. If DF421 is set to 1, the measurement epoch of the GLONASS L2 Phaserange measurements may be unaligned using: Unaligned GLONASS L2 Phaserange = Full GLONASS L2 Phaserange (see DF048) – GLONASS L2 P Code-Phase Bias. A bit pattern equivalent to 8000h (-655.36 m) in this field indicates invalid value (unknown or falling outside DF-range)

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF427	BeiDou Epoch Time (TOW)	0-604,799,999 ms	1 ms	uint30	BeiDou Epoch Time is provided in milliseconds from the beginning of the BeiDou week. Note that the BeiDou week does not begin at the same time as the GPS or Galileo weeks. As a result, the time of week field for a BeiDou message will be 14 seconds less than the time of week for a GPS or Galileo message of the same epoch.
DF428	QZSS Epoch Time (TOW)	0-604,799,999 ms	1 ms	uint30	QZSS Epoch Time is provided in milliseconds from the beginning of the QZSS week.
DF429	QZSS Satellite ID	1-10	1	uint4	A QZSS Satellite ID number from 1 to 10 refer to the PRN code of the QZSS satellite. The ID map to the PRN numbers as follows: 1: 193 2: 194 … 10: 202
DF430	QZSS toc	0-604,784	2^{4}	uint16	SV clock epoch. Unit: seconds See IS-QZSS-PNT-005 4.1.2.3
DF431	QZSS af2	*See Note 1*	2^{-55}	int8	SV clock drift rate correction term. Unit: sec/sec2 See IS-QZSS-PNT-005 4.1.2.3
DF432	QZSS af1	*See Note 1*	2^{-43}	int16	SV clock drift correction term. Unit: sec/sec See IS-QZSS-PNT-005 4.1.2.3
DF433	QZSS af0	*See Note 1*	2^{-31}	int22	SV clock bias correction term. Unit: seconds See IS-QZSS-PNT-005 4.1.2.3
DF434	QZSS IODE	0-255	1	uint8	Orbit information number. Unit: unitless See See IS-QZSS-PNT-005 4.1.2.4
DF435	QZSS Crs	*See Note 1*	2^{-5}	int16	Amplitude of the sine correction term to the orbit radius. Unit: meters
DF436	QZSS Δn	*See Note 1*	2^{-43}	int16	Difference from mean motion calculation at toe. Unit: semi-circles/sec

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF437	QZSS M0	*See Note 1*	2^{-31}	int32	Mean anomaly at toe. Unit: semi-circles
DF438	QZSS Cuc	*See Note 1*	2^{-29}	int16	Amplitude of the cosine harmonic correction term to the argument of latitude. Unit: radians
DF439	QZSS e	0-0.499	2^{-33}	uint32	Eccentricity. Unit: unitless
DF440	QZSS Cus	*See Note 1*	2^{-29}	int16	Amplitude of the sine harmonic correction term to the argument of latitude. Unit: radians
DF441	QZSS $A^{1/2}$	*See Note 1*	2^{-19} $m^{1/2}$	uint32	
DF442	QZSS toe	0-604,784	2^{4}	uint16	Ephemeris epoch (seconds into week). Unit: seconds
DF443	QZSS Cic	*See Note 1*	2^{-29}	int16	Amplitude of the cosine harmonic correction term to the angle of inclination. Unit: radians
DF444	QZSS Ω0	*See Note 1*	2^{-31}	int32	Longitude of ascending node at the beginning of the week. Unit: semi-circles
DF445	QZSS Cis	*See Note 1*	2^{-29}	int16	Amplitude of the sine harmonic correction term to the angle of inclination. Unit: radians
DF446	QZSS i0	*See Note 1*	2^{-31}	int32	Orbit inclination at toe. Unit: semi-circles
DF447	QZSS Crc	*See Note 1*	2^{-5}	int16	Amplitude of the cosine correction term to the orbit radius. Unit: meters
DF448	QZSS ω	*See Note 1*	2^{-31}	int32	Argument of perigee. Unit: semi-circles
DF449	QZSS Ω0 DOT	*See Note 1*	2^{-43}	int24	Rate of right ascension of ascending node (RAAN) difference from reference value. Unit: semi-circles/sec
DF450	QZSS i0-DOT	*See Note 1*	2^{-43}	int14	Change rate in Orbit inclination. Unit: semi-circles/sec

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF451	QZSS Reserved/Ephemeris Status Flag		1	bit(2)	This field corresponds to Bits 71 and 72 in subframe 1 of the QZSS LNAV data (bits 10 and 11 in Word 3). See IS-QZSS-PNT-005 4.1.2.3. MSB(1bit) is reserved. LSB(1bit) defines the Ephemeris Status Flag.
DF452	QZSS Week number	0-1023 weeks	1 week	uint10	QZSS week number. Roll-over every 1024 weeks starting from Midnight on the night of January 5/Morning January 6, 1980
DF453	QZSS URA		N/A	uint4	QZSS Ephemeris accuracy. Unit: unitless See IS-QZSS-PNT-005 4.1.2.3.
DF454	QZSS SV health		1	uint6	SV health. See IS-QZSS-PNT-005 4.1.2.3.
DF455	QZSS TGD	*See Note 1*	2^{-31}	int8	Group delay between SV clock and L1C/A (or L1C/B). Unit: seconds. See IS-QZSS-PNT-005 4.1.2.3.
DF456	QZSS IODC	0-1023	1	uint10	QZSS Issue of Data, Clock Unit: unitless. See IS-QZSS-PNT-005 4.1.2.3.
DF457	QZSS Fit Interval		1	bit(1)	0: curve-fit interval is 2 hours 1: curve-fit is greater than 2 hours
DF488	BDS Satellite ID	1-63		uint6	A BDS Satellite ID number from 1 to 37 refers to the PRN code of the BDS satellite. Satellite ID's higher than 37 are reserved.
DF489	BDS Week Number	0-8191 weeks	1 week	uint13	Roll-over every 8192 weeks starting from 00:00:00 UTC on Jan. 1, 2006 of BDT.
DF490	BDS URAI	0-15	1	bit(4)	User Range Accuracy Index. See BDS-SIS-ICD-2.0 5.2.4.5
DF491	BDS IDOT	$\pm 9.31\times 10^{-10}$ π/s	2^{-43} π/s	int14	Rate of inclination angle. Unit: π/s. See BDS-SIS-ICD-2.0 5.2.4.12.
DF492	BDS AODE	0-31	1	uint5	Age of Data Ephemeris. See BDS-SIS-ICD-2.0 5.2.4.11

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF493	BDS t_{oc}	0-604,792s	8 s	uint17	The reference time of clock parameters. Unit: s. See BDS-SIS-ICD-2.0 5.2.4.10.
DF494	BDS a_2	$\pm 1.38 \times 10^{-17}$ s/s^2	2^{-66} s/s^2	int11	Clock correction parameter. Unit: s/s^2. See BDS-SIS-ICD-2.0 5.2.4.10.
DF495	BDS a_1	$\pm 1.86 \times 10^{-9}$ s/s	2^{-50} s/s	int22	Clock correction parameter. Unit: s/s. See BDS-SIS-ICD-2.0 5.2.4.10.
DF496	BDS a_0	$\pm 9.77 \times 10^{-4}$ s	2^{-33} s	int24	Clock correction parameter. Unit: s. See BDS-SIS-ICD-2.0 5.2.4.10.
DF497	BDS AODC	0-31	1	uint5	Age of Data Clock. See BDS-SIS-ICD-2.0 5.2.4.9.
DF498	BDS C_{rs}	±2048 m	2^{-6} m	int18	Amplitude of sine harmonic correction term to the orbit radius. Unit: m. See BDS-SIS-ICD-2.0 5.2.4.12.
DF499	BDS Δn	$\pm 3.73 \times 10^{-9}$ π/s	2^{-43} π/s	int16	Mean motion difference from computed value. Unit: π/s. See BDS-SIS-ICD-2.0 5.2.4.12.
DF500	BDS M_0	±π	2^{-31} π	int32	Mean anomaly at reference time. Unit: π. See BDS-SIS-ICD-2.0 5.2.4.12.
DF501	BDS C_{uc}	$\pm 6.10 \times 10^{-5}$ rad	2^{-31} rad	int18	Amplitude of cosine harmonic correction term to the argument of latitude. Unit: rad. See BDS-SIS-ICD-2.0 5.2.4.12.
DF502	BDS e	0-0.5	2^{-33}	uint32	Eccentricity. See BDS-SIS-ICD-2.0 5.2.4.12.
DF503	BDS C_{us}	$\pm 6.10 \times 10^{-5}$ rad	2^{-31} rad	int18	Amplitude of sine harmonic correction term to the argument of latitude. Unit: rad.
DF504	BDS $a^{1/2}$	0-8192 m$^{1/2}$	2^{-19} m$^{1/2}$	uint32	Square root of semi-major axis. Unit: m$^{1/2}$. See BDS-SIS-ICD-2.0 5.2.4.12.
DF505	BDS t_{oe}	0-604,792 s	8 s	uint17	Ephemeris reference time. Unit: s. See BDS-SIS-ICD-2.0 5.2.4.12.
DF506	BDS C_{ic}	$\pm 6.10 \times 10^{-5}$ rad	2^{-31} rad	int18	Amplitude of cosine harmonic correction term to the angle of inclination. Unit: rad. See BDS-SIS-ICD-2.0 5.2.4.12.

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF507	BDS Ω_0	$\pm\pi$	$2^{-31}\ \pi$	int32	Longitude of ascending node of orbital of plane computed according to reference time. Unit: π. See BDS-SIS-ICD-2.0 5.2.4.12.
DF508	BDS C_{is}	$\pm 6.10\times10^{-5}$ rad	2^{-31} rad	int18	Amplitude of sine harmonic correction term to the angle of inclination. Unit: rad. See BDS-SIS-ICD-2.0 5.2.4.12.
DF509	BDS i_0	$\pm\pi$	$2^{-31}\ \pi$	int32	Inclination angle at reference time. Unit: π. See BDS-SIS-ICD-2.0 5.2.4.12.
DF510	BDS C_{rc}	±2048 m	2^{-6} m	int18	Amplitude of cosine harmonic correction term to the orbit radius. Unit: m. See BDS-SIS-ICD-2.0 5.2.4.12.
DF511	BDS ω	$\pm\pi$	$2^{-31}\ \pi$	int32	Argument of perigee. Unit: π. See BDS-SIS-ICD-2.0 5.2.4.12.
DF512	BDS $\dot{\Omega}$	$\pm 9.54\times10^{-7}\ \pi$/s	$2^{-43}\ \pi$/s	int24	Rate of right ascension. Unit: π/s. See BDS-SIS-ICD-2.0 5.2.4.12.
DF513	BDS T_{GD1}	±51.2 ns	0.1 ns	int10	Equipment Group Delay Differential. Unit: ns. See BDS-SIS-ICD-2.0 5.2.4.8。
DF514	BDS T_{GD2}	±51.2 ns	0.1 ns	int10	Equipment Group Delay Differential. Unit: ns. See BDS-SIS-ICD-2.0 5.2.4.8。
DF515	BDS SV Health	0 – 1	1	bit(1)	Autonomous satellite health flag. See BDS-SIS-ICD-2.0 5.2.4.6
DF516	NavIC/IRNSS Satellite ID	1-63	1	uint6	PRN code of the NavIC/IRNSS satellite
DF517	NavIC/IRNSS Week Number (WN)	0-1023	1	uint10	(week)
DF518	NavIC/IRNSS Clock bias (a_{fo})	*See Note 1*	2^{-31}	int22	(sec)

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF519	NavIC/IRNSS Clock drift (a_{f1})	*See Note 1*	2^{-43}	int16	(sec/sec)
DF520	NavIC/IRNSS Clock drift rate (a_{f2})	*See Note 1*	2^{-55}	int8	(sec/sec2)
DF521	NavIC/IRNSS SV Accuracy (URA)	*See Note 1*	1	uint4	
DF522	NavIC/IRNSS Time of clock (t_{oc})	*See Note 1*	16	uint16	(sec)
DF523	NavIC/IRNSS Total Group Delay (T_{GD})	*See Note 1*	2^{-31}	int8	(sec)
DF524	NavIC/IRNSS Mean Motion Difference (Δn)	*See Note 1*	2^{-41}	int22	(semi-circles/sec)
DF525	NavIC/IRNSS Issue of Data Ephemeris & Clock (IODEC)	*See Note 1*	1	uint8	-
DF526	NavIC/IRNSS Reserved bits after IODEC		1	uint10	Unused
DF527	NavIC/IRNSS L5 Flag		1	bit(1)	-
DF528	NavIC/IRNSS S Flag		1	bit(1)	-

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF529	NavIC/IRNSS Amplitude of the Cosine Harmonic Correction Term to the Argument of Latitude (C_{uc})	*See Note 1*	2^{-28}	int15	(radians)
DF530	NavIC/IRNSS Amplitude of the Sine Harmonic Correction Term to the Argument of Latitude (C_{us})	*See Note 1*	2^{-28}	int15	(radians)
DF531	NavIC/IRNSS Amplitude of the Cosine Harmonic Correction Term to the Angle of Inclination (C_{ic})	*See Note 1*	2^{-28}	int15	(radians)
DF532	NavIC/IRNSS Amplitude of the Sine Harmonic Correction Term to the Angle of Inclination (C_{is})	*See Note 1*	2^{-28}	int15	(radians)
DF533	NavIC/IRNSS Amplitude of the Cosine Harmonic Correction Term to the Orbit Radius(C_{rc})	*See Note 1*	2^{-4}	int15	(meters)

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF534	NavIC/IRNSS Amplitude of the Sine Harmonic Correction Term to the Orbit Radius (C_{rs})	*See Note 1*	2^{-4}	int15	(meters)
DF535	NavIC/IRNSS Rate of Inclination angle (IDOT)	*See Note 1*	2^{-43}	int14	(semi-circles/sec)
DF536	NavIC/IRNSS Mean Anomaly (M_o)	*See Note 1*	2^{-31}	int32	(semi-circles)
DF537	NavIC/IRNSS Time of ephemeris (t_{oe})	*See Note 1*	16	uint16	(sec)
DF538	NavIC/IRNSS Eccentricity (e)	*See Note 1*	2^{-33}	uint32	dimensionless
DF539	NavIC/IRNSS Square root of Semi major axis ($\sqrt{A}$)	*See Note 1*	2^{-19}	uint32	($\sqrt{meters}$)
DF540	NavIC/IRNSS Long of Ascending Node (Ω_o)	*See Note 1*	2^{-31}	int32	(semi-circles)
DF541	NavIC/IRNSS Argument of perigee (ω)	*See Note 1*	2^{-31}	int32	(semi-circles)

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF542	NavIC/IRNSS Rate of RAAN ($\dot{\Omega}$)	*See Note 1*	2^{-41}	int22	(semi-circles/sec)
DF543	NavIC/IRNSS Inclination (i_0)	*See Note 1*	2^{-31}	int32	(semi-circles)
DF544	NavIC/IRNSS 2 spare bits after IDOT		1	bit(2)	-
DF545	NavIC/IRNSS 2 spare bits after I0		1	bit(2)	-
DF546	NavIC/IRNSS Epoch Time (TOW)	0-604,799,999	1 ms	uint30	NavIC/IRNSS Epoch Time is provided in milliseconds from the beginning of the NavIC/IRNSS week
DF547	Reference Epoch t_0	0-65,535 days	1 day	uint16	Reference epoch t_0 for transformation parameter set given as Modified Julian Day (MJD) Number minus 44244 days
DF548	dX0	± 4194.303 m	0.001 m	int23	Translation in X for Reference Epoch t_0 Refer also to DF156
DF549	dY0	± 4194.303 m	0.001 m	int23	Translation in Y for Reference Epoch t_0 Refer also to DF157
DF550	dZ0	± 4194.303 m	0.001 m	int23	Translation in Z for Reference Epoch t_0 Refer also to DF158
DF551	$R0_1$	± 42,949.67294 ″	0.00002 ″	int32	Rotation around the X-axis for Reference Epoch t_0 Refer also to DF159
DF552	$R0_2$	± 42,949.67294 ″	0.00002 ″	int32	Rotation around the Y-axis for Reference Epoch t_0 Refer also to DF160
DF553	$R0_3$	± 42,949.67294 ″	0.00002 ″	int32	Rotation around the Z-axis for Reference Epoch t_0

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
					Refer also to DF161
DF554	dS0	± 167.77215 ppm	0.00001 ppm	int25	Scale correction for Reference Epoch t_0 Refer also to DF162
DF555	dot dX	± 1.3107 m/yr	0.00002 m/yr	int17	Rate of change of translation in X
DF556	dot dY	± 1.3107 m/yr	0.00002 m/yr	int17	Rate of change of translation in Y
DF557	dot dZ	± 1.3107 m/yr	0.00002 m/yr	int17	Rate of change of translation in Z
DF558	dot R_1	± 0.026214 ″/yr	0.0000004 ″/yr	int17	Rate of change of rotation around the X-axis
DF559	dot R_2	± 0. 026214 ″/yr	0.0000004 ″/yr	int17	Rate of change of rotation around the Y-axis
DF560	dot R_3	± 0. 026214 ″/yr	0.0000004 ″/yr	int17	Rate of change of rotation around the Z-axis
DF561	dot dS	± 0.0016382 ppm/yr	0.0000002 ppm/yr	int14	Rate of change of scale correction
DF562	Service-CRS-Name Counter	0 – 31		uint5	The Service-Name Counter defines the number of characters (bytes) to follow in Service-Name
DF563	Service-CRS-Name			char8(N)	Name of service coordinate-system. (see also Data Field Notes Source/Target Name (DF144, DF146)
DF564	Coordinate Epoch CE	0-655,35	0.01 yr	uint16	"0" indicates current epoch. Otherwise, the coordinate epoch is in decimal year Coordinate Epoch = 1900 + CE Note: CE should be set to "0" for SSR services. Services shall use the CE, if station coordinates in MT1005, 1006 and 1032 are not referenced to the current epoch. It is the responsibility of service provider and rover algorithm to correctly handle station velocities if the Coordinate Epoch is not Current Epoch.

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF565	RTCM-CRS-Name Counter	0 – 31		uint5	The RTCM-CRS-Name Counter defines the number of characters (bytes) to follow in the CRS-Name
DF566	RTCM-CRS-Name			char8(N)	Name of coordinate reference system (CRS) as used in transformation messages (1021,1022 and 1301) as Source-Name/Target-Name or as Service-CRS-Name in Service-CRS message (1300).
DF567	Anchor	0 – 1	N/A	bit(1)	Identifies if the CRS is a global CRS or a CRS fixed to a tectonic plate: 0: global CRS, Plate Number should be set to unknown (0) 1: plate-fixed CRS, tectonic plate should be identified with DF149
DF568	Number of Database-Links	0 – 7	1	uint3	Number of Database-Links to follow. 0: no Database-Link follows
DF569	Database-Link Counter	0 – 31		uint5	The Database-Link Counter defines the number of characters (bytes) to follow in Database-Link
DF570	Database-Link			char8(N)	The Database-Link shall use the name of a database and the name of the category/table separated by a dot, followed by the corresponding code separated by a colon (e.g. EPSG.DATUM:1165 or EPSG.CRS:7789).
DF571	BDS Residuals Epoch Time (TOW)	0-604799 s	1s	uint20	
DF572	BDS Number of Satellite Signals Processed	0-31		uint5	
DF573	Galileo Residuals Epoch Time (TOW)	0-604799 s	1s	uint20	
DF574	Galileo Number of Satellite Signals Processed	0-31		uint5	

© RTCM – Not for reproduction or redistribution

DF #	DF Name	DF Range	DF Resolution	Data Type	Data Field Notes
DF575	QZSS Residuals Epoch Time (TOW)	0-604799 s	1s	uint20	
DF576	QZSS Number of Satellite Signals Processed	0-31		uint5	

Note 1: Effective range is the maximum range attainable with the indicated bit allocation and scale factor.

Table 3.4-2 Lock Time Indicator, Data Fields DF013, DF019, DF043, DF049

Indicator (i)	Minimum Lock Time (s)	Range of Indicated Lock Times
0-23	i	$0 < \text{lock time} < 24$
24-47	$i \cdot 2 - 24$	$24 \leq \text{lock time} < 72$
48-71	$i \cdot 4 - 120$	$72 \leq \text{lock time} < 168$
72-95	$i \cdot 8 - 408$	$168 \leq \text{lock time} < 360$
96-119	$i \cdot 16 - 1176$	$360 \leq \text{lock time} < 744$
120-126	$i \cdot 32 - 3096$	$744 \leq \text{lock time} < 937$
127	---	$\text{lock time} \geq 937$

Note 1 - Determining Loss of Lock: In normal operation, a cycle slip will be evident when the Minimum Lock Time (MLT) has decreased in value. For long time gaps between messages, such as from a radio outage, extra steps should be taken on the rover to safeguard against missed cycle slips.

© RTCM – Not for reproduction or redistribution

Table 3.4-3 SBAS PRN Codes, Data Fields DF009, DF038

SBAS Code	GPS/GLONASS Satellite ID	SBAS Code	GPS/GLONASS Satellite ID	SBAS Code	GPS/GLONASS Satellite ID
120	40	127	47	134	54
121	41	128	48	135	55
122	42	129	49	136	56
123	43	130	50	137	57
124	44	131	51	138	58
125	45	132	52		
126	46	133	53		

Table 3.4-4 Carrier Smoothing Interval of Code Phase, DF008 and DF037

Indicator	Smoothing Interval
000 (0)	*No smoothing*
001 (1)	*< 30 s*
010 (2)	*30-60 s*
011 (3)	*1-2 min*
100 (4)	*2-4 min*
101 (5)	*4-8 min*
110 (6)	*>8 min*
111 (7)	*Unlimited smoothing interval*

© RTCM – Not for reproduction or redistribution

Table 3.4-5 GLONASS Carrier Frequencies in L1 and L2 Bands

Satellite Frequency Channel Indicator	No. of channel	Nominal value of frequency in L1 Band, MHz	Nominal value of frequency in L2 Band, MHz
0	-07	1598.0625	1242.9375
1	-06	1598.6250	1243.3750
2	-05	1599.1875	1243.8125
3	-04	1599.7500	1244.2500
4	-03	1600.3125	1244.6875
5	-02	1600.8750	1245.1250
6	-01	1601.4375	1245.5625
7	00	1602.0	1246.0
8	01	1602.5625	1246.4375
9	02	1603.125	1246.875
10	03	1603.6875	1247.3125
11	04	1604.25	1247.75
12	05	1604.8125	1248.1875
13	06	1605.375	1248.625
14	07	1605.9375	1249.0625
15	08	1606.5	1249.5
16	09	1607.0625	1249.9375
17	10	1607.625	1250.375
18	11	1608.1875	1250.8125
19	12	1608.75	1251.25
20	13	1609.3125	1251.6875

© RTCM – Not for reproduction or redistribution

Table 3.4-6 GLONASS Satellite Frequency Channel Number (DF419)

DF value	No of channel	Nominal value of frequency in L1 Band, MHz	Nominal value of frequency in L2 Band, MHz
0	-7	1598.0625	1242.9375
1	-6	1598.6250	1243.3750
2	-5	1599.1875	1243.8125
3	-4	1599.7500	1244.2500
4	-3	1600.3125	1244.6875
5	-2	1600.8750	1245.1250
6	-1	1601.4375	1245.5625
7	0	1602.0	1246.0
8	1	1602.5625	1246.4375
9	2	1603.125	1246.875
10	3	1603.6875	1247.3125
11	4	1604.25	1247.75
12	5	1604.8125	1248.1875
13	6	1605.375	1248.625
14	Reserved		
15	The Frequency Channel Number is not known, not available, or not applicable (e.g., CDMA-only capable Satellite)		

© RTCM – Not for reproduction or redistribution

3.5 Messages

This section describes the messages. Each message contains a specific set of data fields, sometimes repeated, as in the case where information on several satellites is provided. The data fields are broadcast in the order listed. Multi-byte values are expressed with the most significant byte transmitted first and the least significant byte transmitted last.

3.5.1 Experimental Messages

Message numbers in the range 1-100 shall not be encoded or decoded. These message numbers shall be used exclusively by the SC104 participants for new message development. New messages will be assigned to these message numbers while in development.

3.5.2 Legacy RTK Observable Messages

Message Types 1001-1005 and 1009-1012 are the legacy observable messages supporting only GPS and GLONASS. New implementations should prefer the MSM messages detailed in Section 3.5.12 that support all constellations.

3.5.2.1 GPS RTK Observable Messages

Table 3.5-1 through Table 3.5-5 provide the contents of the GPS real-time kinematic (RTK) messages, which are based on raw data. From these data, valid RINEX files can be obtained, although when the Pseudorange Modulus Ambiguity (DF014, DF044) is not provided, ephemeris and clock information will be required to make the conversion. Consequently, this set of messages offers a high level of interoperability and compatibility with standard surveying practices. If GPS RTK Messages (1001-1004) are used in a Network RTK application, their content representing L1 and L2 Phaseranges might be altered by correcting for antenna phase center variations (see also section 3.1.6). However, the properties of the Phaseranges must be indicated properly with an Antenna Description message. Note: Observations corrected for antenna phase center variations are no longer compatible with the RINEX standard definition.

Table 3.5-1 Contents of the Message Header, Types 1001, 1002, 1003, 1004: GPS RTK Messages

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS
Message Number (e.g., “1001” = 0011 1110 1001)	DF002	uint12	12
Reference Station ID	DF003	uint12	12
GPS Epoch Time (TOW)	DF004	uint30	30
Synchronous GNSS Flag	DF005	bit(1)	1

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS
No. of GPS Satellite Signals Processed	DF006	uint5	5
GPS Divergence-free Smoothing Indicator	DF007	bit(1)	1
GPS Smoothing Interval	DF008	bit(3)	3
TOTAL			***64***

The Type 1001 Message supports single-frequency RTK operation. It does not include an indication of the satellite carrier-to-noise ratio as measured by the reference station.

Table 3.5-2 Contents of the Satellite-Specific Portion of a Type 1001 Message, Each Satellite – GPS Basic RTK, L1 Only

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS
GPS Satellite ID	DF009	uint6	6
GPS L1 Code Indicator	DF010	bit(1)	1
GPS L1 Pseudorange	DF011	uint24	24
GPS L1 Phaserange – L1 Pseudorange	DF012	int20	20
GPS L1 Lock time Indicator	DF013	uint7	7
TOTAL			***58***

The Type 1002 Message supports single-frequency RTK operation and includes an indication of the satellite carrier-to-noise ratio (CNR) as measured by the reference station. Since the CNR does not usually change from measurement to measurement, this Message type can be mixed with the Type 1001, and used primarily when a satellite CNR changes, thus saving broadcast link throughput.

© RTCM – Not for reproduction or redistribution

Table 3.5-3 Contents of the Satellite-Specific Portion of a Type 1002 Message, Each Satellite – GPS Extended RTK, L1 Only

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS
GPS Satellite ID	DF009	uint6	6
GPS L1 Code Indicator	DF010	bit(1)	1
GPS L1 Pseudorange	DF011	uint24	24
GPS L1 Phaserange – L1 Pseudorange	DF012	int20	20
GPS L1 Lock time Indicator	DF013	uint7	7
GPS Integer L1 Pseudorange Modulus Ambiguity	DF014	uint8	8
GPS L1 CNR	DF015	uint8	8
TOTAL			***74***

The Type 1003 Message supports dual-frequency RTK operation but does not include an indication of the satellite carrier-to-noise (CNR) as measured by the reference station.

© RTCM – Not for reproduction or redistribution

Table 3.5-4 Contents of the Satellite-Specific Portion of a Type 1003 Message, Each Satellite – GPS Basic RTK, L1 & L2

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS
GPS Satellite ID	DF009	uint6	6
GPS L1 Code Indicator	DF010	bit(1)	1
GPS L1 Pseudorange	DF011	uint24	24
GPS L1 Phaserange – L1 Pseudorange	DF012	int20	20
GPS L1 Lock time Indicator	DF013	uint7	7
GPS L2 Code Indicator	DF016	bit(2)	2
GPS L2-L1 Pseudorange Difference	DF017	int14	14
GPS L2 Phaserange – L1 Pseudorange	DF018	int20	20
GPS L2 Lock time Indicator	DF019	uint7	7
TOTAL			***101***

© RTCM – Not for reproduction or redistribution

The Type 1004 Message supports dual-frequency RTK operation and includes an indication of the satellite carrier-to-noise (CNR) as measured by the reference station. Since the CNR does not usually change from measurement to measurement, this message type can be mixed with the Type 1003, and used only when a satellite CNR changes, thus saving broadcast link throughput.

Table 3.5-5 Contents of the Satellite-Specific Portion of a Type 1004 Message, Each Satellite – GPS Extended RTK, L1 & L2

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS
GPS Satellite ID	DF009	uint6	6
GPS L1 Code Indicator	DF010	bit(1)	1
GPS L1 Pseudorange	DF011	uint24	24
GPS L1 Phaserange – L1 Pseudorange	DF012	int20	20
GPS L1 Lock time Indicator	DF013	uint7	7
GPS Integer L1 Pseudorange Modulus Ambiguity	DF014	uint8	8
GPS L1 CNR	DF015	uint8	8
GPS L2 Code Indicator	DF016	bit(2)	2
GPS L2-L1 Pseudorange Difference	DF017	int14	14
GPS L2 Phaserange – L1 Pseudorange	DF018	int20	20
GPS L2 Lock time Indicator	DF019	uint7	7
GPS L2 CNR	DF020	uint8	8
TOTAL			***125***

© RTCM – Not for reproduction or redistribution

3.5.2.2 GLONASS RTK Observable Messages

Table 3.5-6 through Table 3.5-10 provide the contents of the GLONASS real-time kinematic (RTK) messages, which are based on raw data. From this data, complete RINEX files can be obtained. As a consequence, this set of messages offers a high level of interoperability and compatibility with standard surveying practices. The service provider using these messages should also transmit an Antenna Reference Point message (Type 1005 or 1006) and an Antenna Descriptor message (Type 1007 or 1008). A provider of combined GPS-GLONASS service should provide completely independent sets of observables. In addition, if the time tags of the GPS and GLONASS RTK data are synchronized, the Synchronous GNSS Flag should be used to connect the entire RTK data block.

Table 3.5-6 Contents of the Message Header, Types 1009 through 1012: GLONASS RTK Messages

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS
Message Number	DF002	uint12	12
Reference Station ID	DF003	uint12	12
GLONASS Epoch Time	DF034	uint27	27
Synchronous GNSS Flag	DF005	bit(1)	1
No. of GLONASS Satellite Signals Processed	DF035	uint5	5
GLONASS Divergence-free Smoothing Indicator	DF036	bit(1)	1
GLONASS Smoothing Interval	DF037	bit(3)	3
TOTAL			***61***

© RTCM – Not for reproduction or redistribution

The Type 1009 Message supports single-frequency RTK operation but does not include an indication of the satellite carrier-to-noise (CNR) as measured by the reference station.

Table 3.5-7 Contents of the Satellite-Specific Portion of a Type 1009 Message, Each Satellite – GLONASS Basic RTK, L1 Only

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS
GLONASS Satellite ID (Satellite Slot Number)	DF038	uint6	6
GLONASS Code Indicator	DF039	bit(1)	1
GLONASS Satellite Frequency Channel Number	DF040	uint5	5
GLONASS L1 Pseudorange	DF041	uint25	25
GLONASS L1 Phaserange – L1 Pseudorange	DF042	int20	20
GLONASS L1 Lock time Indicator	DF043	uint7	7
TOTAL			***64***

The Type 1010 Message supports single-frequency RTK operation and includes an indication of the satellite carrier-to-noise (CNR) as measured by the reference station. Since the CNR does not usually change from measurement to measurement, this message type can be mixed with the Type 1009, and used only when a satellite CNR changes, thus saving broadcast link throughput.

Table 3.5-8 Contents of the Satellite-Specific Portion of a Type 1010 Message, Each Satellite – GLONASS Extended RTK, L1 Only

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS
GLONASS Satellite ID (Satellite Slot Number)	DF038	uint6	6
GLONASS L1 Code Indicator	DF039	bit(1)	1
GLONASS Satellite Frequency Channel Number	DF040	uint5	5
GLONASS L1 Pseudorange	DF041	uint25	25

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS
GLONASS L1 Phaserange – L1 Pseudorange	DF042	int20	20
GLONASS L1 Lock time Indicator	DF043	uint7	7
GLONASS Integer L1 Pseudorange Modulus Ambiguity	DF044	uint7	7
GLONASS L1 CNR	DF045	uint8	8
TOTAL			***79***

The Type 1011 Message supports dual-frequency RTK operation but does not include an indication of the satellite carrier-to-noise (CNR) as measured by the reference station.

Table 3.5-9 Contents of the Satellite-Specific Portion of a Type 1011 Message, Each Satellite – GLONASS Basic RTK, L1 & L2

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS
GLONASS Satellite ID (Satellite Slot Number)	DF038	uint6	6
GLONASS L1 Code Indicator	DF039	bit(1)	1
GLONASS Satellite Frequency Channel Number	DF040	uint5	5
GLONASS L1 Pseudorange	DF041	uint25	25
GLONASS L1 Phaserange – L1 Pseudorange	DF042	int20	20
GLONASS L1 Lock time Indicator	DF043	uint7	7
GLONASS L2 Code Indicator	DF046	bit(2)	2
GLONASS L2-L1 Pseudorange Difference	DF047	int14	14
GLONASS L2 Phaserange – L1 Pseudorange	DF048	int20	20
GLONASS L2 Lock time Indicator	DF049	uint7	7

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS
TOTAL			***107***

The Type 1012 Message supports dual-frequency RTK operation and includes an indication of the satellite carrier-to-noise (CNR) as measured by the reference station. Since the CNR does not usually change from measurement to measurement, this message type can be mixed with the Type 1011, and used only when a satellite CNR changes, thus saving broadcast link throughput.

Table 3.5-10 Contents of the Satellite-Specific Portion of a Type 1012 Message, Each Satellite – GLONASS Extended RTK, L1 & L2

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS
GLONASS Satellite ID (Satellite Slot Number)	DF038	uint6	6
GLONASS L1 Code Indicator	DF039	bit(1)	1
GLONASS Satellite Frequency Channel Number	DF040	uint5	5
GLONASS L1 Pseudorange	DF041	uint25	25
GLONASS L1 Phaserange – L1 Pseudorange	DF042	int20	20
GLONASS L1 Lock time Indicator	DF043	uint7	7
GLONASS Integer L1 Pseudorange Modulus Ambiguity	DF044	uint7	7
GLONASS L1 CNR	DF045	uint8	8
GLONASS L2 Code Indicator	DF046	bit(2)	2
GLONASS L2-L1 Pseudorange Difference	DF047	int14	14
GLONASS L2 Phaserange – L1 Pseudorange	DF048	int20	20
GLONASS L2 Lock time Indicator	DF049	uint7	7

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS
GLONASS L2 CNR	DF050	uint8	8
TOTAL			***130***

3.5.3 System Parameter Messages

The complete list of record announcements summarizes all messages transmitted by a specific reference station.

Table 3.5-11 Contents of the Type 1013 Message, System Parameters

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS
Message Number	DF002	uint12	12
Reference Station ID	DF003	uint12	12
Modified Julian Day (MJD) Number	DF051	uint16	16
Seconds of Day (UTC)	DF052	uint17	17
No. of Message ID Announcements to Follow (N_m)	DF053	uint5	5
Leap Seconds, GPS-UTC	DF054	uint8	8
Message ID #1	DF055	uint12	12
Message #1 Sync Flag	DF056	bit(1)	1
Message #1 Transmission Interval	DF057	uint16	16
Message ID #2	DF055	uint12	12
Message #2 Sync Flag	DF056	bit(1)	1
Message #2 Transmission Interval	DF057	uint16	16
(Repeat until Nm sets)			
TOTAL			***70+29**** N_m

© RTCM – Not for reproduction or redistribution

3.5.4 Stationary Antenna Reference Point Messages

Message Type 1005 (see Table 3.5-12) provides the earth-centered, earth-fixed (ECEF) coordinates of the antenna reference point (ARP) for a stationary reference station. No height above a monument is provided.

Message Type 1006 (see Table 3.5-13) provides all the same information as Message Type 1005, but additionally provides the height of the ARP above a survey monument.

The phase center is not a point in space that can be used as a standard reference. For one thing, it varies with frequency. In addition, the location of L1 phase center is strongly dependent on the antenna calibration method used during the calibration process. Therefore, the location of the L1 phase center may vary between different calibration tables for the same antenna model. Message Types 1005 and 1006 avoid the phase center problem by utilizing the Antenna Reference Point, which is used throughout the International GNSS Service (IGS).

Message Types 1005 and 1006 contain the coordinates of the installed antenna's ARP in Earth-Center-Earth-Fixed (ECEF) coordinates -- datum definitions are not yet supported. The coordinates always refer to a physical point on the antenna, typically the bottom of the antenna mounting surface.

The Quarter Cycle Indicator was introduced to avoid inconsistencies between different approaches by different receiver hardware manufacturers to handle quarter-cycle shifts between carrier phase observations on the same frequency (see Section 3.1.8).

Table 3.5-12 Contents of the Type 1005 Message – Stationary Antenna Reference Point, No Height Information

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS
Message Number	DF002	uint12	12
Reference Station ID	DF003	uint12	12
Reserved	DF001	bit(6)	6
GPS Indicator	DF022	bit(1)	1
GLONASS Indicator	DF023	bit(1)	1
Galileo Indicator	DF024	bit(1)	1
Reference-Station Indicator	DF141	bit(1)	1
Antenna Reference Point ECEF-X	DF025	int38	38

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS
Single Receiver Oscillator Indicator	DF142	bit(1)	1
Reserved	DF001	bit(1)	1
Antenna Reference Point ECEF-Y	DF026	int38	38
Quarter Cycle Indicator	DF364	bit(2)	2
Antenna Reference Point ECEF-Z	DF027	int38	38
TOTAL			***152***

Table 3.5-13 Contents of the Type 1006 Message – Stationary Antenna Reference Point, with Height Information

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS
Message Number	DF002	uint12	12
Reference Station ID	DF003	uint12	12
Reserved	DF001	bit(6)	6
GPS Indicator	DF022	bit(1)	1
GLONASS Indicator	DF023	bit(1)	1
Galileo Indicator	DF024	bit(1)	1
Reference-Station Indicator	DF141	bit(1)	1
Antenna Reference Point ECEF-X	DF025	int38	38
Single Receiver Oscillator Indicator	DF142	bit(1)	1
Reserved	DF001	bit(1)	1
Antenna Reference Point ECEF-Y	DF026	int38	38

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS
Quarter Cycle Indicator	DF364	bit(2)	2
Antenna Reference Point ECEF-Z	DF027	int38	38
Antenna Height	DF028	uint16	16
TOTAL			***168***

3.5.5 Receiver and Antenna Description Messages

Messages 1007, 1008, and 1033 provide information on the reference station antenna and receiver.

Message 1007, described in Table 3.5-14, provides the minimal information on the reference station antenna: an ASCII descriptor of the reference station antenna.

Message 1008 in Table 3.5-15 provides the same information as Message 1007 but additionally includes the antenna serial number, which removes any ambiguity about the model number or production run.

Message 1033 in Table 3.5-16 is an elaboration of Message Type 1008. This message contains information on both the reference station's antenna and receiver type and firmware version. This message serves several purposes:

1. It allows recipients of RTCM data messages to directly generate RINEX files from the data stream. By utilizing the proper IGS syntax for receiver and names, this data as well as serial numbers and firmware versions can be placed in the header of the RINEX observation files.
2. It enables proper identification of the reference station receiver type via the RTCM data stream.

The serial number and firmware version strings are not standardized. They will correspond to the manufacturer's naming convention.

In all cases the receiver type descriptor (DF030) should conform to the 20-character definition used by the IGS as defined in the IGS ANTEX or Naming Convention (rcvr_ant.tab) files. If the receiver type, antenna type, serial numbers, etc., are unknown, the length of the counters should be set to zero, in which case the succeeding descriptor field is not utilized.

Because it has the most complete information, Message 1033 is the preferred option for transmission. Otherwise, Message 1008 is preferred over Message 1007. Only one of 1007, 1008, or 1033 should be transmitted.

© RTCM – Not for reproduction or redistribution

Table 3.5-14 Contents of the Type 1007 Message – Antenna Descriptor

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
Reference Station ID	DF003	uint12	12	
Descriptor Counter N	DF029	uint8	8	N ≤ 31
Antenna Descriptor	DF030	char8(N)	8*N	
Antenna Setup ID	DF031	uint8	8	
TOTAL			***40+8*N***	

Table 3.5-15 Contents of the Type 1008 Message – Antenna Descriptor and Serial Number

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
Reference Station ID	DF003	uint12	12	
Descriptor Counter N	DF029	uint8	8	
Antenna Descriptor	DF030	char8(N)	8*N	N ≤ 31
Antenna Setup ID	DF031	uint8	8	
Serial Number Counter M	DF032	uint8	8	
Antenna Serial Number	DF033	char8(M)	8*M	M ≤ 31
TOTAL			***48+ 8*(M+N)***	

© RTCM – Not for reproduction or redistribution

Table 3.5-16 Contents of the Type 1033 Message – Receiver and Antenna Descriptors

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
Reference Station ID	DF003	uint12	12	
Antenna Descriptor Counter N	DF029	uint8	8	
Antenna Descriptor	DF030	char(N)	8*N	N ≤ 31
Antenna Setup ID	DF031	uint8	8	
Antenna Serial Number Counter M	DF032	uint8	8	
Antenna Serial Number	DF033	char(M)	8*M	M ≤ 31
Receiver Type Descriptor Counter I	DF227	uint8	8	
Receiver Type Descriptor	DF228	char(I)	8*I	I ≤ 31
Receiver Firmware Version Counter J	DF229	uint8	8	
Receiver Firmware Version	DF230	char(J)	8*J	J ≤ 31
Receiver Serial Number Counter K	DF231	uint8	8	
Receiver Serial Number	DF232	char(K)	8*K	K ≤ 31
TOTAL			***72+ 8*(M+N+ I+J+K)***	

3.5.6 Master Auxiliary Concept Network RTK Correction Messages

3.5.6.1 GPS Network RTK Correction Messages

The use of single reference stations to transmit RTK data is limited by the fact that the accuracy and reliability of integer ambiguity resolution deteriorates with increasing distance from the reference station. A powerful solution to this problem is offered by a synchronized network of RTK stations. Networks of reference stations mitigate the distance-dependency of RTK solutions. With such networks, a provider can generate measurement corrections for receivers operating within a large defined region, and this information can be supplied to the user in a standard format. As the current kinematic and high-accuracy Message Types 1001-1012 do not support

© RTCM – Not for reproduction or redistribution

an efficient use of data from multiple reference stations, new approaches must be developed to facilitate the valuable information afforded by networks of reference stations.

The standardization of network information and processing models is also necessary to reduce the size of the network RTK corrections, as well as the satellite-independent error information. A simplified approach of transmitting data from reference station networks to roving users is utilized below in the form of a new message set capable of supporting reference network operations.

Individual reference stations often support more than one network in a large region. A detailed description of how the message set supports these networks is given below. The approach used here provides considerable flexibility for the service provider to support a wide variety of services within range of a large network of reference stations.

The correction difference components have been split into a dispersive and a non-dispersive part. The dispersive correction difference is also called ionospheric correction difference, after its contributor. The opposite, the non-dispersive, is also called ionosphere-free correction difference or geometric correction difference, recognizing that it is not purely due to geometry because of the tropospheric contribution.

The L1 Correction (L1C) and the L2 Correction (L2C) can be determined in general by:

$$L1C_S = s_S - \Phi_{S,1}(t) - \frac{c}{f_1} N_{S,1} + t_{S,1} + A_{S,1}$$, and

$$L2C_S = s_S - \Phi_{S,2}(t) - \frac{c}{f_2} N_{S,2} + t_{S,2} + A_{S,2}$$, with

s_S	=	Computed Geometric Range in meters between the ARP of station S and satellite
$\Phi_{S,1}(t)$	=	Phaserange Measurement in meters for station S, L1 (similarly for L2)
$\frac{c}{f_1} N_{S,1}$	=	Integer Ambiguity part scaled to meters, L1 (similarly for L2). The integer can be arbitrarily chosen for an initial measurement to bring the resulting Phase Correction within range of the field definition. For Network RTK the all-integer ambiguities must be brought onto a common integer level. Therefore, all values per-satellite and per-frequency must be synchronized between reference stations.
$t_{S,1}, t_{S,2}$	=	Receiver clock term for the respective frequency of Phaserange Measurement
$A_{S,1}, A_{S,2}$	=	Antenna Offset and Phase Center Variation Correction for the respective frequency; the service provider must ensure that the antenna phase center corrections do not introduce biases. See, also, Section 3.1.6, “Proper Handling of Antenna Phase Center Variation Corrections”.
f_1	=	L1 carrier frequency

© RTCM – Not for reproduction or redistribution

f_2 = L2 carrier frequency

Satellite and relativistic clock term have been neglected in the given formula. These terms cancel sufficiently in the inter-station single difference. A difference of the clock between both station receivers remains in the correction differences. However, the value common to all correction differences for every master-auxiliary reference station pair can be estimated and removed from the correction differences. These inter-station clock biases are also minimized for typical Network RTK applications. The clock difference term between reference stations in the L1 and L2 correction difference may be treated independently. Therefore, clock effects may influence Ionospheric and geometric correction differences. Nevertheless, this approach is sufficient for general positioning approaches, because residual clock effects are removed in double differences. Proper treatment of antenna phase center corrections is crucial to avoid unrecoverable biases in correction differences: see Section 3.1.6, "Proper Handling of Antenna Phase Center Variation Corrections".

The L1 Correction Difference (L1CD) is calculated as the single-difference of the "Auxiliary Reference Station Carrier Phase Correction" minus "Master Reference Station Carrier Phase Correction".

$$L1CD = L1C_A - L1C_M$$

An alternate way of calculation is to carry out:

$$L1CD = \Delta s_{AM}(t) - \Delta\Phi_{AM,1}(t) - \frac{c}{f_1} \cdot \Delta N_{AM,1} + \Delta t_{AM,1} + \Delta A_{AM,1}$$

$\Delta\Phi_{AM,1}(t)$ = Single-differenced Phaseranges of Auxiliary Reference Station A minus Master Reference Station M

$\Delta s_{AM}(t)$ = Single-differenced slope distances between Satellite and reference station antenna of Auxiliary Reference Station A minus Master Reference Station M

$\frac{c}{f_1} \cdot \Delta N_{AM,1}$ = Single-differenced Integer Ambiguity values of Auxiliary Reference Station A minus Master Reference Station M scaled to meters. In practice only double-differenced integer ambiguities can be fixed due to insufficient modeling of various error sources. The single-differenced integer ambiguities for a particular auxiliary reference station minus master reference station might incorporate an arbitrary integer number. The number is arbitrary but common for all satellites and therefore is observed as a common clock error.

$\Delta t_{AM,1}$ = Estimated single-differenced receiver clock term on L1.

$\Delta A_{AM,1}$ = Single-differenced antenna offset and PCV on L1

Similarly, the L2 Correction Difference (L2CD) is computed as follows:

$$L2CD = \Delta s_{AM}(t) - \Delta\Phi_{AM,2}(t) - \frac{c}{f_2} \cdot \Delta N_{AM,2} + \Delta t_{AM,2} + \Delta A_{AM,2}$$

© RTCM – Not for reproduction or redistribution

Correct integer ambiguity resolution between reference stations is only possible on a double-difference basis. The correct set of double-differenced integer ambiguities is unique for a given data set. A common integer ambiguity level indicates that correction differences are derived from a homogenous solution satisfying the double-difference requirement between all involved reference stations.

Correction differences are typically based on integer-adjusted raw observation data. Certain rules must be observed to preserve the correctness of the double-difference requirement. In particular, introducing a cycle for one specific satellite-station combination must be compensated for by adjusting other satellite-station combinations in order to maintain a homogenous solution.

However, the correction differences are defined as single-differenced values between two reference stations. Therefore, the introduction of a fixed number of cycles for the observations of one satellite for all reference stations throughout the whole network will not show up in the correction differences. Changing all observations for a specific reference station by a fixed number of cycles will change all correction differences. The number of introduced cycles will be absorbed by the clock bias estimation in the rover.

Two subsets of a network might satisfy the requirement of correct double-differenced integer ambiguities, without necessarily being on the same integer ambiguity level since the choices of integers are arbitrary. As soon as a reference station has common integer ambiguity levels with two subsets of a network these two subsets can be joined and brought to the same integer ambiguity level. These two subsets will then form one subnetwork with the same integer ambiguity level. If circumstances are such that not enough satellites with correct fixed integer ambiguities are available for one reference station or a number of reference stations connecting two subsets of a network, it is advisable to treat the subsets separately. Both subsets will be considered to form different subnetworks with different integer ambiguity levels indicated by assigning different subnetwork ID's.

The correct integer ambiguity level L1-L2 widelane indicates that only the L1-L2 widelane is correctly fixed. The individual L1 and L2 integer ambiguities may contain integer offsets. The L1 and the L2 offsets will be the same.

Changing the ambiguity status flag from 3 to 2, 3 to 1, or 2 to 1 without increasing the non-sync count indicates that the previous good guess of the integer ambiguity turned out to be the correct integer and the correctness of the integer has been approved by the networking software.

Unrecoverable cycle slips might occur for permanent reference station applications as well. If the integer ambiguity for the satellite with the cycle slip is on the integer ambiguity level (see also the discussion above on integer ambiguity level), the unrecoverable cycle will cause the loss of the integer ambiguity level for the specific satellite, reference station, and frequency. Correction difference information for this specific combination needs proper flagging in the ambiguity status flag and increasing the non-sync count. The non-sync count must be increased if there is an unrecoverable cycle slip in a correction difference that is not on the integer ambiguity level. Frequent cycle slips may cause problems with the counter's range. The non-sync count should not be increased more than once per minute in order to reduce counter roll-overs. Satellites with cycle slips more frequent than once per minute should not be transmitted.

Arbitrarily fixed, and therefore possibly non-correct integers, provide only sufficient information when the identical integers are used for a certain amount of time. An increase of the non-sync count will be associated with ambiguity status flag of 3. In the continuation of the operation the networking software might prove that the arbitrarily chosen integers are actually on the correct integer ambiguity

© RTCM – Not for reproduction or redistribution

level. Under these circumstances the ambiguity status flag might be changed to the appropriate status of 1 or 2. This is discussed further in Annex A.1

Table 3.5-17 Contents of the Network Auxiliary Station Data Message 1014

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	1014
Network ID	DF059	uint8	8	
Subnetwork ID	DF072	uint4	4	
Number of Auxiliary Stations Transmitted	DF058	uint5	5	0 - 31
Master Reference Station ID	DF060	unit12	12	
Auxiliary Reference Station ID	DF061	uint12	12	
Aux-Master Delta Latitude	DF062	int20	20	
Aux-Master Delta Longitude	DF063	int21	21	
Aux-Master Delta Height	DF064	int23	23	
TOTAL			***117***	

Table 3.5-18 Contents of Header Network RTK -- Messages 1015, 1016 or 1017

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	1015, 1016 or 1017
Network ID	DF059	uint8	8	
Subnetwork ID	DF072	uint4	4	
GPS Epoch Time (GPS TOW)	DF065	uint23	23	

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Multiple Message Indicator	DF066	bit(1)	1	
Master Reference Station ID	DF060	uint12	12	
Auxiliary Reference Station ID	DF061	uint12	12	
# of GPS Sats	DF067	uint4	4	
TOTAL			***76***	

Table 3.5-19 Contents of Data Block for GPS Ionospheric Correction Differences 1015

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GPS Satellite ID	DF068	uint6	6	
GPS Ambiguity Status Flag	DF074	bit(2)	2	
GPS Non Sync Count	DF075	uint3	3	
GPS Ionospheric Carrier Phase Correction Difference	DF069	int17	17	
TOTAL			***28***	

Table 3.5-20 Contents of Data Block for GPS Geometric Correction Differences 1016

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GPS Satellite ID	DF068	uint6	6	
GPS Ambiguity Status Flag	DF074	bit(2)	2	
GPS Non Sync Count	DF075	uint3	3	

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GPS Geometric Carrier Phase Correction Difference	DF070	int17	17	
GPS IODE	DF071	uint8	8	
TOTAL			***36***	

Table 3.5-21 Contents of Data Block for GPS Combined Geometric and Ionospheric Correction Differences

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GPS Satellite ID	DF068	uint6	6	
GPS Ambiguity Status Flag	DF074	bit(2)	2	
GPS Non Sync Count	DF075	uint3	3	
GPS Geometric Carrier Phase Correction Difference	DF070	int17	17	
GPS IODE	DF071	uint8	8	
GPS Ionospheric Carrier Phase Correction Difference	DF069	int17	17	
TOTAL			***53***	

3.5.6.2 GLONASS Network RTK Correction Messages

The principle of GPS Network RTK as described in the previous section can also be adopted for GLONASS Network RTK. A simplified approach of transmitting GLONASS data from reference station networks to roving users is utilized below in the form of a new message set capable of supporting GLONASS reference network operations. The GLONASS carrier phase correction difference messages follow the same format defined for the GPS Network RTK messages and are intended to be utilized by the rover software using the same general processing algorithms.

© RTCM – Not for reproduction or redistribution

The GLONASS L1 carrier phase correction (L1CR) and L2 carrier phase correction (L2CR) can be determined by:

$$L1CR_S = S_S - \Phi_{S,1}(t) - \frac{c}{f_1} N_{S,1} + t_{S,1} + A_{S,1}\text{, and}$$

$$L2CR_S = S_S - \Phi_{S,2}(t) - \frac{c}{f_2} N_{S,2} + t_{S,2} + A_{S,2}\text{, with}$$

S_S	=	Computed geometric range in meters between the ARP of station S and satellite
$\Phi_{S,1}(t)$	=	GLONASS L1 Phaserange measurement in meters for station S, (similarly for L2)
$\frac{c}{f_1} N_{S,1}$	=	GLONASS L1 integer ambiguity part scaled to meters, (similarly for L2).
$t_{S,1}$	=	L1 receiver clock term for the GLONASS Phaserange measurement, (similarly for L2)
$A_{S,1}$	=	GLONASS L1 antenna offset and phase center variation correction (similarly for L2). The service provider shall ensure that the antenna phase center corrections do not introduce biases. (See also Section 3.1.6 "Proper Handling of Antenna Phase Center Variation Corrections")
f_1	=	L1 carrier frequency of the GLONASS satellite signal
f_2	=	L2 carrier frequency of the GLONASS satellite signal

The GLONASS L1 Carrier Phase Correction Difference (L1CDR) is calculated as the single-difference of the "Auxiliary Reference Station Carrier Phase Correction" minus "Master Reference Station Carrier Phase Correction".

$$L1CDR = L1CR_A - L1CR_M$$

An alternate way of calculation is to carry out:

$$L1CDR = \Delta S_{AM}(t) - \Delta\Phi_{AM,1}(t) - \frac{c}{f_1}\Delta N_{AM,1} + \Delta t_{AM,1} + \Delta A_{AM,1}$$

© RTCM – Not for reproduction or redistribution

$\Delta\Phi_{AM,1}(t)$ = Single-differenced GLONASS Phaseranges of Auxiliary Reference Station A minus Master Reference Station M

$\Delta S_{AM}(t)$ = Single-differenced slope distances between satellite and reference station antenna of Auxiliary Reference Station A minus Master Reference Station M

$\frac{c}{f_1}\Delta N_{AM,1}$ = GLONASS single-differenced integer ambiguity values of Auxiliary Reference Station A minus Master Reference Station M scaled to meters. For network RTK, all GLONASS integer ambiguities shall be brought onto a common integer level. Therefore, the single-differenced integer ambiguities for a particular Auxiliary Reference Station minus Master Reference Station might incorporate an arbitrary integer number. Although this number is common for all satellites, it will not be observed as a common clock error or cancel in a double difference due to the GLONASS FDMA signal structure. The service provider shall adjust the absolute magnitude of the GLONASS ambiguity level to minimize inter-frequency biases in the GLONASS double-difference correction differences (see also the description of the GLONASS ambiguity level in this section).

$\Delta t_{AM,1}$ = Estimated single-differenced receiver clock term on L1

$\Delta A_{AM,1}$ = Single-differenced antenna offset and PCV on L1

Similarly, the L2 Carrier Phase Correction Difference (L2CD) is computed as follows:

$$L2CDR = \Delta S_{AM}(t) - \Delta\Phi_{AM,2}(t) - \frac{c}{f_2}\Delta N_{AM,2} + \Delta t_{AM,2} + \Delta A_{AM,2}$$

Like GPS correction differences, the L1 and L2 GLONASS correction differences are factored into ionospheric and geometric components. Satellite and relativistic clock term have been neglected in the given formula. These terms cancel sufficiently in the inter-station single difference. Proper treatment of antenna phase center corrections is crucial to avoid unrecoverable biases in correction differences: see section 3.1.6 “Proper Handling of Antenna Phase Center Variation Corrections”.

A difference of the clock between both station receivers remains in the L1 and L2 correction differences. However, the value is common to all correction differences for a given master-auxiliary reference station pair can be estimated and removed. The clock difference term between reference stations in the GLONASS L1 and L2 correction difference may be treated independently. Therefore, residual clock

© RTCM – Not for reproduction or redistribution

effects may influence GLONASS ionospheric and geometric correction differences. Nevertheless, this approach is sufficient for general positioning schemes, since residual clock effects cancel in double differences.

The GPS correction differences for a given master-auxiliary pair may contain an arbitrary integer bias that cancels in double differences (see also Section 3.5.6 "GPS Network RTK Correction Messages"). However, an arbitrary integer bias influencing GLONASS correction differences may manifest as so-called inter-frequency biases in double differences as a consequence of the GLONASS frequency division multiple access (FDMA) signal structure. For example, approximately 4.3 mm of error will be introduced for every 5 cycles of bias in the L1 ambiguity level if the difference between the frequency channel numbers of the two GLONASS satellite signals involved in the double difference is a maximum (13 for the current GLONASS frequency plan). However, the error is only 0.3 mm for the same integer bias if the frequency channel number difference is a minimum. These apparent inter-frequency biases will not be absorbed by the clock estimation in the rover software or cancel in the double difference and may degrade ambiguity resolution and RTK positioning performance. The Network RTK service provider, through appropriate data processing strategies, shall adjust the absolute magnitude of the L1 and L2 integer ambiguity levels to minimize inter-frequency biases in the GLONASS double-differenced correction differences. In practice, the GLONASS ambiguity level is typically adjusted using algorithms that utilize unambiguous single-differenced pseudorange measurements. The Network RTK service provider should be especially prudent if the network consists of heterogeneous GLONASS receiver makes and models. Interoperability testing has shown that between-station single-difference pseudorange and carrier phase measurements involving mixed receiver types may be influenced by receiver-dependent biases that may introduce significant inter-frequency biases of several centimeters if not treated correctly by the network software.

Table 3.5-22 through Table 3.5-25 provide the contents of the GLONASS Network RTK correction messages.

Table 3.5-22 Contents of Header for GLONASS Network RTK Messages 1037, 1038, or 1039

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS
Message Number	DF002	uint12	12
Network ID	DF059	uint8	8
Subnetwork ID	DF072	uint4	4
GLONASS Network Epoch Time	DF233	uint20	20
Multiple Message Indicator	DF066	bit(1)	1
Master Reference Station ID	DF060	uint12	12
Auxiliary Reference Station ID	DF061	uint12	12
# of GLONASS Data Entries	DF234	uint4	4

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS
TOTAL			***73***

Table 3.5-23 Contents of Data Block for Ionospheric Correction Difference Message 1037

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS
GLONASS Satellite ID (Satellite Slot Number)	DF038	uint6	6
GLONASS Ambiguity Status Flag	DF235	bit(2)	2
GLONASS Non Sync Count	DF236	uint3	3
GLONASS Ionospheric Carrier Phase Correction Difference	DF237	int17	17
TOTAL			***28***

Table 3.5-24 Contents of Data Block for Geometric Correction Difference Message 1038

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS
GLONASS Satellite ID (Satellite Slot Number)	DF038	uint6	6
GLONASS Ambiguity Status Flag	DF235	bit(2)	2
GLONASS Non Sync Count	DF236	uint3	3
GLONASS Geometric Carrier Phase Correction Difference	DF238	int17	17
GLONASS IOD	DF239	bit(8)	8
TOTAL			***36***

© RTCM – Not for reproduction or redistribution

Table 3.5-25 Contents of Data Block for Combined Geometric and Ionospheric Correction Differences Message 1039

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS
GLONASS Satellite ID (Satellite Slot Number)	DF038	uint6	6
GLONASS Ambiguity Status Flag	DF235	bit(2)	2
GLONASS Non Sync Count	DF236	uint3	3
GLONASS Geometric Carrier Phase Correction Difference	DF238	int17	17
GLONASS IOD	DF239	bit(8)	8
GLONASS Ionospheric Carrier Phase Correction Difference	DF237	int17	17
TOTAL			***53***

3.5.7 Unicode Text String Message

Message Type 1029 contains a variable-length text string for any displayable information the service provider may want to transmit to the user. For maximum flexibility, the characters in this message are in the Unicode UTF-8 encoding scheme, in which individual characters, called *code points*, consist of between one to four 1-byte *code units*. See http://www.unicode.org for the Unicode specification and conformance information.

Because the length of the string is known, a terminating NULL must not be included.

Table 3.5-26 Contents of the Unicode Text String, Message Type 1029

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	DF
Reference Station ID	DF003	uint12	12	
Modified Julian Day (MJD) Number	DF051	uint16	16	*(Note 1)*
Seconds of Day (UTC)	DF052	uint17	17	*(Note 1)*

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Number of Character Code Points to Follow	DF138	uint7	7	This represents the number of fully formed Unicode character code points in the message text. It is not necessarily the number of bytes that are needed to represent the character as UTF-8. For some messages it may not be possible to utilize the full range of this field, e.g., where many characters require 3- or 4-byte representations and together will exceed 255 code units.
Number of UTF-8 Code Units (N)	DF139	uint8	8	The length of the message is limited by this field, or possibly by DF138 (see previous note).
UTF-8 Character Code Units	DF140	utf8(N)	8*N	
TOTAL			***72+8*N***	

Note 1 – The time tag used in this message refers to the approximate time of message transmission (the actual time of transmission may be delayed by buffering). If a different time of applicability is required, the service provider may include that time within the Unicode message text.

© RTCM – Not for reproduction or redistribution

The following is an example of the Unicode Text String Message represented in hexadecimal with the UTF-8 code units in bold:

```
D3 00 27 40 50 17 00 84 73 6E 15 1E 55 54 46 2D
38 20 D0 BF D1 80 D0 BE D0 B2 D0 B5 D1 80 D0 BA
D0 B0 20 77 C3 B6 72 74 65 72 ED A3 3B
```

The parameters of the message are

- Message Number = 1029
- Reference Station ID = 23
- Modified Julian Day (MJD) Number = 132
- Seconds of Day (UTC) = 59100
- Number of Characters to Follow = 21
- Number of UTF-8 Code Units = 30

The message text used in the example is “UTF-8 проверка wörter” (UTF-8 check words) without quotes. The Unicode code points and character names for this message are:

0055	LATIN CAPITAL LETTER U	0440	CYRILLIC SMALL LETTER ER
0054	LATIN CAPITAL LETTER T	043A	CYRILLIC SMALL LETTER KA
0046	LATIN CAPITAL LETTER F	0430	CYRILLIC SMALL LETTER A
002D	HYPHEN-MINUS	0020	SPACE
0038	DIGIT EIGHT	0077	LATIN SMALL LETTER W
0020	SPACE	00F6	LATIN SMALL LETTER O WITH DIAERESIS
043F	CYRILLIC SMALL LETTER PE	0072	LATIN SMALL LETTER R
0440	CYRILLIC SMALL LETTER ER	0074	LATIN SMALL LETTER T
043E	CYRILLIC SMALL LETTER O	0065	LATIN SMALL LETTER E
0432	CYRILLIC SMALL LETTER VE	0072	LATIN SMALL LETTER R
0435	CYRILLIC SMALL LETTER IE		

3.5.8 Coordinate Information and Transformation Messages

Further information about coordinate transformations can be found at “OGP Surveying and Positioning Guidance Note Number 7, part 2 - Coordinate Conversions and Transformations including Formulas” (further on referred to as OGP) and EPSG database Version 6.11_2 (further on referred to as EPSG) at https://epsg.org/ or at the European Coordinate Reference System (CRS) website at http://crs.bkg.bund.de/.

© RTCM – Not for reproduction or redistribution

3.5.8.1 Transformation Information

For RTCM data supporting an RTK service, coordinates are measured within the ITRF or a regional realization. Surveyors and other users of RTK services must normally present their results in the coordinates of a local datum. Therefore, coordinate transformations are necessary.

Currently, transformation parameters are calculated and manually transferred to GNSS receivers, a process which can be a source of confusion. Another method is to store models on the GNSS receivers and to use these for the transformation. However, it often happens that revised models are not available in the GNSS receiver, so that users end up utilizing obsolete information.

Users have often expressed their desire to be able to utilize a simpler and more convenient method. By having RTCM messages that contain transformation data and information about the Coordinate Reference Systems, the users of the RTK service can obtain their results in the desired datum without any manual operations. The RTK service providers can then ensure that current information for the computation of the transformations is always used. The convenience of this method will further promote the acceptance of RTK services.

3.5.8.2 Coordinate Reference Systems/Datums

Identification of the CRS used by RTCM services and in the transformation messages is done by the Service-CRS and RTCM-CRS messages.

3.5.8.2.1 Service CRS

The Service-CRS message identifies the Coordinate Reference System (CRS) of the RTCM based service. All datum carrying information in a RTCM data stream such as station coordinates (e.g., Message Types 1005, 1006, 1032) or satellite orbit corrections for RTCM SSR (e.g., Message Types 1057, 1063) refer to the Service-CRS. Only one Service-CRS is allowed in an RTCM data stream.

Coordinate transformation messages (Message Types 1021, 1022 and 1301) use the Source-Name/Target-Name to reference a series of coordinate transformation operations. Multiple pairs of Source/Target transformations are possible. This allows even a chain of subsequent transformations.

Besides the unique identification of the service CRS, the Service-CRS message shall establish a connection to the transformations by linking the Service-CRS-Name to the Source-Name or Target-Name of one of the transformations. The link is established using exactly the same characters for the Service-CRS-Name and the Source-Name or Target-Name. This allows the user to unambiguously identify the CRS and perform required transformation operations to obtain coordinates in the desired Target-CRS.

In general, positioning using GNSS requires the introduction of reference coordinates for the observation epoch (current epoch). The derived rover coordinates are then obtained in the Service-CRS for the current epoch. In case of significant station velocities relative to the reference frame (as defined in the Service-CRS) and if the reference coordinates are not given for the current epoch, then the rover algorithm should transform to the current epoch by applying station velocities obtained from an adequate source. For the same condition of significant station velocities relative to the reference frame and if the rover coordinates shall be reported for a different epoch than

© RTCM – Not for reproduction or redistribution

the current epoch, the velocity of the rover position should be transformed by applying velocities obtained from an adequate source. The velocity transformation may be omitted if the reference station and rover station experience identical velocities and the movement is sufficiently small. It is the responsibility of service provider and rover algorithm to handle station velocities if needed.

The data field Coordinate Epoch (DF564) is used if the Service-CRS coordinates are not in the current epoch. SSR based services shall always set the Coordinate Epoch to current epoch.

The definition of the Service-CRS does not change with time. The Coordinate Epoch is the point of time to which time dependent coordinates of stations are referenced. Different coordinate epochs for coordinates in an RTCM data stream are not supported. It is the responsibility of receiving equipment to perform related transformations.

Services transmitting the Service-CRS message shall ensure that a Service-CRS message is sent within 30 seconds of related coordinate information.

Table 3.5-27 describes the contents of the Service-CRS message.

Table 3.5-27 Contents of Message Type 1300, Service-CRS Message

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
Service-CRS-Name Counter	DF562	uint5	5	N ≤ 31
Service-CRS-Name	DF563	char8(N)	8*N	
Coordinate Epoch	DF564	uint16	16	
TOTAL			***33 + 8*N***	

3.5.8.2.2 RTCM-CRS

The RTCM-CRS message provides additional details for Coordinate Reference Systems (CRS) used in RTCM based services.

The RTCM-CRS message uses a Database-Link as additional information to relate a CRS name (Source-Name, Target-Name or Service-CRS-Name) to a geodetic parameter database (e.g., EPSG or ISO). The Database-Link shall use the name of a database and the name of the category/table separated by a dot, followed by the corresponding ID separated by a colon:

Database.Table:Code

© RTCM – Not for reproduction or redistribution

Database	geodetic parameter database
Table	category/table with unique codes
Code	integer code

The RTCM-CRS is restricted to CRS and Geodetic Datum codes and the corresponding category/table.

To facilitate correct handling of the reference frame, and because RTCM data streams can contain messages with no geocentric coordinates, the RTCM-CRS Database-Link shall contain a Geodetic Datum code implying other consistent coordinate representations. Consistent Geocentric or geodetic CRS codes referring to the same Geodetic Datum can be supplied as additional Database-Links. If only a Geodetic Datum code is provided, then it is implied to apply to any derived CRS. For example, the ITRF2014 Geodetic Datum code implies both the geocentric and geodetic ITRF2014 CRSs, applying to geocentric or geodetic coordinates, respectively.

For each consistent Geodetic Datum/CRS identified through the Source-Name, Target-Name or Service-CRS-Name an individual RTCM-CRS message is required to establish the link to the external databases.

As an example, with the databases "EPSG" or "ISO" (at time of publication, https://epsg.org and https://geodetic.isotc211.org, respectively) and the corresponding geodetic datum registry, a reasonable list of 6 Database-Links (DF570) for a Global SSR service delivered in ITRF2014 would be:

- ISO.DATUM:175 for ISO ITRF2014 datum
- ISO.CRS:425 for ISO ITRF2014 geocentric coordinates CRS
- ISO.CRS:226 for ISO ITRF2014 geodetic coordinates CRS
- EPSG.DATUM:1165 for EPSG ITRF2014 datum
- EPSG.CRS:7789 for EPSG ITRF2014 geocentric coordinates CRS
- EPSG.CRS:7912 for EPSG ITRF2014 geodetic coordinates CRS

Alternatively, only the two Geodetic Datums might be provided, in which case the dependent geocentric and/or geodetic CRS would be inferred:

- ISO.DATUM:175 for ISO ITRF2014 datum
- EPSG.DATUM:1165 for EPSG ITRF2014 datum

An additional example for a regional RTK service delivered in NAD83(2011) is:

- ISO.DATUM:126 for ISO NAD 83 (2011) datum
- ISO.CRS:252 for ISO NAD 83 (2011) geocentric coordinates CRS
- ISO.CRS:239 for ISO NAD 83 (2011) geodetic coordinates CRS
- EPSG.DATUM:1116 for EPSG NAD 83 (2011) datum
- EPSG.CRS:6317 for EPSG NAD 83 (2011) geocentric coordinates CRS

© RTCM – Not for reproduction or redistribution

- EPSG.CRS:6319 for EPSG NAD 83 (2011) geodetic coordinates CRS

Again, just the Geodetic Datums could be provided:

- ISO.DATUM:126 for ISO NAD 83 (2011) datum
- EPSG.DATUM:1116 for EPSG NAD 83 (2011) datum

The structure and content of the RTCM-CRS message is described by Table 3.5-28 and Table 3.5-29.

Table 3.5-28 Header Part of the RTCM-CRS Message 1302

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
RTCM-CRS-Name Counter N	DF565	uint5	5	N ≤ 31
RTCM-CRS-Name	DF566	char8(N)	8*N	
Anchor	DF567	bit(1)	1	
Plate Number	DF149	uint5	5	
Number of Database-Links I	DF568	uint3	3	(No. of Database-Links) * (Database-Link Block) will immediately follow this DF.
TOTAL			***26 + 8*N***	

Table 3.5-29 Database-Link Block of the RTCM-CRS Message 1302

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Database-Link Counter N	DF569	uint5	5	N ≤ 31
Database-Link	DF570	char8(N)	8*N	

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
TOTAL			***I*5+8*M***	***M = Total number of characters N in all I Database-Links***

3.5.8.3 Concatenated Coordinate Operation (ISO 19111)

The change of coordinates from one coordinate reference system to another coordinate reference system follows from a series of coordinate operations consisting of one or more coordinate transformations and/or one or more coordinate conversions. This is called a concatenated coordinate operation.

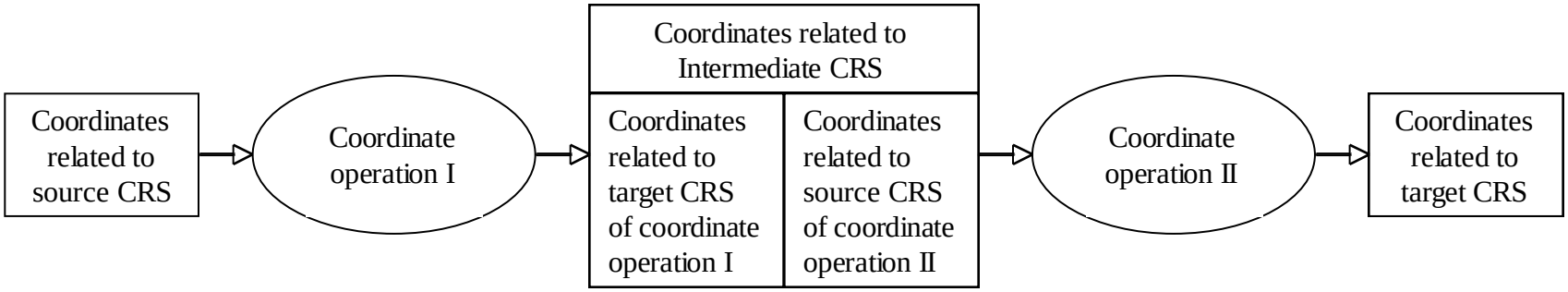

Figure 3.5-1 Steps in the Coordinate Transformation Process

The transformation of coordinates from an ECEF coordinate system to a local coordinate system generally requires several steps, as described in the next few sections.

3.5.8.3.1 Datum transformation

The transformation from the global ECEF (ITRF, ETRF, ...) to the local geodetic datum is generally accomplished by means of a 7-parameter transformation. The ISO19111 specifies the linearized, or approximate, transformation formula. The parameters found in many publications as well as databases like the EPSG or European initiatives, and all the IGS and IERS parameters correspond to this formula. The use of the linearized formula and a respective 7-parameter set is specified by the computation-indicator “0”. The strict formula of a 7-parameter transformation is also often used in practice and is related to finite rotations’ parameterization. The application of the strict formula and a respective set of 7 parameters is specified by the computation-indicator “1” in data field DF150. The 7

© RTCM – Not for reproduction or redistribution

parameters belonging to the indicator "0" and "1", respectively are self-consistent also with respect to the given and different inversion formulas, while the different transformation parameterizations and related parameters themselves can not to be interchanged without a loss of transformation correctness and accuracy.

3.5.8.3.2 Coordinate conversion from geocentric Cartesian representation to ellipsoidal latitude, longitude, and height

This conversion requires the knowledge of the ellipsoidal parameters; for example, semi-major and semi-minor axes.

3.5.8.3.3 Coordinate conversion to plane coordinates

This coordinate conversion uses projection formulas like the Transverse Mercator projection (Gauss-Krüger, UTM, etc.) or other (mostly) conformal projections to obtain 2-D Cartesian plane coordinates (Northing, Easting) and 3.5-D height. The projection step requires the knowledge of the ellipsoidal parameters as well as the parameters for the projection (prime meridian, scale, false northing and easting etc.).

3.5.8.3.4 Height transformation

From ellipsoidal heights to the local, leveling-related height system requires the knowledge of the difference between the ellipsoid and the reference surface for the local height system. This reference surface may be the geoid, quasi-geoid or a similar surface. The representation of such a reference surface includes the vertical datum as well as systematic and stochastic effects in the realization of the local height systems. The height surface can be related either to the global datum and ellipsoid, or to the local datum and ellipsoid.

The height transition from ellipsoidal to physical heights can also be accomplished within the 7-parameter transformation, if the area is very small (see DF152, DF153).

3.5.8.4 3-D Coordinate Transformation Formulas

There are five sets of transformations that will be supported in this Standard:

1. The linear form of the Helmert Transformation
2. The strict form of the Helmert Transformation
3. The abridged Molodenski Transformation
4. The Molodenski-Badekas Transformation
5. The time-dependent linear expression of the Helmert Transformation

These transformations are described in EPSG Guidance Note 7, which is on the website of the "Surveying and Positioning Committee of the International Association of Oil and Gas Producers (OGP)", the former "European Petroleum Survey Group" (EPSG, https://epsg.org/). Figure 3.5-2 shows the geometry of the transformation and the senses of rotation used herein.

© RTCM – Not for reproduction or redistribution

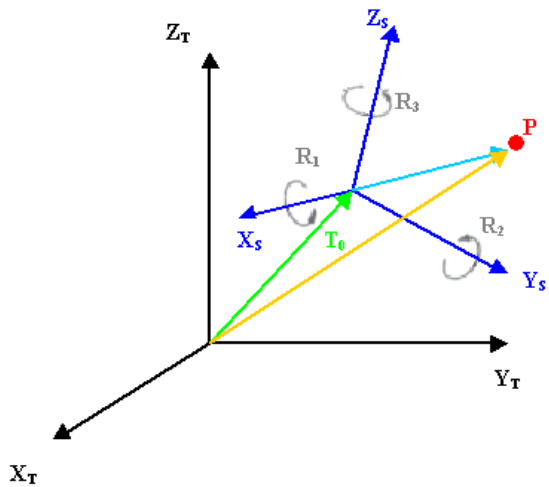

Figure 3.5-2 Definition of Translation and Rotations

3.5.8.4.1 Helmert Transformation, Linear Expression

Reference: OGP 2.4.3.2.2 Coordinate Frame Rotation, EPSG dataset coordinate operation method code 9607

Transformation of coordinates from one geographic coordinate reference system into another (also known as a "datum transformation") is usually carried out as an implicit concatenation of three transformations:

[geographical to geocentric >> geocentric to geocentric >> geocentric to geographic]

The middle part of the concatenated transformation, from geocentric to geocentric, is usually described as a simplified 7-parameter Helmert transformation.

The formula is:

$$\begin{bmatrix} X_T \\ Y_T \\ Z_T \end{bmatrix} = M * \begin{bmatrix} 1 & +R_Z & -R_Y \\ -R_Z & 1 & +R_X \\ +R_Y & -R_X & 1 \end{bmatrix} * \begin{bmatrix} X_S \\ Y_S \\ Z_S \end{bmatrix} + \begin{bmatrix} dX \\ dY \\ dZ \end{bmatrix}. \qquad (3.5\text{-}1)$$

The parameters are defined as:

- R_1, R_2, R_3: Rotations to be applied to the coordinate reference frame. The sign convention is such that a positive rotation

© RTCM – Not for reproduction or redistribution

of the frame about an axis is defined as a clockwise rotation of the coordinate reference frame when viewed from the origin of the Cartesian coordinate system in the positive direction of that axis, that is a positive rotation about the Z-axis only from source coordinate reference system to target coordinate reference system will result in a smaller longitude value for the point in the target coordinate reference system. Although rotation angles may be quoted in any angular unit of measure, the formula as given here requires the angles to be provided in radians.

The formula as given here requires the angles to be provided in radians - conversion from arc seconds (R_1, R_2, R_3) to radians (R_X, R_Y, R_Z):

$$R_X = R_1 * \left(\frac{\pi}{3600 \quad * \quad 180} \right) \tag{3.5-2}$$

$$R_Y = R_2 * \left(\frac{\pi}{3600 \quad * \quad 180} \right) \tag{3.5-3}$$

$$R_Z = R_3 * \left(\frac{\pi}{3600 \quad * \quad 180} \right) \tag{3.5-4}$$

- M: The scale factor to be applied to the position vector in the source coordinate reference system to obtain the correct scale of the target coordinate reference system.

$$M = (1+dS*10^{-6}) \tag{3.5-5}$$

3.5.8.4.2 Helmert Transformation, Strict Formula

The formula is:

$$\begin{bmatrix} X_T \\ Y_T \\ Z_T \end{bmatrix} = \begin{bmatrix} dX \\ dY \\ dZ \end{bmatrix} + M * \mathbf{R} * \begin{bmatrix} X_S \\ Y_S \\ Z_S \end{bmatrix} \tag{3.5-6}$$

The parameters are defined as:

- **R:** The rotation matrix defined by

© RTCM – Not for reproduction or redistribution

$$\mathbf{R} = \mathbf{R}_z \mathbf{R}_y \mathbf{R}_x = \begin{bmatrix} \cos R_3 & \sin R_3 & 0 \\ -\sin R_3 & \cos R_3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos R_2 & 0 & -\sin R_2 \\ 0 & 1 & 0 \\ \sin R_2 & 0 & \cos R_2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos R_1 & \sin R_1 \\ 0 & -\sin R_1 & \cos R_1 \end{bmatrix} \tag{3.5-7}$$

- M : The scale factor to be applied to the position vector in the source coordinate reference system in order to obtain the correct scale of the target coordinate reference system.

$$M = (1+dS*10^{-6}) \tag{3.5-8}$$

To transform in the opposite direction using the same numerical values as for the forward formula use the following strict inverse formula:

$$\begin{bmatrix} X_S \\ Y_S \\ Z_S \end{bmatrix} = \frac{\mathbf{R}^{-1}}{M} \left[\begin{bmatrix} X_T \\ Y_T \\ Z_T \end{bmatrix} - \begin{bmatrix} dX \\ dY \\ dZ \end{bmatrix} \right] \tag{3.5-9}$$

where $\mathbf{R}^{-1}$ is the mathematical inverse of the rotation matrix = the transposed matrix $\mathbf{R}^{\mathbf{T}}$.

3.5.8.4.3 Molodenski Transformation, Abridged

The formula is:

$$\begin{aligned} \varphi_T &= \varphi_S + d\varphi \\ \lambda_T &= \lambda_S + d\lambda \\ h_T &= h_S + dh \end{aligned} \quad , \tag{3.5-10}$$

where:

© RTCM – Not for reproduction or redistribution

$$d\varphi[rad] = \frac{\left(-dX \cdot \sin\varphi_S \cos\lambda_S - dY \cdot \sin\varphi_S \sin\lambda_S + dZ \cdot \cos\varphi_S + \left[(M \cdot a_s / b_s + N \cdot b_s / a_s) \cdot df + (N \cdot e_s^2)/a_s \cdot da\right] \cdot \sin 2\varphi_S / 2\right)}{(M + h_S)}$$

$$d\lambda[rad] = \frac{(-dX \cdot \sin\lambda_S + dY \cdot \cos\lambda_S)}{((N + h_S) \cdot \cos\varphi_S)} \qquad (3.5\text{-}11)$$

$$dh = dX \cdot \cos\varphi_S \cdot \cos\lambda_S + dY \cdot \cos\varphi_S \cdot \sin\lambda_S + dZ \cdot \sin\varphi_S + (df \cdot N \cdot b_s / a_s) \cdot \sin^2\varphi_S - da \cdot a_s / N$$

M and N are the meridian and prime vertical radii of curvature at the given latitude φ_s,

$$M = \frac{a_S \cdot (1 - e_s^2)}{\left(1 - e_s^2 \cdot \sin(\varphi_S)^2\right)^{3/2}} \qquad (3.5\text{-}12)$$

$$N = \frac{a_S}{\sqrt{\left(1 - e_s^2 \cdot \sin(\varphi_S)^2\right)}} \qquad (3.5\text{-}13)$$

$$e_S^2 = (a_s^2 - b_s^2)/a_s^2 \qquad (3.5\text{-}14)$$

da is the difference in the semi-major axes of the target and source ellipsoids [$da = a_T - a_S$] and df is the difference in the flattening of the two ellipsoids [$df = f_t - f_s = 1/(1/f_t) - 1/(1/f_s)$] where $1/f = a/(a - b)$.

The formulas for $d\varphi$ and $d\lambda$ indicate changes in φ_S and λ_S in radians. Finally it holds for the translations of the origins: $dX=X_T - X_S$, $dY=Y_T - Y_S$ and $dZ=Z_T - Z_S$.

3.5.8.4.4 Molodenski-Badekas Transformation

Reference: OGP 2.4.3.3 Molodensky-Badekas 10-parameter transformation, EPSG dataset coordinate operation method code 9636.

To eliminate high correlation between the translations and rotations in the derivation of parameter values for these Helmert transformation methods, instead of the rotations being derived about the geocentric coordinate reference system origin they may be derived at a location within the points used in the determination. Three additional parameters, the coordinates of the rotation point, are then required.

The formula is:

© RTCM – Not for reproduction or redistribution

$$\begin{bmatrix} X_T \\ Y_T \\ Z_T \end{bmatrix} = M * \begin{bmatrix} 1 & +R_Z & -R_Y \\ -R_Z & 1 & +R_X \\ +R_Y & -R_X & 1 \end{bmatrix} * \begin{bmatrix} X_S & -X_P \\ Y_S & -Y_P \\ Z_S & -Z_P \end{bmatrix} + \begin{bmatrix} X_P \\ Y_P \\ Z_P \end{bmatrix} + \begin{bmatrix} dX \\ dY \\ dZ \end{bmatrix} \tag{3.5-15}$$

and the parameters are defined as:

- R_1, R_2, R_3: Rotations to be applied to the coordinate reference frame. The sign convention is such that a positive rotation of the frame about an axis is defined as a clockwise rotation of the coordinate reference frame when viewed from the origin of the Cartesian coordinate system in the positive direction of that axis, that is a positive rotation about the Z-axis only from source coordinate reference system to target coordinate reference system will result in a smaller longitude value for the point in the target coordinate reference system. Although rotation angles may be quoted in any angular unit of measure, the formula as given here requires the angles to be provided in radians. The conversion from arc seconds (R_1, R_2, R_3) to radians (R_X, R_Y, R_Z) is:

$$R_X = R_1 * \left(\frac{\pi}{3600 \quad * \quad 180} \right) \tag{3.5-16}$$

$$R_Y = R_2 * \left(\frac{\pi}{3600 \quad * \quad 180} \right) \tag{3.5-17}$$

$$R_Z = R_3 * \left(\frac{\pi}{3600 \quad * \quad 180} \right) \tag{3.5-18}$$

- (X_P, Y_P, Z_P): Coordinates of the point about which the coordinate reference frame is rotated, given in the source Cartesian coordinate reference system.
- M : The scale factor to be applied to the position vector in the source coordinate reference system in order to obtain the correct scale of the target coordinate reference system.

$$M = (1+dS*10^{-6}) \tag{3.5-19}$$

The Molodensky-Badekas transformation in a strict mathematical sense is not reversible, i.e., in principle the same parameter values cannot be used to execute the reverse transformation. This is because the evaluation point coordinates are in the forward direction source coordinate reference system and the rotations have been derived about this point. They should not be applied about the point having the same coordinate values in the target coordinate reference system, as is required for the reverse transformation. However, in practical application there are exceptions when applied to the approximation of small differences between the geometry of a set of points in two different coordinate reference systems. The typical vector difference in coordinate values is in the order of $6*10^1$ to $6*10^2$ meters,

© RTCM – Not for reproduction or redistribution

whereas the evaluation point on or near the surface of the earth is $6.3*10^6$ meters from the origin of the coordinate systems at the Earth's center. This difference of four or five orders of magnitude allows the transformation in practice to be considered reversible. Note that in the reverse transformation, only the signs of the translations and rotation parameter values are reversed; the coordinates of the evaluation point remain unchanged.

3.5.8.4.5 Helmert Transformation, Time-Dependent Linear Expression

The dynamic earth causes movements of the tectonic plates in the order of centimeters per year. This means that the transformation parameters are not constant. To account for this, different organizations have established regional coordinate reference systems that are tied to the tectonic plates (e.g., ETRS in Europe). For the transformation between two CRS on the same tectonic plate, the transformation parameters may be considered constant for a longer period, as only local movements within the plates are responsible for changes. However, global GNSS (satellite coordinates) are usually described in a global ECEF coordinate system, so an additional transformation step is necessary.

For a service operating in a global earth-fixed CRS (e.g. ITRF), the transformation to a regional plate fixed CRS (e.g., ETRF) can be performed by a time-dependent 15-parameter transformation, Message Type 1301. The time-dependent Helmert transformation (corresponding to MT 1021, computation indicator DF150=0) is applied with the same 3-D Coordinate Transformation formulas defined in formula (3.5-1). Instead of the constant transformation parameters dX, dY, dZ, R_1, R_2, R_3, dS (DF156 – DF162) the corresponding time-dependent transformation parameters $dX(t)$, $dY(t)$, $dZ(t)$, $R_1(t)$, $R_2(t)$, $R_3(t)$, $dS(t)$ (as computed with consistent units from DF548 - DF561, DF547) are used for epoch of transformation t and reference epoch t_0 according to the equations below.

$$dX\ (t) = dX(t_0) + \dot{dX}[t - t_0]$$
$$dY\ (t) = dY(t_0) + \dot{dY}[t - t_0]$$
$$dZ\ (t) = dZ(t_0) + \dot{dZ}[t - t_0]$$

$$R_1\ (t) = R_1(t_0) + \dot{R}_1[t - t_0] \qquad (3.5\text{-}20)$$
$$R_2\ (t) = R_2(t_0) + \dot{R}_2[t - t_0]$$
$$R_3\ (t) = R_3(t_0) + \dot{R}_3[t - t_0]$$

$$dS\ (t) = dS(t_0) + \dot{dS}[t - t_0]$$

with

t epoch of transformation

© RTCM – Not for reproduction or redistribution

t_0	reference epoch for the transformation parameters (DF547)
$dX(t_0), dY(t_0), dY(t_0)$	the translations at the reference epoch t_0 (DF548, DF549, DF550)
$R_1(t_0), R_2(t_0), R_3(t_0)$	the rotations at the reference epoch t_0 (DF551, DF552, DF553)
$dS0(t_0)$	the scale correction at the reference epoch t_0 (DF554)
$\dot{dX}, \dot{dY}, \dot{dZ}$	the translation rates at the reference epoch t_0 (DF555, DF556, DF557)
$\dot{R}_1, \dot{R}_2, \dot{R}_3$	the rotation rates at the reference epoch t_0 (DF558, DF559, DF560)
$\dot{dS}$	the scale correction rate at the reference epoch t_0 (DF561)

The 15-parameter transformation can be used with or without other coordinates transformation messages. A concatenated coordinate operation of the 15-parameter transformation with e.g., a Helmert transformation supports the transformation to a specific regional datum realization.

User should be aware that there may exist intra-plate local movements, which cannot be covered by the 15-parameter transformation from a global to regional/local plate-fixed reference frame.

3.5.8.5 Procedures for Utilizing the Messages

Eight message types are defined in support of the application of coordinate transformations, namely Message Types 1021 through 1027 and 1301. Message Type 1021 provides the basic transformation parameters for the first three sets, while Message Type 1022 provides the information for the fourth set, the Molodenski-Badekas transformation, and Message Type 1301 a time-dependent 15P-transformation. Message Types 1025, 1026 and 1027 define the parameters that support the Lambert Conic Conformal (LCC2SP) projection, the Oblique Mercator (OM) projection, and others. Message Types 1023 and 1024 define the residuals for ellipsoidal and plane grid representations, respectively.

If coordinate transformation messages are supported, the Service Provider should at least send out either Message Type 1021 or 1022 and/or 1301, each of which contains transformation parameters. Either Message Type 1023 or 1024 should be utilized, but not both; similarly, for Message Types, 1025-1027, only one type should be utilized.

The interval between successive Coordinate Transformation Messages is arbitrary, but the following guidelines are provided here. If the communications link is bi-directional,

- 1021 or 1022: Initially send out after 3, 8 and 13 GNSS epochs, each 60 epochs thereafter
- 1023 or 1024: Initially send out after 4, 9 and 14 GNSS epochs, each 60 epochs thereafter
- 1025 *or* 1026 *or* 1027: Initially send out after 5, 10 and 15 GNSS epochs, each 60 epochs thereafter.

For a one-way broadcast link,

© RTCM – Not for reproduction or redistribution

- 1021 or 1022: Each 60 epochs
- 1023 or 1024: 10 epochs after 1021 or 1022
- 1025 *or* 1026 *or* 1027: 20 epochs after 1021 or 1022.

If it is necessary to cover larger areas with a one-way broadcast link, neighbouring grids - for the 'Area of validity of the Helmert/Molodenski transformation' - can be transmitted with Message Type 1023 or 1024.

The Mobile Receiver must perform the following procedures:

- First, the Mobile Receiver should check which Transformation messages are utilized by inspecting the data field DF148, "Utilized Transformation Message Indicator", which is included in Message Types 1021, 1022, and 1301.
- All identified messages should be processed before performing the transformation.
- The Mobile Receiver should process the residual messages using the interpolation technique identified by the Service Provider.
- An estimate of the complete positioning error should be determined using proper error propagation. The contributions of the coordinate transformation are determined by the terms of Helmert/Molodenski error and grid error, respectively.

The Service Provider should observe the following guidelines:

- The Service Provider should utilize a fixed set of Areas of Validity rather than attempt to define the Area of Validity in terms of the location of a particular User.
- If a service is provided to a User in the outer 10% of the Area of Validity the corresponding messages for the adjacent Area should be sent out.
- If the communications link is bi-directional and a User moves into the outer 10% of the Area of Validity, a residual message (1023 or 1024) should be sent out immediately.
    - A new residual message (1023 or 1024) for the neighbouring meshes should be sent out immediately and repeated 5 and 10 epochs later.
    - Type 1021 (or 1022) and 1025 (or 1026 or 1027) messages should be postponed, if they would normally be sent at the same time.
- The grid extensions should be the same within the area of a Service Provider
- The raster points of the grid should not change and should be independent of the rover position. The Service Provider should determine the origin of the raster according to the rover position and then send the appropriate raster.
- The Modified Julian Day (MJD) Number in messages 1023 and 1024 shall be used to define the validity date of transformation information. This is the date when the transformation information data set was created.

Figure 3.5-3 shows the area of validity for the Helmert/Molodenski transformation, with the latitude and longitude coordinates of the origin, and the extent of the area in latitude and longitude.

© RTCM – Not for reproduction or redistribution

If the extent of the area of validity in latitude and longitude is undefined (see DF154, DF155), then the area of validity is global.

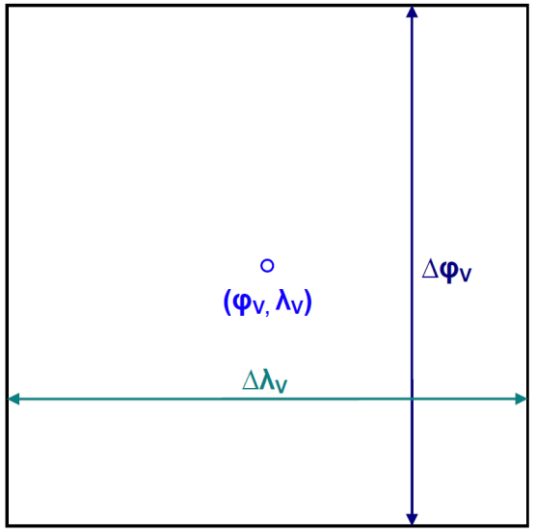

Figure 3.5-3 Area of Validity

Figure 3.5-4 shows the grid definition for the residual messages, i.e., Message Types 1023 and 1024. The Area of Validity is shown in the center, with the eight adjacent Areas surrounding it. The origin of the grid is at the center of the central square (V). The parameters of Message Type 1023 are defined with respect to longitude and latitude, while the parameters of Message Type 1024 are defined with respect to East and North, respectively. The residual messages define the 3-dimensional shifts for each point number.

If the area of validity is undefined (see DF194, DF195 and DF204, DF205), then messages 1023 and 1024 are invalid.

The squares are identified in Roman numerals as follows:

I	II	III
IV	V	VI
VII	VIII	IX

© RTCM – Not for reproduction or redistribution

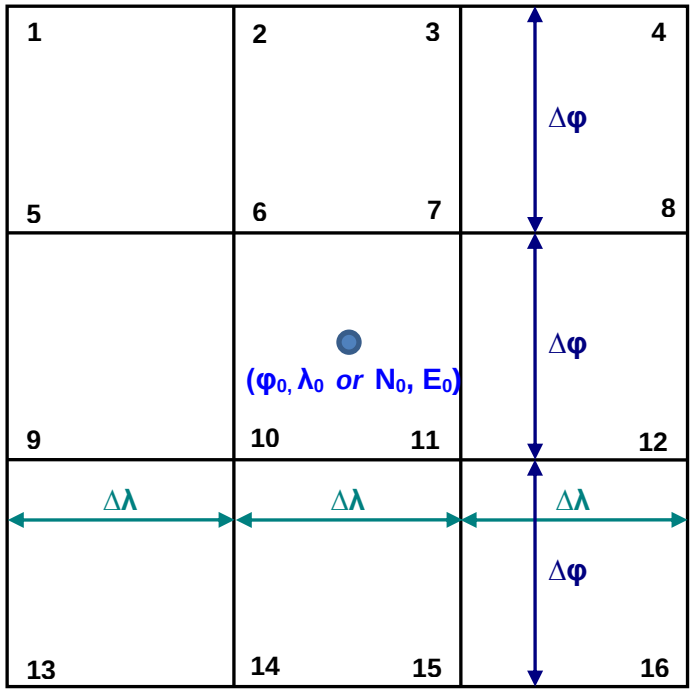

Figure 3.5-4 Grid Definition for Residual Messages

Figures 3.5-5 through 3.5-7 show the raster points used for each interpolation method identified in Message Types 1023 and 1024, namely bi-linear, bi-quadratic, and bi-spline, respectively. The mesh points as provided by messages 1023 and 1024 do not have to overlap with neighboring meshes transmitted; for example, 4 messages might cover overall 8 by 8 mesh points.

In the bi-linear method, the rover can be located anywhere in the shaded area – not just the location shown in Figure 3.5-5; however, it uses only the four surrounding grid points in the interpolation.

© RTCM – Not for reproduction or redistribution

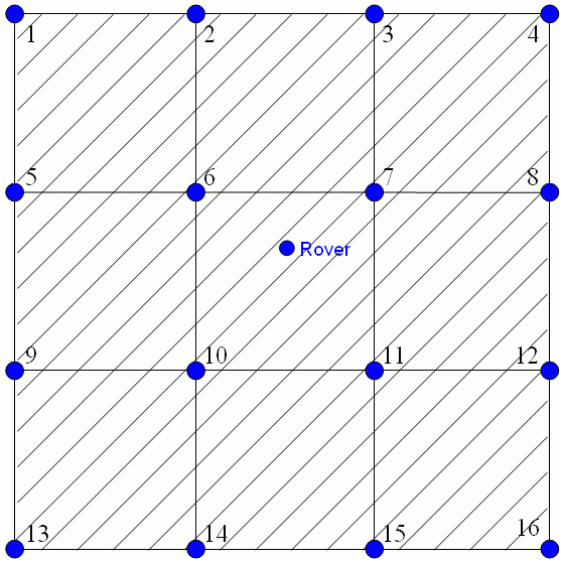

Figure 3.5-5 Raster Points Used for Bi-Linear Interpolation

While it is similar to the bi-linear interpolation, only a section of grid points will be used for the bi-quadratic interpolation. The squares West, North-West and North of the interpolation square are always used for quadratic interpolations. For instance, the grid points in the upper left (1, 2, 3, 5, 6, 7, 9, 10, and 11, squares I, II, IV, and V) are used for interpolation when the rover is within one square, namely the shaded square shown in Figure 3.5-6. For one particular message 1023 or 1024 interpolations can be performed for squares V, VI, VIII, and IX. Note that the squares shown in this example do not necessarily correlate with the content of messages 1023 and 1024.

© RTCM – Not for reproduction or redistribution

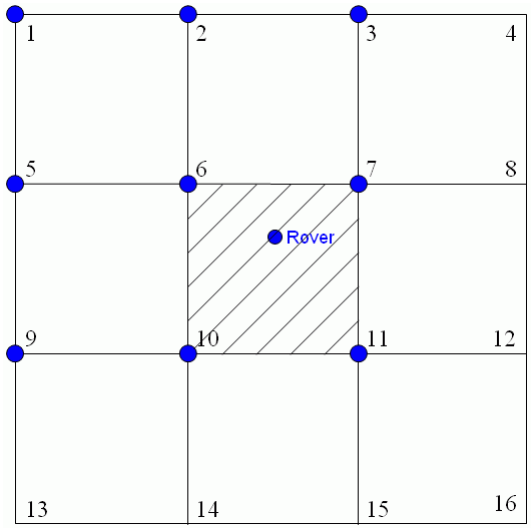

Figure 3.5-6 Raster Points Used for the Bi-quadratic Interpolation.

In the bi-spline method, the rover can be located only in the shaded central square, as indicated in Figure 3.5-7. It uses all 16 grid points in the interpolation.

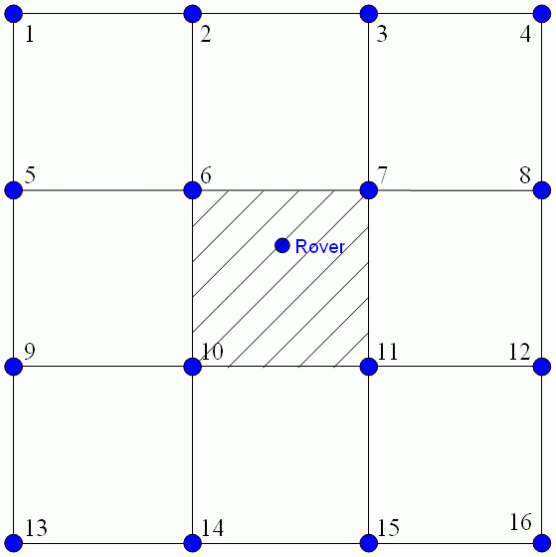

Figure 3.5-7 Raster Points Used for the Bi-Spline Interpolation.

© RTCM – Not for reproduction or redistribution

The sign convention for residual grid representation, i.e., Message Types 1023 and 1024, is different for the horizontal and vertical component. For both, Message Type 1023 defined in longitude and latitude and Message Type 1024 defined in East and North, the horizontal residuals are added and the vertical residuals are subtracted:

Latitude/Northing	= Latitude/Northing	-	Mean Offset (Latitude/Northing)	+	Residual (Grid interpolation)
Longitude/Easting	= Longitude/Easting	-	Mean Offset (Longitude/Easting)	+	Residual (Grid interpolation)
Height	= Height	-	Mean Offset (Height)	-	Residual (Grid interpolation)

3.5.8.6 Contents of the Coordinate Transformation Messages

Table 3.5-30 through Table 3.5-37 describe the contents of the coordinate transformation messages.

Table 3.5-30 Contents of the Helmert / Abridged Molodenski Message Type 1021

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
Source-Name Counter	DF143	uint5	5	
Source-Name	DF144	char8(N)	8*N	
Target-Name Counter	DF145	uint5	5	
Target-Name	DF146	char8(M)	8*M	
System Identification Number	DF147	uint8	8	
Utilized Transformation Message Indicator	DF148	bit(10)	10	
Plate Number	DF149	uint5	5	
Computation Indicator	DF150	uint4	4	
Height Indicator	DF151	uint2	2	
Φ_V - Latitude of Origin, Area of Validity	DF152	int19	19	See Figure 3.5-3
Λ_V – Longitude of Origin, Area of Validity	DF153	int20	20	See Figure 3.5-3
$\Delta\varphi_V$ – N/S Extension, Area of Validity	DF154	uint14	14	See Figure 3.5-3
$\Delta\lambda_V$ – E/W Extension, Area of Validity	DF155	uint14	14	See Figure 3.5-3

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
dX – Translation in X-direction	DF156	int23	23	See Equations 3.5-1, 3.5-6, 3.5-9 & 3.5-15
dY – Translation in Y-direction	DF157	int23	23	See notes for *dX*
dZ – Translation in Z-direction	DF158	int23	23	See notes for *dX*
R_1 – Rotation Around the X-axis	DF159	int32	32	See Section 3.5.10.3.1, para. 3 and Section 3.5.10.3.4, para. 2, and equations 3.5-2, 3.5-3, 3.5-4, 3.5-7, 3.5-15, 3.5-16, 3.5-17, & 3.5-18
R_2 – Rotation Around the Y-axis	DF160	int32	32	See notes for R_1
R_3 – Rotation Around the Z-axis	DF161	int32	32	See notes for R_1
dS – Scale Correction	DF162	int25	25	See equations 3.5-5, 3.5-8 & 3.5-19
add a_S – Semi-major Axis of Source System Ellipsoid	DF166	uint24	24	See Section 3.5.10.2 and final two para's in Section 3.5.10.3.3, and equations 3.5-11 through 3.5-14.
add b_S – Semi-minor Axis of Source System Ellipsoid	DF167	uint25	25	See notes for add a_S
add a_T – Semi-major Axis of Target System Ellipsoid	DF168	uint24	24	See notes for add a_S
add b_T – Semi-minor Axis of Target System Ellipsoid	DF169	uint25	25	See notes for add a_S
Horizontal Helmert/Molodenski Quality Indicator	DF214	uint3	3	
Vertical Helmert/Molodenski Quality Indicator	DF215	uint3	3	

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
TOTAL			***412 + 8*N + 8*M***	

Table 3.5-31 Contents of the Molodenski-Badekas Transformation Message Type 1022

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
Source-Name Counter	DF143	uint5	5	
Source-Name	DF144	char8(N)	8*N	
Target-Name Counter	DF145	uint5	5	
Target-Name	DF146	char8(M)	8*M	
System Identification Number	DF147	uint8	8	
Utilized Transformation Message Indicator	DF148	bit(10)	10	
Plate Number	DF149	uint5	5	
Computation Indicator	DF150	uint4	4	
Height Indicator	DF151	uint2	2	
Φ_V - Latitude of Origin, Area of Validity	DF152	int19	19	See Figure 3.5-3
Λ_V - Longitude of Origin, Area of Validity	DF153	int20	20	See Figure 3.5-3
$\Delta\varphi_V$ – N/S Extension, Area of Validity	DF154	uint14	14	See Figure 3.5-3
$\Delta\lambda_V$ – E/W Extension, Area of Validity	DF155	uint14	14	See Figure 3.5-3
dX – Translation in X-direction	DF156	int23	23	See Equations 3.5-1, 3.5-6, 3.5-9 & 3.5-15
dY – Translation in Y-direction	DF157	int23	23	See notes for *dX*
dZ – Translation in Z-direction	DF158	int23	23	See notes for *dX*

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
R_1 – Rotation Around the X-axis	DF159	int32	32	See Sec. 3.5.10.3.1, para. 3 and Sec. 3.5.10.3.4, para. 2, and equations 3.5-2, 3.5-3, 3.5-4, 3.5-7, 3.5-15, 3.5-16, 3.5-17, & 3.5-18
R_2 – Rotation Around the Y-axis	DF160	int32	32	See notes for R_1
R_3 – Rotation Around the Z-axis	DF161	int32	32	See notes for R_1
dS – Scale Correction	DF162	int25	25	See equations 3.5-5, 3.5-8 & 3.5-19
X_P – X Coordinate for M-B Rotation Point	DF163	int35	35	See equation 3.5-15
Y_P – Y Coordinate for M-B Rotation Point	DF164	int35	35	See notes for X_P
Z_P – Z Coordinate for M-B Rotation Point	DF165	int35	35	See notes for X_P
add a_S – Semi-major Axis of Source System Ellipsoid	DF166	uint24	24	See Sec. 3.5.10.2 and final two para's in Sec. 3.5.10.3.3, and equations 3.5-11 through 3.5-14.
add b_S – Semi-minor Axis of Source System Ellipsoid	DF167	uint25	25	See notes for add a_S
add a_T – Semi-major Axis of Target System Ellipsoid	DF168	uint24	24	See notes for add a_S
add b_T – Semi-minor Axis of Target System Ellipsoid	DF169	uint25	25	See notes for add a_S
Horizontal Helmert/Molodenski Quality Indicator	DF214	uint3	3	
Vertical Helmert/Molodenski Quality Indicator	DF215	uint3	3	
TOTAL			***517 + 8*N + 8*M***	

© RTCM – Not for reproduction or redistribution

Table 3.5-32 Contents of the Residual Message, Ellipsoidal Grid Representation, Message Type 1023

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
System Identification Number	DF147	uint8	8	
Horizontal Shift Indicator	DF190	bit(1)	1	
Vertical Shift Indicator	DF191	bit(1)	1	
φ_0 – Latitude of Origin of Grids	DF192	int21	21	See Figure 3.5-4
λ_0 – Longitude of Origin of Grids	DF193	int22	22	See Figure 3.5-4
$\Delta\varphi$ – N/S Grid Area Extension	DF194	uint12	12	See Figure 3.5-4
$\Delta\lambda$ – E/W Grid Area Extension	DF195	uint12	12	See Figure 3.5-4
Mean $\Delta\varphi$ – Mean Latitude Offset	DF196	int8	8	
Mean $\Delta\lambda$ – Mean Longitude Offset	DF197	int8	8	
Mean ΔH – Mean Height Offset	DF198	int15	15	
Three shifts for 16 grid points (i=1,16)			**16*(9+9+9)**	
$\delta\varphi_i$ – Latitude Residual	DF199	int9		See Figure 3.5-4
$\delta\lambda_i$ – Longitude Residual	DF200	int9		See Figure 3.5-4
δh_i – Height Residual	DF201	int9		See Figure 3.5-4
Horizontal Interpolation Method Indicator	DF212	uint2	2	See Figure 3.5-5 through Figure 3.5-7
Vertical Interpolation Method Indicator	DF213	uint2	2	See Figure 3.5-5 through Figure 3.5-7
Horizontal Grid Quality Indicator	DF216	uint3	3	
Vertical Grid Quality Indicator	DF217	uint3	3	
Modified Julian Day (MJD) Number	DF051	uint16	16	'0' = unknown creation date
TOTAL			***578***	

© RTCM – Not for reproduction or redistribution

Table 3.5-33 Contents of the Residual Message, Plane Grid Representation, Message Type 1024

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
System Identification Number	DF147	uint8	8	
Horizontal Shift Indicator	DF190	bit(1)	1	
Vertical Shift Indicator	DF191	bit(1)	1	
N_0 – Northing of Origin	DF202	int25	25	See Figure 3.5-4
E_0 – Easting of Origin	DF203	uint26	26	See Figure 3.5-4
ΔN – N/S Grid Area Extension	DF204	uint12	12	See Figure 3.5-4
ΔE – E/W Grid Area Extension	DF205	uint12	12	See Figure 3.5-4
Mean ΔN – Mean Local Northing Offset	DF206	int10	10	
Mean ΔE – Mean Local Easting Offset	DF207	int10	10	
Mean Δh – Mean Local Height Offset	DF208	int15	15	
Three shifts for 16 grid points (i=1,16)			**16*(9+9+9)**	
δN_i – Residual in Local Northing	DF209	int9		See Figure 3.5-4
δE_i – Residual in Local Easting	DF210	int9		See Figure 3.5-4
δh_i – Residual in Local Height	DF211	int9		See Figure 3.5-4
Horizontal Interpolation Method Indicator	DF212	uint2	2	See Figure 3.5-5 through Figure 3.5-7
Vertical Interpolation Method Indicator	DF213	uint2	2	See Figure 3.5-5 through Figure 3.5-7
Horizontal Grid Quality Indicator	DF216	uint3	3	
Vertical Grid Quality Indicator	DF217	uint3	3	
Modified Julian Day (MJD) Number	DF051	uint16	16	'0' = unknown creation date
TOTAL			***590***	

© RTCM – Not for reproduction or redistribution

Table 3.5-34 Contents of the Projection Message Type 1025 (Projection Types except LCC2SP, OM)

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
System Identification Number	DF147	uint8	8	
Projection Type	DF170	uint6	6	
LaNO – Latitude of Natural Origin	DF171	int34	34	See EPSG dataset coordinate operation
LoNO – Longitude of Natural Origin	DF172	int35	35	See EPSG dataset coordinate operation
add SNO – Scale Factor at Natural Origin	DF173	uint30	30	See EPSG dataset coordinate operation Ignore if projection = CS
FE – False Easting	DF174	uint36	36	See EPSG dataset coordinate operation
FN – False Northing	DF175	int35	35	See EPSG dataset coordinate operation
TOTAL			***196***	

Table 3.5-35 Contents of the Projection Message 1026 (Projection Type LCC2SP - Lambert Conic Conformal (2 SP))

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
System Identification Number	DF147	uint8	8	
Projection Type	DF170	uint6	6	

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
LaFO – Latitude of False Origin	DF176	int34	34	See EPSG dataset coordinate operation
LoFO – Longitude of False Origin	DF177	int35	35	See EPSG dataset coordinate operation
LaSP1 – Latitude of Standard Parallel No. 1	DF178	int34	34	See EPSG dataset coordinate operation
LaSP2 – Latitude of Standard Parallel No. 2	DF179	int34	34	See EPSG dataset coordinate operation
EFO – Easting of False Origin	DF180	uint36	36	See EPSG dataset coordinate operation
NFO – Northing of False Origin	DF181	int35	35	See EPSG dataset coordinate operation
TOTAL			***234***	

Table 3.5-36 Contents of the Projection Message 1027 (Projection Type OM - Oblique Mercator)

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
System Identification Number	DF147	uint8	8	
Projection Type	DF170	uint6	6	
Rectification Flag	DF182	bit(1)	1	
LaPC – Latitude of Projection Center	DF183	int34	34	See EPSG dataset coordinate operation

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
LoPC – Longitude of Projection Center	DF184	int35	35	See EPSG dataset coordinate operation
AzIL – Azimuth of Initial Line	DF185	uint35	35	See EPSG dataset coordinate operation
Diff ARSG – Difference, Angle from Rectified to Skew Grid	DF186	int26	26	See EPSG dataset coordinate operation
Add SIL – Scale factor on Initial Line	DF187	uint30	30	See EPSG dataset coordinate operation
EPC – Easting at Projection Center	DF188	uint36	36	See EPSG dataset coordinate operation
NPC – Northing at Projection Center	DF189	int35	35	See EPSG dataset coordinate operation
TOTAL			***258***	

Table 3.5-37 Contents of the Time-Dependent Linear Expression Helmert Transformation Message 1301

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
Source-Name Counter N	DF143	uint5	5	N ≤ 31
Source-Name	DF144	char8(N)	8*N	
Target-Name Counter M	DF145	uint5	5	M ≤ 31
Target-Name	DF146	char8(M)	8*M	
System Identification Number	DF147	uint8	8	
Utilized Transformation Message	DF148	bit(10)	10	

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Reference Epoch t_0	DF547	uint16	16	
dX0	DF548	int23	23	
dY0	DF549	int23	23	
dZ0	DF550	int23	23	
$R0_1$	DF551	int32	32	
$R0_2$	DF552	int32	32	
$R0_3$	DF553	int32	32	
dS0	DF554	int25	25	
dot dX	DF555	int17	17	
dot dY	DF556	int17	17	
dot dZ	DF557	int17	17	
dot R_1	DF558	int17	17	
dot R_2	DF559	int17	17	
dot R_3	DF560	int17	17	
dot dS	DF561	int14	14	
TOTAL			***362 + 8*N + 8*M***	

© RTCM – Not for reproduction or redistribution

3.5.9 Network RTK Residual Error Messages

3.5.9.1 Background

The Network RTK concept is based on having a plurality of GNSS reference stations continuously connected via data links to a control center. One or more computers at the control center continuously gather the information from all receivers and create network error models such as ionospheric, geometric (including troposphere and orbit) models and site-specific multipath error models in real time. One approach providing Network RTK services is described in Section 3.5.6 and utilizes Message Types 1014 through 1017. Another approach is to have the rover working in the network area send its approximate position to the control center via a bi-direction data link. The control center can then generate an "optimal" data stream derived from the physical reference station data and the network error models at the approximate rover position, and then sends the data stream to the rover via the same data link. This technique creates reference station data for a new invisible, unoccupied station. Hereafter this will be referred to as a "non-physical reference station".

Note: A Non-Physical or Computed Reference Station is typically calculated based on information from a network of reference stations. Different approaches have been established over the years. The Non-Physical or Computed Reference Stations are sometimes trademarked and may not be compatible. Examples of these names are "Virtual Reference Stations", "Pseudo-Reference Stations", and "Individualized Reference Stations".

3.5.9.2 Network RTK Residual Error Information

The Network RTK correction information provided to the rover can be considered as interpolated corrections between the reference stations in the RTK network. Depending on the actual conditions of the atmosphere, this interpolation is not perfect. A residual interpolation error must be expected. With sufficient redundancy in the RTK network, the network server process can provide an estimate for residual interpolation errors. Such quality estimates may be used by the rover to optimize the performance of RTK solutions. The values may be considered by the rover as a priori estimates only, with sufficient tracking data available the rover might be able to judge residual geometric and ionospheric errors itself.

The method of interpolating corrections and quality information for the rover position is proprietary and different solutions may be expected from different network service providers. It is the responsibility of the network server software provider to ensure that the residual information is representative of the actual situation.

Network RTK residual error information should be transmitted every 10-60 seconds.

The standard deviation for the residuals in the messages below may refer to the physical or the non-physical reference station position (indicated by the physical reference station flag in the messages 1005 or 1006) allowing maximum flexibility for the network system.

Different scenarios are:

- The interpolation of the corrections for the rover location is performed by the network processing software resulting in a

© RTCM – Not for reproduction or redistribution

non-physical reference station. This approach requires the rover to transmit via NMEA GGA its position to the network processing control center and, hence, bidirectional communication is required. In this scenario the information in the Network RTK Residual Error Messages can be estimated together with the corrections transmitted to the rover. The Network RTK Residual Error Messages will be referenced to the non-physical reference station.

- The interpolation of the correction is performed by the rover using data from the Network RTK Messages. The rover then has the option to use the Network RTK Residual Error Messages instead of self-computed values. In this case the Network RTK Residual Error Messages are referenced to the closest master or auxiliary station to the rover. This approach requires bidirectional communication, as the rover's position is needed to determine the closest reference station.

3.5.9.3 Contents of the Network RTK Residual Error Messages

Table 3.5-38 Header Part of the GPS Network RTK Residual Message Type 1030

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
GPS Residuals Epoch Time (TOW)	DF224	uint20	20	
Reference Station ID	DF003	uint12	12	May be the ID of a physical or non-physical station
N-Refs	DF223	uint7	7	
GPS Number of Satellite Signals Processed	DF006	uint5	5	
TOTAL			***56***	

Table 3.5-39 Satellite-Specific Part of the GPS Network RTK Residual Message Type 1030

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GPS Satellite ID	DF009	uint6	6	
s_{oc}	DF218	uint8	8	
s_{od}	DF219	uint9	9	
s_{oh}	DF220	uint6	6	
s_{Ic}	DF221	uint10	10	

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
s_{Id}	DF222	uint10	10	
TOTAL			***49***	

Table 3.5-40 Header Part of the GLONASS Network RTK Residual Message Type 1031

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
GLONASS Residuals Epoch Time (tk)	DF225	uint17	17	
Reference Station ID	DF003	uint12	12	May be the ID of a physical or non-physical station
N-Refs	DF223	uint7	7	
GLONASS Number of Satellite Signals Processed	DF035	uint5	5	
TOTAL			***53***	

Table 3.5-41 Satellite-Specific Part of the GLONASS Network RTK Residual Message Type 1031

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GLONASS Satellite ID	DF038	uint6	6	
s_{oc}	DF218	uint8	8	
s_{od}	DF219	uint9	9	
s_{oh}	DF220	uint6	6	

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
s_{Ic}	DF221	uint10	10	
s_{Id}	DF222	uint10	10	
TOTAL			***49***	

Table 3.5-42 Header Part of the BDS Network RTK Residual Message Type 1303

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
BDS Residuals Epoch Time (TOW)	DF571	uint20	20	
Reference Station ID	DF003	uint12	12	May be the ID of a physical or non-physical station
N-Refs	DF223	uint7	7	
BDS Number of Satellite Signals Processed	DF572	uint5	5	
TOTAL			***56***	

Table 3.5-43 Satellite-Specific Part of the BDS Network RTK Residual Message Type 1303

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
BDS Satellite ID	DF488	uint6	6	
s_{oc}	DF218	uint8	8	
s_{od}	DF219	uint9	9	
s_{oh}	DF220	uint6	6	
s_{Ic}	DF221	uint10	10	

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
s_{Id}	DF222	uint10	10	
TOTAL			***49***	

Table 3.5-44 Header Part of the Galileo Network RTK Residual Message Type 1304

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
Galileo Residuals Epoch Time (TOW)	DF573	uint20	20	
Reference Station ID	DF003	uint12	12	May be the ID of a physical or non-physical station
N-Refs	DF223	uint7	7	
Galileo Number of Satellite Signals Processed	DF574	uint5	5	
TOTAL			***56***	

Table 3.5-45 Satellite-Specific Part of the Galileo Network RTK Residual Message Type 1304

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Galileo Satellite ID	DF252	uint6	6	
s_{oc}	DF218	uint8	8	
s_{od}	DF219	uint9	9	
s_{oh}	DF220	uint6	6	
s_{Ic}	DF221	uint10	10	
s_{Id}	DF222	uint10	10	

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
TOTAL			***49***	

Table 3.5-46 Header Part of the QZSS Network RTK Residual Message Type 1305

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
QZSS Residuals Epoch Time (TOW)	DF575	uint20	20	
Reference Station ID	DF003	uint12	12	May be the ID of a physical or non-physical station
N-Refs	DF223	uint7	7	
QZSS Number of Satellite Signals Processed	DF576	uint5	5	
TOTAL			***56***	

Table 3.5-47 Satellite-Specific Part of the QZSS Network RTK Residual Message Type 1305

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
QZSS Satellite ID	DF429	uint4	4	
s_{oc}	DF218	uint8	8	
s_{od}	DF219	uint9	9	
s_{oh}	DF220	uint6	6	
s_{Ic}	DF221	uint10	10	
s_{Id}	DF222	uint10	10	
TOTAL			***47***	

© RTCM – Not for reproduction or redistribution

3.5.9.4 Physical Reference Station Position Messages

Message Type 1032 (see Table 3.5-48) is like Message Type 1005, but provides the earth-centered, earth-fixed (ECEF) coordinates of the antenna reference point (ARP) for the real (or "physical") reference station used. No height above a monument is provided.

Message Type 1032 contains the coordinates of the ARP in the GNSS coordinate system Earth-Center-Earth-Fixed (ECEF) coordinates; local datums are not supported. The coordinates always refer to a physical point at the bottom of the antenna.

Message Type 1032 is used in case of the non-physical reference station approach to allow the rover to refer baseline vectors to a physical reference rather than to a non-physical reference without any connection to a physical point. This feature is mainly desired for post-processing in office software applications.

Table 3.5-48 Physical Reference Station Position Information Message Type 1032

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS
Message Number	DF002	uint12	12
Non-Physical Reference Station ID	DF003	uint12	12
Physical Reference Station ID	DF226	uint12	12
Reserved	DF001	bit(6)	6
Physical reference station ARP ECEF-X	DF025	int38	38
Physical reference station ARP ECEF-Y	DF026	int38	38
Physical reference station ARP ECEF-Z	DF027	int38	38
TOTAL			***156***

3.5.10 State Space Messages

The principle of the state space concept is to provide information on the status of individual GNSS error sources. Therefore, the term State Space Representation (SSR) is used. Differential GNSS and RTK operation using corrections and/or raw measurements from

© RTCM – Not for reproduction or redistribution

single or multiple reference stations must be distinguished from SSR. Differential positioning using “observation space” to transport information is called Observation Space Representation (OSR).

The GNSS state vector of SSR consists of the following basic parameters:

- satellite orbit errors
- satellite clock errors
- satellite signal biases (delays of codes and carrier phases within satellite hard- and software)
- ionospheric delay parameters
- tropospheric delay parameters
- quality indicators for state parameters

RTCM SSR messages are used to disseminate the state space information, which is combined with individual tracked GNSS data of a user (rover) to improve its positioning. Already a sub-set of state space parameters can be sufficient for meaningful applications. Different levels of accuracy are supported depending on the provided SSR messages and the corresponding properties.

The actual application scenario for a GNSS rover is flexible and may vary. The state space information can be converted into observation space for a rover position and can thus be used to correct the rover observables for more accurate positioning. Then, the application is a conventional OSR process for the rover. Alternatively, the state information can directly be used in the rover's processing or adjustment model.

The state space approach is a commonly applied concept. The PPP (Precise Point Positioning) concept uses mainly precise satellite orbit and clock parameters derived consistently from global networks of reference stations as well as global atmospheric models to perform single station positioning. Accuracies in the sub-meter or even sub-decimeter level can be achieved with dual-frequency receivers on a global scale.

For RTK applications, the SSR concept has been demonstrated in an indirect mode. Some network RTK software is based on state space models and the network RTK services are derived from the state information. Without RTCM SSR messages, the state space information is transformed to observation space and represented by observation-based RTCM messages.

3.5.10.1 Technical Background and Detail

3.5.10.1.1 SSR Orbit Correction Message

© RTCM – Not for reproduction or redistribution

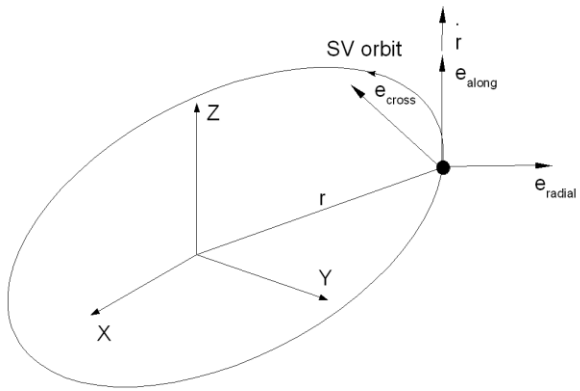

Figure 3.5-8 Radial, along-track and cross-track orbit components

The Orbit Correction Message contains the parameters for orbit corrections $\delta\boldsymbol{O}$ in radial, along-track and cross-track component. These orbit corrections are used to compute a satellite position correction $\delta\boldsymbol{X}$, to be combined with satellite position $\boldsymbol{X}_{broadcast}$ calculated from broadcast ephemeris. The sign definition of the correction is

$$\boldsymbol{X}_{orbit} = \boldsymbol{X}_{broadcast} - \delta\boldsymbol{X} \tag{3.12-1}$$

with

$\boldsymbol{X}_{orbit}$	satellite position corrected by SSR Orbit Correction message
$\boldsymbol{X}_{broadcast}$	satellite position computed according to corresponding GNSS ICD from broadcast ephemeris parameter set identified by IOD/IODE in SSR Orbit Correction message
$\delta\boldsymbol{X}$	satellite position correction

The satellite position correction $\delta\boldsymbol{X}$ is computed according to

$$\boldsymbol{e}_{along} = \frac{\dot{\boldsymbol{r}}}{|\dot{\boldsymbol{r}}|} \tag{3.12-2}$$

$$\boldsymbol{e}_{cross} = \frac{\boldsymbol{r} \times \dot{\boldsymbol{r}}}{|\boldsymbol{r} \times \dot{\boldsymbol{r}}\,|} \tag{3.12-3}$$

$$\boldsymbol{e}_{radial} = \boldsymbol{e}_{along} \times \boldsymbol{e}_{cross} \tag{3.12-4}$$

$$\delta\boldsymbol{X} = [\boldsymbol{e}_{radial} \quad \boldsymbol{e}_{along} \quad \boldsymbol{e}_{cross}]\,\delta\boldsymbol{O} \tag{3.12-5}$$

© RTCM – Not for reproduction or redistribution

with

$\boldsymbol{r} = \boldsymbol{X}_{broadcast}$	satellite broadcast position vector
$\dot{\boldsymbol{r}} = \dot{\boldsymbol{X}}_{broadcast}$	satellite broadcast velocity vector
$\boldsymbol{e}_i$	direction unit vector, i = {radial, along, cross}
$\delta\boldsymbol{O}$	orbit correction vector

Note: The radial vector according to formula (3.12-4) has to be used for the computation and should not be confused with a radial vector of a circular orbit.

The complete orbit correction vector δO is computed from the individual correction terms and their velocities:

$$\delta\boldsymbol{O} = \begin{bmatrix} \delta O_{radial} \\ \delta O_{along} \\ \delta O_{cross} \end{bmatrix} + \begin{bmatrix} \delta \dot{O}_{radial} \\ \delta \dot{O}_{along} \\ \delta \dot{O}_{cross} \end{bmatrix} (t - t_0) \tag{3.12-6}$$

with

t	time
t_0	reference time obtained from SSR Orbit Correction message
$\delta O_i, \delta \dot{O}_i$	orbit correction terms from SSR Orbit message, i = {radial, along, cross}

The reference time for the velocity term is computed from the GNSS epoch time (GPS: DF385, GLONASS: DF386) plus half the SSR Update Interval (DF391). Exception is SSR Update Interval “0”, which uses the GNSS Epoch time as reference time.

Orbit representation requires the definition of a coordinate reference system. For global services, the coordinate system should be related to the ITRS. For regional services, a reference system related to the tectonic plate of the region is often used. The messages (1057, 1060, 1063, 1066) allow orbits transformed from ITRF to a global coordinate system close to ITRF (e.g., ETRF, NAD, JGD2000). Then, it is not necessary for the rover to perform the corresponding transformation. A regional datum is indicated by the Satellite Reference Datum flag, while the actual coordinate reference system will be identified by the configured stream of the service provider.

3.5.10.1.2 SSR Clock Correction Message

The Clock Message contains the parameters to compute the clock correction δC applied to the broadcast satellite clock. The polynomial representation describes the clock differences for a certain period. The sign definition of the corrections is

$$t_{satellite} = t_{broadcast} - \frac{\delta C}{Speed\ of\ light} \tag{3.12-7}$$

© RTCM – Not for reproduction or redistribution

with

$t_{broadcast}$	satellite time computed according to corresponding GNSS ICD from broadcast clock parameters, identified by IOD/IODE of corresponding SSR Orbit Correction message
$t_{satellite}$	satellite time corrected by SSR Clock Correction message
δC	clock correction obtained from SSR Clock Correction message

The polynomial is computed according to

$$\delta C = C_0 + C_1(t - t_0) + C_2(t - t_0)^2 \tag{3.12-8}$$

with

t	time
t_0	reference time obtained from SSR Clock Correction message
C_i	polynomial coefficients from SSR Clock Correction message, i = {0, 1, 2}

The reference time for the polynomial terms is computed from the GNSS epoch time (GPS: DF385, GLONASS: DF386) plus half the SSR Update Interval (DF391). Exception is SSR Update Interval "0", which uses the Epoch time as reference time.

Note: The relativity correction has to be applied for GPS according to IS-GPS-200D to compute $t_{broadcast}$. Referring to paragraph 20.3.3.3.3.1 the relativistic correction term $\Delta \boldsymbol{t}_T$ is

$$\Delta \boldsymbol{t}_T = -\frac{\mathbf{2}\,\boldsymbol{r} \cdot \dot{\boldsymbol{r}}}{c^2}$$

with vectors $\boldsymbol{r}$ and $\dot{\boldsymbol{r}}$ computed from broadcast ephemeris.

The relativistic effects are already accounted for in the broadcast clock parameters for GLONASS.

Satellite clocks are determined from ionosphere-free signals derived from observations used by the service provider. Such observations are affected by delays introduced in the satellite hardware (code biases). For example, GPS broadcast clocks are referenced to the ionosphere-free linear combination of the P codes on L1 and L2, ignoring any code biases of these signals. For SSR, the selection of signals used to generate the satellite clock corrections and the treatment of code biases are left to the service provider. The service provider shall ensure a consistent transmission of clock and code bias parameters. A rover must then consistently apply the code biases and clock corrections.

© RTCM – Not for reproduction or redistribution

3.5.10.1.3 SSR Code Bias Message

The Code Bias Message uses a Signal and Tracking Mode Indicator to describe the actual signal properties. The Signal and Tracking Mode Indicator maps the RINEX 3/4 observation types into a more compact storage scheme of integer indices. The RINEX 3/4 observation types based on type (code, carrier, etc.), band (L1, L2, etc.) and attribute (tracking mode) are required to account for the future variety of signal tracking.

The Code Biases reported in the SSR Code Bias Message must be added to the pseudorange measurements of the corresponding code signal to get corrected pseudoranges. Any code biases transmitted in the broadcast messages (e.g., the GPS group delay differential, T_{GD}, IS-GPS 200D) are not applied at all.

The Code Bias Message contains absolute values, but also enables the alternative use of differential code biases by setting one of the biases to zero.

The provider shall support as many signals as possible and must report code biases which are zero. A rover can consistently use signals for which a code bias is transmitted. It is not reliable for a rover to use a signal without retrieving a corresponding code bias from the data stream.

Satellite code biases, especially for GLONASS, may depend on the receiver type used to track the signals. The SSR stream provider shall correct the bias level to be consistent with a certain receiver type and shall indicate this receiver type by sending a corresponding Message Type 1033 Receiver and Antenna Descriptors.

3.5.10.1.4 SSR High-Rate Clock Message

To support higher resolution of the clock state and higher update rates, a High-Rate Clock SSR message is used. Both constituents, Clock Message and High-Rate Clock Message, define the complete state of the satellite clock. The High-Rate Clock correction is added to the corresponding clock correction.

3.5.10.1.5 SSR Combined Orbit and Clock Correction Message

A Combined Orbit and Clock Correction Message is provided, which reduces the bandwidth and can easily maintain the consistency of orbit and clock data. The Combined Orbit and Clock Correction Message requires that orbit and clock have the same SSR Update Interval.

3.5.10.1.6 SSR URA Message

Clock and radial orbit state parameters are correlated. A SSR User Range Accuracy (URA) is used as one single statistical indicator to describe the quality of both parameters. The SSR User Range Accuracy is transmitted in a SSR URA message. A special formula is used to enable high resolution for small numbers and low resolution for large numbers.

© RTCM – Not for reproduction or redistribution

3.5.10.1.7 Consistency of Data and Processing

A state space parameter may consist of different constituents disseminated in different SSR messages. The use of different SSR messages is intentional to support different applications, update rates and accuracy requirements. Additional SSR messages will consequently add additional resolution and positioning accuracy. This creates the need to know the consistency of the SSR parameters for an application. Only a consistent set of SSR parameters can build up a complete and accurate correction. The consistency of SSR data becomes more important with increasing resolution provided by additional messages.

Generally, the continuous chronology of messages can be used to check the consistency, but in real-time applications messages may be lost or delayed. A consistency parameter is also the IOD (IODE for GPS) for orbit in the correction messages, which is used to obtain consistency in the computation and applications of RTCM messages. A similar requirement exists for state space parameters distributed over several SSR messages.

The combination of SSR data depends on the consistent update of the state parameters as correlations of the parameters exist. It is of advantage, if an SSR message contains all relevant information on the state parameter update without any dependency on other messages. The consistency of data is enabled through the SSR Update Interval. The meaning of SSR Update Interval is different from the transmission rate of SSR messages. The validity interval of SSR parameters can be longer than the SSR Update Interval.

There are several SSR Update Intervals defined, which all start at the start of a day in the GPS time scale (time: 00:00:00). The SSR message contains the GNSS epoch time (DF385, DF386). The consistency of the SSR parameters can be verified from this information. A change of an SSR Update Interval is only allowed if the consistency of SSR data is maintained, i.e., at the end of a SSR Update Interval. The rover is then capable of combining all relevant SSR parameters consistently and it secures the combination of state parameters from different messages.

Note: To align GPS and GLONASS SSR Update Intervals, the leap second time difference must be considered for GLONASS.

Figure 3.5-9 Sketch of Concept of SSR Update Interval to Ensure the Consistency of Data for Five Different SSR Parameters. shows a sketch of the state parameter influence on the Range for a certain location. The updates of five different SSR Range errors are displayed over time. Code biases are state parameters with a static character or a very low update rate. Typically orbit and clock parameters do have different update rates. In the example, the same SSR Update Interval is used for the two different SSR parameters orbit and troposphere. The High-Rate Clock will have the highest update rate.

© RTCM – Not for reproduction or redistribution

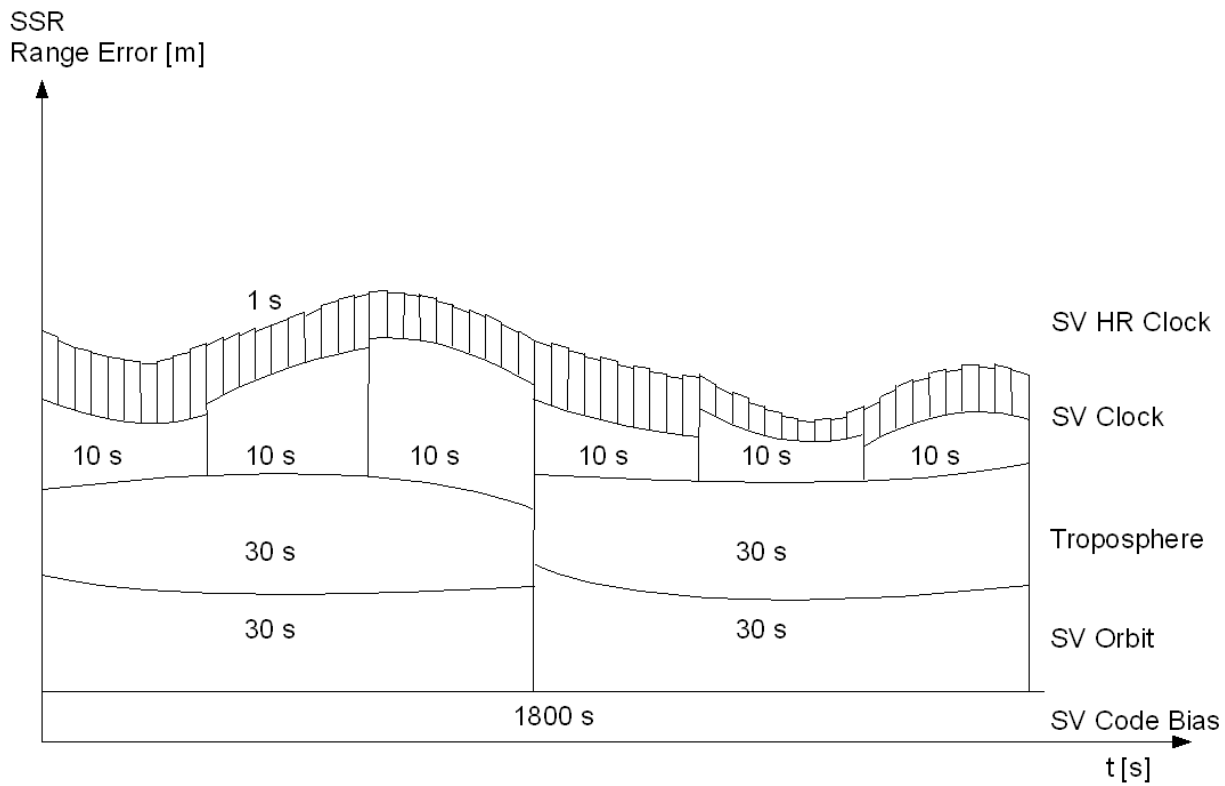

Figure 3.5-9 Sketch of Concept of SSR Update Interval to Ensure the Consistency of Data for Five Different SSR Parameters.

IOD information of broadcast clocks are redundant information and are not required in addition to the orbit IOD. That is, the SSR Orbit and Clock corrections refer to the same IOD. The service provider should refer to the latest set of broadcast messages, which are generally also received in real-time by a GNSS rover. However, it is recommended to delay the use of the latest broadcast message for a period of 60 seconds, measured from the time of complete reception of ephemeris and clock parameters, in order to accommodate rover applications to obtain the same set of broadcast orbital and clock parameters. It is left to the service provider to send the RTCM Satellite Ephemeris Data (e.g., Message Types 1019 for GPS and 1020 for GLONASS) before switching to a new IOD. It is not allowed to send corrections for more than one IOD for the same orbit Update Interval. The consistency of SSR Orbit and Clock messages is maintained through the SSR Update Interval.

© RTCM – Not for reproduction or redistribution

The computation of satellite positions and satellite clock offsets from broadcast data must be consistent. The GLONASS ICD refers to two different algorithms. For SSR, the GLONASS broadcast ephemeris computation "Simplify of algorithm for re-calculation of ephemeris to current time" must be used (see GLONASS ICD Version 5.1, A.3.1.2).

Note: Please consider two corrections in the GLONASS ICD Version 5.1. In annex A.3.1.2, the sign of the 4th term in dV_y/dt correctly reads $-2\omega V_x$ instead of $+2\omega V_x$ and for dV_z/dt the term in parentheses correctly reads $\left(3 - \frac{5z^2}{r^2}\right)$ instead of $\left(1 - \frac{5z^2}{r^2}\right)$.

Note: The numerical integration of satellite coordinates from broadcast ephemeris should be performed with sufficient accuracy. For instance, the Runge-Kutta method of 4th order with the step of numerical integration that does not exceed 30 seconds can be used to obtain a precision of 0.1 mm.

The SSR Update Interval serves to uniquely identify all consistent parameters from different messages, which can be used to compute consistent corrections at one epoch.

3.5.10.2 GPS SSR Orbit Correction, Clock Correction and Code Bias Messages

Table 3.5-49 to Table 3.5-61 describe the GPS SSR messages including a SSR URA quality message.

Table 3.5-49 Header Part of the SSR GPS Orbit Correction Message 1057

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
GPS Epoch Time 1s	DF385	uint20	20	
SSR Update Interval	DF391	bit(4)	4	
Multiple Message Indicator	DF388	bit(1)	1	
Satellite Reference Datum	DF375	bit(1)	1	
IOD SSR	DF413	uint4	4	
SSR Provider ID	DF414	uint16	16	
SSR Solution ID	DF415	uint4	4	
No. of Satellites	DF387	uint6	6	
TOTAL			***68***	

© RTCM – Not for reproduction or redistribution

Table 3.5-50 Satellite-Specific Part of the SSR GPS Orbit Correction Message 1057

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GPS Satellite ID	DF068	uint6	6	
GPS IODE	DF071	uint8	8	
Delta Radial	DF365	int22	22	
Delta Along-Track	DF366	int20	20	
Delta Cross-Track	DF367	int20	20	
Dot Delta Radial	DF368	int21	21	
Dot Delta Along-Track	DF369	int19	19	
Dot Delta Cross-Track	DF370	int19	19	
TOTAL			***135***	

Table 3.5-51 Header Part of the SSR GPS Clock Correction Message 1058

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
GPS Epoch Time 1s	DF385	uint20	20	
SSR Update Interval	DF391	bit(4)	4	
Multiple Message Indicator	DF388	bit(1)	1	
IOD SSR	DF413	uint4	4	
SSR Provider ID	DF414	uint16	16	
SSR Solution ID	DF415	uint4	4	
No. of Satellites	DF387	uint6	6	
TOTAL			***67***	

© RTCM – Not for reproduction or redistribution

Table 3.5-52 Satellite-Specific Part of the of the SSR GPS Clock Correction Message 1058

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GPS Satellite ID	DF068	uint6	6	
Delta Clock C0	DF376	int22	22	
Delta Clock C1	DF377	int21	21	
Delta Clock C2	DF378	int27	27	
TOTAL			***76***	

Table 3.5-53 Header Part of the SSR GPS Satellite Code Bias Message 1059

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
GPS Epoch Time 1s	DF385	uint20	20	
SSR Update Interval	DF391	bit(4)	4	
Multiple Message Indicator	DF388	bit(1)	1	
IOD SSR	DF413	uint4	4	
SSR Provider ID	DF414	uint16	16	
SSR Solution ID	DF415	uint4	4	
No. of Satellites	DF387	uint6	6	(No. of Satellites) * (SV Block) will immediately follow this DF.
TOTAL			***67***	

** SV Block consists of Satellite-Specific Part immediately followed by Code-specific part for the corresponding satellite.*

© RTCM – Not for reproduction or redistribution

Table 3.5-54 Satellite-Specific Part of the of the SSR GPS Satellite Code Bias Message 1059

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GPS Satellite ID	DF068	uint6	6	
No. of Code Biases Processed	DF379	uint5	5	The Code-Specific Parts for all processed Code Biases are directly following the Satellite-Specific Part of the corresponding satellite. (No. of Code Biases Processed) * (Code-Specific Part) will immediately follow this DF.
TOTAL			***11***	

Table 3.5-55 Code-Specific Part of the of the SSR GPS Satellite Code Bias Message 1059

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GPS Signal and Tracking Mode Indicator	DF380	uint5	5	
Code Bias	DF383	int14	14	
TOTAL			***19***	

Table 3.5-56 Header Part of the SSR GPS Combined Orbit and Clock Correction Message 1060

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
GPS Epoch Time 1s	DF385	uint20	20	
SSR Update Interval	DF391	bit(4)	4	

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Multiple Message Indicator	DF388	bit(1)	1	
Satellite Reference Datum	DF375	bit(1)	1	
IOD SSR	DF413	uint4	4	
SSR Provider ID	DF414	uint16	16	
SSR Solution ID	DF415	uint4	4	
No. of Satellites	DF387	uint6	6	
TOTAL			***68***	

Table 3.5-57 Satellite-Specific Part of the SSR GPS Combined Orbit and Clock Correction Message 1060

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GPS Satellite ID	DF068	uint6	6	
GPS IODE	DF071	uint8	8	
Delta Radial	DF365	int22	22	
Delta Along-Track	DF366	int20	20	
Delta Cross-Track	DF367	int20	20	
Dot Delta Radial	DF368	int21	21	
Dot Delta Along-Track	DF369	int19	19	
Dot Delta Cross-Track	DF370	int19	19	
Delta Clock C0	DF376	int22	22	
Delta Clock C1	DF377	int21	21	
Delta Clock C2	DF378	int27	27	
TOTAL			***205***	

© RTCM – Not for reproduction or redistribution

Table 3.5-58 Header Part of the SSR GPS URA Message 1061

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
GPS Epoch Time 1s	DF385	uint20	20	
SSR Update Interval	DF391	bit(4)	4	
Multiple Message Indicator	DF388	bit(1)	1	
IOD SSR	DF413	uint4	4	
SSR Provider ID	DF414	uint16	16	
SSR Solution ID	DF415	uint4	4	
No. of Satellites	DF387	uint6	6	
TOTAL			***67***	

Table 3.5-59 Satellite-Specific Part of the SSR GPS URA Message 1061

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GPS Satellite ID	DF068	uint6	6	
SSR URA	DF389	bit(6)	6	
TOTAL			***12***	

Table 3.5-60 Header Part of the SSR GPS High-Rate Clock Correction Message 1062

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GPS Epoch Time 1s	DF385	uint20	20	
SSR Update Interval	DF391	bit(4)	4	
Multiple Message Indicator	DF388	bit(1)	1	
IOD SSR	DF413	uint4	4	
SSR Provider ID	DF414	uint16	16	
SSR Solution ID	DF415	uint4	4	
No. of Satellites	DF387	uint6	6	
TOTAL			***67***	

Table 3.5-61 Satellite-Specific Part of the SSR GPS High-Rate Clock Message 1062

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GPS Satellite ID	DF068	uint6	6	
High-Rate Clock Correction	DF390	int22	22	
TOTAL			***28***	

3.5.10.3 GLONASS SSR Orbit Correction, Clock Correction and Code Bias Messages

Table 3.5-62 to Table 3.5-74 describe the GLONASS SSR messages including a SSR URA quality message.

Table 3.5-62 Header Part of the SSR GLONASS Orbit Correction Message 1063

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
GLONASS Epoch Time 1s	DF386	uint17	17	
SSR Update Interval	DF391	bit(4)	4	

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Multiple Message Indicator	DF388	bit(1)	1	
Satellite Reference Datum	DF375	bit(1)	1	
IOD SSR	DF413	uint4	4	
SSR Provider ID	DF414	uint16	16	
SSR Solution ID	DF415	uint4	4	
No. of Satellites	DF387	uint6	6	
TOTAL			***65***	

Table 3.5-63 Satellite-Specific Part of the SSR GLONASS Orbit Correction Message 1063

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GLONASS Satellite ID	DF384	uint5	5	
GLONASS IOD	DF392	bit(8)	8	
Delta Radial	DF365	int22	22	
Delta Along-Track	DF366	int20	20	
Delta Cross-Track	DF367	int20	20	
Dot Delta Radial	DF368	int21	21	
Dot Delta Along-Track	DF369	int19	19	
Dot Delta Cross-Track	DF370	int19	19	
TOTAL			***134***	

© RTCM – Not for reproduction or redistribution

Table 3.5-64 Header Part of the SSR GLONASS Clock Correction Message 1064

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
GLONASS Epoch Time 1s	DF386	uint17	17	
SSR Update Interval	DF391	bit(4)	4	
Multiple Message Indicator	DF388	bit(1)	1	
IOD SSR	DF413	uint4	4	
SSR Provider ID	DF414	uint16	16	
SSR Solution ID	DF415	uint4	4	
No. of Satellites	DF387	uint6	6	
TOTAL			***64***	

Table 3.5-65 Satellite-Specific Part of the of the SSR GLONASS Clock Correction Message 1064

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GLONASS Satellite ID	DF384	uint5	5	
Delta Clock C0	DF376	int22	22	
Delta Clock C1	DF377	int21	21	
Delta Clock C2	DF378	int27	27	
TOTAL			***75***	

© RTCM – Not for reproduction or redistribution

Table 3.5-66 Header Part of the SSR GLONASS Satellite Code Bias Message 1065

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
GLONASS Epoch Time 1s	DF386	uint17	17	
SSR Update Interval	DF391	bit(4)	4	
Multiple Message Indicator	DF388	bit(1)	1	
IOD SSR	DF413	uint4	4	
SSR Provider ID	DF414	uint16	16	
SSR Solution ID	DF415	uint4	4	
No. of Satellites	DF387	uint6	6	(No. of Satellites) * (SV Block) will immediately follow this DF.
TOTAL			***64***	

** SV Block consists of Satellite Specific Part immediately followed by Code specific part for the corresponding satellite.*

Table 3.5-67 Satellite-Specific Part of the of the SSR GLONASS Satellite Code Bias Message 1065

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GLONASS Satellite ID	DF384	uint5	5	
No. of Code Biases Processed	DF379	uint5	5	The Code Specific Parts for all processed Code Biases are directly following the Satellite Specific Part of the corresponding satellite. (No. of Code Biases Processed) * (Code Specific Part) will immediately follow this DF.
TOTAL			***10***	

© RTCM – Not for reproduction or redistribution

Table 3.5-68 Code-Specific Part of the of the SSR GLONASS Satellite Code Bias Message 1065

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GLONASS Signal and Tracking Mode Indicator	DF381	uint5	5	
Code Bias	DF383	int14	14	
TOTAL			***19***	

Table 3.5-69 Header Part of the SSR GLONASS Combined Orbit and Clock Correction Message 1066

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
GLONASS Epoch Time 1s	DF386	uint17	17	
SSR Update Interval	DF391	bit(4)	4	
Multiple Message Indicator	DF388	bit(1)	1	
Satellite Reference Datum	DF375	bit(1)	1	
IOD SSR	DF413	uint4	4	
SSR Provider ID	DF414	uint16	16	
SSR Solution ID	DF415	uint4	4	
No. of Satellites	DF387	uint6	6	
TOTAL			***65***	

Table 3.5-70 Satellite-Specific Part of the SSR GLONASS Combined Orbit and Clock Correction Message 1066

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GLONASS Satellite ID	DF384	uint5	5	

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GLONASS IOD	DF392	bit(8)	8	
Delta Radial	DF365	int22	22	
Delta Along-Track	DF366	int20	20	
Delta Cross-Track	DF367	int20	20	
Dot Delta Radial	DF368	int21	21	
Dot Delta Along-Track	DF369	int19	19	
Dot Delta Cross-Track	DF370	int19	19	
Delta Clock C0	DF376	int22	22	
Delta Clock C1	DF377	int21	21	
Delta Clock C2	DF378	int27	27	
TOTAL			***204***	

Table 3.5-71 Header Part of the SSR GLONASS URA Message 1067

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
GLONASS Epoch Time 1s	DF386	uint17	17	
SSR Update Interval	DF391	bit(4)	4	
Multiple Message Indicator	DF388	bit(1)	1	
IOD SSR	DF413	uint4	4	
SSR Provider ID	DF414	uint16	16	
SSR Solution ID	DF415	uint4	4	
No. of Satellites	DF387	uint6	6	

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
TOTAL			***64***	

Table 3.5-72 Satellite-Specific Part of the SSR GLONASS URA Message 1067

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GLONASS Satellite ID	DF384	uint5	5	
SSR URA	DF389	bit(6)	6	
TOTAL			***11***	

Table 3.5-73 Header Part of the SSR GLONASS High-Rate Clock Correction Message 1068

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
GLONASS Epoch Time 1s	DF386	uint17	17	
SSR Update Interval	DF391	bit(4)	4	
Multiple Message Indicator	DF388	bit(1)	1	
IOD SSR	DF413	uint4	4	
SSR Provider ID	DF414	uint16	16	
SSR Solution ID	DF415	uint4	4	
No. of Satellites	DF387	uint6	6	
TOTAL			***64***	

© RTCM – Not for reproduction or redistribution

Table 3.5-74 Satellite-Specific Part of the SSR GLONASS High-Rate Clock Correction Message 1068

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GLONASS Satellite ID	DF384	uint5	5	
High-Rate Clock Correction	DF390	int22	22	
TOTAL			***27***	

Table 3.5-49 through Table 3.5-57 and Table 3.5-62 through Table 3.5-69 provide the contents of the basic GPS and GLONASS SSR orbit, clock and code bias messages, which support GNSS Precise Point Positioning (PPP) for dual-frequency receiver applications. The orbit and clock messages contain data to be combined with the corresponding values obtained from the satellites broadcast message. The supported broadcast messages are currently the GPS navigation message (NAV data, D(t), IS-GPS-200D) and GLONASS-M navigation message (GLONASS ICD Version 5.1).

All messages may be split over several messages using the Multiple Message Indicator. It increases the number of, e.g., satellites to be transmitted, but it also enables the receiver to begin immediately processing of data after decoding the first message of a multiple message type.

3.5.11 FKP Network RTK Correction Messages

With the concept of FKP (from the German word: “Flächenkorrekturparameter” = Area Correction Parameters) horizontal gradients of distance-dependent errors like ionosphere, troposphere and orbits are derived from a network of GNSS reference stations and transmitted to a rover together with raw or correction data of a corresponding reference station. The rover may use the gradients to compute the influence of the distance-dependent errors for its own position.

It can be used as a stand-alone technique or in combination with a non-physical reference station. The gradient message thus can be used as follows:

- In connection with a physical reference station the gradients are applicable in the neighborhood of the physical station.
- In connection with a non-physical reference station the gradient messages are defined with respect to the non-physical reference station, so that the gradients are applicable in the neighborhood of the non-physical station.

© RTCM – Not for reproduction or redistribution

The horizontal gradients serve as a first-order approximation of the geometric and ionospheric errors for rovers that are located apart or moving away from the non-physical or physical reference station position. Table 3.5-75 and Table 3.5-77 define the headers for the GPS and GLONASS FKP Gradient messages, respectively, and Table 3.5-76 and Table 3.5-78 define the satellite-specific data.

Network FKP gradient information should be typically transmitted every 10-60 seconds.

The horizontal gradients (FKP) are a linear approximation of the geometric and ionospheric errors in the neighborhood of the physical or virtual reference station. The geometric gradient contains non-dispersive (orbit and troposphere residual) errors while the ionospheric gradient contains the dispersive errors. The horizontal gradients are defined on a surface parallel to the ellipsoid at the height of the physical or non-physical reference station. Hence, troposphere effects caused by differences in station heights and station location must be corrected before computation of the geometric gradient by using a standard troposphere model. The following relations shall be used to apply the gradient corrections. The geographical coordinates φ_R and λ_R are the ellipsoidal coordinates of the corresponding reference station and the rover has the coordinates φ, λ. The corrections are then:

$$\delta\rho_0 = 6.37 \cdot (N_0(\varphi - \varphi_R) + E_0(\lambda - \lambda_R)\cos(\varphi_R))$$
$$\delta\rho_I = 6.37 \cdot H \cdot (N_I(\varphi - \varphi_R) + E_I(\lambda - \lambda_R)\cos(\varphi_R))$$

with:

N_0	the gradient in north-south direction for the geometric (ionosphere-free) signal in [ppm]
E_0	the gradient in east-west direction for the geometric (ionosphere-free) signal in [ppm]
N_I	the gradient in north-south direction for the ionospheric signal in [ppm] (influence on GPS L1 frequency)
E_I	the gradient in east-west direction for the ionospheric signal in [ppm] (influence on GPS L1 frequency)
φ, λ	the ellipsoidal coordinates of the rover in radians
φ_R, λ_R	the ellipsoidal coordinates of the reference station in radians
H	$H= 1 +16 (0.53 –E/\pi)^3$
E	the elevation angle of the satellite at the rover position in radians
$\delta\rho_0$	the distance-dependent error for the geometric (ionosphere free) signal [m]
$\delta\rho_I$	the distance-dependent error for the ionospheric signal [m]

The distance-dependent error $\delta\rho_{\phi,f}$ for a carrier phase measurement Φ on a signal with frequency f can be computed in [m] by:

$$\delta\rho_{\phi,f} = \delta\rho_0 + \left(\frac{f_1}{f}\right)^2 \delta\rho_I$$

where f_1 is the GPS L1 frequency (1575.42 MHz).

The Phaserange measurement P_f of the rover is corrected by

© RTCM – Not for reproduction or redistribution

$$P_{corr,f} = P_f - \delta\rho_{\phi,f}$$

The distance-dependent error $\delta\rho_{r,f}$ for a Pseudorange measurement *r* on a signal with frequency *f* can be computed in [m] by

$$\delta\rho_{r,f} = \delta\rho_0 - \left(\frac{f_1}{f}\right)^2 \delta\rho_I$$

and the Pseudorange measurement PR_f of the rover is corrected by

$$PR_{corr,f} = PR_f - \delta\rho_{r,f}$$

The gradient corrections are applied to the reference station data. Standard single-base rover algorithm can then be utilized. No further modifications are necessary.

The gradients can, for instance, be computed from three Non-Physical Reference Stations located at the reference station itself as well as at a position 1000 m north and 1000 m east on the same height. Assuming that the clocks of all three stations are identical and the carrier phase ambiguities have been resolved, the north-south respectively east-west gradients can be derived from the observation differences of the geometric and ionospheric observables between the physical or non-physical reference station and the non-physical station north respectively east. For the geometric gradient a standard troposphere model has to be applied before computing the difference. It is recommended to use the Niell mapping function and the Saastamoinen troposphere zenith delay model. However, it is noted, that the geometric gradients are not significantly affected by changes of the troposphere model, therefore it is not required to use the same model in the rover algorithm. The obtained geometric and ionospheric signal differences in millimeters can be treated as gradients in ppm. Refer to 3.5.6 GPS Network RTK Messages and 3.5.3 Antenna Description Messages for details on a Non-Physical or Computed Reference Station.

Table 3.5-75 through Table 3.5-78 describe the GPS and GLONASS FKP Network RTK message types.

Table 3.5-75 Header Part of the GPS Network FKP Gradient Message 1034

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
Reference Station ID	DF003	uint12	12	May be the ID of a physical or non-physical station
GPS FKP Epoch Time (TOW)	DF240	uint20	20	

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
No. of GPS Satellite Signals Processed	DF006	uint5	5	
TOTAL			***49***	

Table 3.5-76 Satellite-Specific Part of the GPS Network FKP Gradient Message 1034

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GPS Satellite ID	DF009	uint6	6	
GPS Issue of data ephemeris (IODE)	DF071	bit(8)	8	Issue Of Data (GPS broadcast) Ephemeris to reference the geometric gradients
N0: Geometric gradient (North)	DF242	int12	12	
E0: Geometric gradient (East)	DF243	int12	12	
NI: Ionospheric gradient (North)	DF244	int14	14	
EI: Ionospheric gradient (East)	DF245	int14	14	
TOTAL			***66***	

Table 3.5-77 Header Part of the GLONASS Network FKP Gradient Message 1035

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
Reference Station ID	DF003	uint12	12	May be the ID of a physical or non-physical station
GLONASS FKP Epoch Time	DF241	uint17	17	
No. of GLONASS Satellite Signals Processed	DF035	uint5	5	
TOTAL			***46***	

© RTCM – Not for reproduction or redistribution

Table 3.5-78 Satellite-Specific Part of the Network FKP GLONASS Gradient Message 1035

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GLONASS Satellite ID	DF038	uint6	6	
GLONASS Issue Of Data (IOD)	DF392	bit(8)	8	Issue Of Data of GLONASS ephemeris to reference the geometric gradients
N0: Geometric gradient (North)	DF242	int12	12	
E0: Geometric gradient (East)	DF243	int12	12	
NI: Ionospheric gradient (North)	DF244	int14	14	
EI: Ionospheric gradient (East)	DF245	int14	14	
TOTAL			***66***	

3.5.12 Multiple Signal Messages Observable Messages

3.5.12.1 Background

The Multiple Signal Messages (MSM) allow generation of GNSS receiver observables in a universal manner for all constellations and signals.

MSMs are designed to cover the following:

- Maximum compatibility with RINEX-3/4
- Replace legacy RTCM-3 messages (MT1001-1004 and MT1009-1012), transmitting the most essential information in a generalized form, incorporating new signals and GNSS.
- Universality for all existing and future GNSS signals
- Compactness of presentation
- No ambiguity in interpretation
- Simplicity of generation/decoding
- Flexibility and scalability

The message organization is equally well suited for:

- Fully deployed GNSS with each satellite transmitting the same set of signals.
- GNSS transition period when different satellites transmit different sets of signals.

© RTCM – Not for reproduction or redistribution

The similar nature of the observables for each of the currently known GNSSs (both operational and planned) allows all observables for each GNSS to be presented in a universal format. The generic MSM structure is described first using generic field (MT) numbers. The definitions for each of the GNSS follows, which includes definitions for the GNSS-specific fields.

The MSMs are split into compact messages and full messages, similar to the current RTCM approach (e.g., 1003 and 1004, or 1011 and 1012). The proposed messages have the following intended applications:

Message type	Intended Application
MSM1	Conventional and advanced DGNSS
MSM2	Conventional RTK modes
MSM3	
MSM4	
MSM5	Storing data in a complete set of RINEX observables
MSM6	RTK with extended resolution. Realtime Network data streaming.
MSM7	Transmission of a complete set of RINEX observations with extended resolution

MSM2 contains only Phaserange observables, which allows for more flexibility when working with low-bandwidth data links and/or high-rate transmissions. For example, the reference station can send MSM2 with a rate as high as possible, while from time-to-time inserting MSM3 or MSM4 messages to provide additionally pseudorange and carrier-to-noise ratio (CNR) data, which are usually not needed at a high rate.

The key features of the messages are as follows:

- Effective identification of satellites and their signals by introducing satellite and signal masks
- Effective bit consumption for ‘transition GNSS periods’ by introducing cell masks
- Effective decomposition of observables by introducing the rough/fine range concept
- Effective scalability of different observables by introducing observation blocks (with own internal looping) which can be effectively inserted into or removed from message body
- Expressing primary observables (Pseudorange and Phaserange) and all their components (Integer number of milliseconds, RoughRange, FinePseudorange and FinePhaserange) for all bands and all signals in the same units (milliseconds). In MSM it is assumed that the speed of light is c = 299,792,458 meters per second.

One of the most important data fields (DF) in MSMs is the signal mask. It is a bitset indicating which signals from a given GNSS are available from at least one of the multitude of tracked satellites. Each bit in the signal mask is representative of a specific GNSS signal.

© RTCM – Not for reproduction or redistribution

The definition of the signal mask bits is different for each GNSS and is provided for each GNSS separately. Some of the currently defined signals (in RINEX-3.01 definition) have been omitted to make the table clearer. There are several reserved bits in the signal mask, which could be used for these signals in future revisions.

Messages MSM1 through MSM5 are standard-precision messages and messages MSM6 and MSM7 are high-precision messages. MSM6 and MSM7 contain the same fields as MSM4 and MSM5, respectively, but with a higher resolution.

3.5.12.2 Integration with Other Existing Messages

The MSMs are intended to replace the legacy RTCM-3 observation messages (MT1001-1004 and MT1009-1012). MSMs utilize a Multiple Message Bit (DF393), while legacy RTCM-3 messages utilize a Synchronous GNSS Message Flag (DF005) for similar purposes. These DFs allow the epoch to be compiled independently using either MSM or legacy messages. The legacy and MSM messages are not intended to be mixed. Decoding software shall not attempt to mix data from legacy messages and MSMs. The standard does not restrict providers from sending legacy and MSM messages in the same stream, though such operation is not recommended. In the case where legacy messages and MSM are being transmitted over the same data stream, encoding software shall ensure that the intended services are achieved with either MSM or legacy messages without the need to mix observations from the different sets.

The Multiple Message Bit (DF393) allows compiling a complete GNSS observation epoch when the data of the epoch is spread over sequential messages. Some of the legacy messages are intended to be used together with the MSM observation messages: Reference position MT 1005/1006, receiver/antenna identification information MT 1033, Network corrections MT 1014-1017, GNSS ephemeris data MT 1019,1020 etc.

MSM is intended for the following applications:

- Transmitting raw data for DGNSS/RTK applications, as well as for raw data logging for further archiving or post-processing.
- Transmitting raw data from both physical and non-physical reference stations; The Reference-Station Indicator (DF141), used in MT1005 and MT1006 applies to MSM messages.
- Transmitting or recording of raw data from static or moving receivers. If a receiver is not static, it is suggested to first transmit position messages for each epoch, followed by messages containing observables.
- Transmitting raw data for more than one GNSS at an epoch. In this case different GNSSs may be controlled by the same clock, or by independent clocks; The Single Receiver Oscillator Indicator (DF142), used in MT1005 and MT1006 is applied to MSMs as well.

Clock steering may (or may not) be applied. If for some reason clock steering is applied to each GNSS independently, then the above-mentioned Receiver Oscillator Indicator (DF142) should be set to “0”, indicating different clocks, even if the original unsteered raw observables were controlled by the same clock.

The Synchronous GNSS Message Flag (DF005) is NOT applied to MSMs.

© RTCM – Not for reproduction or redistribution

The Quarter Cycle Indicator (DF364), used in MT1005 and 1006 is NOT applied to MSMs.

3.5.12.3 Generic GNSS MSM description

The seven types of MSM messages, differing in content, range, and resolution, are described in Table 3.5-79.

Table 3.5-79 MSM Message Types

MSM Type	Message Name	No. of Bits (upper bound)	Notes
MSM1	Compact GNSS Pseudoranges	169+Nsat*(10+16*Nsig) 1353 for Nsat=16, Nsig=4	Modulo 1 ms Pseudoranges for most GNSS signals. Recommended as DGNSS reference data. Only data with a steered clock should be transmitted in this message, which shall be indicated in DF411 (Clock Steering Indicator). If data from multiple GNSSs are transmitted and the difference between the clocks for these GNSS systems exceeds 0.25 ms (modulo 1 second), then this message type shall not be used.
MSM2	Compact GNSS Phaseranges	169+Nsat*(10+28*Nsig) 2121 for Nsat=16, Nsig=4	Modulo 1 ms phase-ranges for most GNSS signals. Only data with a steered clock should be transmitted in this message, which shall be indicated in DF411 (Clock Steering Indicator). If data for multiple GNSSs are transmitted and the difference between the clocks for these GNSS exceeds 0.25 ms (modulo 1 second), then this message type shall not be used.

© RTCM – Not for reproduction or redistribution

MSM Type	Message Name	No. of Bits (upper bound)	Notes
MSM3	Compact GNSS Pseudoranges and Phaseranges	169+Nsat*(10+43*Nsig) 3081 for Nsat=16, Nsig=4	Modulo 1 ms Pseudoranges and Phaseranges for most GNSS signals. Only data with a steered clock should be transmitted in this message, which shall be indicated in DF411 (Clock Steering Indicator). If data for multiple GNSS are transmitted and the difference between the clocks for these GNSS exceeds 0.25 ms (modulo 1 second), then this message type shall not be used.
MSM4	Full GNSS Pseudoranges and Phaseranges plus CNR	169+Nsat*(18+49*Nsig) 3593 for Nsat=16, Nsig=4	Full Pseudoranges, Phaseranges and CNR (carrier-no-noise ratio) for most GNSS signals
MSM5	Full GNSS Pseudoranges, Phaseranges, PhaserangeRate and CNR	169+Nsat*(36+64*Nsig) 4841 for Nsat=16, Nsig=4	Full Pseudoranges, Phaseranges, CNR and PhaserangeRate for most GNSS signals. Recommended for RINEX data generation.
MSM6	Full GNSS Pseudoranges and Phaseranges plus CNR (high resolution)	169+Nsat*(18+66*Nsig) 4681 for Nsat=16, Nsig=4	Twin of MSM4 but with higher observables resolution

© RTCM – Not for reproduction or redistribution

MSM Type	Message Name	No. of Bits (upper bound)	Notes
MSM7	Full GNSS Pseudoranges, Phaseranges, PhaserangeRate and CNR (high resolution)	169+Nsat*(36+81*Nsig) 5929 for Nsat=16, Nsig=4	Twin of MSM5 but with higher observables resolution

Notes:

- In Table 3.5-79, Nsat refers to the number of GNSS Satellites; Nsig refers to the number of different signals included in the transmitted message
- The table provides an upper bound of bit consumption for what one could call a 'complete' GNSS constellation. Actual throughput would be lower, especially during the 'transition period' when not all GNSS satellites transmit all (Nsig) signals.
- Specific numbers are also provided as a reference for the case of Nsat=16 and Nsig=4.
- Raw data without clock steering shall not be used for messages that do not transmit "The number of integer milliseconds".
- MSM1, MSM2 and MSM3 shall not be used for each GNSS where the difference between the clocks for the GNSS exceeds 0.25 ms (modulo 1 second). MSM4, MSM5, MSM6 or MSM7 shall be used in these cases.
- Table 3.5-80 gives an overview of the difference between MSM message types in the sense of compactness of message vs. availability of data

Table 3.5-80 Contents of the MSM Message Types

	DATA FIELD	MSM1	MSM2	MSM3	MSM4	MSM5	MSM6	MSM7
Satellite	The number of integer milliseconds in GNSS Satellite rough ranges				DF397	DF397	DF397	DF397
	Extended Satellite Information					*		*

© RTCM – Not for reproduction or redistribution

	GNSS Satellite rough ranges modulo 1 millisecond	DF398	DF398	DF398	DF398	DF398	DF398	DF398
	GNSS Satellite rough Phaserange Rates					DF399		DF399
Signal Data	GNSS signal fine Pseudoranges	DF400		DF400	DF400	DF400	DF405[1]	DF405[1]
	GNSS signal fine Phaserange data		DF401	DF401	DF401	DF401	DF406[1]	DF406[1]
	GNSS Phaserange Lock Time Indicator		DF402	DF402	DF402	DF402	DF407[1]	DF407[1]
	Half-cycle ambiguity indicator		DF420	DF420	DF420	DF420	DF420	DF420
	GNSS signal CNRs				DF403	DF403	DF408[1]	DF408[1]
	GNSS signal fine Phaserange Rates					DF404		DF404

[1] With extended resolution
* GNSS-specific field

Notes:

- Different MSMs are just different methods of packing the same data, providing that the clock-steering indicator is the same for these messages. This in particular does not require that the RoughRange be the same but implies that the reconstructed Pseudorange and Phaserange would be the same for a given satellite-signal on a given epoch, regardless of which specific MSM was used to encode these observables.
- Data packed into MSM1-3 and MSM4-7 may differ only if clock steering is not applied for MSM4-7.
- Having the same clock-steering status, the difference between the reconstructed Pseudorange, Phaserange and CNR from the standard-precision messages (MSM1-5) and the high-precision messages (MSM6-7) shall differ only in resolution.
- GNSS Phaserange Lock Time Indicator in high-precision messages (MSM6 and MSM7) has higher resolution and wider range, compared to the same indicator in standard-precision messages (MSM2-5).
- The integer number of ambiguities applied to Phaserange at initialization shall be the same across all MSM messages, regardless of whether or not a steered clock is used.
- Complete Pseudorange, Phaserange and PhaserangeRate for each signal (i) of given satellite can be restored as follows:

For standard-precision messages:

$$\text{Pseudorange}(i) = c/1000 \times (\text{Nms} + \text{Rough_range}/1024 + 2^{-24} \times \text{Fine_Pseudorange}(i)), \text{ meter,}$$

© RTCM – Not for reproduction or redistribution

Phaserange(i) = c/1000 × (Nms + Rough_range/1024 + 2^{-29} × Fine_Phaserange(i)), meter,
PhaserangeRate (i) = Rough_ PhaserangeRate + 0.0001*Fine_ PhaserangeRate (i), meter/sec,

For high-precision messages:

Pseudorange(i) = c/1000 × (Nms + Rough_range/1024 + 2^{-29} × Fine_Pseudorange (i)), meter,
Phaserange(i) = c/1000 × (Nms + Rough_range/1024 + 2^{-31} × Fine_Phaserange(i)), meter,
PhaserangeRate (i) = Rough_ PhaserangeRate + 0.0001*Fine_ PhaserangeRate (i), meter/sec,

where c is speed of light (meter/sec).

3.5.12.3.1 Phase Alignment

Carrier phases tracked on different signal channels or modulation bands of the same frequency may differ in phase by 1/4 (e.g., GPS: P/Y-code-derived L2 phase vs. L2C-based phase) or, in some exceptional cases, by other fractional parts of a cycle. It is known that such cycle shifts may vary between different manufacturers as well as different hardware or firmware releases. In order to facilitate data processing, phases packed in MSMs must be consistent across all satellites of a satellite system and a specific frequency within continuous MSM transmission with respect to such cycle shifts.

The phase observations shall be shifted by the respective fraction of a cycle, either directly by the receiver or by a correction program or the MSM conversion program, prior to MSM stream generation, to align them to the reference signal. The reference signal shall be selected according to Table 3.5-81. The reference signal may or may not be tracked.

Notes:

1. While aligning the GPS L2 Phaserange encoding software shall ignore FlexPower status (when the phases of the L2S/L2L/L2X can be changed on the satellite).
2. B2 is a BDS-2 signal and only present on BDS-2 satellites, while B2b is a BDS-3 signal and only present on BDS-3 satellites. The phase alignment between these two signals is unknown and should be treated as independent.

Table 3.5-81 Reference Signal for Phase Alignment

System	Frequency Band	Frequency [MHz]	Reference Signal (RINEX Observation Code)
GPS	L1	1575.42	L1C
	L2	1227.60	L2P
	L5	1176.45	L5I
GLONASS	G1	1602+k*9/16	L1C
	G2	1246+k*7/16	L2C

© RTCM – Not for reproduction or redistribution

Galileo	E1	1575.42	L1B
	E5A	1176.45	L5I
	E5B	1207.140	L7I
	E5(A+B)	1191.795	L8I
	E6	1278.75	L6B
SBAS	L1	1575.42	L1C
	L5	1176.45	L5I
QZSS	L1	1575.42	L1C
	L2	1227.60	L2S
	L5	1176.45	L5I
BeiDou	B1	1561.098	L2I
	B2	1207.140	L7I (BDS-2)
	B3	1268.52	L6I
	B2a	1176.45	L5D
	B2b	1207.140	L7D (BDS-3)
	B1C	1575.42	L1D
NavIC/IRNSS	L5	1176.45	L5A
	S	2492.028	L9A

3.5.12.3.2 Lock Time Indicator

The GNSS Phaserange Lock Time Indicator is used in MSM to indicate intervals of Phaserange continuity and to indicate possible cycle slips, loss of lock or some other internal reinitializations. Each case of Phaserange non-continuity shall be indicated by encoding software by resetting this Lock Time Indicator to zero.

Table 3.5-82 GNSS Phaserange Lock Time Indicator (DF402)

Indicator (i)	Minimum Lock Time (ms)	Range of Indicated Lock Times (t) (ms)
0	0	$0 \leq t < 32$
1	32	$32 \leq t < 64$
2	64	$64 \leq t < 128$

© RTCM – Not for reproduction or redistribution

Indicator (i)	Minimum Lock Time (ms)	Range of Indicated Lock Times (t) (ms)
3	128	$128 \leq t < 256$
4	256	$256 \leq t < 512$
5	512	$512 \leq t < 1024$
6	1024	$1024 \leq t < 2048$
7	2048	$2048 \leq t < 4096$
8	4096	$4096 \leq t < 8192$
9	8192	$8192 \leq t < 16384$
10	16384	$16384 \leq t < 32768$
11	32768	$32768 \leq t < 65536$
12	65536	$65536 \leq t < 131072$
13	131072	$131072 \leq t < 262144$
14	262144	$262144 \leq t < 524288$
15	524288	$524288 \leq t$

Decoding software shall compare the "Minimum Lock Time" reconstructed from the "Lock Time Indicator" at the given epoch with the one reconstructed on the previous epoch. If the "Minimum Lock Time" at the given epoch is smaller than the one at the previous epoch, it is an indication that Phaserange experienced some discontinuity, which caused Lock Time Indicator to be reset to zero. At the same time, if the reconstructed "Minimum Lock Time" at a given epoch is the same or larger than at the previous epoch, it does not necessarily guarantee continuous tracking between two epochs. Some values may clearly indicate Loss Of Continuity (LOC), some values may indicate "potential LOC", some values may indicate "Continuous Tracking", while some values may indicate defect in encoding software.

An algorithm for detecting LOC using the GNSS Phaserange Lock Time Indicator is provided in Figure 3.5-10.

p = Minimum Lock Time in milliseconds, as reconstructed at previous epoch
n = Minimum Lock Time in milliseconds, as reconstructed at current epoch
dt = Time Interval in milliseconds between current epoch and previous epoch

if (p > n)	LOC

© RTCM – Not for reproduction or redistribution

if (p = n)	and (dt ≥ p)	LOC
if (p = n)	and (dt < p)	ok, tracking was continuous
if (p < n)	and (dt ≥ (2n-p))	LOC
if (p < n)	and (n < dt < (2n-p))	LOC (possible)
if (p < n)	and (dt ≤ n)	ok, tracking was continuous

Figure 3.5-10 Loss of Lock detection algorithm

For high-precision messages (MSM6 and MSM7) the GNSS Phaserange Lock Time Indicator with extended range and resolution is used (DF407). It has the same meaning, but due to higher resolution and range ensures higher confidence with less uncertainty. When reliable LOC detection is not possible, the decoding software must assume a discontinuity in Phaserange even when potentially tracking could have been continuous.

Table 3.5-83 GNSS Phaserange Lock-Time Indicator with Extended Range and Resolution (DF407)

Indicator (i)	Supplementary coefficient (k)	Minimum Lock Time (ms)	Range of Indicated Lock Times (t) (ms)
0 – 63	1	i	0 ≤ t < 64
64 – 95	2	2 • i – 64	64 ≤ t < 128
96 – 127	4	4 • i – 256	128 ≤ t < 256
128 – 159	8	8 • i – 768	256 ≤ t < 512
160 – 191	16	16 • i – 2048	512 ≤ t < 1024
192 – 223	32	32 • i – 5120	1024 ≤ t < 2048
224 – 255	64	64 • i – 12288	2048 ≤ t < 4096
256 – 287	128	128 • i – 28672	4096 ≤ t < 8192
288 – 319	256	256 • i – 65536	8192 ≤ t < 16384
320 – 351	512	512 • i – 147456	16384 ≤ t < 32768
352 – 383	1024	1024 • i – 327680	32768 ≤ t < 65536
384 – 415	2048	2048 • i – 720896	65536 ≤ t < 131072
416 – 447	4096	4096 • i – 1572864	131072 ≤ t < 262144
448 – 479	8192	8192 • i – 3407872	262144 ≤ t < 524288

© RTCM – Not for reproduction or redistribution

Indicator (i)	Supplementary coefficient (k)	Minimum Lock Time (ms)	Range of Indicated Lock Times (t) (ms)
480 – 511	16384	16384 • i – 7340032	524288 ≤ t < 1048576
512 – 543	32768	32768 • i – 15728640	1048576 ≤ t < 2097152
544 – 575	65536	65536 • i – 33554432	2097152 ≤ t < 4194304
576 – 607	131072	131072 • i – 71303168	4194304 ≤ t < 8388608
608 – 639	262144	262144 • i – 150994944	8388608 ≤ t < 16777216
640 – 671	524288	524288 • i – 318767104	16777216 ≤ t < 33554432
672 – 703	1048576	1048576 • i – 671088640	33554432 ≤ t < 67108864
704	2097152	2097152 • i – 1409286144	67108864 ≤ t
705 – 1023	Reserved		

An algorithm for detecting LOC using the Lock Time Indicator with extended range and resolution is provided in Figure 3.5-10. The figure does not consider some combinations for p, d, and dt that are possible.

p = Minimum Lock Time in milliseconds, as reconstructed at previous epoch
n = Minimum Lock Time in milliseconds, as reconstructed at current epoch
a = Supplementary Coefficient, as reconstructed at previous epoch
b = Supplementary Coefficient, as reconstructed at current epoch
dt = Time Interval in milliseconds between current epoch and previous epoch

if (p > n)			LOC
if (p = n)	and (dt ≥ a)		LOC
if (p = n)	and (dt < a)		ok, tracking was continuous
if (p < n)	and (b > p)	and (dt ≥ (n+b-p))	LOC
if (p < n)	and (b > p)	and (n < dt < (n+b-p))	LOC (possible)
if (p < n)	and (b > p)	and (dt ≤ n)	ok, tracking was continuous
if (p < n)	and (b ≤ p)	and (dt > n)	LOC
if (p < n)	and (b ≤ p)	and (dt ≤ n)	ok, tracking was continuous

Figure 3.5-11 Loss of Lock detection algorithm for extended range and resolution lock time indicator

© RTCM – Not for reproduction or redistribution

Table 3.5-84 describes how to compute the GNSS Phaserange Lock Time Indicator (DF402) from a continuous lock time reported by a GNSS receiver.

Table 3.5-84 Computing GNSS Phaserange Lock-Time Indicator from Actual Lock Time (DF407)

Range of Indicated Lock Times (t) (ms)	Indicator (i) computation from Lock Time (t)
$0 \leq t < 64$	t
$64 \leq t < 128$	64 + (t – 64) / 2
$128 \leq t < 256$	96 + (t – 128) / 4
$256 \leq t < 512$	128 + (t – 256) / 8
$512 \leq t < 1024$	160 + (t – 512) / 16
$1024 \leq t < 2048$	192 + (t – 1024) / 32
$2048 \leq t < 4096$	224 + (t – 2048) / 64
$4096 \leq t < 8192$	256 + (t – 4096) / 128
$8192 \leq t < 16384$	288 + (t – 8192) / 256
$16384 \leq t < 32768$	320 + (t – 16384) / 512
$32768 \leq t < 65536$	352 + (t – 32768) / 1024
$65536 \leq t < 131072$	384 + (t – 65536) / 2048
$131072 \leq t < 262144$	416 + (t – 131072) / 4096
$262144 \leq t < 524288$	448 + (t – 262144) / 8192
$524288 \leq t < 1048576$	480 + (t – 524288) / 16384
$1048576 \leq t < 2097152$	512 + (t – 1048576) / 32768
$2097152 \leq t < 4194304$	544 + (t – 2097152) / 65536
$4194304 \leq t < 8388608$	576 + (t – 4194304) / 131072

© RTCM – Not for reproduction or redistribution

Range of Indicated Lock Times (t) (ms)	Indicator (i) computation from Lock Time (t)
8388608 ≤ t < 16777216	608 + (t – 8388608) / 262144
16777216 ≤ t < 33554432	640 + (t – 16777216) / 524288
33554432 ≤ t < 67108864	672 + (t – 33554432) / 1048576
67108864 ≤ t	704

For the Lock-Time Indicator with extended range and resolution (DF407) one may use the following algorithm to compute this indicator based on actual tracking time:

```
// Input : t - Tracking Time in [ms]
// Output: i - Lock Time Indicator DF407
int GetIndex( int t )
{
  if(        0 <= t && t < 64        ) return (       t                  );
  if(       64 <= t && t < 128       ) return ( 64 + (t-64      )/2      );
  if(      128 <= t && t < 256       ) return ( 96 + (t-128     )/4      );
  if(      256 <= t && t < 512       ) return (128 + (t-256     )/8      );
  if(      512 <= t && t < 1024      ) return (160 + (t-512     )/16     );
  if(     1024 <= t && t < 2048      ) return (192 + (t-1024    )/32     );
  if(     2048 <= t && t < 4096      ) return (224 + (t-2048    )/64     );
  if(     4096 <= t && t < 8192      ) return (256 + (t-4096    )/128    );
  if(     8192 <= t && t < 16384     ) return (288 + (t-8192    )/256    );
  if(    16384 <= t && t < 32768     ) return (320 + (t-16384   )/512    );
  if(    32768 <= t && t < 65536     ) return (352 + (t-32768   )/1024   );
  if(    65536 <= t && t < 131072    ) return (384 + (t-65536   )/2048   );
  if(   131072 <= t && t < 262144    ) return (416 + (t-131072  )/4096   );
  if(   262144 <= t && t < 524288    ) return (448 + (t-262144  )/8192   );
  if(   524288 <= t && t < 1048576   ) return (480 + (t-524288  )/16384  );
  if(  1048576 <= t && t < 2097152   ) return (512 + (t-1048576 )/32768  );
  if(  2097152 <= t && t < 4194304   ) return (544 + (t-2097152 )/65536  );
  if(  4194304 <= t && t < 8388608   ) return (576 + (t-4194304 )/131072 );
  if(  8388608 <= t && t < 16777216  ) return (608 + (t-8388608 )/262144 );
  if( 16777216 <= t && t < 33554432  ) return (640 + (t-16777216)/524288 );
```


© RTCM – Not for reproduction or redistribution

```
  if( 33554432 <= t && t < 67108864  ) return (672 + (t-33554432)/1048576);
  if( 67108864 <= t                  ) return (704                        );
  return 1023; // will never happen
}
```

3.5.12.3.3 General Message Structure

Each MSM consists of 3 blocks, listed in Table 3.5-85.

Table 3.5-85 Content of an MSM Message, and Sequence of Blocks

Block type	Comment
Message Header	Contains all information about satellites and signals that are transmitted in this message
Satellite Data	Contains all satellite data that is common for all signals of a given satellite (e.g., Rough Range)
Signal Data	Contains all signal data that is specific for each signal (e.g., Fine Phaserange)

All data fields are grouped by data type, rather than by satellite or signal. That is, if more than one data field is transmitted in a Satellite Data block then the first data field for all available satellites is packed, followed by the second data field for all available satellites, etc. Similarly, if more than one Signal Data field is to be transmitted then the first data field for each available satellite/signal combination is packed, followed by the second data field, again for each available satellite and signal. This packing order is called “internal looping”.

The standard reserves a possibility for potential future MSM extensions. These extensions may be introduced by adding data to the end of a message. This flexibility leads to the following considerations:

- Actual message length (as decoded from message header) may not match (may be greater than) the minimal required message length (as computed based upon the message contents).
- Decoding software shall skip (ignore) any unexpected data at the end of the message. The presence of such extra data shall be considered as normal and shall not result in raising warning flag.
- Encoding software shall NOT use this expansion feature for proprietary data, nor shall an encoder add any extra information to the end of an MSM.

3.5.12.3.4 Message Header Description

Table 3.5-86 describes the MSM header.

© RTCM – Not for reproduction or redistribution

Table 3.5-86 Content of MSM Header

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message number	DF002	uint12	12	
Reference station ID	DF003	uint12	12	
GNSS Epoch Time	Specific for each GNSS	uint30	30	Specific for each GNSS
Multiple Message Bit	DF393	bit(1)	1	
IODS – Issue of Data Station	DF409	uint3	3	
Reserved	DF001	bit(7)	7	
Clock Steering Indicator	DF411	uint2	2	
External Clock Indicator	DF412	uint2	2	
GNSS Divergence-free Smoothing Indicator	DF417	bit(1)	1	
GNSS Smoothing Interval	DF418	bit(3)	3	
GNSS Satellite Mask	DF394	bit(64)	64	Specific for each GNSS
GNSS Signal Mask	DF395	bit(32)	32	Specific for each GNSS
GNSS Cell Mask	DF396	bit(X)	X (X≤64)	
TOTAL			**169+X**	

Notes:

- The size of the cell mask is not fixed but is determined after decoding the satellite and signal masks. The size of the cell mask is X=Nsat*Nsig, where Nsat is the number of satellites (the number of bits, which are set to 1 in the satellite mask), and Nsig is the number of Signals (the number of bits, which are set to 1 in signal mask).
- The limit X≤64 is selected to guarantee the full size of MSM7 (the largest MSM) fits into a single RTCM-3 transport frame. Preliminary estimates of the full size of MSM7 under an X≤64 condition do not exceed 5865 bits, which is about half of the maximum permissible size for any RTCM-3 message.
- In most real time applications, data to be transmitted will fit the limit of X≤64 (e.g., Nsat≤16, Nsig≤4), therefore all data for a given GNSS should be able to be generated within a single RTCM-3 transmission most of the time.

© RTCM – Not for reproduction or redistribution

- In cases where there are many satellites and signals for a given GNSS, it is the responsibility of the encoding software to ensure that the "X≤64" rule by dividing the complete observation set among several messages. For example, if Nsat=14 and Nsig=6 (i.e., up to 14*6=84 observables), then the encoding software must use 2 separate transmissions. For example:
  The first for 7 Satellites and 6 signals
  The second for the remaining 7 Satellites and 6 signals
  In this case, the multiple message bit must be set accordingly. For more details, see section "Multiple MSM output" below.

3.5.12.3.5 Satellite Data Description

Satellite Data is transmitted only for those satellites for which the corresponding bit is set to 1 in Satellite Mask (DF394). Thus, in the tables below, Nsat refers to the number of satellites, i.e., the number of bits set to 1 in the Satellite Mask. Each data field is repeated Nsat times (internal looping is used). The order of looped data corresponds to the order of bits in the Satellite Mask.

Table 3.5-87 Content of Satellite Data for MSM1, MSM2, MSM3

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GNSS Satellite rough ranges modulo 1 millisecond	DF398	uint10 (Nsat times)	10*Nsat	
TOTAL			**10*Nsat**	

Table 3.5-88 Content of Satellite Data for MSM4 and MSM6

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
The number of integer milliseconds in GNSS Satellite rough ranges	DF397	uint8 (Nsat times)	8*Nsat	
GNSS Satellite rough ranges modulo 1 millisecond	DF398	uint10 (Nsat times)	10*Nsat	
TOTAL			**18*Nsat**	

© RTCM – Not for reproduction or redistribution

Table 3.5-89 Content of Satellite Data for MSM5 and MSM7

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
The number of integer milliseconds in GNSS Satellite rough ranges	DF397	uint8 (Nsat times)	8*Nsat	
Extended Satellite Information	Specific for each GNSS	uint4 (Nsat times)	4*Nsat	Specific for each GNSS
GNSS Satellite rough ranges modulo 1 millisecond	DF398	uint10 (Nsat times)	10*Nsat	
GNSS Satellite rough PhaserangeRates	DF399	int14 (Nsat times)	14*Nsat	
TOTAL			**36*Nsat**	

3.5.12.3.6 Message Types Signal Data Description

Signal data is transmitted only for those signal-satellite combinations, for which the corresponding bit is set to 1 in Cell Mask (DF396). Thus, in the tables below, Ncell refers to the number of signal data blocks, i.e., the number of bits set to 1 in the Cell Mask. Each data field is repeated Ncell times (internal looping is used). The order of the looped data corresponds to the order of bits in the Cell Mask.

Table 3.5-90 Content of Signal Data for MSM1

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GNSS signal fine Pseudoranges	DF400	int15 (Ncell times)	15*Ncell	
TOTAL			**15*Ncell**	

© RTCM – Not for reproduction or redistribution

Table 3.5-91 Content of Signal Data for MSM2

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GNSS signal fine Phaserange data	DF401	int22 (Ncell times)	22*Ncell	
GNSS Phaserange Lock-Time Indicator	DF402	uint4 (Ncell times)	4*Ncell	
Half-cycle ambiguity indicator	DF420	bit(1) (Ncell times)	1*Ncell	
TOTAL			**27*Ncell**	

Table 3.5-92 Content of Signal Data for MSM3

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GNSS signal fine Pseudoranges	DF400	int15 (Ncell times)	15*Ncell	
GNSS signal fine Phaserange data	DF401	int22 (Ncell times)	22*Ncell	
GNSS Phaserange Lock-Time Indicator	DF402	uint4 (Ncell times)	4*Ncell	
Half-cycle ambiguity indicator	DF420	bit(1) (Ncell times)	1*Ncell	
TOTAL			**42*Ncell**	

© RTCM – Not for reproduction or redistribution

Table 3.5-93 Content of Signal Data for MSM4

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GNSS signal fine Pseudoranges	DF400	int15 (Ncell times)	15*Ncell	
GNSS signal fine Phaserange data	DF401	int22 (Ncell times)	22*Ncell	
GNSS Phaserange Lock-Time Indicator	DF402	uint4 (Ncell times)	4*Ncell	
Half-cycle ambiguity indicator	DF420	bit(1) (Ncell times)	1*Ncell	
GNSS signal CNRs	DF403	uint6 (Ncell times)	6*Ncell	
TOTAL			**48*Ncell**	

Table 3.5-94 Content of Signal Data for MSM5

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GNSS signal fine Pseudoranges	DF400	int15 (Ncell times)	15*Ncell	
GNSS signal fine Phaserange data	DF401	int22 (Ncell times)	22*Ncell	
GNSS Phaserange Lock-Time Indicator	DF402	uint4 (Ncell times)	4*Ncell	
Half-cycle ambiguity indicator	DF420	bit(1) (Ncell times)	1*Ncell	
GNSS signal CNRs	DF403	uint6 (Ncell times)	6*Ncell	

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GNSS signal fine PhaserangeRates	DF404	int15 (Ncell times)	15*Ncell	
TOTAL			**63*Ncell**	

Table 3.5-95 Content of Signal Data for MSM6

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GNSS signal fine Pseudoranges with extended resolution	DF405	int20 (Ncell times)	20*Ncell	
GNSS signal fine Phaserange data with extended resolution	DF406	int24 (Ncell times)	24*Ncell	
GNSS Phaserange Lock-Time Indicator with extended range and resolution.	DF407	uint10 (Ncell times)	10*Ncell	
Half-cycle ambiguity indicator	DF420	bit(1) (Ncell times)	1*Ncell	
GNSS signal CNRs with extended resolution	DF408	uint10 (Ncell times)	10*Ncell	
TOTAL			**65*Ncell**	

© RTCM – Not for reproduction or redistribution

Table 3.5-96 Content of Signal Data for MSM7

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GNSS signal fine Pseudoranges with extended resolution	DF405	int20 (Ncell times)	20*Ncell	
GNSS signal fine Phaserange data with extended resolution	DF406	int24 (Ncell times)	24*Ncell	
GNSS Phaserange Lock-Time Indicator with extended range and resolution.	DF407	uint10 (Ncell times)	10*Ncell	
Half-cycle ambiguity indicator	DF420	bit(1) (Ncell times)	1*Ncell	
GNSS signal CNRs with extended resolution	DF408	uint10 (Ncell times)	10*Ncell	
GNSS signal fine PhaserangeRates	DF404	int15 (Ncell times)	15*Ncell	
TOTAL			**80*Ncell**	

3.5.12.3.7 Multiple MSM Output

Multiple MSM messages may be transmitted for a given physical epoch. To indicate the "End of Epoch" DF393 (MSM Multiple Message Bit) has been introduced. This DF is the same for all GNSS and should be set to 1 if more MSMs for same or another GNSS follows for given physical epoch and given Reference Station ID. To effectively utilize this bit, the decoding software should be aware of all possible GNSS transmitted in the stream. Since new MSM messages for more GNSS are expected to be designed in the future, the message range MT1070 – MT1229 is reserved exclusively for MSM.

It is guaranteed that the 55th bit (counting from one) represents the MSM Multiple Message Bit. This enables the decoding software to detect the end of epoch without knowledge of the content or format of any new MSM for new GNSS.

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message number	DF002		12	
			12	
			30	
Multiple Message Bit	DF393	Bit(1)	1	
The rest of the message…				

Each MSM has two size limitations:

- The size of Cell Mask is limited to 64 bits
- The size of entire message is limited to 1023 bytes

If at least one of the limitations is reached then a provider shall split the MSM message for the given GNSS into a set of two or more sequential, complementary messages, with proper care of the Multiple Message Bit.

The standard does not put any restrictions on message splitting. Generally, splitting on satellite-by-satellite basis is the most effective way of splitting because it does not require duplication of RoughRange (i.e., DF397, DF398). At the same time, in case of severe divergence of Pseudorange observables for the same satellite (due to receiver or satellite hardware bias or due to any other cause), DF-range for Fine Pseudorange may be insufficient to pack all Pseudorange observables. In this case encoding software should choose either not to output the Pseudorange for some signals on a given epoch, or to transmit them in a separate MSM message. Having *a priori* information about large bias between Pseudorange on different signals for given GNSS, encoding software may choose to consider the less efficient (in terms of bandwidth) signal-by-signal MSM splitting.

3.5.12.4 GPS Messages Description

For GPS all seven MSM messages are introduced.

3.5.12.4.1 Message Types Summary

Table 3.5-97 lists the MSM message types for GPS.

© RTCM – Not for reproduction or redistribution

Table 3.5-97 GPS Message Types

Message Type	Message Name	MSM type
1071	Compact GPS Pseudoranges	MSM1
1072	Compact GPS Phaseranges	MSM2
1073	Compact GPS Pseudoranges and Phaseranges	MSM3
1074	Full GPS Pseudoranges and Phaseranges plus CNR	MSM4
1075	Full GPS Pseudoranges, Phaseranges, PhaserangeRate and CNR	MSM5
1076	Full GPS Pseudoranges and Phaseranges plus CNR (high resolution)	MSM6
1077	Full GPS Pseudoranges, Phaseranges, PhaserangeRate and CNR (high resolution)	MSM7

3.5.12.4.2 GPS Specific Data Fields

For GPS satellites the following clarifications apply:

- GPS Epoch Time (TOW) DF004 is used for GNSS Epoch Time.
- GPS PRNs are mapped to satellite mask IDs according to Table 3.5-98
- GPS signals are mapped to signal mask IDs according to Table 3.5-99
- Extended satellite information is reserved for future use.

Table 3.5-98 GPS Satellite ID mapping

Satellite ID in Satellite Mask (DF394)	GPS Satellite PRN
1	1
2	2
…	…
63	63
64	Reserved

© RTCM – Not for reproduction or redistribution

Table 3.5-99 GPS Signal ID mapping

Signal ID in Signal Mask (DF395)	Frequency Band	Signal	RINEX code	Notes
1				Reserved
2	L1	C/A	1C	
3	L1	P	1P	
4	L1	Z-tracking or similar	1W	
5-7				Reserved
8	L2	C/A	2C	
9	L2	P	2P	
10	L2	Z-tracking or similar	2W	
11-14				Reserved
15	L2	L2C(M)	2S	
16	L2	L2C(L)	2L	
17	L2	L2C(M+L)	2X	
18-21				Reserved
22	L5	I	5I	
23	L5	Q	5Q	
24	L5	I+Q	5X	
25-29				Reserved
30	L1	L1C-D	1S	
31	L1	L1C-P	1L	
32	L1	L1C-(D+P)	1X	

Notes:

- For legacy messages DF016 provided information about signal type. Table 3.5-100 describes the matching between DF016 and the GPS signal mask for MSM

© RTCM – Not for reproduction or redistribution

Table 3.5-100 Matching Between DF016 and GPS MSM Signal ID

DF016	GPS MSM signal ID
0	8 (L2C) or 15 (L2S) or 16 (L2L) or 17 (L2X)
1	9 (L2P)
2	10 (L2W)
3	10 (L2W)

3.5.12.5 GLONASS Messages Description

For GLONASS all seven MSM messages are introduced. The GLONASS Code-Phase Bias Message described in Section 3.5.13 is also related to the transmission of GLONASS observables.

3.5.12.5.1 Message Types Summary

Table 3.5-101 lists the MSM message types for GLONASS.

Table 3.5-101 GLONASS Message Types

Message Type	Message Name	MSM type
1081	Compact GLONASS Pseudoranges	MSM1
1082	Compact GLONASS Phaseranges	MSM2
1083	Compact GLONASS Pseudoranges and Phaseranges	MSM3
1084	Full GLONASS Pseudoranges and Phaseranges plus CNR	MSM4
1085	Full GLONASS Pseudoranges, Phaseranges, PhaserangeRate and CNR	MSM5
1086	Full GLONASS Pseudoranges and Phaseranges plus CNR (high resolution)	MSM6
1087	Full GLONASS Pseudoranges, Phaseranges, PhaserangeRate and CNR (high resolution)	MSM7

3.5.12.5.2 GLONASS Specific Data Fields

For GLONASS satellites the following clarifications apply:

- GNSS Epoch Time for GLONASS MSM's is a composite field as presented in Table 3.5-102
- GLONASS slot numbers are mapped to Satellite Mask IDs according to Table 3.5-103
- GLONASS signals are mapped to Signal Mask IDs according to Table 3.5-104

© RTCM – Not for reproduction or redistribution

- GLONASS Satellite Frequency Channel Number (DF419) is used as extended satellite information in the Satellite Data Description of MSM5 and MSM7

Table 3.5-102 GNSS Time for GLONASS MSMs

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GLONASS Day Of Week	DF416	uint3	3	
GLONASS Epoch Time	DF034	uint27	27	

Table 3.5-103 GLONASS Satellite ID mapping

Satellite ID in Satellite Mask (DF394)	GLONASS Satellite slot number
1	1
2	2
…	…
24	24
25–64	Reserved

Table 3.5-104 GLONASS Signal ID mapping

Signal ID in Signal Mask (DF395)	Frequency Band	Signal	RINEX code	Notes
1				Reserved
2	G1	C/A	1C	
3	G1	P	1P	
4-7				Reserved
8	G2	C/A	2C	
9	G2	P	2P	
10-32				Reserved

© RTCM – Not for reproduction or redistribution

Notes:

- In MSM1, MSM2, and MSM3 the number of integer milliseconds in the GNSS Satellite Rough Range (DF397) is not transmitted. For odd frequency channel numbers, 1 millisecond does not contain integer number of GLONASS wavelengths. The decoding software shall not utilize GLONASS Phaserange without restoring the number of integer milliseconds in GNSS Satellite rough range, as it may introduce half cycle error.

3.5.12.6 Galileo Messages Description

For Galileo, all seven MSM messages are introduced.

3.5.12.6.1 Message Types Summary

Table 3.5-105 lists the MSM message types for Galileo.

Table 3.5-105 Galileo Message Types

Message Type	Message Name	MSM type
1091	Compact Galileo Pseudoranges	MSM1
1092	Compact Galileo Phaseranges	MSM2
1093	Compact Galileo Pseudoranges and Phaseranges	MSM3
1094	Full Galileo Pseudoranges and Phaseranges plus CNR	MSM4
1095	Full Galileo Pseudoranges, Phaseranges, PhaserangeRate and CNR	MSM5
1096	Full Galileo Pseudoranges and Phaseranges plus CNR (high resolution)	MSM6
1097	Full Galileo Pseudoranges, Phaseranges, PhaserangeRate and CNR (high resolution)	MSM7

3.5.12.6.2 Galileo-Specific Data Fields

For Galileo satellites the following clarifications apply:

- Galileo Epoch Time (TOW) DF248 is used for GNSS epoch time.
- Galileo PRNs are mapped to satellite mask IDs according to Table 3.5-106
- Galileo signals are mapped to signal mask IDs according to Table 3.5-107
- Extended satellite information is reserved for future use

© RTCM – Not for reproduction or redistribution

Table 3.5-106 Galileo Satellite ID mapping

Satellite ID in Satellite Mask (DF394)	Galileo Satellite PRN
1	1
2	2
…	…
50	50
51–64	Reserved

Table 3.5-107 Galileo Signal ID mapping

Signal ID in Signal Mask (DF395)	Frequency Band	Signal	RINEX code	Notes
1				Reserved
2	E1	C no data	1C	
3	E1	A	1A	
4	E1	B I/NAV OS/CS/SoL	1B	
5	E1	B+C	1X	
6	E1	A+B+C	1Z	
7				Reserved
8	E6	C	6C	
9	E6	A	6A	
10	E6	B	6B	
11	E6	B+C	6X	
12	E6	A+B+C	6Z	
13				Reserved
14	E5B	I	7I	
15	E5B	Q	7Q	
16	E5B	I+Q	7X	

© RTCM – Not for reproduction or redistribution

17				Reserved
18	E5(A+B)	I	8I	
19	E5(A+B)	Q	8Q	
20	E5(A+B)	I+Q	8X	
21				Reserved
22	E5A	I	5I	
23	E5A	Q	5Q	
24	E5A	I+Q	5X	
25–32				Reserved

3.5.12.7 SBAS Messages Description

For SBAS, all seven MSM messages are introduced.

3.5.12.7.1 Message Types Summary

Table 3.5-108 lists the MSM message types for SBAS.

Table 3.5-108 SBAS Message Types

Message Type	Message Name	MSM type
1101	Compact SBAS Pseudoranges	MSM1
1102	Compact SBAS Phaseranges	MSM2
1103	Compact SBAS Pseudoranges and Phaseranges	MSM3
1104	Full SBAS Pseudoranges and Phaseranges plus CNR	MSM4
1105	Full SBAS Pseudoranges, Phaseranges, PhaserangeRate and CNR	MSM5
1106	Full SBAS Pseudoranges and Phaseranges plus CNR (high resolution)	MSM6
1107	Full SBAS Pseudoranges, Phaseranges, PhaserangeRate and CNR (high resolution)	MSM7

3.5.12.7.2 SBAS Specific Data Fields

For SBAS satellites the following clarifications apply:

© RTCM – Not for reproduction or redistribution

- GPS Epoch Time (TOW) DF004 is used for GNSS Epoch Time.
- SBAS PRNs are mapped to satellite mask IDs according to Table 3.5-109
- SBAS signals are mapped to signal mask IDs according to Table 3.5-110
- Extended satellite information is reserved for future use.
- Satellite ID mapping is not defined for SBAS signals transmitted by QZSS (L1-SAIF). This mapping may be included in the future.

Table 3.5-109 SBAS Satellite ID mapping

Satellite ID in Satellite Mask (DF394)	Satellite PRN
1	120
2	121
…	…
39	158
40-64	Reserved

Table 3.5-110 SBAS Signal ID Mapping

Signal ID in Signal Mask (DF395)	Frequency Band	Signal	RINEX code	Notes
1				Reserved
2	L1	C/A	1C	
3-21				Reserved
22	L5	I	5I	
23	L5	Q	5Q	
24	L5	I+Q	5X	
25-32				Reserved

© RTCM – Not for reproduction or redistribution

3.5.12.8 QZSS Messages Description

For QZSS all seven MSM messages are introduced.

3.5.12.8.1 Message Types Summary

Table 3.5-111 lists the MSM message types for QZSS.

Table 3.5-111 QZSS Message Types

Message Type	Message Name	MSM type
1111	Compact QZSS Pseudoranges	MSM1
1112	Compact QZSS Phaseranges	MSM2
1113	Compact QZSS Pseudoranges and Phaseranges	MSM3
1114	Full QZSS Pseudoranges and Phaseranges plus CNR	MSM4
1115	Full QZSS Pseudoranges, Phaseranges, PhaserangeRate and CNR	MSM5
1116	Full QZSS Pseudoranges and Phaseranges plus CNR (high resolution)	MSM6
1117	Full QZSS Pseudoranges, Phaseranges, PhaserangeRate and CNR (high resolution)	MSM7

3.5.12.8.2 QZSS Specific Data Fields

For QZSS satellites the following clarifications apply:

- QZSS Epoch Time (TOW) DF428 is used for GNSS Epoch Time.
- QZSS PRNs are mapped to satellite mask IDs according to Table 3.5-112
- QZSS signals are mapped to signal mask IDs according to Table 3.5-113
- Extended satellite information is reserved for future use.

Table 3.5-112 QZSS Satellite ID mapping

Satellite ID in Satellite Mask (DF394)	Satellite PRN
1	193

© RTCM – Not for reproduction or redistribution

Satellite ID in Satellite Mask (DF394)	Satellite PRN
2	194
…	…
10	202
11-64	Reserved

S

Table 3.5-113 QZSS Signal ID Mapping

Signal ID in Signal Mask (DF395)	Frequency Band	Signal	RINEX code	Notes
1				Reserved
2	L1	C/A	1C	
3-8				Reserved
9	L6	S	6S	
10	L6	L	6L	
11	L6	S+L	6X	
12-14				Reserved
15	L2	L2C(M)	2S	
16	L2	L2C(L)	2L	
17	L2	L2C(M+L)	2X	
18-21				Reserved
22	L5	I	5I	
23	L5	Q	5Q	
24	L5	I+Q	5X	
25-29				Reserved
30	L1	L1C(D)	1S	
31	L1	L1C(P)	1L	
32	L1	L1C(D+P)	1X	

© RTCM – Not for reproduction or redistribution

3.5.12.9 BeiDou Messages Description

For BeiDou all seven MSM messages are introduced.

© RTCM – Not for reproduction or redistribution

3.5.12.9.1 Message Types Summary

Table 3.5-114 lists the MSM message types for BeiDou.

Table 3.5-114 BeiDou Message Types

Message Type	Message Name	MSM type
1121	Compact BeiDou Pseudoranges	MSM1
1122	Compact BeiDou Phaseranges	MSM2
1123	Compact BeiDou Pseudoranges and Phaseranges	MSM3
1124	Full BeiDou Pseudoranges and Phaseranges plus CNR	MSM4
1125	Full BeiDou Pseudoranges, Phaseranges, PhaserangeRate and CNR	MSM5
1126	Full BeiDou Pseudoranges and Phaseranges plus CNR (high resolution)	MSM6
1127	Full BeiDou Pseudoranges, Phaseranges, PhaserangeRate and CNR (high resolution)	MSM7

3.5.12.9.2 BeiDou-Specific Data Fields

For BeiDou satellites, the following clarifications apply:

- BeiDou Epoch Time (TOW) DF427 is used for GNSS Epoch Time.
- BeiDou PRNs are mapped to satellite mask IDs according to Table 3.5-115
- BeiDou signals are mapped to signal mask IDs according to
- Table 3.5-116
- Extended satellite information is reserved for future use.

Table 3.5-115 BeiDou Satellite ID Mapping

Satellite ID in Satellite Mask (DF394)	**Satellite PRN**
1	1
2	2
…	…
63	63
64	Reserved

© RTCM – Not for reproduction or redistribution

Table 3.5-116 BeiDou Signal ID Mapping

Signal ID in Signal Mask (DF395)	Frequency Band	Signal	RINEX code	Notes
1				Reserved
2	B1	I	2I	
3	B1	Q	2Q	
4	B1	I+Q	2X	
5-7				Reserved
8	B3	I	6I	
9	B3	Q	6Q	
10	B3	I+Q	6X	
11-13				Reserved
14	B2	I	7I	
15	B2	Q	7Q	
16	B2	I+Q	7X	
17-21				Reserved
22	B2a	D	5D	
23	B2a	P	5P	
24	B2a	D+P	5X	
25	B2b	I	7D	
26-29				Reserved
30	B1C	D	1D	
31	B1C	P	1P	
32	B1C	D+P	1X	

3.5.12.10 NavIC/IRNSS messages description

For NavIC/IRNSS, all seven MSM messages are introduced.

3.5.12.10.1 Message Types Summary

Table 3.5-117 lists the MSM message types for NavIC/IRNSS.

© RTCM – Not for reproduction or redistribution

Table 3.5-117 NavIC/IRNSS Message Types

Message Type	Message Name	MSM type
1131	Compact NavIC/IRNSS Pseudoranges	MSM1
1132	Compact NavIC/IRNSS Phaseranges	MSM2
1133	Compact NavIC/IRNSS Pseudoranges and Phaseranges	MSM3
1134	Full NavIC/IRNSS Pseudoranges and Phaseranges plus CNR	MSM4
1135	Full NavIC/IRNSS Pseudoranges, Phaseranges, PhaserangeRate and CNR	MSM5
1136	Full NavIC/IRNSS Pseudoranges and Phaseranges plus CNR (high resolution)	MSM6
1137	Full NavIC/IRNSS Pseudoranges, Phaseranges, PhaserangeRate and CNR (high resolution)	MSM7

3.5.12.10.2 NavIC/IRNSS-Specific data fields

For NavIC/IRNSS satellites the following clarifications apply:

- NavIC/IRNSS Epoch Time DF546 is used for GNSS Epoch Time.
- NavIC/IRNSS PRNs are mapped to satellite mask IDs according to Table 3.5-118.
- NavIC/IRNSS signals are mapped to signal mask IDs according to Table 3.5-119.
- Extended satellite information is reserved for future use.

Table 3.5-118 NavIC/IRNSS Satellite ID Mapping

Satellite ID in Satellite Mask (DF394)	**NavIC/IRNSS Satellite PRN**
1	1
2	2
…	…
14	14
15…64	Reserved

© RTCM – Not for reproduction or redistribution

Table 3.5-119 NavIC/IRNSS Signal ID Mapping

Signal ID in Signal Mask (DF395)	Frequency Band	Signal	RINEX code	Notes
1 – 7				Reserved
8	S	SPS – S	9A	
9 – 21				Reserved
22	L5	SPS – L5	5A	
23-32				Reserved

3.5.13 GLONASS Bias Information Message

In general, GLONASS Pseudorange and Phaserange measurements can be affected by different receiver-dependent biases. An offset of the so-called code-phase bias (CPB) between the reference and rover receivers will manifest itself as frequency-dependent biases in the corresponding GLONASS double-difference Phaserange measurements. The magnitude of these inter-frequency biases will adversely affect ambiguity resolution when mixed receiver types are in use.

Message type 1230, described in Table 3.5-120, provides information that enables compensation for the first-order inter-frequency Phaserange biases introduced by the reference receiver code-phase bias. Higher-order frequency-dependent biases may still be present in the reference Phaserange data; however, the magnitude of this class of bias is typically insignificant for ambiguity resolution.

Table 3.5-120 Contents of GLONASS L1 and L2 Code-Phase Biases Message 1230

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
Reference Station ID	DF003	uint12	12	
GLONASS Code-Phase bias indicator	DF421	bit(1)	1	
Reserved	DF001	bit(3)	3	
GLONASS FDMA signals mask	DF422	bit(4)	4	
GLONASS L1 C/A Code-Phase Bias	DF423	int16	16	
GLONASS L1 P Code-Phase Bias	DF424	int16	16	
GLONASS L2 C/A Code-Phase Bias	DF425	int16	16	

© RTCM – Not for reproduction or redistribution

GLONASS L2 P Code-Phase Bias	DF426	int16	16	
TOTAL			32+16*N	N corresponds to the number of bits set to 1 in DF422.

Notes:

- MT1230 with valid CPB values for all GLONASS FDMA signals available in the data stream (MT1009-1012 and MT1081-1088) is mandatory for full GLONASS and combined GPS/GLONASS RTK interoperability, including GLONASS and combined GPS/GLONASS network RTK services. For example, if a reference station broadcasts GLONASS observables related to GLONASS L1 C/A and GLONASS L2 C/A signals, MT1230 shall include DF423 and DF425. MT1230 may include information about a greater number of GLONASS FDMA signals than those available in the data stream that includes GLONASS observables.
- To provide a full GLONASS RTK service that is also backwards compatible with GLONASS observable messages 1009-1012 and proprietary methods of processing the content of those messages, the encoder shall not apply the contents of MT1230 to GLONASS observation data *a priori*, if MT1230 is broadcast in the data stream that includes MT 1009-1012.

Transmitted Configuration	Any of 1081-1087	Any of 1009-1012	DF421
MSM only	Transmitted	Not transmitted	Can be 0 or 1
Legacy only	Not transmitted	Transmitted	Shall always be 0
MSM + Legacy	Transmitted	Transmitted	Shall always be 0

- The content of message 1033 is commonly used in practice by the decoder to first identify and then compensate for the expected GLONASS receiver-dependent biases present in observation messages 1009-1012. Although this method has proven relatively reliable, 1033 was not developed specifically for this purpose and has several limitations. Message 1230 has been developed to improve GLONASS interoperability. Therefore, if a data stream contains both messages 1230 and 1033 then the content of 1230 takes precedence over the expected GLONASS receiver biases indicated by the receiver description in 1033.
- For MAC Network RTK services, MT 1230 defines Code-Phase Biases related to the Master Reference Station. Therefore, the content of DF003 (Reference Station ID) in MT 1230 shall correspond to the content of DF060 (Master Reference Station ID).
- GLONASS Code-Phase Biases are transmitted only for those GLONASS FDMA signals, for which the corresponding bit is set to 1 in GLONASS FDMA Signals Mask (DF422).

© RTCM – Not for reproduction or redistribution

- The standard reserves a possibility for potential future MT1230 extensions. These extensions may be introduced by adding data to the end of the message.

3.5.14 Ephemeris Messages

The ephemeris messages contain satellite ephemeris information.

The ephemeris messages can be used to help user receivers to quickly acquire satellites. For example, if the user receiver has access to a wireless service with these messages, then rather than waiting until one satellite has been acquired and its almanac data processed it can utilize the ephemeris information immediately.

These messages could also be broadcast if an anomaly in new ephemeris data set is detected that would require the differential reference station to base corrections on the previous good satellite ephemeris. This would allow the user equipment just entering the differential system to utilize the corrections being broadcast for that ephemeris and would support the use of the satellite for differential navigation even though the satellite ephemeris was in error. It is anticipated that this message type would be broadcast every 2 minutes or so while this condition persisted. The schedule would be maintained until the satellite broadcast was corrected, or until the satellite dropped below the coverage area of the reference station.

3.5.14.1 GPS Ephemeris

The GPS LNAV ephemeris message contains GPS satellite ephemeris information for the L1 C/A signal.

The data fields in the GPS ephemeris message are direct copies of fields in the GPS navigation data. They have the same number of bits, scale factor, and units as defined in the GPS signal specification[2]. The names of the data fields are also kept as close as possible to those defined in the signal specification for cross-referencing.

[2] Section 20.3.3.3.3, “Subframe 1” and Section 20.3.3.4, “Subframes 2 and 3.”, Global Positioning Systems Directorate Systems Engineering & Integration (2018)

© RTCM – Not for reproduction or redistribution

Table 3.5-121 Contents of GPS Satellite Ephemeris Data, Message Type 1019

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	1019
GPS Satellite ID	DF009	uint6	6	
GPS Week Number	DF076	uint10	10	See GPS signal specification. The GPS Week Number is referenced to the GPS Time Of Week (TOW) when the ephemeris data was transmitted. In periods near the week rollover this week number may differ by ±1 from the week number associated with the t_{oc} and t_{oe}. Additional information or steps must be used to determine the t_{oc}/t_{oe} week number.
GPS SV ACCURACY	DF077	uint4	4	See GPS signal specification
GPS CODE ON L2	DF078	bit(2)	2	See GPS signal specification
GPS IDOT	DF079	int14	14	See GPS signal specification
GPS IODE	DF071	uint8	8	See GPS signal specification
GPS t_{oc}	DF081	uint16	16	See GPS signal specification
GPS a_{f2}	DF082	int8	8	See GPS signal specification
GPS a_{f1}	DF083	int16	16	See GPS signal specification
GPS a_{f0}	DF084	int22	22	See GPS signal specification
GPS IODC	DF085	uint10	10	See GPS signal specification
GPS C_{rs}	DF086	int16	16	See GPS signal specification
GPS Δn (DELTA n)	DF087	int16	16	See GPS signal specification

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GPS M_0	DF088	int32	32	See GPS signal specification
GPS C_{uc}	DF089	int16	16	See GPS signal specification
GPS Eccentricity (e)	DF090	uint32	32	See GPS signal specification
GPS C_{us}	DF091	int16	16	See GPS signal specification
GPS $(A)^{1/2}$	DF092	uint32	32	See GPS signal specification
GPS t_{oe}	DF093	uint16	16	See GPS signal specification
GPS C_{ic}	DF094	int16	16	See GPS signal specification
GPS Ω_0 $(OMEGA)_0$	DF095	int32	32	See GPS signal specification
GPS C_{is}	DF096	int16	16	See GPS signal specification
GPS i_0	DF097	int32	32	See GPS signal specification
GPS C_{rc}	DF098	int16	16	See GPS signal specification
GPS ω (Argument of Perigee)	DF099	int32	32	See GPS signal specification
GPS OMEGADOT (Rate of Right Ascension)	DF100	int24	24	See GPS signal specification
GPS t_{GD}	DF101	int8	8	See GPS signal specification
GPS SV HEALTH	DF102	uint6	6	See GPS signal specification
GPS L2 P data flag	DF103	bit(1)	1	See GPS signal specification
GPS Fit Interval	DF137	bit(1)	1	See GPS signal specification
TOTAL			***488***	

3.5.14.2 GLONASS Ephemeris

The GLONASS ephemeris message contains GLONASS satellite ephemeris information for the L1C signal.

© RTCM – Not for reproduction or redistribution

All data fields in the GLONASS ephemeris message have the same number of bits, scale factor, and units as defined in the GLONASS signal specification[3]. The names of the data fields are also kept as close as possible to those defined in the signal specification for cross-referencing.

Table 3.5-122 Contents of GLONASS Satellite Ephemeris Data, Message Type 1020

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	1020
GLONASS Satellite ID (Satellite Slot Number)	DF038	uint6	6	
GLONASS Satellite Frequency Channel Number	DF040	uint5	5	
GLONASS almanac health (C_n word)	DF104	bit(1)	1	See GLONASS signal specification
GLONASS almanac health availability indicator	DF105	bit(1)	1	See GLONASS signal specification
GLONASS P1	DF106	bit(2)	2	See GLONASS signal specification
GLONASS t_k	DF107	bit(12)	12	See GLONASS signal specification
GLONASS MSb of B_n word	DF108	bit(1)	1	See GLONASS signal specification
GLONASS P2	DF109	bit(1)	1	See GLONASS signal specification
GLONASS t_b	DF110	uint7	7	See GLONASS signal specification
GLONASS $x_n(t_b)$, first derivative	DF111	intS24	24	See GLONASS signal specification
GLONASS $x_n(t_b)$	DF112	intS27	27	See GLONASS signal specification
GLONASS $x_n(t_b)$, second derivative	DF113	intS5	5	See GLONASS signal specification
GLONASS $y_n(t_b)$, first derivative	DF114	intS24	24	See GLONASS signal specification
GLONASS $y_n(t_b)$	DF115	intS27	27	See GLONASS signal specification

[3] Section 4.4, "Immediate information and ephemeris parameters", Russian Institute of Space Device Engineering (2008)

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
GLONASS $y_n(t_b)$, second derivative	DF116	intS5	5	See GLONASS signal specification
GLONASS $z_n(t_b)$, first derivative	DF117	intS24	24	See GLONASS signal specification
GLONASS $z_n(t_b)$	DF118	intS27	27	See GLONASS signal specification
GLONASS $z_n(t_b)$, second derivative	DF119	intS5	5	See GLONASS signal specification
GLONASS P3	DF120	bit(1)	1	See GLONASS signal specification
GLONASS $\gamma_n(t_b)$	DF121	intS11	11	See GLONASS signal specification
GLONASS-M P	DF122	bit(2)	2	See GLONASS signal specification
GLONASS-M l_n (third string)	DF123	bit(1)	1	See GLONASS signal specification
GLONASS $\tau_n(t_b)$	DF124	intS22	22	See GLONASS signal specification
GLONASS-M $\Delta\tau_n$	DF125	intS5	5	See GLONASS signal specification
GLONASS E_n	DF126	uint5	5	See GLONASS signal specification
GLONASS-M P4	DF127	bit(1)	1	See GLONASS signal specification
GLONASS-M F_T	DF128	uint4	4	See GLONASS signal specification
GLONASS-M N_T	DF129	uint11	11	See GLONASS signal specification
GLONASS-M M	DF130	bit(2)	2	See GLONASS signal specification
Availability of Additional Data	DF131	bit(1)	1	See data field notes
GLONASS N^A	DF132	uint11	11	See GLONASS signal specification
GLONASS τ_c	DF133	intS32	32	See GLONASS signal specification
GLONASS-M N_4	DF134	uint5	5	See GLONASS signal specification
GLONASS-M τ_{GPS}	DF135	intS22	22	See GLONASS signal specification
GLONASS-M l_n (fifth string)	DF136	bit(1)	1	See GLONASS signal specification

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Reserved		bit(7)	7	
TOTAL			***360***	

3.5.14.3 Galileo Ephemerides

The Galileo ephemeris messages contain Galileo satellite ephemerides, accuracy and health status (signal and data) information broadcast by the satellites as F/NAV or I/NAV information.

All data fields in the Galileo ephemeris messages have the same number of bits, the same scale factor and units as defined in the Galileo signal specification[4]. The names of the data fields are also kept as close as possible to those defined in signal specification for cross-referencing.

Table 3.5-123 Contents of Galileo F/NAV Satellite Ephemeris Data, Message Type 1045

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
Galileo Satellite ID	DF252	uint6	6	The Galileo SVID parameter is coded with 6 bits. However, the maximum constellation which can be accommodated within the I/NAV and F/NAV frames is 36 satellites (3 planes of 12 satellites each). The range for the satellite ID (SVID) is defined as 0-63 in OS SIS ICD. See Galileo Signal Specification Section 5.1.9.1. Note: SVID = 0 is used in the broadcast almanac data to indicate unused almanac entries. SVID values 1 to 36

[4] The Galileo ephemerides messages are defined in the OS SIS ICD, in Section 4.2.4, “F/NAV Page Contents” and Section 4.3.5, “I/NAV Word Types”. See Galileo Signal Specification.

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
				are defined in the OS SIS ICD. Higher values are reserved for future use
Galileo Week Number (WN)	DF289	uint12	12	See Galileo signal specification The Galileo Week Number is referenced to the Galileo Time Of Week (TOW) when the ephemeris data was transmitted. In periods near the week rollover this week number may differ by ±1 from the week number associated with the t_{oc} and t_{oe}. Additional information or steps must be used to determine the t_{oc}/t_{oe} week number.
Galileo IODnav	DF290	uint10	10	See Galileo signal specification
Galileo SV SISA (SIS Accuracy)	DF291	uint8	8	See Galileo signal specification
Galileo $\dot{i}$ (IDOT)	DF292	int14	14	See Galileo signal specification
Galileo t_{oc}	DF293	uint14	14	See Galileo signal specification
Galileo a_{f2}	DF294	int6	6	See Galileo signal specification
Galileo a_{f1}	DF295	int21	21	See Galileo signal specification
Galileo a_{f0}	DF296	int31	31	See Galileo signal specification
Galileo C_{rs}	DF297	int16	16	See Galileo signal specification
Galileo Δn	DF298	int16	16	See Galileo signal specification
Galileo M_0	DF299	int32	32	See Galileo signal specification
Galileo C_{uc}	DF300	int16	16	See Galileo signal specification
Galileo e	DF301	uint32	32	See Galileo signal specification
Galileo C_{us}	DF302	int16	16	See Galileo signal specification
Galileo $a^{1/2}$	DF303	uint32	32	See Galileo signal specification
Galileo t_{oe}	DF304	uint14	14	See Galileo signal specification

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Galileo C_{ic}	DF305	int16	16	See Galileo signal specification
Galileo Ω_0	DF306	int32	32	See Galileo signal specification
Galileo C_{is}	DF307	int16	16	See Galileo signal specification
Galileo i_0	DF308	int32	32	See Galileo signal specification
Galileo C_{rc}	DF309	int16	16	See Galileo signal specification
Galileo ω	DF310	int32	32	See Galileo signal specification
Galileo $\dot{\Omega}_0$ (OMEGADOT)	DF311	int24	24	See Galileo signal specification
Galileo $BGD_{E5a/E1}$	DF312	int10	10	See Galileo signal specification
Galileo NAV Signal Health Status (OSHS)	DF314	bit(2)	2	See Galileo signal specification
Galileo NAV Data Validity Status (OSDVS)	DF315	bit(1)	1	See Galileo signal specification
Reserved	DF001	bit(7)	7	
TOTAL			***496***	***62 bytes***

Table 3.5-124 Contents of Galileo I/NAV Satellite Ephemeris Data, Message Type 1046

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
Galileo Satellite ID	DF252	uint6	6	
Galileo Week Number	DF289	uint12	12	See Galileo signal specification The Galileo Week Number is referenced to the Galileo Time Of Week (TOW) when the ephemeris data was transmitted. In

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
				periods near the week rollover this week number may differ by ±1 from the week number associated with the t_{oc} and t_{oe}. Additional information or steps must be used to determine the t_{oc}/t_{oe} week number.
Galileo IODnav	DF290	uint10	10	See Galileo signal specification
Galileo SISA Index (E1,E5b)	DF286	uint8	8	See Galileo signal specification
Galileo $\dot{i}$ (IDOT)	DF292	int14	14	See Galileo signal specification
Galileo t_{oc}	DF293	uint14	14	See Galileo signal specification
Galileo a_{f2}	DF294	int6	6	See Galileo signal specification
Galileo a_{f1}	DF295	int21	21	See Galileo signal specification
Galileo a_{f0}	DF296	int31	31	See Galileo signal specification
Galileo C_{rs}	DF297	int16	16	See Galileo signal specification
Galileo Δn	DF298	int16	16	See Galileo signal specification
Galileo M_0	DF299	int32	32	See Galileo signal specification
Galileo C_{uc}	DF300	int16	16	See Galileo signal specification
Galileo e	DF301	uint32	32	See Galileo signal specification
Galileo C_{us}	DF302	int16	16	See Galileo signal specification
Galileo $a^{1/2}$	DF303	uint32	32	See Galileo signal specification
Galileo t_{oe}	DF304	uint14	14	See Galileo signal specification
Galileo C_{ic}	DF305	int16	16	See Galileo signal specification
Galileo Ω_0	DF306	int32	32	See Galileo signal specification
Galileo C_{is}	DF307	int16	16	See Galileo signal specification
Galileo i_0	DF308	int32	32	See Galileo signal specification
Galileo C_{rc}	DF309	int16	16	See Galileo signal specification

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Galileo ω	DF310	int32	32	See Galileo signal specification
Galileo $\dot{\Omega}$ (OmegaDot)	DF311	int24	24	See Galileo signal specification
Galileo $BGD_{E5a/E1}$	DF312	int10	10	See Galileo signal specification
Galileo $BGD_{E5b/E1}$	DF313	int10	10	See Galileo signal specification
Galileo E5b Signal Health Status	DF316	bit(2)	2	See Galileo signal specification
Galileo E5b Data Validity Status	DF317	bit(1)	1	See Galileo signal specification
Galileo E1-B Signal Health Status	DF287	bit(2)	2	See Galileo signal specification
Galileo E1-B Data Validity Status	DF288	bit(1)	1	See Galileo signal specification
Reserved	DF001	bit(2)	2	
TOTAL			***504***	***63 Bytes***

3.5.14.4 QZSS Ephemeris

The QZSS ephemeris message contains QZSS satellite ephemeris information for the L1 C/A signal.

All data fields in the QZSS ephemeris message have the same number of bits, scale factor, and units as defined in the QZSS signal specification[5]. The names of the data fields are also kept as close as possible to those defined in signal specification document for cross-referencing.

[5] Section 4.1.2.3, "Subframe 1 (SV Clock)", Section 4.1.2.4, "Subframe 2 (Ephemeris 1)", and Section 4.1.2.3, "Subframe 2 (Ephemeris 2)", Cabinet Office, Government of Japan (2018)

© RTCM – Not for reproduction or redistribution

Table 3.5-125 Contents of QZSS Satellite Ephemeris Data, Message Type 1044

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	
QZSS Satellite ID	DF429	uint4	4	
QZSS t_{oc}	DF430	uint16	16	See QZSS signal specification
QZSS a_{f2}	DF431	int8	8	See QZSS signal specification
QZSS a_{f1}	DF432	int16	16	See QZSS signal specification
QZSS a_{f0}	DF433	int22	22	See QZSS signal specification
QZSS IODE	DF434	uint8	8	See QZSS signal specification
QZSS C_{rs}	DF435	int16	16	See QZSS signal specification
QZSS Δn_0	DF436	int16	16	See QZSS signal specification
QZSS M_0	DF437	int32	32	See QZSS signal specification
QZSS C_{uc}	DF438	int16	16	See QZSS signal specification
QZSS e	DF439	uint32	32	See QZSS signal specification
QZSS C_{us}	DF440	int16	16	See QZSS signal specification
QZSS $A^{1/2}$	DF441	uint32	32	See QZSS signal specification
QZSS t_{oe}	DF442	uint16	16	See QZSS signal specification
QZSS C_{ic}	DF443	int16	16	See QZSS signal specification
QZSS Ω_0	DF444	int32	32	See QZSS signal specification
QZSS C_{is}	DF445	int16	16	See QZSS signal specification
QZSS i_0	DF446	int32	32	See QZSS signal specification
QZSS C_{rc}	DF447	int16	16	See QZSS signal specification
QZSS ω_n	DF448	int32	32	See QZSS signal specification

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
QZSS Ω DOT	DF449	int24	24	See QZSS signal specification
QZSS i_0-DOT	DF450	int14	14	See QZSS signal specification
QZSS Codes on L2 Channel	DF451	bit(2)	2	See QZSS signal specification
QZSS Week number	DF452	uint10	10	See QZSS signal specification The QZSS Week Number is referenced to the QZSS Time Of Week (TOW) when the ephemeris data was transmitted. In periods near the week rollover this week number may differ by ±1 from the week number associated with the t_{oc} and t_{oe}. Additional information or steps must be used to determine the t_{oc}/t_{oe} week number.
QZSS URA	DF453	uint4	4	See QZSS signal specification
QZSS SV health	DF454	uint6	6	See QZSS signal specification
QZSS T_{GD}	DF455	int8	8	See QZSS signal specification
QZSS IODC	DF456	uint10	10	See QZSS signal specification
QZSS Fit Interval	DF457	bit(1)	1	See QZSS signal specification
TOTAL			***485***	

3.5.14.5 BeiDou Ephemeris

The BeiDou ephemeris message contains the BeiDou satellite ephemeris information from the B1I signal.

All data fields have the same number of bits, the same scale factor and units defined in the BeiDou signal specification[6]. The names of the data fields are also kept as close as possible to those defined in signal specification for cross-referencing.

[6] Section 5.2.4, "D1 NAV Message Content and Algorithm" and Section 5.3.3, "D2 NAV Message Content and Algorithm", China Satellite Navigation Office (2019)

© RTCM – Not for reproduction or redistribution

Table 3.5-126 Contents of BDS Satellite Ephemeris Data, Message Type 1042

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	1042
BDS Satellite ID	DF488	uint6	6	
BDS Week Number	DF489	uint13	13	See BeiDou signal specification The BeiDou Week Number is referenced to the BeiDou Time Of Week (TOW) when the ephemeris data was transmitted. In periods near the week rollover this week number may differ by ±1 from the week number associated with the t_{oc} and t_{oe}. Additional information or steps must be used to determine the t_{oc}/t_{oe} week number.
BDS SV URAI	DF490	bit(4)	4	See BeiDou signal specification
BDS IDOT	DF491	int14	14	See BeiDou signal specification
BDS AODE	DF492	uint5	5	See BeiDou signal specification
BDS t_{oc}	DF493	uint17	17	See BeiDou signal specification
BDS a_2	DF494	int11	11	See BeiDou signal specification
BDS a_1	DF495	int22	22	See BeiDou signal specification
BDS a_0	DF496	int24	24	See BeiDou signal specification
BDS AODC	DF497	uint5	5	See BeiDou signal specification
BDS C_{rs}	DF498	int18	18	See BeiDou signal specification
BDS Δn	DF499	int16	16	See BeiDou signal specification
BDS M_0	DF500	int32	32	See BeiDou signal specification
BDS C_{uc}	DF501	int18	18	See BeiDou signal specification
BDS e	DF502	uint32	32	See BeiDou signal specification
BDS C_{us}	DF503	int18	18	See BeiDou signal specification

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
BDS $a^{1/2}$	DF504	uint32	32	See BeiDou signal specification
BDS t_{oe}	DF505	uint17	17	See BeiDou signal specification
BDS C_{ic}	DF506	int18	18	See BeiDou signal specification
BDS Ω_0	DF507	int32	32	See BeiDou signal specification
BDS C_{is}	DF508	int18	18	See BeiDou signal specification
BDS i_0	DF509	int32	32	See BeiDou signal specification
BDS C_{rc}	DF510	int18	18	See BeiDou signal specification
BDS ω	DF511	int32	32	See BeiDou signal specification
BDS $\dot{\Omega}$	DF512	int24	24	See BeiDou signal specification
BDS T_{GD1}	DF513	int10	10	See BeiDou signal specification
BDS T_{GD2}	DF514	int10	10	See BeiDou signal specification
BDS SV HEALTH	DF515	bit(1)	1	See BeiDou signal specification
TOTAL			511	

3.5.14.6 NavIC/IRNSS Ephemeris

The NavIC/IRNSS ephemeris message contains satellite ephemeris information for the L5 signal.

All data fields have the same number of bits, the same scale factor and units defined in NavIC/IRNSS signal specification[7]. The names of the data fields are also kept as close as possible to those defined in signal specification for cross-referencing.

[7] Section 6.1.1, “Subframe 1 Data Format”, Section 6.1.2, “”, and Section 6.2.1, “Subframe1 & Subframe 2”, Indian Space Research Organization Satellite Centre (2017)

© RTCM – Not for reproduction or redistribution

Table 3.5-127 Contents of NavIC/IRNSS Satellite Ephemeris Message Type 1041

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Message Number	DF002	uint12	12	1041
NavIC/IRNSS Satellite ID	DF516	uint6	6	
Week Number (WN)	DF517	uint10	10	See NavIC/IRNSS signal specification The NavIC/IRNSS Week Number is referenced to the NavIC/IRNSS Time Of Week (TOW) when the ephemeris data was transmitted. In periods near the week rollover this week number may differ by ±1 from the week number associated with the t_{oc} and t_{oe}. Additional information or steps must be used to determine the t_{oc}/t_{oe} week number.
Clock bias (a_{fo})	DF518	int22	22	See NavIC/IRNSS signal specification
Clock drift (a_{f1})	DF519	int16	16	See NavIC/IRNSS signal specification
Clock drift rate (a_{f2})	DF520	int8	8	See NavIC/IRNSS signal specification
SV Accuracy (URA)	DF521	uint4	4	See NavIC/IRNSS signal specification
Time of clock (t_{oc})	DF522	uint16	16	See NavIC/IRNSS signal specification
Total Group Delay (T_{GD})	DF523	int8	8	See NavIC/IRNSS signal specification
Mean Motion Difference (Δn)	DF524	int22	22	See NavIC/IRNSS signal specification

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Issue of Data Ephemeris & Clock (IODEC)	DF525	uint8	8	See NavIC/IRNSS signal specification
Reserved bits after IODEC	DF526	uint10	10	See NavIC/IRNSS signal specification
L5 Flag	DF527	bit(1)	1	See NavIC/IRNSS signal specification
S Flag	DF528	bit(1)	1	See NavIC/IRNSS signal specification
Amplitude of the Cosine Harmonic Correction Term to the Argument of Latitude (C_{uc})	DF529	int15	15	See NavIC/IRNSS signal specification
Amplitude of the Sine Harmonic Correction Term to the Argument of Latitude (C_{us})	DF530	int15	15	See NavIC/IRNSS signal specification
Amplitude of the Cosine Harmonic Correction Term to the Angle of Inclination (C_{ic})	DF531	int15	15	See NavIC/IRNSS signal specification
Amplitude of the Sine Harmonic Correction Term to the Angle of Inclination (C_{is})	DF532	int15	15	See NavIC/IRNSS signal specification
Amplitude of the Cosine Harmonic Correction Term to the Orbit Radius (C_{rc})	DF533	int15	15	See NavIC/IRNSS signal specification
Amplitude of the Sine Harmonic Correction Term to the Orbit Radius (C_{rs})	DF534	int15	15	See NavIC/IRNSS signal specification
Rate of Inclination angle (IDOT)	DF535	int14	14	See NavIC/IRNSS signal specification

© RTCM – Not for reproduction or redistribution

DATA FIELD	DF NUMBER	DATA TYPE	NO. OF BITS	NOTES
Mean Anomaly (M_0)	DF536	int32	32	See NavIC/IRNSS signal specification
Time of ephemeris (t_{oe})	DF537	uint16	16	See NavIC/IRNSS signal specification
Eccentricity (e)	DF538	uint32	32	See NavIC/IRNSS signal specification
Square root of Semi major axis ($\sqrt{A}$)	DF539	uint32	32	See NavIC/IRNSS signal specification
Long of Ascending Node (Ω_o)	DF540	int32	32	See NavIC/IRNSS signal specification
Argument of perigee (ω)	DF541	int32	32	See NavIC/IRNSS signal specification
Rate of RAAN ($\dot{\Omega}$)	DF542	int22	22	See NavIC/IRNSS signal specification
Inclination (i_0)	DF543	int32	32	See NavIC/IRNSS signal specification
2 spare bits after IDOT	DF544	bit(2)	2	See NavIC/IRNSS signal specification
2 spare bits after I0	DF545	bit(2)	2	See NavIC/IRNSS signal specification
TOTAL			482	

3.6 Proprietary Messages

Message types from 4001 through 4095 are reserved for proprietary use. The first field in the proprietary messages is the 12-bit message number. Content following this is implementation defined. The overall message structure must follow that defined by the transport layer (see Section 4).

© RTCM – Not for reproduction or redistribution

Each organization may be assigned by RTCM one message type for proprietary use. Organizations are free to define sub-types of messages according to their own conventions, but they must all utilize their assigned message type. Contact RTCM to acquire a new proprietary type.

Table 3.6-1 lists the assigned proprietary message types at the time of publishing the current revision of this standard. Additional proprietary numbers may be assigned and in existence between revisions. An up-to-date list of proprietary message types can be found on the RTCM website. See the DIFFERENTIAL GLOBAL NAVIGATION SATELLITE (DGNSS) STANDARDS section at www.rtcm.org under the Publications tab.

Table 3.6-1 Assigned Proprietary Message Types

Message Type	Organization	Contact
4095	Spectra Geospatial	www.spectrageospatial.com
4094	Trimble Navigation Ltd.	www.trimble.com
4093	NovAtel Inc.	www.novatel.com
4092	Leica Geosystems	www.leica-geosystems.com
4091	Topcon Positioning Systems	www.topconpositioning.com
4090	Geo++	www.geopp.de
4089	Septentrio Satellite Navigation	www.septentrio.com
4088	IfEN GmbH	www.ifen.com
4087	Fugro	www.fugro.com
4086	inPosition GmbH	www.inposition.ch
4085	European Union Agency for the Space Programme (EUSPA)	www.euspa.europa.eu
4084	Geodetics, Inc.	www.geodetics.com
4083	German Aerospace Center, Institute of Communications and Navigation (DLR)	www.dlr.de/kn/en/desktopdefault.aspx/tabid-2204/3257_read-19445/

© RTCM – Not for reproduction or redistribution

Message Type	Organization	Contact
4082	Cooperative Research Centre for Spatial Information	www.crcsi.com.au
4081	Seoul National University GNSS Lab	gnss.snu.ac.kr/nav
4080	NavCom Technology, Inc.	navcomtech.com
4079	SubCarrier Systems Corp. (SCSC)	www.use-SNIP.com
4078	ComNav Technology Ltd.	www.comnavtech.com
4077	Hemisphere GNSS Inc.	www.hemispheregnss.com
4076	International GNSS Service (IGS)	www.igs.org
4075	Alberding GmbH	www.alberding.eu
4074	Unicore Communications Inc.	www.unicorecomm.com
4073	Mitsubishi Electric Corp.	www.mitsubishielectric.com
4072	u-blox AG	www.u-blox.com
4071	Wuhan Navigation and LBS	www.wnlbs.com
4070	Wuhan MengXin Technology	www.wh-mx.com
4069	VERIPOS Ltd	www.veripos.com
4068	Qianxun Spatial Intelligence Inc.	http://www.qxwz.com/en
4067	China Transport telecommunications & Information Center	www.cttic.cn
4066	Lantmäteriet	www.lantmateriet.se
4065	ALLYSTAR	www.allystar.com
4064	NTLab	www.ntlab.com
4063	CHCNAV	www.chcnav.com
4062	Swift Navigation Inc.	www.swiftnav.com

© RTCM – Not for reproduction or redistribution

Message Type	Organization	Contact
4061	Zhejiang GeeSpace Technology Co., Ltd.	www.geespace.com
4060	ALES Corp.	ales-corp.co.jp
4059	Natural Resources Canada	www.nrcan.gc.ca
4058	Anello Photonics	www.anellophotonics.com
4057	Sixents Technology	www.sixents.com
4056	China Dayou Positioning Intelligence	www.dayoupi.cn
4055	Korea Research Institute of Ships & Ocean engineering (KRISO)	www.kriso.re.kr
4054	GEODNET	geodnet.com
4053	Qualcomm Technologies, Inc.	www.qualcomm.com
4052	Furuno Electric Co., LTD.	www.furuno.com
4051	Hi-Target	en.hi-target.com.cn
4050	STMicroelectronics SRL	www.st.com
4049	CM Intelligent Mobility Network Co., Ltd.	https://pnt.10086.cn/v1/mallportal/#/home
4048	Point One Navigation	www.pointonenav.com
4047-4001	RESERVED	

© RTCM – Not for reproduction or redistribution

4 TRANSPORT LAYER

4.1 Description

The transport layer defines the frame architecture for sending or receiving Version 3 messages. The purpose of defining this layer is to ensure that Version 3 data can be properly decoded by applications. The frame is mandatory from this respect, but it is not required throughout the transmission of the data. Providers may package the messages into a separate frame structure that best suits the transmission medium. The data set would need to have this frame structure re-established before transfer to the application. For high-integrity applications, it would be up to the provider to demonstrate that adequate integrity is maintained in the process of disassembling and reassembling the transport layer frame structure. The basic frame structure consists of a fixed preamble, a message length definition, a message, and a 24-bit Cyclic Redundancy Check (CRC) for high data transfer integrity.

The structure of the frame format is shown in Table 4.1-1.

Table 4.1-1 Frame Structure

Preamble	Reserved	Message Length	Variable Length Data Message	CRC
8 bits	6 bits	10 bits	Variable length, integer number of bytes	24 bits
11010011	Not defined – set to 000000	Message length in bytes	0-1023 bytes	Qualcomm definition CRC-24Q

The Preamble is a fixed 8-bit sequence.

The next six bits are reserved and should be set to zero by the Transport Layer Control for all messages. The mobile user receiver should ignore these bits and not assume they will always be set to zero. In future versions these bits may contain the version number of the standard.

The Variable Length Messages are those defined in Chapter 3. If the data link requires short messages to maintain a continuous stream of data, the message length may be set to "0", providing a "filler" message of 48 bits in length, because the data message length will be zero.

This standard uses the Qualcomm CRC algorithm. Twenty-four bits of CRC parity will provide protection against burst as well as random errors with a probability of undetected error $\leq 2^{-24} = 5.96\times10^{-8}$ for all channel bit-error probabilities ≤ 0.5. The CRC operates on the sequence of bits beginning with the preamble, through to the end of the Variable Length Message Field, using a seed of 0. The sequence of 24 bits $(p_1,p_2,...,p_{24})$ is generated from the sequence of information bits $(m_1,m_2,...,m_{8N})$, where N is the total number of bytes in the sequence consisting of the message plus preamble and Message Length Definition Parameter. This is accomplished by means of a code that is generated by the polynomial

© RTCM – Not for reproduction or redistribution

$$g(X) = \sum_{i=0}^{24} g_i X^i$$

where

g_i = 1 for i = 0, 1, 3, 4, 5, 6, 7, 10, 11, 14, 17, 18, 23, 24
= 0 otherwise

This code is called CRC-24Q (Q for Qualcomm Corporation). The generator polynomial of this code is in the following form (using binary polynomial algebra):

$$g(X) = (1+X)p(X)$$

where *p(X)* is the primitive and irreducible polynomial

$$p(X) = X^{23} + X^{17} + X^{13} + X^{12} + X^{11} + X^{9} + X^{8} + X^{7} + X^{5} + X^{3} + 1$$

When, by the application of binary polynomial algebra, the above g(X) is divided into $m(X)X^{24}$, where the information sequence m(X) is expressed as

$$m(X) = m_k + m_{k-1}X + m_{k-2}X^2 + \cdots + m_1X^{k-1}$$

the result is a quotient and a remainder R(X) of degree < 24. The bit sequence formed by this remainder represents the parity check sequence. Parity bit p_i, for any i from 1 to 24, is the coefficient of X^{24-i} in R(X).

This code has the following characteristics:

1) It detects all single-bit errors per code word.

2) It detects all double-bit error combinations in a codeword because the generator polynomial *g(X)* has a factor of at least three terms.

3) It detects any odd number of errors because *g(X)* contains a factor *1+X*.

4) It detects any burst error for which the length of the burst is ≤ 24 bits.

5) It detects most large error bursts with length greater than the parity length r = 24 bits. The fraction of error bursts of length b > 24 that are undetected is:

a) $2^{-24} = 5.96 \times 10^{-8}$, if b > 25 bits.
b) $2^{-23} = 1.19 \times 10^{-7}$, if b = 25 bits.

As noted earlier, the reference station should insert zeros in all reserved fields, and for messages whose lengths that don't line up with byte boundaries, zeros should be used for undefined bits to fill out the last unfilled byte.

4.2 Example

The following is a Hex-ASCII example of a Message Type 1005 (Stationary Antenna Reference Point, No Height Information).

© RTCM – Not for reproduction or redistribution

```
D3 00 13 3E D7 D3 02 02 98 0E DE EF 34 B4 BD 62
AC 09 41 98 6F 33 36 0B 98
```

The parameters for this message are:

- Reference Station Id = 2003
- GPS Service supported, but not GLONASS or Galileo
- ARP ECEF-X = 1114104.5999 meters
- ARP ECEF-Y = -4850729.7108 meters
- ARP ECEF-Z = 3975521.4643 meters

© RTCM – Not for reproduction or redistribution

5 DATA LINK LAYER

The Data Link Layer defines how the RTCM 10403 message data stream is encoded on the Physical Layer. This may also include flow control, packetization, encryption, or additional error checking.

It is up to the service provider to determine how to define this layer as appropriate to the application.

© RTCM – Not for reproduction or redistribution

6 PHYSICAL LAYER

The Physical Layer defines how the RTCM 10403 message data is conveyed at the electrical and mechanical level – for example: beacons, MSK; UHF, VHF Modems; DARC FM Subcarrier, Satellite links, fixed cable.

It is up to the service provider to determine how to define this layer as appropriate to the application.

© RTCM – Not for reproduction or redistribution

© RTCM – Not for reproduction or redistribution

This page intentionally left blank.

ANNEX A
(INFORMATIVE)
SUGGESTIONS AND EXAMPLES FOR NETWORK OPERATION

The following two sections provide additional guidance and information for Vendors, Service Providers and Mobile Users. Annex A.1 provides guidance to Service Providers on the selection of master and auxiliary stations, and to Users on how to best use the information in the messages. Annex A.2 provides a scheduling example that demonstrates one method of utilizing the Synchronous Message Flags and Multiple Message Indicators to support operation with multiple GNSSs.

A.1 How to Use Network ID's and Subnetwork ID's

Figure A.1-1 gives an example of a network containing 23 reference stations identified by Network ID. In general, the network will consist of only one subnetwork identified by Subnetwork ID. Every one of these 23 reference stations can be used as a Master Reference Station, and all reference stations within a certain range around the Master Reference Station might be used as Auxiliary Reference Stations. The figure shows 3 different Master Reference Stations (11 <red>, 14 <blue>, and 17 <purple>) and their associated radii. The radii are defining in this example the range holding all associated Auxiliary Reference Stations for the designated Master Reference Station. However, other ways of selecting Auxiliary Reference Stations for a designated Master Reference Station are possible, and the selection process might have to be adapted to the specific environment of the permanent networking application.

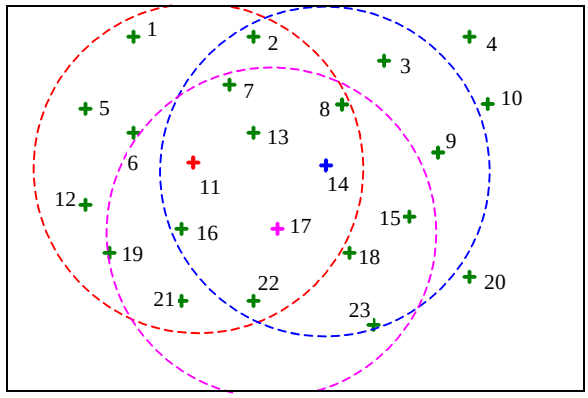

Figure A.1-1. Network Example

Master Reference Station M11 has the Auxiliary Reference Stations A1, A2, A5, A6, A7, A8, A12, A13, A14, A16, A17, A19, A21, and A22. Master Reference Station M14 has the Auxiliary Reference Stations A2, A3, A7, A8, A9, A11, A13, A15, A16, A17, A18, A22, and A23. Master Reference Station M17 has the Auxiliary Reference Stations A7, A8, A11, A13, A14, A15, A16, A18, A19, A21, A22, and A23. All three Master Reference Stations (M11, M14, and M17) are also Auxiliary Reference Stations (A11, A14, and A17) for each other. Under the assumption that all Master Reference Stations are on a common Integer Ambiguity Level, a handover to the next Master Reference Station data stream would be possible without reinitialization of the integer ambiguities as used in the rover application. However, Integer Ambiguity Leveling is not mandatory for Master Reference Stations. Normally, all stations would be available to a network processing facility, and a networking provider could combine all stations in one network (e.g., 122). When all Reference Stations are brought to a joint Ambiguity Level, a common Subnetwork ID 0 would be assigned.

An example of individual data streams may be for Master Reference Stations M11, M14, and M17 respectively as a sequence of information of Network ID (122), Subnetwork ID (0), Master Reference Station ID, Auxiliary Reference Station ID, …:

© RTCM – Not for reproduction or redistribution

122, 0, M11, A1, …
122, 0, M11, A2, …
122, 0, M11, A5, …
122, 0, M11, A6, …
122, 0, M11, A7, …
122, 0, M11, A8, …
122, 0, M11, A12, …
122, 0, M11, A13, …
122, 0, M11, A14, …
122, 0, M11, A16, …
122, 0, M11, A17, …
122, 0, M11, A19, …
122, 0, M11, A21, …
122, 0, M11, A22, …

122, 0, M14, A2, …
122, 0, M14, A3, …
122, 0, M14, A7, …
122, 0, M14, A8, …
122, 0, M14, A9, …
122, 0, M14, A11, …
122, 0, M14, A13, …
122, 0, M14, A15, …
122, 0, M14, A16, …
122, 0, M14, A17, …
122, 0, M14, A18, …
122, 0, M14, A22, …
122, 0, M14, A23, …

122, 0, M17, A7, …
122, 0, M17, A8, …
122, 0, M17, A11, …
122, 0, M17, A13, …
122, 0, M17, A14, …
122, 0, M17, A15, …
122, 0, M17, A16, …
122, 0, M17, A18, …
122, 0, M17, A19, …
122, 0, M17, A21, …
122, 0, M17, A22, …
122, 0, M17, A23, …

Message streams such as those in the example above might be transmitted on separate data links or jointly on one data link. Note that the current standard document recommends disseminating only one Master Reference Station per data link (See section 3.1.5).

As long as all reference stations are on the same integer ambiguity level, a hand-over to another Master Reference Station is possible as long as the new Master Reference Station was already an Auxiliary Reference Station for the previous Master Reference Station. In this example, user equipment could operate using the red Master Reference Station with fixed integer ambiguities on the rover. When roving it might be desirable to switch to the pink or blue Master Reference Station and its associated Auxiliary Reference Stations. Note that with the current recommendation of only one Master Reference Station per data link, switching Master Reference Stations requires that different data links be used.

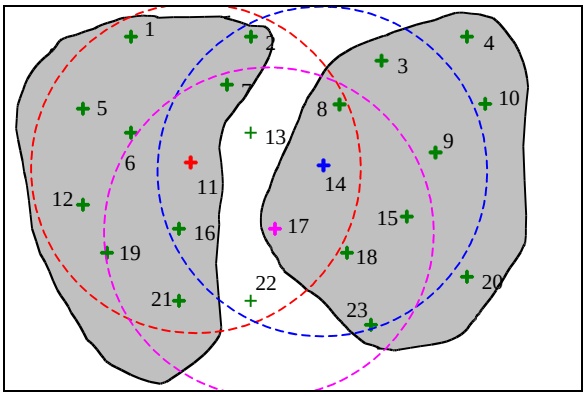

Figure A.1-2. Example of Multiple Solutions

There are situations where a homogeneous integer ambiguity solution might fall apart. For instance, some stations in the center could have communication problems with the central computation facility, or their observations might become unavailable for some other reason. The example given in **Figure A.1-2** shows 2 independent homogenous solutions with separated integer ambiguity levels (gray shaded areas). The circles marking the initial area with all the associated Auxiliary Reference Stations for particular Master Reference Stations span both shaded areas.

Since the correction differences (with Ambiguity Status Flag = 1, i.e., correct Integer Ambiguity Level for L1 and L2) may be generated only between Master Reference Station and its Auxiliary Reference Stations on the same integer ambiguity level, the example data streams will be different:

© RTCM – Not for reproduction or redistribution

122, 1, M11, A1, …	122, 2, M14, A3, …	122, 2, M17, A8, …
122, 1, M11, A2, …	122, 2, M14, A8, …	122, 2, M17, A14, …
122, 1, M11, A5, …	122, 2, M14, A9, …	122, 2, M17, A15, …
122, 1, M11, A6, …	122, 2, M14, A15, …	122, 2, M17, A18, …
122, 1, M11, A7, …	122, 2, M14, A17, …	122, 2, M17, A23, …
122, 1, M11, A12, …	122, 2, M14, A18, …	
122, 1, M11, A16, …	122, 2, M14, A23, …	
122, 1, M11, A19, …		
122, 1, M11, A21, …		

Note: It is possible to form other Correction Differences as well, but their Ambiguity Status Flag will be different from "1".

The reaction of the user system may be manifold and is strongly dependent on the processing strategy implemented in the user system itself. The simplest strategy is to use only homogenous information of Master Reference Station-Auxiliary Reference Stations combinations referring to the identical epoch. In this case, the implementation and handling within the user system is quite straightforward. The user system will first receive the homogenous ones of any of the Master Reference Station-Auxiliary Reference Station combinations and may correct its observations. When a second data stream, with a different Master Reference Station, becomes more suitable, the user system may switch to the new data stream and probably needs only to ensure that no jump occurs in its results due to the switch of the Master Reference Station. However, this again is dependent on the designated strategy of the user system's processing.

When a homogenous solution with a common integer ambiguity level (as given in **Figure A.1-1**) breaks apart (resulting in a setup as given in **Figure A.1-2**) the reaction of the user system might depend on the area of actual operation. Within the white area, the zone where the reference stations are no longer part of any solution, the user system's only chance is to use the remaining information and try an extrapolation. The resulting positioning will probably be degraded under this circumstance.

If the user system is operating in one of the gray areas and is using a Master Reference Station-Auxiliary Reference Station combination outside its gray area, the user system may continue without major interruption but eventually with less flexibility in modeling the information for proper mitigation of biases.

If the user system is operating in one of the gray areas but is utilizing a Master Reference Station-Auxiliary Reference Station combination of the opposite gray area, the user system probably will have to change to a Master Reference Station-Auxiliary Reference Station combination in its area of operation. Ultimately this will probably result in a reset of the user system's integer ambiguities and the user will have to reinitialize.

As stated above, a processing strategy using only information of the latest available epoch is probably the simplest strategy possible. More sophisticated processing strategies may involve the usage of the information of Master Reference Station-Auxiliary Reference Station combinations of different Master Reference Stations and/or different observation epochs. The information of the Network RTK messages have the potential to be used within another centralized networking solution. In the future a further application might be a complete network solution on a user receiver. The status information of the network calculation transmitted in the form of the data fields Network ID and Subnetwork ID provide sufficient proof to indicate to the user system that it is secure or non-secure to continue operation with the new set of information. At the time, the complete description of all these potential

© RTCM – Not for reproduction or redistribution

applications is not possible due to the number of variations. The experienced developer in the field of network RTK will understand the meaning of the Network ID and Subnetwork ID for the envisioned applications. Therefore, further details are left open and will need further description in case the given description is considered ambiguous.

A.2 A Scheduling Example for RTK Networks

It is anticipated that RTK services in the future will transmit data for more than one satellite system concurrently. Such facilities already exist for non-network RTK service, using GPS and GLONASS data all referenced to the same time epoch. Such operation supports the mixing of measurements from different GNSSs. As Galileo comes online, it will initially be used in conjunction with GPS and GLONASS, so RTK services utilizing multiple GNSSs will be common. The Synchronous Message Flags and Multiple Message Indicator in the RTK messages are designed to support multiple GNSS operation.

However, not all rover receivers will be designed to receive and process information from more than one GNSS, and the RTCM standard should enable rovers to know when relevant information has been completely transmitted for a particular epoch. Otherwise, such user receivers would have to wait until data for the next epoch has been received before they could begin processing. Such delays are neither desirable nor necessary.

The Synchronous Message Flag supports the use of two or more GNSSs. It is set to “1” if data from another GNSS will follow, and it is set to “0” if there are no other services or if this data set is for the last GNSS for that epoch.

With networks, it might be necessary to transmit data for different Auxiliary Reference Stations at different times but referenced to a single epoch. For example, if there are 5 Auxiliary Reference Stations and one particular message is sent for one auxiliary station every epoch, it will take 5 epochs to provide a complete set of information. The Multiple Message Indicator is set to “0” for the last Auxiliary Reference Station in the specific message set (e.g., 1015 or 1016), and the others are set to “1”.

With more than one GNSS, the data stream will contain the Auxiliary Reference Station data first for one satellite system, then the other.

How these rules are applied is demonstrated in Table A.2-1 below for a combined GPS and GLONASS service. The Epoch is the reference time in seconds. GPS 1004 refers to the GPS Master Reference Station reference epoch, and 1004 SMF refers to the value of the Synchronous Message Flag in the 1004 message. GLONASS 1012 refers to the GLONASS Master Reference Station reference epoch, and 1012 SMF refers to the corresponding GLONASS flag. For ease of explanation, at each epoch Network RTK data for one Auxiliary Reference Station and one GNSS is transmitted. Actual service implementations may send several messages per epoch, but the explained principle is readily apparent. GPS 1015 refers to the reference epoch for the GPS Auxiliary Reference Station Iono message, and the corresponding MMI refers to the value of the Multiple Message Indicator in the 1015 message; similarly for the 1016 Geometric message. The corresponding GLONASS values are then given, where 1037 refers to the GLONASS Iono message, and similarly for 1038.

© RTCM – Not for reproduction or redistribution

Table A.2-1 Example Showing the Use of SMF's and MMI's

	GPS		GLONASS		GPS Network RTK				GLONASS Network RTK			
Epoch	1004	SMF	1012	SMF	1015	MMI	1016	MMI	1037	MMI	1038	MMI
1	1	1	1	0	1	1	1	1				
2	2	1	2	0	1	1	1	1				
3	3	1	3	0	1	1	1	1				
4	4	1	4	0	1	1	1	1				
5	5	1	5	0	1	1	1	1				
6	6	1	6	0	1	1	1	1				
7	7	1	7	0	1	1	1	1				
8	8	1	8	0	1	1	1	1				
9	9	1	9	0	1	0	1	0				
10	10	1	10	0					1	1	1	1
11	11	1	11	0					1	1	1	1
12	12	1	12	0					1	1	1	1
13	13	1	13	0					1	1	1	1
15	15	1	15	0	15	1	15	1				
16	16	1	16	0	15	1	15	1				
17	17	1	17	0	15	1	15	1				
18	18	1	18	0	15	1	15	1				
19	19	1	19	0	15	1	15	1				
20	20	1	20	0	15	1	15	1				
21	21	1	21	0	15	1	15	1				
22	22	1	22	0	15	1	15	1				
23	23	1	23	0	15	0	15	0				
24	24	1	24	0					15	1	15	1
25	25	1	25	0					15	1	15	1
26	26	1	26	0					15	1	15	1
27	27	1	27	0					15	1	15	1
28	28	1	28	0					15	0	15	0
29	29	1	29	0								

It can be seen that all the GPS and GLONASS Auxiliary Reference Station messages are referenced to Epoch 1 until the entire set of data has been broadcast, after which the data is referenced to Epoch 15. Note that the GPS SMF's are all "1", indicating that the GLONASS message is to follow, while the GLONASS SMF's are all "0", indicating that there is not a third GNSS represented in the data stream. The final MMI for each GNSS and message type is set to "0", indicating that the complete set of network corrections have been transmitted, and the rover can proceed immediately to apply them.

The extension of this technique to three or more GNSSs is readily apparent.

© RTCM – Not for reproduction or redistribution

© RTCM – Not for reproduction or redistribution

This page intentionally left blank.

ANNEX B
(INFORMATIVE)
PATENTS

B.1 Policy

It is the intent of the Radio Technical Commission for Maritime Services (RTCM) to develop standards that do not require the use of patented technologies as the only means to comply with a standard (essential patents).

B.2 Essential Patented Technologies

RTCM has not been informed of any patented technologies that must be used in order to design or produce equipment covered by this standard.

B.3 Non-essential Patented Technologies

RTCM has been informed of the patented technologies listed in Table B.1 that may be used in the design and production of systems covered by this standard. However, RTCM believes it is possible to design and produce systems covered by this standard without using any of the patented technologies listed in the table.

Table B.1 is provided for the information of users of this standard. RTCM takes no responsibility for the accuracy or completeness of this list.

Table B.1 – Non-essential patents

Issuing Country	Patent Number	Patent Holder	Brief Description (See Note)
USA	6469663 B1	Hemisphere [8] 8444 N. 90th Street, Suite 120 Scottsdale, AZ 85258, USA	Method and system for GPS and WAAS carrier phase measurements for relative positioning
USA	8694250 B2	Trimble Navigation Limited [8] P.O. Box 3642 Sunnyvale, CA 94088 USA	Processing multi-GNSS data from mixed-type receivers

Note: Do not rely on this summary as a complete description of the technology. See patent documents for a full description.

[8] Patent holder or related organization is a member of RTCM and Special Committee 104.

© RTCM – Not for reproduction or redistribution